NOTES OF
STELLAR PHYSICS

version 1.1

Giacomo Pannocchia

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For those who come after

PREFACE

> *In a sense, you might say I'm getting quite fond of writing these kind of notes.*

Not long ago, it occurred to me how cool it is when someone unexpectedly releases a very detailed and all-comprehensive version of their notes, especially when dealing with a course that has a handful of topics often interacting together in unpredictable, yet fascinating, ways.

These notes will be mainly based on *my* own notes of the lectures by Professor Scilla Degl'Innocenti during the academic year 2025-2026. However, since I take little to no pride in my messy notes, I'll be often using some of the references you can find on the course catalogue page or that have been cited during class. Either way, I'll do my best keeping track of them all in the bibliography of this humble collection.

You can report errors (whatever their nature might be) and suggestions for additions at g.pannocchia3@studenti.unipi.it or through whatever convoluted way (conventional or not) you prefer[1].

You can find all the notes I've redacted here.

Without further ado, we'd better not lose much more time on a preface and get started with it.

> *There was Eru, the One, who in Arda is called Ilúvatar; and he made first the Ainur* [...] *But for a long while they sang only each alone, or but few together, while the rest hearkened; for each comprehended only that part of the mind of Ilúvatar from which he came, and in the understanding of their brethren they grew but slowly.*
> *Yet ever as they listened they came to deeper understanding, and increased in unison and harmony.*

Ainulindalë, "The music of the Ainur",
Silmarillion, J. R. R. Tolkien

1 I'd like, however, not to see my house stormed by homing pidgeons.

CONTENTS

Part I

STELLAR PROPERTIES

1 ASTROPHYSICS' BUILDING BLOCKS

1.1 INTRODUCTION

As per the name of this chapter, we're going to introduce a series of important quantities and concepts that are going to be particularly useful later on. For this first introductory part I'm going to follow [8], §1.

For expressing astrophysical measurements, it should be clear that the units that we customarily use for our everyday life wouldn't be doing a really good job. For example, most of the times, it isn't particularly convenient to have masses expressed in grams or kilograms, for the masses we'll be usually concerned with are so large that saying that a star weighs 10^{33} g isn't particularly insightful, since we don't have an everyday-life object to compare it with.

What often happens in astrophysics is that masses are usually expressed in terms of *solar masses* $M_\odot$, which has value

$$M_\odot = 1.99 \cdot 10^{33}\,\mathrm{g} \tag{1}$$

For example Vega, a bright white star in the Lyra constellation, has a mass roughly estimated to $M_{\mathrm{Vega}} = 2.15^{+0.10}_{-0.15}\,M_\odot$. The masses of most stars lie within a relatively narrow range from $0.1 - 20\,M_\odot$.

Stars are formed in clusters, which are predominantely composed of neutral Hydrogen. Thanks to a whole series of mechanisms interplaying in said clusters, it's often possible to assume that stars born within a given cluster are approximately coeval and have a similar chemical composition. What we can't assume, however, is them having the same masses.

To describe how the masses of stars are distributed within a cluster, it is customary to invoke a IMF, an *initial mass function*, which is essentially an empirical function that describes the mass distribution of stars for a given cloud.

Similarly, it isn't particularly convenient to express distances with meters, centimeters, kilometers or, God forbids, miles and feet. For relatively "close" objects, it is often used the *astronomical unit* (AU)

$$1\,\mathrm{AU} = 1.50 \cdot 10^{13}\,\mathrm{cm} \tag{2}$$

which is the average distance of the Earth from the Sun. Although this is a rather useful quantity, we're not still up there. We have to kick it up a notch once more. As the Earth moves around the Sun, the "nearby" stars seem to change their position just slightly with respect to the faraway stars, which appear to be fixed in place.

This phenomenon is called *parallax*; it is not particularly different from what you experience when observing your own finger first with one eye and then with the other.

Consider the construction shown in Fig.1.

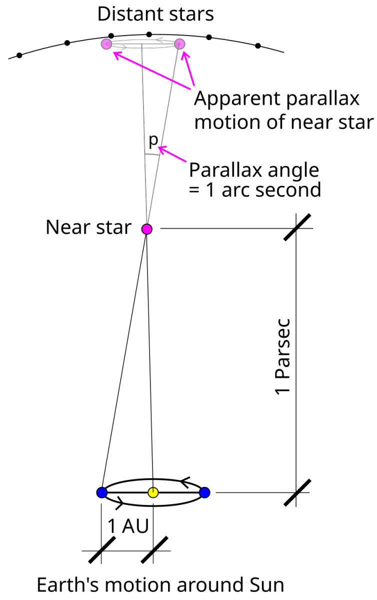

Figure 1: Defintion of parsec. Credits: Wikipedia.

Let us consider a star on the polar axis of the Earth's orbit at a distance d from the center of the orbit. The angle p in the picture is half of the angle by which this star appears to shift with the annual motion of the Earth on the distant stars' plane.

With simple geometrical considerations, if we assume Earth's orbit to be circular[1], then we can write the following relation

$$\tan(p) = \frac{1\,\mathrm{AU}}{d} \approx p$$

if we assume d to be much larger than an astronomic unit (which for all relevant scenarios is a condition that is well satisfied).

We can then define the *parsec* (pc) as the distance the star has to be so that its geometrical parallax turns out to be 1 arcsec. Hence

$$d = \frac{3.09 \cdot 10^{18}\,\mathrm{cm}}{p\,[\mathrm{arcsec}]} \implies 1\,\mathrm{pc} = 3.09 \cdot 10^{18}\,\mathrm{cm} \tag{3}$$

Obviously, for even larger distances the standard units are the *kiloparsec* (roughly the measure of galactic sizes), the *megaparsec* (the typical measure of galactic distances) and the *gigaparsec* (the typical measure of the observable Universe).

1.2 RELEVANT QUANTITIES FOR RADIATIVE TRANSFER

Although some books often start their description of radiatively relevant quantities from the definition of *monochromatic energy* and *monochromatic intensity*, I found that it is most misleading, since, in all but a few cases, what we experimentally measure are fundamentally *fluxes*.

1 Actually, it is not. But Earth's orbital eccentricity is roughly $e = 0.017$, which is sufficiently small to let us approximate it as circular.

We shall then consider the *monochromatic flux* F_ν ($\mathrm{erg\,s^{-1}\,Hz^{-1}\,cm^{-2}}$) produced by some source passing through a small area $\mathrm{d}A$ located somewhere in space.

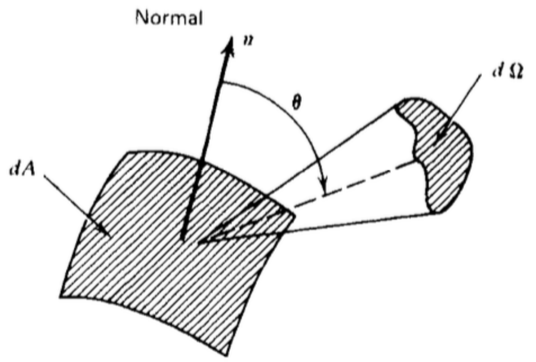

Figure 2: Schematic geometrical representation of the system.
Credits: G. Rybicki, A. Lightman [20].

If we call $\hat{k}$ the propagation direction of the flux and $\hat{n}$ the unit vector emerging from the surface $\mathrm{d}A$, it's easy to get convinced that what is actually passing through the surface is something proportional to $F_\nu(\hat{k}\cdot\hat{n})$.

From the monochromatic flux we can define the *bolometric flux*, which is just the monochromatic flux integrated over all frequencies (or wavelengths)

$$F = \int_0^{+\infty} F_\nu \,\mathrm{d}\nu = \int_0^{+\infty} F_\lambda \,\mathrm{d}\lambda \tag{4}$$

This also tells us how to convert a flux per unit frequency to a flux per unit wavelength

$$F_\nu \,\mathrm{d}\nu = F_\lambda \,\mathrm{d}\lambda$$

It should be clear that, despite being experimentally sensible for us to use the flux, we're losing much information sticking with it, namely directional information.

We consider then the amount of radiation $E_\nu \,\mathrm{d}\nu$ passing through the same area in time $\mathrm{d}t$ and solid angle $\mathrm{d}\Omega$. Hence we can write

$$\mathrm{d}E_\nu \,\mathrm{d}\nu = I_\nu(\mathbf{r}, t, \hat{k})(\hat{k}\cdot\hat{n})\,\mathrm{d}t\,\mathrm{d}\Omega\,\mathrm{d}A\,\mathrm{d}\nu \tag{5}$$

where the quantity $I_\nu(\mathbf{r}, t, \hat{k})$ is called the *specific monochromatic intensity*. If $I_\nu(\mathbf{r}, t, \hat{k})$ is specified for all directions at every point in a certain region of spacetime, then we'd have a complete prescription of the radiation field we intend on studying.

Capitalizing on the blatant similarities with distribution functions, we can evaluate the moments of the monochromatic intensity.

Definition 1.2.1. *Monochromatic mean intensity J_ν*

$$J_\nu = \frac{1}{4\pi}\int_\Omega I_\nu \,d\Omega = \frac{c}{4\pi} U_\nu$$

with U_ν the total energy density of radiation. Note that J_ν is essentially just an average of the monochromatic intensity over all solid angles.

Definition 1.2.2. *Monochromatic flux* $\vec{F}_\nu$

$$\vec{H}_\nu = \frac{1}{4\pi}\int_\Omega I_\nu(\hat{k})\hat{k}\, d\Omega \implies \frac{1}{4\pi}F_\nu = \vec{H}_\nu \cdot \hat{n}$$

I haven't explicitly proved the last equality, but it shouldn't be hard for you to convince yourself (or prove it yourself) that it is indeed true.

Definition 1.2.3. *Monochromatic radiation pressure* p_ν

The monochromatic pressure is defined starting from the different directions correlations of the monochromatic intensity

$$K_\nu^{ij} = \frac{1}{4\pi}\int_\Omega I_\nu(\hat{k})k^i k^j\, d\Omega$$

The pressure in particular is usually expressed as

$$P_\nu = \frac{1}{c}\int_\Omega I_\nu(\hat{k})\cos^2\theta\, d\Omega$$

where $\cos^2\theta = (\hat{k}\cdot\hat{n})^2$.

1.2.1 Blackbody radiation

Even at an undergraduate level, we're all fairly familiar with *blackbody radiation*. The easiest way to deduce the expression for the energy density of photons in *thermal equilibrium* (STE) inside a cavity is by the means of statistical mechanics.

Remember the Bose-Einstein distribution ($\mu = 0$)

$$n = \frac{1}{\exp(h\nu/kT) - 1}$$

and the phase space density of states (per unit volume)

$$\rho(\nu)\,\mathrm{d}\nu = \frac{4\pi h g \nu^3}{c^3}\,\mathrm{d}\nu$$

from which deducing the expression from internal energy is straightforward. Recalling $g = 2$ is the quantum degeneracy of photons, a simple moltiplication of the previous expressions yields

$$U_\nu\,\mathrm{d}\nu = \frac{8\pi h}{c^3}\frac{\nu^3}{\exp(h\nu/kT) - 1}\,\mathrm{d}\nu$$

Since blackbody radiation is isotropic (it depends only on the absolute temperature T), the definition of mean monochromatic intensity yields

$$\boxed{B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{\exp(h\nu/kT) - 1}} \tag{6}$$

It's important to notice that, in principle, such a fundamental result holds only in *strict thermodynamic equilibrium* (STE), but we'll soon see how to generalize this formulation for less "restrictive" environments.

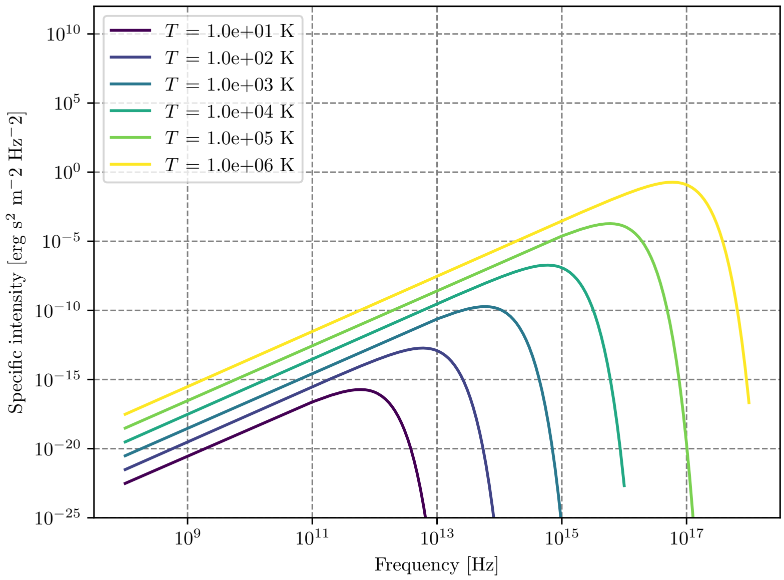

Figure 3: Blackbody frequency spectrum.

An incredible number of important results descends from (6), and it may be worthwhile to cite at least some of them, starting from Stefan-Boltzmann law. We'll use the following result without proving it

$$\int_0^{+\infty} B_\nu(T)\,\mathrm{d}\nu = \frac{2h}{c^2}\frac{\pi^4}{15}\left(\frac{kT}{h}\right)^4$$

Computing the bolometric flux and the bolometric energy density by integrating over all frequencies using what we've just written down, you find the following

$$U(T) = aT^4 \quad F(T) = \sigma_{SB}T^4$$

Clearly the two constants a and σ_{SB} cannot be independent, and are actually related by the integral we've previously calculated. Using for example[2]

$$F(T) = \pi \int_0^{+\infty} B_\nu(T)\,\mathrm{d}\nu$$

you can easily find out that the *Stefan-Boltzmann constant* is equal to

$$\sigma_{SB} = \frac{2\pi^5 k^4}{15c^2h^3}$$

and the relation with a is simply $\sigma_{SB} = ac/4$.

The equation

$$F(T) = \frac{2\pi^5 k^4}{15c^2h^3}T^4 \tag{7}$$

is what is usually known as the *Stefan-Boltzmann law*.

2 The emergent flux from an isotropically emitting surface (such as a blackbody) is $\pi \cdot$ brightness which is none other than the specific intensity.

Let us now consider two different regimes for eq.6: $h\nu/kT \ll 1$ and $h\nu/kT \gg 1$. The first yields what is commonly known as the Rayleigh-Jeans Law which is, sadly, pretty much relevant only for radioastronomy.

Since

$$\exp\left(\frac{h\nu}{kT}\right) = 1 + \frac{h\nu}{kT} + o\left(\frac{h\nu}{kT}\right)^2$$

the blackbody radiation assumes the much simpler form of

$$B_\nu^{RJ} = \frac{2\nu^2}{c^2}kT$$

Another important results is achieved in the opposite regime, when the exponential term is rather larger than unity

$$B_\nu^{W} = \frac{2h\nu^3}{c^2}\exp\left(-\frac{h\nu}{kT}\right)$$

This expression is known as Wien's Law.

1.2.2 Characteristic Temperatures

Starting from the blackbody spectrum we can give various definitions of temperature, that are going to be more and less useful when dealing with stars.

Brightness Temperature

One way of characterizing brightness (specific intensity) at a certain frequency is to give the temperature of the blackbody having the same brightness at that frequency. That is, for any value I_ν we define $T_b(\nu)$ by the relation

$$I_\nu = B_\nu(T_b)$$

this way of specifying brightness has the advantage of being closely connected with the physical properties of the emitter, and has the simple unit (K).

This procedure is used especially in radio astronomy, where the Rayleigh-Jeans law is usually applicable.

$$T_b = \frac{c^2}{2\nu k}I_\nu \quad h\nu/kT \ll 1 \tag{8}$$

Note that the uniqueness of the definition of brightness temperature relies on the monotonicity property of Planck's law. Also note that, in general, the brightness temperature is a function of ν.

Color Temperature

Often a spectrum is measured to have a shape more or less of blackbody form, but not necessarily of the proper absolute value. For example, by measuring F_ν, from an unresolved source we cannot find I_ν, unless we know the distance to the source and its physical size.

By fitting the data to a blackbody curve without regard to vertical scale, a color temperature T_c, is obtained. Often the "fitting" procedure is nothing more than estimating the peak of the spectrum and applying Wien's displacement law to find a temperature

$$\lambda_{\text{peak}} = \frac{b}{T} \tag{9}$$

with b a known constant of value[3]

$$b = 2.897771955 \cdot 10^{-3}\,\text{m} \cdot \text{K} \tag{10}$$

Effective Temperature

The effective temperature of a source T_{eff} is derived from the total amount of flux, integrated over all frequencies, radiated at the source. We obtain T_{eff} by equating the actual flux F to the flux of a blackbody at temperature T_{eff}

$$F = \int I_\nu \cos\theta \, \mathrm{d}\nu \, \mathrm{d}\Omega := \sigma T_{eff}^4 \tag{11}$$

1.3 MAGNITUDE SCALES

When coming down to determining the properties of a star, there are two quantities that shine a little brighter than the others: The luminosity and the flux.

The luminosity of a star is generally defined as the power emitted by a given star over time (potentially over a given wavelength) and its expression is given by the relation

$$L = -\frac{\mathrm{d}E}{\mathrm{d}t} \quad \text{erg} \cdot \text{s}^{-1} \tag{12}$$

As we'll see, stars often emit isotropically in space. It goes without saying, however, that an observer at distance d from that star won't be able to collect all the light that is emitted. What we can measure is a flux, the fraction of energy that impinges on our detector. The flux is related to the luminosity through the following relation

$$\phi = \frac{L}{4\pi r^2} \quad \text{erg} \cdot \text{s}^{-1} \cdot \text{cm}^{-2} \tag{13}$$

The flux, however, doesn't tell us much about the characteristics of the star, differently from the luminosity. But since what we can measure are fluxes, it is then crucial that we find a way to accurately measure distances so to infer the luminosity of an object.

Since ancient times, mankind has always tried to measure the brightness of stars. The efforts of those that came before us are nowadays crystallized in a quantity that is still in use: The *magnitude*.

The concept of magnitude is strictly related to the way the human eye was believed to perceive the differences in intensities.

3 The reported value is the one presented in the most recent release of the Particle Data Group [18]. The value is calculated from exact values, and is thus exact to the number of decimal places given.

Hypparchus, back in ancient Greece, classified all stars into six classes according to their apparent brightness. These classes went from class I, the brightest stars, to class VI, the less bright. According to Hypparchus, the brightest and faintest stars had an apparent brightness that is in the ratio of 100.

Let us consider a star with flux F_* and some reference flux F_0. We define the *apparent magnitude* as

$$m_* = x \log_{10}\left(\frac{F_*}{F_0}\right)$$

Calculating the difference in apparent magnitude between the brigthest and faintest stars and making use of the aforementioned characteristics, we can find the correct values for the constant x and define the apparent magnitude as

$$m_* = -2.5 \log_{10}\left(\frac{F_*}{F_0}\right) \tag{14}$$

Obviously, unless a bolometer is used, you can't perform a measurement covering all wavelengths. What you're going to measure is actually a convolution of the flux emitted by the star (as a function of the wavelength) and the transmission curve of the detector you're using

$$S_{det} = \int_0^\infty \mathrm{d}\lambda\, F(\lambda) T(\lambda)$$

More often than not, it is customary to measure magnitudes in given wavelengths' ranges, from which a "bolometric" magnitude can be inferred by adding a *bolometric correction* that is given from stellar atmosphere models.

Similarly, we can define the *absolute magnitude* as the magnitude a star would have if it was located 10 pc away.

Using the fact that $F \propto r^{-2}$, we find a relation for the absolute magnitude M

$$m - M = 5 \log_{10}\left(\frac{d}{10\,[\mathrm{pc}]}\right) \tag{15}$$

The difference $m - M$ is often called *distance module* (DM) and is used as an indirect way to express a distance.

For classifying stars according to their color it's often used the so-called UBV photometric system (from *Ultraviolet*, *Blue*, *Visual*), also called the Johnson system (or Johnson-Morgan system), which makes use of filters so that the mean wavelengths of response functions (at which magnitudes are measured to great precision) are 364 nm for U, 442 nm for B, 540 nm for V.

Given this setup, we can easily calculate the magnitudes, say, in the Blue and Visible light-bands, so that the relative magnitudes are

$$m_B - m_V = -2.5 \log_{10}\left(\frac{F_B}{F_V}\right) := B - V$$

where $B - V$ is said *color index*.

When passing through interstellar dust[4] the intensity of radiation gets modified. The dust that makes the ISM absorbs light in the V-UV bands to

4 The interstellar dust, which is usually referred to as ISM, consists in small particles of silicates, graphite, amorphous carbon, ices and organical mixtures of size $10^{-2} - 1\,\mu$m

later re-emit it as InfraRed (IR) radiation or scatters the "short" wavelength, producing a net effect that essentially "increases" the wavelength of the incoming radiation and decreases its intensity.

These two phenomena are known as *reddening* (do not confuse it with *redshift*) and *extinction*.

The absolute magnitude in the, say, blue and visible bands, will thus be modified

$$M_V = M_V^0 + A_V$$
$$M_B = M_B^0 + A_B$$

where the quantities with the "0" superscript are the un-reddened magnitudes. We can define the new color index

$$B - V = (B - V)^0 + E[B - V]$$

where $E[B - V] = A_B - A_V$ is the extinction. Within our galaxy, the ratio of reddening to the extinction is a well-known quantity

$$R = \frac{A}{E[B - V]} \approx 3.0$$

1.4 ELEMENTS ABUNDANCES

Observing the spectrum emitted by stars we might expect, a bit naively, to observe a blackbody spectrum, but that isn't quite the case. The radiation field produced by stars is only approximately equal to a blackbody function in the inner regions of stars.

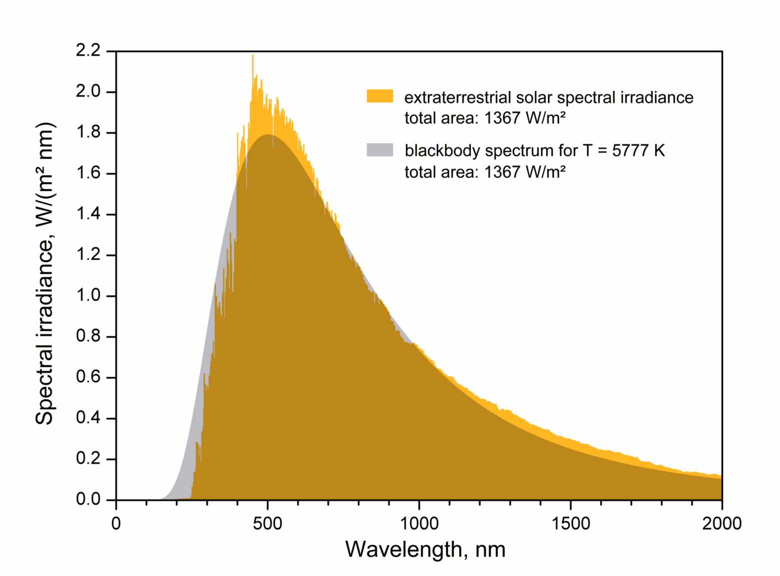

Figure 4: Deviation from blackbody behavior for the solar spectrum. Credits: Unknown source.

Although a tad annoying for the characterization of the radiation field, this deviation from ideal behavior is what allows us to infer the chemical composition of stars!

Deviation from blackbody behavior is in fact usual to take the form of *emission* and *absorption* lines in the spectrum, which tend to stick out like sore thumbs. Clearly, different elements (and different species of the same element) will produce very different lines, allowing us to reconstruct the chemical composition of the photosphere, which is the region where the photons we observe are coming from.

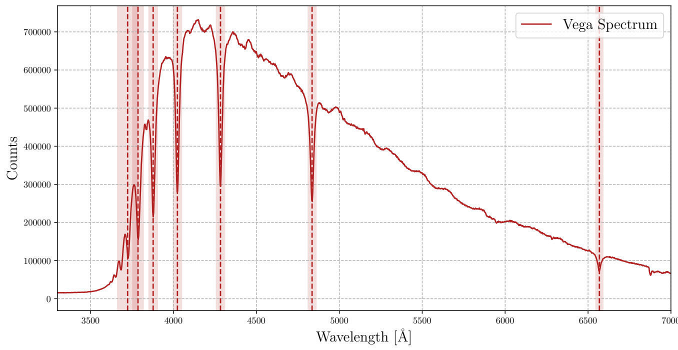

Figure 5: Absorption lines in the spectrum of Vega as measured with an Alpy600 spectrograph. Credits: Group 8.

For our purposes, we'll be interested in the *original* chemical composition of a star rather than its actual, or future, value, so the one that is shed of the modifications induced by ongoing nuclear reactions.

At this point, you may be objecting: Isn't the effect of nuclear reactions relevant only in the inner regions of stars?

As often happens in Physics, the correct answer is "might be." Can we take for granted that the chemical composition we see in the photosphere is the same as the original one?

Generally, no. As we'll see, stars in a certain range of masses and in some evolutionary stage of their life may develope convective or radiative envelopes, and the answer to our question we can find once we determine which of the two is going to win.

1.4.1 Universal Curve of abundances

A matter we're going to be most interested in discussing is the provenience of the various (chemical) elements we observe on Earth and all around the Universe and *how* they are produced. For example, we now know that the almost entirety of Hydrogen was produced during the *Primordial Nucleosynthesis* along with a bit of Helium.

Metals, on the other hand, are entirely produced by *Stellar Nucleosynthesis*. That's because in the early stages of the formation of the Universe, temperature dropped too fast below the critical threshold that allows the relevant nuclear reactions to produce efficiently Helium and metals. On top of that, light metals (Lithium, Beriullium and Boron) are generally destroyed for the sake of producing lighter elements like Helium (see Chapter 4). Fig.6 presents a comparison of chemical composition of the solar photosphere with that of CI chondrites.

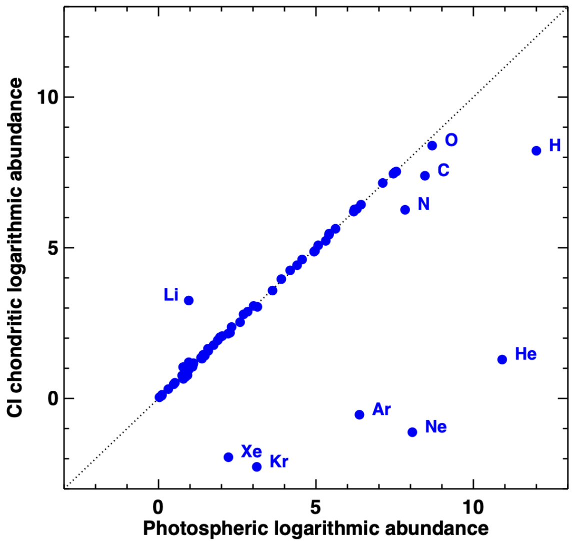

Figure 6: Comparison bewteen the recommended abundances by [2] in present-day solar photosphere and CI chondrites. Most elements agree well, with the exception of the highly volatile elements, which are depleted in meteorites, and Li, which has been largely destroyed in the Sun due to nuclear burning. Credits: [2].

Since elements are (on average) produced essentially via the same procedures, we may expect to measure a constant ratio between the various abundances of elements. Consider for example Fig.7.

First of all: As anticipated, light metals have a relatively low abundance, which is consistent with them being burned for the sake of producing lighter elements (Hydrogen and Helium) which have above-average abundance.

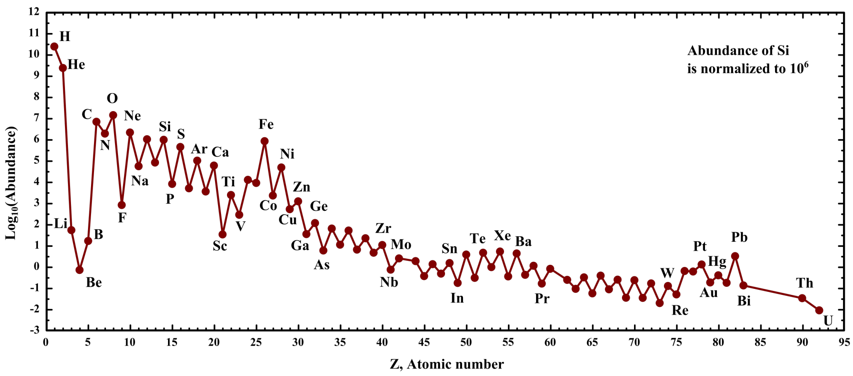

Figure 7: Universal curve of abundances, normalized on the relative abundance of Silicon. It represents the average value in the galactic disk of chemical abundances in stars. Credits: Wikipedia, but see also [2].

Similarly, the heavier elements which are produced by α-particles capture have roughly the same relative abundance, dropping off only when crossing the peak of Iron. The formation of elements with $Z > 26$ can happen only with proton/neutron capture on the nucleus, which is a fairly rare process that can occur only in the late evolutionary stages of massive stars.

As we have anticipated somewhere above, to model the evolution of a star, it's important to know the original chemical composition of the cluster the star was born from. You can think of it as "setting" the starting conditions for your system. In particular, it is of crucial importance to determine the primordial abundance of Helium.

For this purpose, blue galaxies hosting very bright, massive stars have been observed, where it's possible to study the recombination lines of Helium in metal-poor H_{II} regions. A very recent study [3] determined up to a precision of 0.5% the primordial Helium abundance

$$Y_p = 0.2458 \pm 0.0013$$

If, for whatever reason, we want to determine the abundance of non-primordial Helium, we have to consider stellar models. That is because, usually, stars presenting He-lines aren't particularly trustworthy due to sedimentation of Helium, which essentially causes the superficial abundance not to reflect the original one.

To create a stellar model for this purpose, we consider solar-mass stars, whose original metal abundance has been corrected for diffusion[5], and an unknown abundance of Helium. Then you let the system evolve until the present-day age of the Sun and you cross-check if you've obtained the same luminosity $L_\odot$ and radius $R_\odot$. In particular the luminosity is going to be closely related to the Helium abundance.

Metallicity and Enrichment ratio

Usually, the total abundance of Helium is expressed in terms of the metallicity of a given star

$$Y_X = Y_p + \frac{\Delta Y}{\Delta Z} Z_* \tag{16}$$

We can also define the $[\mathrm{Fe} \mid \mathrm{H}]$ as

$$[\mathrm{Fe} \mid \mathrm{H}] = \log_{10}\left[\frac{(N_{\mathrm{Fe}}/N_{\mathrm{H}})_*}{(N_{\mathrm{Fe}}/N_{\mathrm{H}})_\odot}\right] = \log_{10}\left[\frac{(Z/X)_*}{(Z/X)_\odot}\right] \tag{17}$$

where we've defined $N_i = X_i/A_i m_{\mathrm{H}}$. From this definition is obvious then that if $[\mathrm{Fe} \mid \mathrm{H}] = 0$, the abundance of Iron is equal to that of the Sun. Doing a slight manipulation[6] we can also write

$$[\mathrm{Fe} \mid \mathrm{H}] \approx \log_{10}\left[\frac{(\frac{Z}{1-Y})_*}{(\frac{Z}{X})_\odot}\right] = \log_{10}(Z_*) - \log_{10}(1 - Y_*) - \log_{10}\left(\frac{Z}{X}\right)_\odot \tag{18}$$

which can be inverted to obtain an expression of the form $\log_{10}(Z_*) \sim [\mathrm{Fe} \mid \mathrm{H}]^{-1.85}_{-1.77}$, where we're assuming $Y \sim Y_\odot \approx 0.24$. The subscript comes from the estimate of [17], while the superscript comes from the estimate of [2] of the solar metallicity.

Similarly, we can define the *alpha-enhancement* as

$$[\alpha \mid \mathrm{Fe}] = \log_{10}\left[\frac{(N_\alpha/N_{\mathrm{Fe}})_*}{(N_\alpha/N_{\mathrm{Fe}})_\odot}\right] \tag{19}$$

which, on average, evaluates to $[\alpha \mid \mathrm{Fe}] \sim 0.3$ for stars in galactic halos.

5 Don't worry, we're going to see this later on much more in detail.
6 We're making use of $X + Y + Z = 1$.

1.4.2 Initial Mass Function (IMF)

As we mentioned somewhere above, stars are born in clusters. During the arduous process that is stacking bit after bit of Hydrogen (along with Helium and whatever the star may found in the cluster) to build its own body, it's clear that we cannot reasonably expect all stars to be born with the same masses, just like you wouldn't expect your brother/sister to weigh just as much as you did when you were born.

What we experimentally observe is rather a *distribution of mass* in the cluster, influenced by every little fluctuation occuring in a cluster.

It is worth to notice, however, that the distribution we usually refer to as IMF (*Initial Mass Function*) is not the *Present Day Mass Function*, which gives the statistic of the masses of stars as seen at present day. That the two distributions must be different is evident if you consider more massive stars: These, that would appear in the IMF, *would not* appear in the PDMF, because odds are they are most likely dead. Using stellar evolution models, however, it is possible to pass from a given IMF to the PDMF.

Evaluating the mass distribution of stars right away is a burdensome task, that, for all but a few rare cases[7], proves to be quite impossible. For that reason, it is customary to study the distribution in luminosity, or, equivalently, the distribution in given magnitude bands.

In fact, with a basic application of the chain rule, you can write

$$\frac{\mathrm{d}N}{\mathrm{d}M_\nu} = \frac{\mathrm{d}N}{\mathrm{d}m} \cdot \frac{\mathrm{d}m}{\mathrm{d}M_\nu} \tag{20}$$

Unable to directly measure our quantity of interest $\mathrm{d}N/\mathrm{d}m$, we can circumvent the problem by calculating the number of stars in a given luminosity range and calculating the relation between mass and luminosity, which is going to be one of the many jobs of stellar models.

Once you're finally done, you'll probably find a relation of the form

$$\frac{\mathrm{d}N}{\mathrm{d}m} \propto m^{-\alpha} \quad (\alpha > 0) \tag{21}$$

so less massive stars will be favored with respect to heavier stars.

For example, at present day, the IMF that is customarily into use is that found by [16]

$$M \in (0.08, 0.5)\, M_\odot \to \alpha = 1.3 \pm 0.3 \tag{22}$$

$$M > 0.5\, M_\odot \to \alpha \approx 2.3 \tag{23}$$

Note that Kroupa's IMF is in good accord with Salpeter's IMF [22], which was one of the first proposals for a IMF.

For the moment, although we know that it shouldn't actually be the case, we're observing a *constant* IMF in galaxies, clusters and so on. Just recently, we're discovering that, luckily for us, our prediction of non-constancy is verified when you take into account regions with a high star formation rate and not too low metallicity.

7 For example, one of the few systems we're able to "accurately" determine the mass of is that of spectroscopic binaries.

1.4.3 Stellar Trivia

In this subsection I'm going to anticipate some of the subjects that are going to be the main interest of many among the following chapters.

Starting from dimensional considerations only (which, however, requires that you know the equations describing stellar structures), you can easily show that more massive stars are hotter and have a higher luminosity, hence them having a shorter lifespan. You may argue that more massive stars have more "fuel" to burn, thus "cancelling out" the effect of a higher luminosity.

Although that is technically true, it doesn't take into account the fact that (effective) temperature is positively correlated with the mass of a star. That means that more massive stars are generally hotter, which in turn pumps up the efficiency of nuclear reactions[8]. This means that more fuel gets burned to balance the collapse caused by gravitational pulls.

This phenomenon can be generally summarized under the name of *stellar thermostate*: When a star expands, its density decreases, and so its temperature; the efficiency of nuclear reactions thus decreases until its evenly balanced by gravitational collapse. Conversely, if the gravitational pull is stronger, the star contracts and gets hotter, the efficiency of the nuclear reactions ramps up along with the luminosity, halting the collapse.

This inspires our first, important deduction: If in a cluster stars are predominantely blue, the cluster must be young! Since more massive stars have a shorter lifespan, if the cluster was older, then we wouldn't be able to observe such stars.

It's important to distinguish two kinds of clusters: *Globular clusters* are generally old clusters found in galactic halos; they have little gas, no (recent) star formation and are generally arranged spherically symmetrically around the GC. In globular clusters, it would be logical to expect a lower metallicity, for a lower number of generations of stars might have followed one another in that region.

On the other hand *open clusters* are now found in the galactic plane[9]; they have much more gas available and there's continuous star formation. Generally, open clusters house a smaller number of stars, therefore allowing us to see only stars still in the *main sequence* stage[10]. In open clusters, it's sensible to expect a higher metallicity (with respect to globular clusters).

As a further classification, astrophysicists are used to distinguish two (three) populations of stars, depending on their characteristics and the cluster they're found in:

- *PopI Stars*: These are typically young, blue stars, which are very bright and with a high metallicity content. They're found in open clusters.

8 Nuclear reactions in stars can be shown to present a power-law dependency on the temperature, scaling at least as T^4.

9 Originally, they are likely to have formed in the halo, but their elliptic motion intersecting the GP have experienced numerous interactions with other stars and objects that have brought their trajectory to ultimately lie on the GP.

10 The main sequence stage is an evolutionary stage of stars during which Hydrogen is burned in the core.

- *PopII Stars*: These are typically old, reddish stars, which aren't particularly bright and have a low metallicity content. They're found in globular clusters.

- *PopIII Stars*: Allegedly, this population of stars should be the first one to form after the Big Bang. As so, its metallicity content should be virtually zero. This would imply that they should have large masses[11] and be very bright. As you might imagine, PopIII star wouldn't live for long.

Thick Disk

Along with the halo, the buldge and the *thin* disk, there's another structure in galaxies that is worth mentioning: The *thick disk*.

As the name suggests, the thick disk has a scale height that is greater than that of its thinner counterpart ($H_{\text{thick}} \sim 1\,\text{kpc}$ opposed to $H_{\text{thin}} \sim 0.1\,\text{kpc}$) but a shorter scale length ($l_{\text{thick}} \sim 3 - 4\,\text{kpc}$ opposed to $l_{\text{thin}} \sim 8 - 15\,\text{kpc}$). Like the halo, it is one of the more ancient structures, and presents characteristics that are hybrid between the halo and the thin disk, especially if you consider its rotational velocity and its metallicity content.

It is difficult to trace a sharp boundary between the thick disk and the halo/thin disk; usually considerations are made taking into account the kinematic of the disk.

Dwarf Galaxies

Dwarf galaxies are, unexpectedly, low-mass galaxies, in a range that can span from the typical mass of globular clusters ($10^5\,M_\odot$) to that of more "standard" galaxies ($10^9 - 10^{11}\,M_\odot$). An example of dwarf galaxies are the Large and Small Magellanic Clouds, that are both satellites of the MW.

One way to indentify dwarf galaxies is by observing the *streams* they leave in their wake as they move around their host galaxy, which are essentially groups of stars that get left behind when the smaller cloud gets gravitationally captured. The presence of streams is usually reflected into zones of (sensibly) different metallicity.

Moreover, it has been observed that the α-enhancement in dwarf galaxies is smaller than that of the MW's halo, making it quite improbable that they can make a relevant contribution.

On a concluding note, I think it's worth to mention at least the existence of Ultra-Faint Dwarf Galaxies (UFDs), which are a class of galaxies that contain from a few hundred to one hundred thousand stars, making them the faintest galaxies in the Universe.

UFDs resemble globular clusters in appearance but have very different properties. For once, UFDs contain a significant amount of dark matter and are more (spatially) extended.

UFDs are the most dark matter-dominated systems known to present day.

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11 In fact, they should be massive enough to leave behind, after their death, an intermediate mass black hole.

2 STELLAR STRUCTURES

2.1 STELLAR EQUILIBRIUM EQUATIONS

In this chapter we're going to introduce and describe the so-called *stellar equilibrium equations*, which account for the physical mechanisms that are interplaying within stellar structures, and compare it with what real-life observations tell us.

The goal of stellar structure equations is making possible to predict instant after instant in every point of the star the value of each physical and chemical properties, so that once an initial condition is set, we're going to be virtually able to accurately determine the evolution of those properties.

On a first approximation, we can consider stars to be autogravitating mass conglomerates, for which the gravitational pull is evenly balanced by the pressure gradients (and possibly other forces) that set in within the star. Since gravity is a *central force*, we can well expect the equilibrium configuration to be that of a sphere. This greatly simplifies the problem at hand, since a three-dimensional configuration is reduced to a one-dimensional configuration by means of symmetry.

The approximation of spherical symmetry allows us to formally decompose the star in a series of *shells* of infinitesimal width $\mathrm{d}r$, so that the physical properties of the star are approximately constant within said shell. The quantities we'll be interested in considering are the pressure $p(r)$, the density $\rho(r)$, the temperature $T(r)$, the various relative abundances $X_i(r)$, the mass $M(r)$ (that is considered to be the mass contained in the sphere of radius r) and the luminosity $L(r)$ (the energy per unit time entering the shell at distance r from the center of the star).

Given this notation, the mass found inside radius $r + \mathrm{d}r$ should be $M(r) + \mathrm{d}M$, where $\mathrm{d}M$ can be identified with the mass of the infinitesimal spherical shell. Said $\rho(r)$ the density, then the mass of this shell is

$$\mathrm{d}M = 4\pi r^2 \rho \, \mathrm{d}r$$

which yields our first stellar structure equation

$$\boxed{\frac{\mathrm{d}M}{\mathrm{d}r} = 4\pi r^2 \rho(r)} \tag{24}$$

Note that (24) is essentially a restating of the continuity equation of mass.

Let's stop for a moment to consider how sensible are the two approximations we're implementing in our description, that are the request of spherical symmetry and that only pressure gradients are balancing the gravitational pull. The former is generally met by the vast majority of stars, while for the latter it might be necessary to use some more care, especially when dealing with (highly) rotating stars, forcing us to add centrifugal forces to the picture[1].

Generally, if the centrifugal acceleration at the surface is too high with respect to the surface gravitational acceleration, the necessity of modifying our equations to keep track of centrifugal forces may arise.

For this purpose, we define the *critical angular velocity* Ω_c so that

$$\Omega_c^2 \sim \frac{GM_*}{R_*^3} \implies v_c \sim \sqrt{\frac{GM_*}{R_*}} \tag{25}$$

Note that depending on the mass and the radius of the star we're considering, critical velocities might get as high as $v_c \sim 300 - 600\,\mathrm{km/s}$; for comparison, the equatorial velocity of the Sun is $v_\odot^{\mathrm{eq}} \sim 2\,\mathrm{km/s}$.

To result in non-negligible effects, it is sufficient that the angular velocity gets as high as $\Omega_{\mathrm{sup}} \gtrsim 0.5\Omega_c$; if, instead, $\Omega_{\mathrm{sup}} \gtrsim 0.7\Omega_c$ you start observing deformations in the structure of the star, but these are fairly rare. As you may easily guess, these are problems that concern with the more massive stars, getting more and more relevant the bigger the star gets.

About 20% of the stars with mass in the range $10M_\odot \lesssim M \lesssim 19M_\odot$ (so typically stars of spectral class B or O) have velocities of order

$$150 < v_{\sin i} < 300 - 400\,\mathrm{km/s}$$

Note that through Doppler effect, we're able to measure the velocity of objects along the line of sight, but in principle, the rotation axis of the object we're observing may be tilted of an angle i with respect to the observer, hence the notation/factor $\sin i$.

Increasing the mass, we find that about 30% of the stars with mass in the range $20M_\odot \lesssim M \lesssim 40M_\odot$ have roughly the rotational velocities of the order of the ones we've written above.

To "detect" the presence and relevance of rotation it is customary to perform a spectroscopic analysis of the star. In fact, it is clear that the presence of rotation issues the arising of convection motion within the star, which in turn implies a "mixing" of the star's contents.

In principle, if we observe main sequence stars, the chemical composition of the surface should be roughly equal to the starting composition[2], for temperature isn't high enough to start the nuclear processing on the surface. But, in presence of strong mixing mechanisms, we may observe an excess of Helium and Nitrogen that are brought to the surface from the inner regions of the star (where nuclear reactions are at least partially active).

1 Note that in the presence of rotation, the equilibrium configuration is no longer spherically symmetric, since rather than the gravitational potential, we have now to consider a baroclyinic potential (gravitational+centrifugal).

2 If we neglect the effects of *sedimentation*, that is. The characteristic times are however too short for sedimentation to occur, so it should be probably safe neglecting it.

The Effect of Magnetic Fields (anticipation)

In principle, even particularly strong magnetic fields may influence the surface composition and structure of the star. This, however, requires

$$\beta = \frac{p_{\text{gas}}}{p_{\text{mag}}} \sim 1 \text{ with } \beta = \frac{8\pi p}{B^2}$$

that in turn would mean having magnetic fields as high as $B \approx 3000 - 5000\,\text{G}$. In general they are not relevant, but there exist very rare cases such that of *magnetic stars* of class A_p or B_p, where the subscript "p" is to indicate particular lines in the spectrum of such stars. For comparison

$$\langle B \rangle_\odot \sim 0.5 - 4\,\text{G vs } \langle B \rangle_{MS} \sim 1 - 30\,\text{kG}$$

Magnetic fields that strong are assured to influence the external layers of such stars, inducing mixings that lead to a modification of the surface abundances.

Another class of magnetic stars is that of *magnetic white dwarves* that, although very rare, can reach magnetic fields of order $10^6 - 10^9\,\text{G}$. Similarly, in *magnetars* (magnetic neutron stars), magnetic fields can reach intensities of $10^{14} - 10^{15}\,\text{G}$.

Note that in principle we're able to detect the presence of strong magnetic fields by studying the surface oscillation of the star, performing what's called *asteroseismology*.

2.1.1 Free Fall Time

Our initial claim was that for the majority of their life, stars are in *hydrostatic equilibrium*, in a configuration where pressure gradients are evenly matching the collapse forces. But can we be so sure that a star is, as a matter of fact, in equilibrium?

We know, for example, that the Sun has been stable on timescales of at least a couple hundred years. On top of that, looking at the Milky Way we can find billions of stars with characteristics similar to those of the Sun but in different stages of their life, so that we might have an idea of the whole life cycle of our dearest yellow dwarf.

From those observations, we learn that the Sun must have been stable for a time of the order of millions of years. Of course, to make sure it is indeed in hydrostatic equilibrium, we should compare with the *dynamical timescales* the observation timescales over which we can safely claim that its characteristics (e.g., the luminosity and the radius) haven't changed.

Let's consider a spherically symmetric star with constant density. In presence of gravity alone, we can write

$$\ddot{r} = -\frac{GM}{r^2}$$

which can be immediately integrated to yield

$$\begin{aligned} \frac{1}{2}\dot{r}^2 &= \frac{GM}{r} + \text{const.} \\ &= \frac{GM}{r} - \frac{GM}{R} \end{aligned}$$

where we have set the initial conditions for $t = 0$: $r = R$ and $\dot{r} = 0$.

The free-fall timescale can be thus obtained by committing a mathematical atrocity, but I (and, for extension, you too) don't care[3]

$$\begin{aligned}\tau_{ff} &= \int_0^{t_{ff}} \mathrm{d}t = \int_R^0 \frac{1}{\dot{r}}\,\mathrm{d}r \\ &= \int_0^R \frac{\mathrm{d}r}{\sqrt{2GM\left(\frac{1}{r} - \frac{1}{R}\right)}} = \frac{\pi}{2\sqrt{2}}\sqrt{\frac{R^3}{GM}}\end{aligned}$$

thus, if we use $R^3/GM = 3/4\pi G\rho$

$$\tau_{ff} = \sqrt{\frac{3\pi}{32G\rho}} \approx 1.63\,\mathrm{Myr} \cdot \left(\frac{10^3\,\mathrm{cm}^{-3}}{\mu n}\right)^{1/2} \tag{26}$$

For the Sun, the free-fall timescale amounts to roughly $\tau_{ff} \sim 30\,\mathrm{min} - 1\,\mathrm{h}$ depending on the calculation that has been used. So we can safely claim that the Sun is definitely not collapsing. What if, instead, it is exploding?[4]

To see that, we can calculate the explosion time, sound time, or dynamical time as

$$\tau_{\mathrm{sound}} = \frac{R}{c_s} \approx 0.5\,\mathrm{Myr}\left(\frac{R}{0.1\,\mathrm{pc}}\right)\left(\frac{0.2\,\mathrm{km\,s^{-1}}}{c_s}\right) \tag{27}$$

If $\tau_{ff} > \tau_{\mathrm{sound}}$ the system returns to a stable equilibrium as the pressure forces can overcome gravity. Conversely, for $\tau_{ff} < \tau_{\mathrm{sound}}$ gravitational collapse takes place.

2.1.2 Hydrostatic Equilibrium

In the last section we showed that is fair for us to assume that stars are in *hydrostatic equilibrium* for most of their lives. We now turn to write an equation (the second of four) that will allow to characterize our star.

Consider a star in which the only relevant forces acting on a spherical shell between r and $r + \mathrm{d}r$ and transverse area $\mathrm{d}A$ are the gravitational pull and the pressure forces (both due to the gas and photons). It's not difficult to show that the balance condition for the mass of the small element is

$$-\mathrm{d}p\mathrm{d}A - \frac{GM_r}{r^2}\rho\mathrm{d}r\mathrm{d}A = 0$$

from which

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{GM_r\rho}{r^2}} \tag{28}$$

A quick look at (24) and (28) will convince you that we still have more free variables (functions of the radial coordinate) than equations. This is one of the many cases in which the *closure problem* pops out. To solve it we're gonna need to introduce an equation of state (EoS)

$$\boxed{p = p(\rho, T, X, Y, X_i)} \tag{29}$$

3 It is actually less atrocious than I'm making it sound like.
4 Read: expanding outwards.

where the various quantities have their obvious meaning. A particularly simple configuration (that we'll have the pleasure to look somewhere in the following) is the one where we're allowed to take a *politropic equation of state*

$$p = K\rho^{\gamma} \tag{30}$$

with K some constant and γ some power that depends on the particular gas and its degenerate state.

Let us now take a look at another particularly simple EoS, that is the equation of state of perfect (non degenerate) gas

$$p = nkT = \frac{\rho(r)}{\mu(r)m_{\mathrm{H}}}kT(r) \tag{31}$$

with $\mu(r)$ the *mean molecular weight*, which clearly depends on the elements present in the star and their ionization state, which can (hopefully) inferred from considering Saha's equation (143). In general it's not easy to determine the exact form of $\mu(r)$ so it is customary to consider the easy scenario of a *completely ionized* star[5].

If we consider X the relative abundance of Hydrogen, Y the relative abundance of Helium and X_i the relative abundance of the metal with i so that[6] $A > 4$. If we consider the number of ions and electrons that complete ionization generates:

- Hydrogen: #ions X/m_{H}, #electrons X/m_H;
- Helium: #ions $Y/4m_{\mathrm{H}}$, #electrons $2Y/4m_{\mathrm{H}}$;
- Metal i-th: #ions X_i/A_im_{H}, #electrons Z_iX_i/A_im_{H}

so that the total number of particles (per unit volume) is just

$$n = \left[2X + \frac{3Y}{4} + \sum_{i=4}^{N}\left(\frac{X_i}{A_i} + \frac{Z_iX_i}{A_i}\right)\right]\frac{\rho}{m_{\mathrm{H}}} \tag{32}$$

If we consider the following approximation for (32)

$$\frac{X_i}{A_i} \ll 1 \qquad \frac{Z_i}{A_i} \sim \frac{1}{2}$$

we find

$$\mu^{-1} \approx 2X + \frac{3Y}{4} + \frac{1}{2}\sum_{i=4}^{N}X_i = 2X + \frac{3Y}{4} + \frac{Z}{2} \tag{33}$$

where we've introduced the *metallicity* Z.

We can also consider the molecular weight restricted to ions only

$$\mu_i^{-1} = X + \frac{Y}{4} + \sum_{i=4}^{N}\frac{X_i}{A_i}$$

5 People tend to say that a star is completely ionized when its temperature is at least $T > 10^6$ K, although that's a completely arbitrary lower limit. For instance, heavier elements like Iron, which have electrons to spare, are completely ionized at much higher temperatures.

6 Sorry $^4_3$He, you're too less abundant to be considered. But your time will come soon.

and consider an average charge for the mixture of elements $\bar{Z}$, so that $n_e = \bar{Z} n_i \approx \bar{Z}\rho/\mu_i m_\mathrm{H}$. From this we can define the mean molecular weight of electrons as

$$\frac{\mu_i}{\bar{Z}} \to n_e = \frac{\rho}{\mu_e m_\mathrm{H}}$$

If we plug this all back in the EoS for the perfect gas

$$p_\mathrm{gas} = nkT = (n_e + n_i)kT = \frac{\rho k T}{m_\mathrm{H}}\left(\frac{1}{\mu_i} + \frac{1}{\mu_e}\right)$$

we conclude

$$\boxed{\frac{1}{\mu} = \frac{1}{\mu_i} + \frac{1}{\mu_e}} \tag{34}$$

2.1.3 Temperature Gradients

As you probably know, in the inner regions of stars holds the so called *Local Thermodynamic Equilibrium* (for friends, LTE): In these regions there is, as the name suggests, punctual and *local* thermodynamic equilibrium between matter, which will be then well described by a Maxwell-Boltzmann distribution, and radiation, which can be described by (6).

This fact should also suggest you that, since we're able to observe temperature gradients within stars, it is not possible for stars to be in *stric thermodynamic equilibrium* (STE) and thus emit like perfect blackbodies with fixed temperature.

Let us try to give an order of magnitude estimate of the temperature gradient within, say, the Sun

$$\langle \frac{\mathrm{d}T}{\mathrm{d}r} \rangle = \frac{T_c - T_S}{R_\odot} \approx 10^{-4}\,\mathrm{K\,cm^{-1}}$$

which means that over the average mean free path of a photon in the inner regions of stars ($\lambda \sim 0.1 - 1\,\mathrm{cm}$), the difference in temperature will be

$$\Delta T = \langle \frac{\mathrm{d}T}{\mathrm{d}r} \rangle \cdot \lambda \approx 10^{-5} - 10^{-4}\,\mathrm{K}$$

which is very small. But to actually check if the assumption of LTE is valid (if at all sensible) we shall consider the variation in internal energy of the photons. We know that in thermal equilibrium $U = aT^4$, so

$$\frac{\mathrm{d}U}{U} = 4\frac{\mathrm{d}T}{T} \sim 10^{-12} - 10^{-11}$$

which means the deviation from isotropic (blackbody) behavior is negligibly small over the mean free path of photons, therefore allowing us to assume the LTE configuration.

We know, either through direct observation or because you've read the Appendix about radiative transfer, that wherever is a temperature gradient there is also a radiative flux (along with a convective motion and electronic conduction)!

On a first approximation we may write

$$\left.\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_\mathrm{rad} \propto F(r)\bar{\kappa}$$

where F is the flux we've defined in (4) and $\bar{\kappa}$ is some average *opacity*[7] that sums up the various interaction channels between matter and radiation. Note that in general it is frequency dependent[8].

In the equation we've written above we're assuming that the transport of energy is carried out by radiation alone, which is essentially the temperature gradient that we'd observe if all the flux was transported by radiation.

Taking a closer look to the micro-physics involved in absorption phenomena we learn that it is possible to link the opacity to the mean free path of electrons

$$\lambda_\sigma = (n\langle\sigma\rangle)^{-1} = (\rho\bar{\kappa})^{-1}$$

If, on the other hand, there were another channel available for transporting energy on top of radiation, we may be writing

$$F(r) = \frac{L(r)}{4\pi r^2} = F_{\text{rad}} + F_{\text{conv}}$$

where we've assumed the presence of, say, relevant convective motions within the star. It should go without saying that the temperature gradient will no longer be the gradient induced by radiative transport, but will rather be an *ambient gradient*, which we might express as

$$\left.\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_{\text{amb}} \propto \bar{\kappa}F_{\text{rad}}$$

Clearly, since now radiation is transporting only a part of the total flux

$$\left|\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_{\text{amb}} < \left|\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_{\text{rad}}$$

If we define with ϕ the flux of photons per unit time and unit surface, and approximate the number of interactions in an infinitesimal distance $\mathrm{d}r$ as $\mathrm{d}r/\lambda$, with λ the mean free path, we can calculate the transferred momentum

$$\delta p = \phi\frac{\mathrm{d}r}{\lambda}\frac{h\nu}{c} = \frac{F(r)}{\lambda c}\mathrm{d}r$$

But the variation in (linear) momentum is equal to the variation of pressure $\delta p = \delta P_{\text{rad}}$

$$\frac{\mathrm{d}P_{\text{rad}}}{\mathrm{d}r} = \frac{F(r)}{\lambda c}$$

If we recall that $P_{\text{rad}} = U/3 = aT^4/3$ and correct the sign of the expression (to address the mismatch of signs between the flux and temperature gradient), we finally yield

$$F(r) = -\lambda ca\frac{4}{3}T^3\frac{\mathrm{d}T}{\mathrm{d}r}$$

Recalling that $\lambda = (\rho\bar{\kappa})^{-1}$ and that $\sigma_B = ac/4$ we can write our third stellar structure equation

$$\boxed{F(r) = -\frac{16}{3}\frac{\sigma_B T^3}{\rho\bar{\kappa}}\frac{\mathrm{d}T}{\mathrm{d}r}} \tag{35}$$

7 Opacity is measured in $\mathrm{cm^2\,g^{-1}}$.
8 For a more rigorous definition see either the next chapter or [20].

It is often useful to flip the equation

$$\left.\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_{\mathrm{rad}} = -\frac{3\bar{\kappa}\rho}{4ac}\frac{1}{T^3}\frac{L(r)}{4\pi r^2}$$

so that in general, if $F_{\mathrm{rad}} \neq F_{\mathrm{tot}}$, we can make a slight manipulation and write

$$\left.\frac{\mathrm{d}T(r)}{\mathrm{d}r}\right|_{\mathrm{amb}} = -\frac{3\bar{\kappa}\rho}{4ac}\frac{1}{T^3}F_{\mathrm{rad}}$$

If you want to see a slight more rigorous derivation of (35), you may check the last part of the "Radiative Transfer" Appendix.

2.1.4 Energy Production

Let us consider what happens if the stellar atmosphere was not constantly replenished with energy from the inner layers. How long would it take for the star to deplete completely the energy stored in the atmosphere? The average density of the outer layers of the atmosphere is $\rho_{\odot}^{\mathrm{ext}} \sim 10^{-7}\,\mathrm{g\,cm^{-3}}$. We can thus give an estimate of the mean free path $\lambda_j \sim (\rho\bar{\kappa})^{-1} \sim 100 - 300\,\mathrm{km}$.

We've made the (usually decent) approximation of opacity being dominated by photoelectric effect on H^- ions, $\bar{\kappa} \approx 0.3\,\mathrm{cm^2\,g^{-1}}$. Note that in these conditions, the width of the atmosphere is of the same order of the mean free path of photons.

We know the solar flux $F_{\odot} \sim 6 \cdot 10^{10}\,\mathrm{erg\,cm^{-2}\,s^{-1}}$, as well as its temperature and pressure $T_{eff} \sim 6000\,\mathrm{K}$, $p \approx 10^5\,\mathrm{dyne\,cm^{-2}}$. The average number of particles is $n \sim 10^{17}\,\mathrm{cm^{-3}}$, which can be integrated over the vertical direction to find an average number of photons per unit surface.

The mean energy of photons can be expressed as $E_p = 3kT/2 \sim 10^{-12}\,\mathrm{erg}$. The energy per unit surface is thus $E_s \approx 10^{12}\,\mathrm{erg\,cm^{-2}}$. Finally, we find that the energy stored in the atmosphere would be depleted in roughly

$$\tau = \frac{E_s}{F} \approx 15\,\mathrm{s}$$

In such a short time, we would be able to discern immediately the changes in color in the atmosphere. Similarly, through the random-walk approximation, we may find an estimate of the time it takes photons to escape the inner regions of stars.

Random Walks

Consider a photon emitted in an infinite, homogeneous scattering region. It travels a displacement $\vec{r}_1$ before being scattered, then travels in a new direction over a displacement $\vec{r}_2$ before being scattered, and so on. After N free paths, the total displacement will look like

$$\vec{R} = \sum_{i=1}^{N} \vec{r}_i$$

The average total displacement will be identically null. Therefore, we must evaluate the mean square of the total displacement, which, in the assumption of independent and isotropic scattering, must be expressed as

$$l_*^2 = \langle \vec{R}^2 \rangle = \sum_{i=1}^{N} \langle \vec{r}_i^2 \rangle = \sum_{i=1}^{N} l_i^2$$

The quantity l_* is the root-mean-square net displacement of the photon, while l^2 denotes the mean square of the free path of a photon, which within a factor of order unity, it's simply the mean free path of a photon.

Since the mean square displacements of each scattering iteration have no reason to be different, we'll just write

$$l_* = \sqrt{N} l$$

Said L the linear dimension of the object photons are trying to "escape" from, the number of scatterings events required to actually escape from regions of large optical depth is roughly determined by setting $l_* \approx L$. Then $N \sim L^2/l^2$.

This way, we can find the number of interactions needed to travel through a solar radius $R_\odot$ if the average displacement is a mean free path λ

$$\Delta x = \sqrt{z}\lambda \implies z \approx \left(\frac{R_\odot}{\lambda}\right)^2 = 1.96 \cdot 10^{22}$$

If an interaction takes, on average, $\tau = 10^{-8}$ s, the time needed to perform all those interactions is $\Delta t \approx 10\,\mathrm{Myr}$. This usually goes by the name of *thermal timescale*.

If we consider the average density of a star and a mean opacity independent of temperature and density, we can write

$$\bar{\rho} = \frac{M}{\frac{4\pi}{3} R^3}$$

but for stars with mass in the range $M_\odot \lesssim M \lesssim 10M_\odot$, mass scales linearly with the radius $M \propto R$. So

$$\lambda = (\bar{\kappa}\rho)^{-1} \propto R^2 \to \sqrt{z} \propto R^{-1} \implies \Delta t_{th} \propto R^{-2}$$

This is telling us that increasing the mass of a star (and thus the radius) has the surprising consequence of a (very) reduced lifespan.

We now present the last equation of stellar structure, that relates the luminosity gradient to the rate of production of energy

$$\boxed{\frac{\mathrm{d}L(r)}{\mathrm{d}r} = 4\pi r^2 \rho(r)\epsilon(r)} \tag{36}$$

where $\epsilon(r)$ is the "energy production rate", that is usually decomposed in three contributions

$$\epsilon(r) = \epsilon_g(r) + \epsilon_N(r) - \epsilon_\nu(r)$$

The easier to address is the term $-\epsilon_\nu$, which expresses the energy lost to neutrinos that are produced where there's no nuclear reactions. These neutrinos, produced via various channels, are often called *thermal neutrinos*. The main channels of production are summed up below:

$$\begin{aligned} e^- + (Z, A) \text{ or } e^- \to e^- + (Z, A) \text{ or } e^- + \nu_e + \bar{\nu}_e \quad &\text{Bremsstrahlung} \\ \gamma + e^- \to e^- + \nu_e + \bar{\nu}_e \quad &\text{Photoproduction} \\ \gamma \to e^- + e^- \to +\nu_e + \bar{\nu}_e \quad &\text{Pair annihilation} \\ \gamma_{\text{plasmon}} \to \nu_x + \bar{\nu}_x \quad &\text{Neutrinic oscillations} \\ (Z, A) + e^- \to (Z-1, A) + \nu_e \quad &\text{Neutralization} \end{aligned}$$

Please note that the subject of neutrinic oscillations is rather technical and I know nothing of it to make this surely intriguing subject justice. The only thing I know is that the collective oscillations in a plasma, *plasmons*[9], may decay into one of the three leptonic pairs of neutrinos.

The next one to address is the contribution given by *gravitational energy* ϵ_g. Consider the 1st principle of thermodynamics

$$\mathrm{d}Q = \mathrm{d}U + p\mathrm{d}V \quad \mathrm{d}Q_m = \mathrm{d}U_m + p\mathrm{d}\left(\frac{1}{\rho}\right) \tag{37}$$

where the subscript m in the second equation means "per unit mass". We may then define $\epsilon_g = -\mathrm{d}Q_m/\mathrm{d}t$. Invoking entropy, this becomes

$$\epsilon_g = -\frac{\mathrm{d}Q_m}{\mathrm{d}t} = -T\frac{\mathrm{d}S_m}{\mathrm{d}t} = -T\left[\partial_p S)_T \frac{\mathrm{d}p}{\mathrm{d}t} + \partial_T S_m)_p \frac{\mathrm{d}T}{\mathrm{d}t}\right]$$

which, once we introduce the *specific heat capacity at constant pressure* $c_p = T\partial_T S_m)_p$ and the heat exchanged during an isothermal transformation $E_T = -T\partial_p S)_T$, we may finally write

$$\epsilon_g = -\frac{\mathrm{d}Q_m}{\mathrm{d}t} = E_T\dot{p} - c_p\dot{T} \tag{38}$$

Since time derivatives are involved, it should be clear that we are in the need of having two different models to describe the structure of a star in different times. To do so, it is necessary to solve *mesh by mesh* the stellar structure equations. It is then of crucial importance knowing how abundances change within a given mesh. Those who work with this stuff usually linearize the function describing the evolution of the abundance of the i-th element as such

$$X_{i,k}^{\text{new}} = X_{i,k}^{\text{old}} + \frac{\mathrm{d}X_i}{\mathrm{d}t}\Delta t \tag{39}$$

where the subscript k refers to a given mesh. The time interval Δt is often chosen so that in each mesh the physical variables and/or the abundances do not vary more than a predetermined value.

9 Not to be confused with the omonimous biscuits.

2.2 VIRIAL THEOREM

For stellar structures in hydrostatic equilibrium, for which only pressure and gravitational forces are present, the following *virial theorem* holds

$$2\langle K\rangle_{\text{trasl}} + \langle\Omega\rangle = \frac{1}{2}\ddot{I} \approx 0 \tag{40}$$

This strikingly simple equation allows us to find some enlightning relations. Let us then consider a monoatomic perfect gas for which the equipartition theorem grants us $\langle K\rangle = 3kT/2 = E_{th}$. The virial theorem thus implies

$$E_{th} = -\frac{\Omega}{2} > 0 \implies E_{tot} = \frac{\Omega}{2}$$

Note that under the assumption of spherical symmetry, the gravitational energy can be found evaluating the following integral

$$\Omega = -\int_0^{R_*} \frac{GM_r}{r} 4\pi\rho r^2 \,\mathrm{d}r < 0$$

From the former expression, we can calculate the luminosity

$$L = -\frac{\mathrm{d}E_{tot}}{\mathrm{d}t} = -\frac{1}{2}\frac{\mathrm{d}\Omega}{\mathrm{d}t} = \frac{\mathrm{d}E_{th}}{\mathrm{d}t} > 0$$

From this we're learning that from the gravitational contraction, half the energy is dissipated through radiation, the other half is used to "fuel" the thermal content of the star (i.e., the star gets hotter).

What if we relax the condition of the gas being monoatomic? The virial theorem (40) still holds, but in general we can't say $\langle K\rangle = E_{th}$. The traslational (average) kinetic energy is still $\langle K\rangle = 3kT/2$, but now $E_{th} = lkT/2$, where l is the number of degrees of freedom of the single gas element. It isn't particularly convenient to work with the degrees of freedom, so we're going to convert it to a more thermodynamically acceptable quantity, the adiabatic index γ. It is possible to show that

$$\gamma = \frac{C_p}{C_V} = 1 + \frac{2}{l}$$

This way we can cast the virial theorem in an alternative form

$$3(\gamma - 1)E_{th} + \Omega \approx 0$$

The total energy can be calculated with a direct substitution

$$E_{tot} = E_{th} + \Omega = \frac{3\gamma - 4}{3(\gamma - 1)}\Omega$$

from which

$$\Delta E_{tot} = \frac{3\gamma - 4}{3(\gamma - 1)}\Delta\Omega$$

$$\Delta E_{th} = -\frac{\Delta\Omega}{3(\gamma - 1)}$$

Since $\Delta\Omega$ is necessarily smaller than 0 (if the star contracts), we thus obtain a stability criterion for the star: A star can be stable if $\gamma > 4/3$. If such condition is not met, all the energy released by gravitational collapse is converted into thermal energy, thus resulting in instabilities.

We can also extend the virial theorem to different equations of state if we allow for the translational kinetic energy to be written as $\langle K \rangle = \xi E_{th}$. Clearly, we have to assume that the coefficient ξ is constant over the whole stellar structure. Repeating the same calculations as before (*mutatis mutandis*, that is), virial theorem yields

$$E_{tot} = \frac{\xi - 1}{\xi}\Omega$$

and the stability condition therefore becomes $\xi > 1$.

2.2.1 Thermal Timescale Revisited

A few pages ago we made use of the random walk approximation to find the time that it takes on average to a photon to scatter its way out of the inner region of a star. We called that time *thermal timescale*. It is the aim of this section to show that it is indeed possible to find an equivalent formulation for that timescale starting from the virial theorem.

Let us recall (40)

$$2\langle K \rangle + \langle \Omega \rangle \approx 0$$

For the sake of simplicity, we're going to assume a monoatomic gas, so that $\langle K \rangle = E_{th}$. We can assume the following expression for the gravitational potential

$$\Omega = -q\frac{GM^2}{R}$$

with q that depends on the specific density profile that is given. For a homogeneous star, for example, $q = 3/5$. We showed that the luminosity of the star will be such that

$$L = -\frac{1}{2}\dot{\Omega} = -\frac{q}{2}\frac{GM^2}{R^2}\dot{R}$$

Separating variables and integrating, we find

$$\tau_{KH} = \frac{q}{2}\frac{GM^2}{RL} \tag{41}$$

which is called *Kelvin-Helmoltz timescale*. In short, it represents the time during which the stellar structure is able to sustain itself radiating away the energy generated from the gravitational shrinking.

Note that we're implicitly assuming that the luminosity doesn't change while the star collapses, which is in general a very bad approximation.

Under the approximation of homogeneous density $q = 3/5$, the Kelvin-Helmoltz timescale is

$$\tau_{KH} \approx 10\,\mathrm{Myr}\left(\frac{M}{M_\odot}\right)^2\left(\frac{R_\odot}{R}\right)\left(\frac{L_\odot}{L}\right)$$

which is much greater than the other dynamical times, but still not quite as large as it should be to match the estimated lifespan of the Sun (that it's also connected to the fairly long half-life of some radionuclides we observe here on Earth).

2.3 QUALITATIVE STELLAR STRUCTURE RELATIONS

Now that we have the virial theorem at our disposal, we might think of looking for very approximated expressions that can give us an idea of how physical properties in a star are interconnected. All quantities will be interpreted as *averaged* quantities.

Let us start with density and temperature. If we know M and R, then

$$\Omega = -q\frac{GM^2}{R}$$

with q a constant given by the profile density. We can now use the virial theorem and invert the relation between mass, density and radius

$$E_{th} \sim \frac{3kT}{2}\frac{M}{\mu m_{\mathrm{H}}} \quad \rho = \frac{M}{\frac{4\pi}{3}R^3}$$

finding

$$\boxed{T \propto M^{2/3}\rho^{1/3}} \tag{42}$$

This means that two stars with different masses must have different densities to achieve the same temperature! This will then reflect onto smaller stars having the higher densities.

Upon a first approximation, the luminosity L does not depend on the energy production rate ϵ. If we butcher the equation of hydrostatic equilibrium and plug in the proportionalities of the density

$$p_C \propto \frac{M^2}{R^4}$$

where p_C is the central pressure. We can make the further assumption of a perfect gas and negligible radiation pressure, so that

$$\boxed{T_C \propto \frac{p_C\mu}{\rho} \propto \mu\frac{M}{R}} \tag{43}$$

If we make the assumption of exclusively radiative transport, we find that

$$\frac{\mathrm{d}T}{\mathrm{d}r} \propto -\mu M R^{-2} \to L \propto -\frac{T^3}{\bar{\kappa}\rho}\frac{\mathrm{d}T}{\mathrm{d}r}R^2$$

We then make the (very wrong, both conceptually and experimentally) approximation of constant opacity, thus finding

$$\boxed{L \propto \mu^4 M^3} \tag{44}$$

Finally, we can find a relation between luminosity and density

$$\frac{\mathrm{d}L}{\mathrm{d}r} = 4\pi r^2\rho\epsilon \to \frac{L}{R^3} \propto \rho\epsilon \to \rho \propto \frac{L}{R^3}\frac{1}{\epsilon}$$

but we showed that $R \propto M/T$, so

$$\rho \propto \frac{T^3}{\epsilon}$$

without some kind of prescription for the energy production rate, this is as far as we can get. Fortunately for us, for main sequence stars that still burn Hydrogen, it is given (as we'll see) the following expression

$$\epsilon \propto \rho^m T^n$$

with $m = 1$ both in the case of energy production through the ppI chain and the CN-NO chain. The two of them, however, have really different dependencies on the temperature: The ppI scales as $\epsilon \propto T^4$, while the CN-NO as $\epsilon \propto T^{15-17}$. Plugging in the aforementioned expression, we finally find

$$\boxed{\rho^{m+1} \propto T^{3-n}} \tag{45}$$

2.4 POLITROPIC STRUCTURES

A simple way of solving stellar structure equations is by killing off one of the variables, namely temperature. This isn't always possible, but there exist some special cases in which pressure can be assumed to be dependent only on density. This is the case of the so-called *politropic structures*. Politropic structures are described by the following equation of state

$$p(\rho) = K\rho(r)^{1+\frac{1}{n}} = K\rho(r)^\Gamma \tag{46}$$

where n is the *index of the politropic*. Consider the equation for hydrostatic equilibrium (28) and twist it so that

$$\frac{\mathrm{d}p}{\mathrm{d}r}\frac{r^2}{\rho} = -GM_r$$

This we can derive with respect to the radial coordinate, finding

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{\mathrm{d}p}{\mathrm{d}r}\frac{r^2}{\rho}\right) = -G\frac{\mathrm{d}M_r}{\mathrm{d}r} = -4\pi G\rho r^2$$

where we made use of (24). Dividing by r^2 we finally obtain

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{\mathrm{d}p}{\mathrm{d}r}\frac{r^2}{\rho}\right) = -4\pi G\rho \tag{47}$$

which is still the same equation, but with a *crispier* form that will make a little easier the discussion we'll do in the next few paragraphs.

Lane-Emden equation

Assume for example a density profile of the form

$$\rho(r) = \rho_c\phi^n(r)$$

with ϕ some adimensional density profile with the following characteristics: $\phi(0) = 1$ and $\partial_r\phi|_{r=0} = 0$. Substituting in the politropic equation (46) we find that the pressure is simply

$$p(r) = K\rho_c^{1+\frac{1}{n}}\phi^{n+1} = p_C\phi^{n+1}$$

In terms of an adimensional radius $r = \lambda\xi$, with λ constant depending only on n

$$\lambda = \left[\frac{K\rho_c^{1/n-1}(n+1)}{4\pi G}\right]^{1/2}$$

(47) takes a much simpler and elegant form

$$\boxed{\frac{1}{\xi^2}\partial_\xi(\xi^2\partial_\xi\phi) = -\phi^n} \tag{48}$$

which is the general *Lane-Emden equation*. Solutions for this equation have been calculated numerically in Fig.8.

Note that solving (48) gives us two things: The profile density ϕ and the adimensional radius ξ_0 for which $\phi(\xi_0) = 0$, which is essentially the radius of the star.

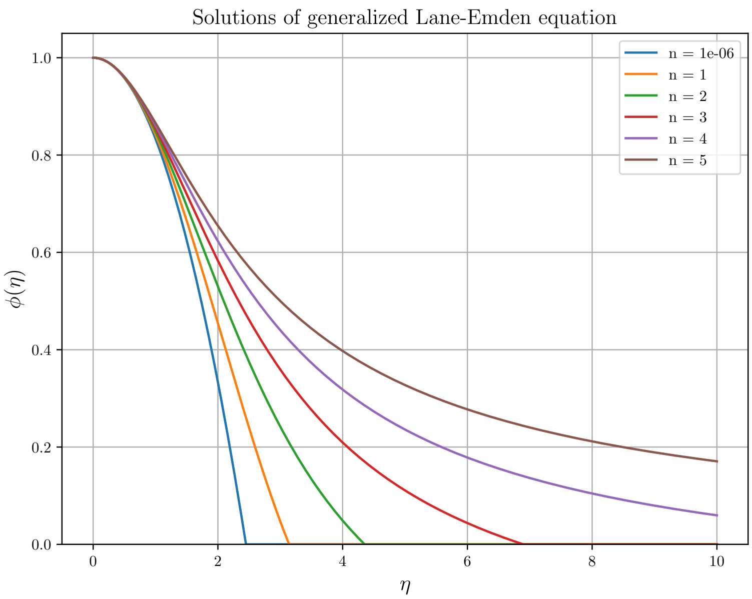

Figure 8: Numerical solutions to the Lane-Emden equation.

Since we know quite a few things already, we may ask ourselves if it is possible to calculate the central density of the star as well. Assuming that we know from some other observations the total mass of the star, we can indeed calculate ρ_c. In fact, we can simply integrate the equation of continuity (24) up to the radius of the star

$$M = \int_0^R 4\pi r^2 \rho \,\mathrm{d}r = 4\pi\lambda^3\rho_c \int_0^{\xi_0} \xi^2\phi^n \,\mathrm{d}\xi \tag{49}$$

The value of that integral is actually fixed by the original equation itself to $-\xi_0^2\partial_\xi\phi|_{\xi_0}$. So we can promptly invert the equation above and find the central density of the star. It goes without saying that once we've completely specified the density profile we'll have automatically determined the pressure profile as well by means of mere substitution.

Politropic Integration for the Solar Structure

In the following section we'll try to integrate analitically the stellar structure equations for the Sun assuming a politropic EoS (46). Before we begin, let us make the following approximation

$$\beta = \frac{p_{\text{gas}}}{p_{\text{tot}}} \sim \text{const.}$$

where $p_{\text{tot}} = p_{\text{gas}} + p_{\text{rad}}$. This assumption follows from the conditions of hydrostatic equilibrium and radiative transfer, by means of which it's possible to prove that

$$1 - \beta \sim \bar{\kappa}(r)\epsilon$$

This is the so called *Eddington Approximation*: Since $\epsilon \propto \rho T^4$ (at least as far as Hydrogen burning in main sequence stars is concerned) and $\bar{\kappa} \propto \rho^{\alpha} T^{-3.5}$ (Kramers' law), their product will be approximately constant[10].

Moreover

$$p_{\text{tot}} = p = p_{\text{gas}} + p_{\text{rad}} = \frac{\rho k T}{\mu m_{\text{H}}} + \frac{1}{3} a T^4$$

Let us use the expression for β to remove the temperature, so to write a politropic equation

$$\frac{p_{\text{rad}}}{p_{\text{gas}}} = \frac{1-\beta}{\beta} = \frac{1}{3}\frac{aT^3}{\rho}\mu m_{\text{H}}$$

so that

$$T = \left[\frac{1-\beta}{\beta}\frac{3\rho}{\mu m_{\text{H}} a}\right]^{1/3}$$

This way, if we substitute in the equation for the pressure, we find

$$p = \left[\frac{3}{a}\left(\frac{k}{\mu m_{\text{H}}}\right)^4 \frac{1-\beta}{\beta^4}\right]^{1/3} \rho^{4/3} \tag{50}$$

which is a politropic equation as requested. With reference to (46), this relation would have a politropic index $n = 3$. Note that in this case, since we don't know β, we don't know the multiplicative factor, nor we have the absolute certainty of it being constant. For the moment, we'll consider a molecular weight averaged over the whole stellar structure[11].

We still don't know the multiplicative constant, but we know both the mass and the radius of the Sun. We've shown in (49) that if we know both the mass and the radius of a star we're able to infer its central density

$$M = M(\beta, \rho_C) = 4\pi\lambda^3 \rho_c (-\xi^2 \partial_\xi \phi)_{\xi_0} = M_\odot$$

with

$$\lambda = \left[\frac{K\rho_c^{1/n-1}(n+1)}{4\pi G}\right]^{1/2}$$

10 Note that this is only approximately true for Sun-like stars which are burning Hydrogen in the Main Sequence stage.

11 Note that generally the solar structure is divided into two regions, one where Hydrogen burning has happened (*solar interior*: $X \sim 0.34$, $Y \sim 0.64$, $Z \sim 0.02$ and $\mu \approx 0.85$, assuming complete ionization) and one where it has not (*external envelope*: $X \sim 0.70$, $Y \sim 0.28$, $Z \sim 0.02$ and $\mu \approx 0.62$, assuming complete ionization).

Moreover, we know that $R_\odot = R(\beta, \rho_C) = \lambda\xi_0$, thus we have enough equations to solve for. Behold

$$\rho_C = \frac{M_\odot}{4\pi R_\odot^3} \frac{\xi_0^3}{(-\xi^2 \partial_\xi \phi)_{\xi_0}}$$

So if we plug in the necessary solar parameters and look up to some table to find the values for $\xi_0 \approx 6.9$ and $-\xi^2\partial_\xi\phi|_{\xi_0} \approx 2.02$ for a politropic index $n = 3$, we'd find

$$\rho_C \sim 75\,\mathrm{g\,cm^{-3}}$$

which is about half of the best estimate we'd find using less approximate models, so everything considered we've done a good job.

At this point we can equate the expression for K coming from the definition of the parameter λ and that coming from the multiplicative factor of (50)

$$K = \frac{\lambda^2 4\pi G}{(1+n)\rho_c^{1/n-1}} \qquad K_\odot = \left[\frac{3}{a}\left(\frac{k}{\mu m_{\mathrm{H}}}\right)^4 \frac{1-\beta}{\beta^4}\right]^{1/3}$$

thus allowing us to solve for β (since we've calculated ρ_C and we can deduce the value of $\lambda = R_\odot/\xi_0$). The solution is not exactly trivial, and is shown in (51)

$$\frac{1-\beta}{\mu^4\beta^4} = \frac{a\pi}{48} \frac{G^3 M_\odot^2 m_{\mathrm{H}}^4}{k^4(-\xi^2\partial_\xi\phi)_{\xi_0}} = A \tag{51}$$

which you can use also for Sun-like stars. Given what we've found earlier, we can substitute and find $\beta \approx 0.998$, meaning that most of the pressure is due to raw gas pressure (and not from radiation); the value of β coming from evolution models is $\beta \sim 0.999$, which is in great agreement with our present approximated result.

We can also calculate the central pressure $p_C \sim 1.2 \cdot 10^{17}\,\mathrm{dyne/cm^2}$ and the central temperature $T_C \sim 1.6 \cdot 10^7\,\mathrm{K}$.

Similarly to what we've done in (49), we can calculate the mass within a sphere of radius r. The expression is almost identical

$$M = \int_0^r 4\pi r^2 \rho \,\mathrm{d}r = 4\pi\lambda^3\rho_c \int_0^\xi \xi^2\phi^n \,\mathrm{d}\xi = 4\pi\lambda^3\rho_c(-\xi^2\partial_\xi\phi) \tag{52}$$

2.5 HOMOLOGOUS MODELS: SELF-SIMILARITY

Let's start with a simple consideration: Different politropic structures, *with the same politropic index n*, differ only for a scaling factor. This is not particularly difficult to show. Consider

$$\frac{\rho(\xi)}{\rho'(\xi)} = \frac{\rho_C}{\rho'_C}$$

and the same thing holds for the pressure, mass and so on

$$\frac{m(\xi)}{m'(\xi)} = \frac{M}{M'} \qquad \frac{r(\xi)}{r'(\xi)} = \frac{R}{R'}$$

All we have done was writing down the various politropic equations (and what descends from them) and *scale* them, finding out that you can jump from one another by simply rescaling an already existent solution. That is a notable simplification if you think about it.

What are, then, in general the properties a star must have to be self-similar? The answer you can find in the following Table.

Self-Similarity

Suppose to have a quantity dependent on two independent variables $Q(x, y)$. Characteristic scales $Q_0(x)$ and $\delta(x)$ are defined for Q, y respectively. Then the scaled variables are just

$$\xi = \frac{y}{\delta(x)} \quad \tilde{Q}(\xi, x) = \frac{Q(x, \xi)}{Q_0(x)}$$

If the scaled dependent variable is independent of x, i.e.

$$\exists \hat{Q}(\xi) : \tilde{Q}(\xi, x) = \hat{Q}(\xi)$$

then $Q(x, y)$ is self-similar. The interested reader can refer, for example, to [19].

Let's make a few other definitions, starting from $\eta = m(r)/M(R)$

$$\frac{r(\eta)}{r'(\eta)} = z \qquad \frac{P(\eta)}{P'(\eta)} = p$$
$$\frac{T(\eta)}{T'(\eta)} = t \qquad \frac{L(\eta)}{L'(\eta)} = s$$
$$\frac{\rho(\eta)}{\rho'(\eta)} = d$$

Given these definitions, we'll be interested in finding out how these scaling factors depend on the variations in mass and chemical composition. We'll then define

$$\frac{M(\eta)}{M'(\eta)} = x \quad \frac{\mu(\eta)}{\mu'(\eta)} = y$$

Tragically, we have to make another approximation: We have to assume that also the opacity and the energy production rate are self-similar

$$\frac{\epsilon(\eta)}{\epsilon'(\eta)} = \varepsilon \quad \frac{\bar{\kappa}(\eta)}{\bar{\kappa}'(\eta)} = k$$

This is a very restrictive limitation to the possible self-similar stellar structures: We've already seen that for the energy production rate the greater the mass, the greater the temperature and the greater gets the energy production rate, so we can't really compare much different masses. A similar thing holds for the opacity. This works kinda well for stars that have just entered the main sequence, when nuclear reactions haven't really started.

We can write all the structure equations in a self-similar fashion

$$\frac{\mathrm{d}p}{\mathrm{d}\eta} = \frac{\mathrm{d}p}{\mathrm{d}r}\frac{\mathrm{d}r}{\mathrm{d}\eta} = \frac{\mathrm{d}p}{\mathrm{d}r}\frac{M}{4\pi\rho r^2}$$

so that plugging this in the hydrostatic equilibrium condition yields

$$\frac{dp}{d\eta} = -\frac{G}{4\pi}\frac{M^2\eta}{r^4}$$

for both the structures we're considering (*mutatis mutandis*, that is). We can extend this calculation to all the other equations and start using all the self-similar relations we've written above. This way we can write

$$\begin{aligned}
\frac{dp'}{d\eta} &= -\frac{G}{4\pi}\frac{M'^2\eta}{r'^4}\left(\frac{x^2}{z^4 p}\right) \\
\frac{dr'}{d\eta} &= \frac{M'}{4\pi\rho' r'^2}\left(\frac{x}{z^3 d}\right) \\
\frac{dT'}{d\eta} &= -\frac{3}{64a\pi^2 c}\frac{\bar{\kappa}' M' L'(\eta)}{r'^4 T'^3}\left(\frac{kxs}{z^4 t^4}\right) \\
\frac{dL'}{d\eta} &= M'\epsilon'\left(\frac{\varepsilon x}{s}\right)
\end{aligned}$$

but since the stellar structure equations must hold also for the second star (the one with the primed superscript), all the terms within parentheses must be equal to 1!

$$1 = \frac{x^2}{z^4 p} \qquad 1 = \frac{x}{z^3 d} \tag{53}$$

$$1 = \frac{kxs}{z^4 t^4} \qquad 1 = \frac{\varepsilon x}{s} \tag{54}$$

and we also have the equation of state, so we have enough equations to solve this[12]. It is possible to show that the system of equations can be solved with solutions given by

$$\begin{aligned}
R:\ z &= x^{z_1} y^{z_2} \qquad & L:\ s &= x^{s_1} y^{s_2} \\
P:\ p &= x^{p_1} y^{p_2} \qquad & T:\ t &= x^{t_1} y^{t_2}
\end{aligned}$$

thus reducing the equations to equations for the powers of x and y, whose sum must necessarily be equal to zero (by simple means of dimensional analysis). You can read an example of solution right below (or in any book that deals with self-similar solutions for stellar structures)

$$z_1 = \frac{1}{2}(1+A) \quad A = \left[\frac{4\delta(1+a+\lambda\alpha)}{\nu+b-4-\lambda\delta} + 4\alpha - 3\right]^{-1}$$

2.6 CONVECTIVE TRANSPORT

Matter inside the stars is never at rest, but usually the movements of the gas elements are small, random perturbations around their equilibrium positions. Under certain conditions, these small random perturbations can

12 Note that the EoS isn't enough to completely specify and solve the system. We have seven variables and only four equations, meaning we're in the need of three other relations. These can be found in the EoS $\rho \propto p^\alpha T^{-\delta}\mu^{-\phi}$, the energy production rate $\epsilon \propto \rho^\lambda T^\nu$ and the opacity $\bar{\kappa} \propto p^a T^b$.

trigger large-scale motions that involve sizable fractions of the total stellar mass. These large-scale motions are called *convection*. Hot elements may rise to the top, thereby transporting energy from the hottest to the coolest regions, where they cool and then fall down as cold material.

In order to include convective transport in the stellar model computation we'll need to find a criterion for the onset of convection, and the expression of the temperature gradient in a convection zone. Note that the treatment of convection in the stellar interior is extremely complicated and needs the introduction of various approximations. This stems from the fact that the flow of gas in a stellar convective region is turbulent, in the sense that the velocity and all the other properties of the flow vary in a random and chaotic way.

The random nature of turbulence precludes computation based on a complete description of the motion of all the fluid particles based on a solution of the Navier-Stokes equations for the stellar fluid.

In the following we're going to adapt an extremely simple model in which the gas flow is made of gas elements with a certain characteristic size, the same in all dimensions, that move by a certain mean free path before dissolving. All gas elements have the same physical properties at a given distance r from the center. We're going to assume that the "bubble" always remains

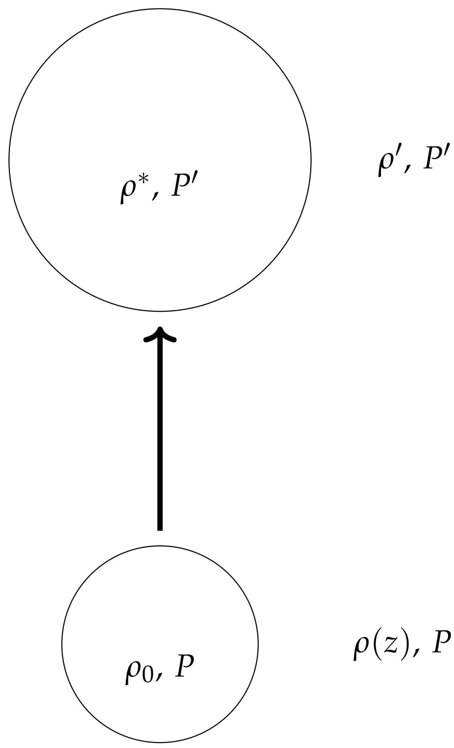

Figure 9: Vertical displacement of a blob of gas in a stratified atmosphere.

in *pressure equilibrium* as it moves, which is a condition that is generally met for subsonic flows for which

$$\tau_d \sim \frac{l}{c_s} \ll \frac{l}{v_{\text{bubble}}}$$

For example, in the Sun's convective region, which extends approximately to 30% of the total solar radius, the sound speed is roughly $2 - 5 \cdot 10^2$ km/s, while the convective velocity is of order of a few meters per second. Note that only in the outskirts of the star the convective velocity gets high enough to give a little trouble to our initial assumption. With some simple computations we are able to show that the time needed to roughly move across the convective region is of the order of a week, while the dynamic time is much smaller. Under these assumptions, convective motions are unlikely to disturb too much the hydrostatic equilibrium condition.

Moreover, we're going to assume that the bubble moves *adiabatically*, so that there's no energy exchange with the surrounding fluid. As a further approximation we'all assume (fort the moment) that the chemical composition of the star μ is constant along the bubble's motion.

With reference to Fig.9, we can expand to first order the temperature of the bubble and that of the surrounding fluid

$$T_b' = T_0 + \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\mathrm{ad}} l \quad T_1 = T_0 + \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\mathrm{rad}} l$$

we'all assume both the temperature gradients to be constant across a small displacement. To achieve the onset of covection, it's necessary that the bubble is *lighter* than the surrounding fluid, so that it will start floating upwards because of buyoancy. This is equivalent to requiring

$$(\rho_{amb} - \rho_{ad})g > 0$$

This condition we can in principle transform in a condition for temperature gradients, but to do so, we're in the need of some equation of state. Before taking a closer look at the various configurations, let us write down a few useful expansions

$$\begin{aligned} T_a &= T_0 + \partial_r T|_{\mathrm{rad}} l \\ T_b &= T_0 + \partial_r T|_{\mathrm{amb}} l \\ \mu_a &= \mu + \partial_r \mu|_{\mathrm{amb}} l \\ \mu_b &= \mu + \partial_r \mu|_{\mathrm{b}} l = \mu \end{aligned}$$

In the last equation we assumed that the chemical composition of the bubble remains constant as it moves[13].

Perfect Gas

Let us assume a perfect gas EoS, so that

$$\frac{\mathrm{d}\rho}{\rho} = \frac{\mathrm{d}p}{p} + \frac{\mathrm{d}T}{T} + \frac{\mathrm{d}\mu}{\mu}$$

If we assume that the chemical composition of the background fluid is constant, we're left with a relation that connects only the variation in temperature and the variation in density, since we've assumed pressure equilibrium along the whole displacement of the bubble. Thus, if we request $\mathrm{d}\rho > 0$

$$-\frac{1}{T}\partial_r T|_{\mathrm{rad}} + \frac{1}{T}\partial_r T|_{\mathrm{b}} > 0$$

13 This clearly is a possibly faulty approximation, because as the bubble moves, instabilities will surely arise, producing mixing between the bubble and the surrounding fluid.

which we can convert into gradients with respect to the pressure simply multiplying by $\mathrm{d}r/\mathrm{d}p$. Taking the absolute values

$$\left|\frac{\mathrm{d}\ln T}{\mathrm{d}\ln p}\right|_{\mathrm{rad}} > \left|\frac{\mathrm{d}\ln T}{\mathrm{d}\ln p}\right|_{\mathrm{ad}} \rightarrow \nabla_{\mathrm{rad}} > \nabla_{\mathrm{ad}} \tag{55}$$

which is known as *Schwarzschild criterion*. We have defined with ∇_x the gradient of temperature with respect to pressure. In the presence of convection we've seen that

$$\left.\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{\mathrm{amb}} = -\frac{3}{4ac}\frac{\bar{\kappa}\rho}{T^3}F_{\mathrm{rad}}(r)$$

but the total flux does not completely overlap with the radiative flux $F_{\mathrm{tot}} = F_{\mathrm{rad}} + F_{\mathrm{conv}}$.

Note that if we relax the perfect gas condition

$$\rho = \rho(p, T, \mu) \rightarrow \alpha = \frac{\partial \ln \rho}{\partial \ln p} \quad \delta = \frac{\partial \ln \rho}{\partial \ln T} \quad \phi = \frac{\partial \ln \rho}{\partial \ln \mu}$$

we obtain a condition like

$$\frac{\mathrm{d}\rho}{\rho} = -\delta\frac{\mathrm{d}T}{T} + \phi\frac{\mathrm{d}\mu}{\mu}$$

which if we assume $\mathrm{d}\mu = 0$ bears just the same result as before.

Perfect Gas with Non-Uniform Chemical Composition

As per our previous discussions, we can write

$$\frac{\Delta\rho}{\rho} = -\frac{(T_{\mathrm{amb}} - T_{\mathrm{b}})}{T} + \frac{(\mu_{\mathrm{amb}} - \mu_{\mathrm{b}})}{\mu} > 0$$

which in terms of the gradients we've defined earlier becomes

$$\left|\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{\mathrm{rad}} > \left|\frac{\mathrm{d}T}{\mathrm{d}r}\right|_{\mathrm{amb}} + \frac{T}{\mu}\left|\frac{\mathrm{d}\mu}{\mathrm{d}r}\right|_{\mathrm{amb}}$$

Now we can make the same trick of multiplying for the spatial pressure gradient so to obtain

$$\nabla_{\mathrm{rad}} > \nabla_{\mathrm{ad}} + \nabla_{\mu} \tag{56}$$

which is called *Ledoux Criterion*. Note that in this case it's much harder to prompt the onset of convection (the radiative gradient must be greater than that predicted by the Schwarzschild criterion). I think it won't surprise you that we can generalize this condition to a non-perfect gas, thus finding

$$\nabla_{\mathrm{rad}} > \nabla_{\mathrm{ad}} + \frac{\phi}{\mu}\nabla_{\mu}$$

2.6.1 The Overshooting Problem

The *overshooting problem* can be quite a nuisance if we want to calculate the spatial extension of the convective region of the star. To put it simply, in the convective region, the blob of gas will rise with non-zero velocity which, in turn, won't be magically disappearing as soon as it crosses the non-convective region. This causes the convective region to formally "expand" outwards. This can potentially impact significantly on the permanency of a star in the Main Sequence stage. If we want to address the increasing in size of the convective region, it is customary to introduce a new parameter β_{ov} so that

$$l_{ov} = \beta_{ov} H_p$$

where H_p is the scale height of the pressure profile.

To find the ambient temperature gradient, we're in the need of recalling the 1st principle of thermodynamic

$$\delta Q = c_p \rho V (T_0' - T_1) = C_p \Delta T \tag{57}$$

The temperature difference can be expressed in terms of derivatives with respect to pressure

$$\Delta T = \left(\frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\mathrm{amb}} - \frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\mathrm{ad}} \right) \Delta p$$

where we can define the *superadiabaticity* δ_{ad}

$$\delta_{\mathrm{ad}} = \left(\frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\mathrm{amb}} - \frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\mathrm{ad}} \right) \tag{58}$$

Since the thermal capacity is very large in the stellar interiors, we can assume the superadiabaticity to be vanishing

$$\frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\mathrm{amb}} \sim \frac{\mathrm{d}T}{\mathrm{d}p}\bigg|_{\mathrm{ad}}$$

Note that in the stellar external envelopes, we can't assume the same principle to hold. But the problem is partially mitigated by the fact that in the most external layers there's roughly no energy production. This means that convection will influence the effective temperature, but not the luminosity.

To find the radiative gradient, we first have to find F_{rad} and, consequently, F_{conv}. In order to do so, we're going to implement somewhat "empirical" models, that tells us that $F_{\mathrm{conv}} \propto l$. The mean free path l we'll parametrize as

$$l = \alpha_{ML} H_p$$

where α_{ML} is the *mixing length parameter*. We're not assured that such parameter is the same in different stars, or even in the same star but in different evolutionary stages.

We know that in order for convection to settle

$$\nabla_{\mathrm{rad}} > \nabla_{\mathrm{ad}}$$

and so that energy can be transported

$$\nabla_{\mathrm{amb}} \geq \nabla_{\mathrm{ad}}$$

In short

$$\boxed{\nabla_{\mathrm{ad}} \lesssim \nabla_{\mathrm{amb}} < \nabla_{\mathrm{rad}}}$$

This tells us that the adiabatic gradient is the maximum gradient that can set into a given structure (assuming $\delta_{\mathrm{ad}} \approx 0$). What we still have to do is find the convective gradient in the external regions.

2.6.2 Mixing Length Theory

In order to find what we're looking for (an approximate description of the convective gradient) we're going to need some simplifying assumptions:

1. The theory behind mixing length is going to be *local*, so that this criterion can be applied on a "mesh by mesh" basis to check its stability;
2. The mean free path of the bubble we can parametrize as $l = \alpha_{ML} H_p$;
3. Small variations in temperature and density (with respect to the surrounding environment);
4. The variations induced by different shapes and sizes of the bubbles are negligible (this is usually not true, but their contribution enters as a second order correction).

First and foremost, let us define the *pressure scale height* H_p as the distance the bubble has to travel so that pressure drops to e^{-1} of its original value, in formulas

$$\frac{1}{H_p} = -\frac{1}{p}\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{\mathrm{d}\ln p}{\mathrm{d}r}$$

If we assume that all contribution to the pressure comes from the gas (and so we neglect other contributions, like radiation pressure), we can plug in the equation for hydrostatic equilibrium and find

$$H_p = \frac{p}{g\rho}$$

where g is the local acceleration of gravity.

Recall when we have first introduced the 1st Principle of Thermodynamics (57). We assumed that the temperature difference was proportional to the difference between the adiabatic gradient and the ambient temperature gradient. It's clear that if the two gradients were the same, no heat would be released and no energy transport is possible. As we've mentioned before, when convection involves layers closer to the stellar surface, we cannot make the the simplifications we've used in last section.

In this case we need to find a more complex expression for ∇_{conv}. We start by enforcing the condition that at a given layer within the convective region the total energy flux is simply the sum of the flux carried by radiation and convection

$$F_{\text{tot}} = F_{\text{rad}} + F_{\text{conv}} \qquad F_{\text{rad}} = -\frac{4acT^3}{3\bar{\kappa}\rho}\frac{\mathrm{d}T}{\mathrm{d}r} \tag{59}$$

where $\mathrm{d}T/\mathrm{d}r$ is the the actual temperature gradient of the convecting region. So far so good, there's nothing new here.

Taken that we refer with the subscript "u" to rising matter and "d" to downwards moving matter, we can write an expression for the amount of stuff that is either rising or descending

$$\rho_u \sigma_u v_u \qquad \rho_d \sigma_d v_d$$

where σ is the surface area of the convection column. The net transport of energy will then be

$$2F_{\text{conv}} = \rho_u v_u c_p T_u + \frac{1}{2}\rho_u v_u v_u^2 - \left[\rho_d v_d c_p T_d + \frac{1}{2}\rho_d v_d v_d^2\right]$$

Note that in order to preserve hydrostatic equilibrium, the amount of falling material must be the same as the rising one, otherwise the star would either dissolve or concentrate all the mass in the interior. To avoid this, the only solution is that after a distance l of order H_p the bubble stops and invert its motion. This reflects in requiring

$$\rho_u \sigma_u v_u = \rho_d \sigma_d v_d$$

so that if we assume little differences in density and equal cross-sections for the convective columns, the condition simply becomes $v_d \sim v_u = v$.

By neglecting the energy losses during the bubble's displacement (adiabatic assumption), the convective flux transported by bubbles that move *upwards* with a velocity v is then given by

$$F_{\text{conv}} = \frac{1}{2}\rho v c_p \left[\left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\text{ad}} - \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)\right] l$$

The factor $1/2$ accounts for the fact that at each layer approximately half of the matter is rising and half is moving downwards. We've already expanded T_u and T_d. Assuming an "average" temperature T_{amb} between the temperatures of the two columns, we can write

$$T_u - T_d = 2\Delta T = 2(T_u - T_{\text{amb}}) \rightarrow \Delta T(s) = \left[\left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\text{ad}} - \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)\right] s$$

where s is an average displacement which we can conveniently take as $s = l/2$, so that everything is consistent with the equation presented above. Performing the usual trick, we can transform all the gradient in logarithmic gradients (∇), so that

$$F_{\text{conv}} = \frac{1}{2}\rho v c_p \frac{T}{p}\frac{\mathrm{d}p}{\mathrm{d}z}\left[\nabla_{\text{amb}} - \nabla_{\text{ad}}\right] l$$

where $\mathrm{d}z = -\mathrm{d}r$ is the depth. We can substitute $\mathrm{d}p/\mathrm{d}z$ from the equation of hydrostatic equilibrium obtaining the following expression that is all nice and dandy (*said Lying Odysseus*)

$$\nabla_{\text{amb}} = \frac{F_{\text{tot}} + \frac{l}{2}\rho v c_p \frac{T}{p}\frac{\mathrm{d}p}{\mathrm{d}z}\nabla_{\text{ad}}}{\frac{l}{2}\rho v c_p \frac{T}{p}\frac{\mathrm{d}p}{\mathrm{d}z} + \frac{T^3}{\bar{\kappa}\rho}\frac{4ac}{3}\frac{T}{p}\frac{\mathrm{d}p}{\mathrm{d}z}} \tag{60}$$

where we've used the decomposition of the total flux and the expression for the radiative flux (59).

What we are still lacking is a prescription for the velocity of the convective elements, which, however, we can conveniently obtain from the kinetic energy theorem ("work-energy principle")[14]: The net force per unit volume

14 The interested reader can refer to either [9] or [21] for a more detailed discussion.

f_r acting on the convective element at the layer r is given by $f_r = -g\Delta\rho(r) = -g\rho\Delta T/T$, meaning the work done per unit volume in moving the bubble through a distance Δr is

$$W(\Delta r) = -\left(\frac{1}{2}\right) g\Delta\rho(\Delta r)\Delta r$$

which we can average over a distance $l/2$. Moreover, it is customary to express $\langle W(\Delta r)\rangle$ in terms of the average speed of the convective elements as

$$\langle W(\Delta r)\rangle = \frac{1}{2}\rho v^2$$

Equating the two expressions we finally obtain

$$\boxed{v^2 = \left(\frac{1}{2}\right)\frac{g}{T}\left(\frac{l}{2}\right)^2\left[\left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{\mathrm{amb}} - \left(\frac{\mathrm{d}T}{\mathrm{d}r}\right)_{ad}\right]} \tag{61}$$

Note that the factor 1/2 within parentheses was introduced by [21] when performing the integral of the force per unit volume. It is still not clear to me where that comes from, but I think it may be to address for some sort of average value for the density.

2.7 OPACITY

As we might have mentioned somewhere in the above, the physical property known as *opacity* is the collection of all the mechanisms of interaction between radiation and matter. A few important mechanisms are listed below:

- Scattering $e^-\gamma$: Since the average energy of particles in a star is generally such that $kT \ll m_e c^2 \approx 0.511\,\mathrm{MeV}$, the prevalent scattering channel is the *Thomson scattering*;
- *Inverse Bremsstrahlung*: also known as "braking radiation" is a channel of interaction that leaves an unbound particle unbound (Free-Free);
- *Photoionization*: it's a type of interaction between a photon and a bound electron, that gets freed from a given atom. So, it's a type of Bound-Free interaction;
- *Level Jump*: radiative excitation/de-excitation of an electron, that is made jump from an energetic level to another. In this sense, this is a Bound-Bound interaction.

It won't come as a surprise that the correct calculation of opacity is made all the more difficult when we're not assured a complete-ionization scenario or if we can't exclude the presence of dust grains (or grains in general) in the stellar structure.

In general, the total opacity is the sum of all the contributions from the various channels which, of course, will also be frequency dependent

$$\kappa_\nu = (\kappa_{bb} + \kappa_{bf} + \kappa_{bf})\left[1 - \exp\left(-\frac{h\nu}{kT}\right)\right] + \kappa_{sc} \tag{62}$$

If you recall, in our stellar structure equations enters an opacity averaged over all frequencies, that is generally known as *Rosseland Mean Opacity*. A possible way to obtain it is presented at the end of Appendix A (156).

We're going to dedicate the next few sections to obtaining analitical expressions for the most common sources of opacity, under the assumption of being the dominating contribution.

2.7.1 H^- opacity

The opacity due to Bound-Free or Free-Free interactions with the H^- ion is dominant, for example, in the solar photosphere. In order for this channel to grow relevant, it's clear that we're going to need plenty of Hydrogen and free electrons (which will be typically coming from ionized metals).

If we want to exclude the possibility of other sources of opacity being more relevant, we'll have to work with temperatures in the range $3000 - 6000\,\mathrm{K}$ and with a metallicity Z in the range $10^{-3} - 3 \cdot 10^{-3}$. Similarly, we have to require a low density $10^{-10} < \rho < 10^{-5}\,\mathrm{g/cm^3}$ to avoid what's called *collision induced absorption*, which becomes more and more relevant at low temperatures and high densities, as can happen in main sequence stars with masses $M \leq 0.5 M_\odot$ or in the atmosphere of some white dwarves.

It is possible to show that the opacity due to interactions with the H^- ion is given by

$$\bar{\kappa}_{\mathrm{H}^-} \sim 2.5 \cdot 10^{-31} \left(\frac{Z}{0.02}\right) \rho^{1/2} T^9\,\mathrm{cm^2 g^{-1}} \tag{63}$$

while for the collision induced absorption

$$\bar{\kappa}_{\mathrm{CIA}} = k_2 \rho^2 \tag{64}$$

where k_2 is a constant.

Note that if temperature drops particularly low ($T \lesssim 1400 - 1800\,\mathrm{K}$) we can observe the formation of *grains*, which are another important source of opacity.

2.7.2 Thomson Scattering

We know that in your typical star temperature is such that

$$kT \ll m_e c^2 \approx 0.511\,\mathrm{MeV}$$

We have formerly defined the opacity as

$$\rho \kappa_\nu = n \sigma_\nu \tag{65}$$

where n is the numeric density of the particles that interact with photons. Particle physics (or your Physics III classes, if you prefer) tells us that scattering Thomson is a particular type of *isotropic* and *frequency independent* scattering channel, which is automatically translated into the expression for its cross-section

$$\sigma_T = \frac{8\pi}{3} \left(\frac{e^2}{m_e c^2}\right)^2 = 0.66\,\mathrm{b} \tag{66}$$

where the b stands for *Barns* ($10^{-24}\,\mathrm{cm}^2$) a unit that is particularly used in particle physics. The opacitive channel of scattering Thomson benefits from high temperatures (more electrons since matter's more likely to be ionized) and not too high densities (other channels become dominant). As a rule of thumb, you might want to remember that the more dominant opacitive channels are the level jump, closely followed by photoionization, and then the others.

A while back ago, we've defined the number density of electrons in the case of total ionization (32)

$$n_e = \frac{\rho}{m_\mathrm{H}} \sum_{i=1}^{N} \frac{Z_i X_i}{A_i} \approx \frac{\rho}{m_\mathrm{H}} \left(X + \frac{Y}{2} + \frac{Z}{2} \right) = \frac{\rho}{m_\mathrm{H}} \left(\frac{X+1}{2} \right) \tag{67}$$

where the last equality was obtained exploiting the condition $X + Y + Z = 1$.

We can therefore plug (67) into (65), obtaining

$$\bar{\kappa}_T = \kappa_T = \frac{8\pi}{3} \left(\frac{e^2}{m_e c^2} \right)^2 \frac{X+1}{2m_\mathrm{H}} \tag{68}$$

Note that in general, for higher temperatures ($T > 10^8\,\mathrm{K}$), the Compton scattering probability grows non-negligible. Compton scattering is an anisotropic, anelastic scattering mechanism

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta) \tag{69}$$

Since the effect produced by Compton scattering is small (but sometimes not negligible), people used to correct the Thomson scattering cross-section as such

$$\sigma_{T,\,\mathrm{corr}} = \left[1 + 0.35 \left(\frac{T}{10^9\,\mathrm{K}} \right)^{1/2} + 0.73 \left(\frac{T}{10^9\,\mathrm{K}} \right) \right]^{-1} \sigma_T$$

2.7.3 Inverse Bremsstrahlung

Another channel that is particularly important when BB and BF channels are quiescent[15], is that of inverse Bremsstrahlung[16]. Particle physicists usually manage to convince us that the *differential cross-section* (read: velocity dependent) for inverse Bremsstrahlung is described by

$$\frac{\mathrm{d}\sigma_{FF}}{\mathrm{d}v_e} = \frac{4\pi}{3\sqrt{3}hc} \frac{Z^2 e^6}{m_e^2} \frac{1}{v_e} \frac{1}{\nu^3} n_e f(v_e) \bar{g}_{ff}(\nu, v_e) \tag{70}$$

where here Z is the electrical charge of the ion interacting with electrons, $f(v_e)$ is some velocity distribution for electrons (typically a Maxwellian distribution) and $\bar{g}_{ff}$ is the *Gaunt factor*, a fancy numerical coefficient that we will approximate as roughly constant and of order unity in the following. Note that since there isn't an energy threshold for this opacity channel, in principle we have to take into account all possible species that might be contributing.

15 Oh my, what a fancy word I've found.
16 Remember the capital "B" and the two "s", it's a German word (well, two, actually).

If we take an average value for the velocity

$$\langle\frac{1}{v_e}\rangle = \left(\frac{2m_e}{\pi kT}\right)^{1/2} \propto \left(\frac{m_e}{T}\right)^{1/2}$$

then we can consider an average cross section

$$\langle\sigma_{FF}\rangle = \text{const}\cdot\frac{Z^2 n_e}{(kT)^{1/2}}\frac{\bar{g}_{ff}}{\nu^3}$$

Recalling the definition of opacity (65)

$$\begin{aligned}\kappa_{\nu,ff}\rho &= \sum_{A=1}^{N} n(A, Z_A)\langle\sigma_{FF}\rangle \\ &= \sum\frac{X_A\rho}{Am_H}\langle\sigma_{FF}\rangle\end{aligned}$$

To the cross section only the more abundant species will participate (so Hydrogen and Helium). Thus

$$\kappa_{\nu,FF} \sim \frac{X}{m_{\mathrm{H}}}\langle\sigma_{FF}(1,\nu)\rangle + \frac{Y}{4m_+}\langle\sigma_{FF}(2,\nu)\rangle$$

If we assume completely ionized matter (67) and notice that

$$\sigma \propto T^{-1/2}\nu^{-3} \to T^{-3.5}$$

then after computing the Rosseland mean opacity, we're left with

$$\bar{\kappa}_{FF} \sim 4\cdot 10^{22}(X+Y)(1+X)\rho T^{-3.5} \tag{71}$$

2.7.4 Photoionization

Not surprisingly, for photoionization mechanisms to arise, it's necessary the presence of atoms with at least one bounded electron. Differently from the channels we've taken a look so far, photoionization is a threshold process. For it to happen, photons must have an energy at least equal to the ionization potential of the impinged atom. In general

$$h\nu = \chi_n + \frac{p_e^2}{2m_e}$$

with

$$\chi_n = \frac{2\pi^2 m_e e^4}{hn^2}(Z')^2$$

where Z' is the effective charge of the ion. Unless we're particularly happy of doing a lousy job, we should, in principle, consider all relevant energy levels of the ions, the ionization degree of the species and the number of species available. In short: There are many things you might want to keep track of.

What you would find out, were you actually to do the computations (and do not make any mistakes in the process), is that the photoionization channel has the same density and temperature dependencies as the Inverse

Bremsstrahlung one, but with a greater multiplicative coefficient, thus making it usually more relevant.

Assuming a prevalence of hydrogenoid atoms[17], it is possible to prove that the cross section for photoionization is given by

$$\sigma_{bf} = \frac{64\pi^4 m_e e^{10} (Z')^4 g}{3\sqrt{3} c h^6 n^5 \nu^3} \tag{72}$$

where all symbols have their customary meaning. Said N_A the numeric density of the atomic species A

$$N_A = \frac{X_A \rho}{A m_{\mathrm{H}}}$$

we can define a numeric density expressing the average number of electrons for each atom A which are bounded to an energy state n. This quantity, which we're calling $N_{A,n}$ is given by

$$N_{A,n} = \frac{n_e n^2 h^3 \exp(\chi_n / kT)}{2(2\pi m_e kT)^{3/2}}$$

Given these definitions, we can write down the expression for the opacity

$$\kappa_{bf}(\nu)\rho = \sum_{A,n} \sigma_{bf}(\nu) \frac{X_A \rho}{A m_{\mathrm{H}}} N_{A,n}$$

in which we're going to consider only elements with $A \geq 6$, for Hydrogen and Helium are completely ionized at the temperatures we're considering and are thus unable to participate to the opacity. So, we can write the numeric density of electrons as

$$N_e \sim \frac{\rho}{m_{\mathrm{H}}} \left(\frac{X+1}{2} \right)$$

so that plugging in all the various definitions we obtain

$$\kappa_{bf}(\nu) = \sum_{A,n} \frac{4\pi^2 e^2 g}{3\sqrt{3} c \nu^3} \left(\frac{\chi_n^2}{m_e} \right) \frac{kT}{kT} \frac{X_A \rho (1+X)}{A m_{\mathrm{H}}^2} \frac{nh \exp(\chi_n / kT)}{(2\pi m_e kT)^{3/2}}$$

where we've used the definition of the ionization potential χ_n to eliminate the effective ion charge. Summing over A gives us the metallicity Z, thus leaving us with

$$\kappa_{bf}(\nu) = \sum_{n} \frac{4\pi^2 e^2 g}{3\sqrt{3} c \nu^3} \left(\frac{\chi_n^2}{m_e} \right) \frac{kT}{kT} \frac{Z \rho (1+X)}{A m_{\mathrm{H}}^2} \frac{nh \exp(\chi_n / kT)}{(2\pi m_e kT)^{3/2}}$$

If we consider only the energy level $n = 1$ we may find that the opacity, leaving behind all numerical factors, is a function of

$$\kappa_{bf}(\nu) \propto \left[\frac{\chi_n}{kT} \exp\left(\chi_n / kT\right) \left(\frac{kT}{h\nu} \right)^3 \right] Z(1+X)\rho T^{-3.5}$$

17 That is atoms of given Z that through processes we're not even interested in have been left with just one electron.

and the term within squared parentheses is approximately of order 1, since terms with $h\nu \sim kT$ will be weighed more when performing the Rosseland mean. If we restore all numerical factors and perform the aforementioned Rosseland mean, you'd find

$$\bar{\kappa}_{bf} \sim 4 \cdot 10^{25}\, Z(1+X)\rho T^{-3.5} \tag{73}$$

which, as anticipated, has the same temperature and density dependencies as the free-free contribution.

Level Jump

With *level jump* we refer to the absorption of radiation by an electron bound to an ion that causes the electron to move from one bound state to a more energetic one. After the transition, the electron will have to move down to the original bound state, for the ion has to return in equilibrium with the surroundings. The photon emitted during this de-excitation will have the same energy of the absorbed one, but it is re-emitted in a random direction.

This process necessarily involves photons of certain frequencies, corresponding to the energy differences between the initial and final bound state of the electron. Bound-bound processes can become important contributors to the total opacity for temperatures below $\sim 10^6\,$K.

Since a formal discussion of bound-bound opacitive mechanisms would have to make use of Einstein's coefficients (and we would end up not finding a nice analitical solution, but rather statistics for the process) we'll skip right to the next opacitive channel.

2.7.5 Electron Conduction

As we might expect from its name, electron conduction is a *diffusive process*, in the sense that it can be described by a diffusive equation of the form

$$\partial_t n = D_{\text{cond}} \partial_r^2 n \quad D_{\text{cond}} = \frac{\lambda_e \langle v_e \rangle}{3}$$

If electrons are non-degenerate, the cross section for conductive mechanisms is much larger than, say, the photoionization cross section. In fact, the mean free path of electrons is much smaller than that of photons

$$\lambda_{\text{cond}} \propto \frac{1}{n_i \sigma_{\text{cond}}} \ll \lambda_\gamma$$

But things change rather dramatically if the gas becomes degenerate: The cross section for the process decreases and thus the mean free path for electrons raises considerably. For the purposes of our discussion, let us recall the Fermi momentum

$$p_F = \left(\frac{3 n_e h^3}{8\pi}\right)^{1/3} \qquad n_e = \frac{\rho}{\mu_e m_{\text{H}}} \tag{74}$$

If diffusion is isotropic, we can assume the generalized Fick law to hold, and so

$$F(r) = -\frac{1}{3}\langle v_e \rangle \lambda_e \frac{\mathrm{d}U(r)}{\mathrm{d}r} = -\frac{1}{3}\langle v_e \rangle \lambda_e \frac{\mathrm{d}U}{\mathrm{d}T}\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{1}{3}\langle v_e \rangle \lambda_e c_V \frac{\mathrm{d}T}{\mathrm{d}r}\bigg|_{\text{amb}} \tag{75}$$

where we've introduced c_V the constant volume specific heat. We'll call

$$k_{\text{cond}} = \frac{1}{3}\langle v_e \rangle \lambda_e c_V \tag{76}$$

the thermal conduction coefficient. To proceed further we have to determine the Coulombian cross section

$$m_e v_e^2 \approx \frac{Z_C e^2}{b}$$

where b is the impact parameter. Inverting last equation and defining a conduction cross section $\sigma_{\text{cond}} \approx \pi b^2$ yields

$$\sigma_{\text{cond}} \propto \frac{\langle Z_C^2 \rangle}{\langle v_e^4 \rangle}$$

Since the gas is degenerate, the electrons that will have the larger contribution are those with momentum approximately equal to the Fermi momentum (74), and thus

$$v_e \approx \frac{p_F}{m_e} \propto \left(\frac{\rho_e}{\mu_e}\right)^{1/3}$$

hence

$$\sigma_{\text{cond}} \propto \left(\frac{\mu_e}{\rho_e}\right)^{4/3}$$

Now we can finally calculate the mean electronic free path

$$\lambda_e = \frac{1}{n_i \sigma_{\text{cond}}} = \frac{\mu_i m_{\text{H}}}{\rho} \left(\frac{\rho_e}{\mu_e}\right)^{4/3} \frac{1}{\langle Z_C^2 \rangle}$$

Moreover, we're in the need of an expression for the specific heat

$$c_V = \frac{8\pi^3 m_e k^2 T p_F}{3h^3} \propto \left(\frac{\rho_e}{\mu_e}\right)^{1/3} T$$

At last we've finally found an expression for the thermal conduction coefficient

$$k_{\text{cond}} \propto \rho \frac{\mu_i}{\mu_e^2} \frac{T}{\langle Z_C^2 \rangle}$$

From this we can then define an opacity coefficient for thermal electronic conduction

$$\kappa_{\text{cond}} := \frac{4acT^3}{3k_{\text{cond}}\rho} \approx 4 \cdot 10^{-8} \left(\frac{\mu_e^2}{\mu_i}\right) \left(\frac{T}{\rho}\right)^2 \langle Z_C^2 \rangle \tag{77}$$

Note that with this we can define a *total transported flux*

$$F_{\text{tot}} = F_{\text{cond}} + F_{\text{rad}} = -\frac{4acT^3}{3\rho}\left(\frac{1}{\bar{\kappa}} + \frac{1}{\bar{\kappa}_{\text{cond}}}\right) = -\frac{4acT^3}{3\rho\bar{\kappa}_{\text{tot}}}$$

The opacitive coefficient within parentheses is the one that is customarily found in professional opacitive tables.

3 STELLAR MODELS

3.1 INTRODUCTION

In the last chapter we've managed to write down the four (five, if you consider the EoS) equations through which we're able to describe stellar structures. For the sake of completeness (and not to make you flip your pages back and forth), I'll write them all down below

$$\begin{aligned}
\frac{dM_r}{dr} &= 4\pi r^2 \rho(r) \\
\frac{dP_r}{dr} &= -\frac{GM_r}{r^2} \\
F(r) &= -\frac{16}{3}\frac{\sigma_B T^3}{\rho\bar{\kappa}}\frac{dT}{dr} \\
\frac{dL(r)}{dr} &= 4\pi r^2 \rho(r)\epsilon(r) \\
p &= p(\rho, T, X, Y, X_i)
\end{aligned}$$

Those who are used to working with stellar structure equations are often solving them (numerically, of course) using different independent variables in the stead of the radial distance r.

For example, in the inner regions of stars, where the mass isn't saturated, we may use the mass M_r and express all the derivatives consequently. When the mass is saturated we have to switch to another variable, for example the pressure p (the associated region is the *sub-atmosphere*). As we move to the outskirts of the star, all relevant quantities will be saturated to their boundary value, and we have to switch to the *optical depth* to parametrize the stellar structure equations (this is the case for the *photosphere*).

The interested reader may refer to [8], §3.3, to have an idea of how the boundary conditions are chosen and linked together.

Let us now take a closer look at how the deed is done.

3.2 FITTING METHOD FOR STELLAR STRUCTURES INTEGRATION

In principle, you could use an atmosphere model to describe more or less accurately the atmosphere of a star. However, it may also happen that you *don't want* to use an atmosphere model but rather use an approximate model. In the present section, we're going to discuss a little bit of how this can be done. Clearly, these rough calculation will hold, if approximately, only in small main sequence stars, which have "simpler" atmospheres.

The procedure here described follows closely the discussion by [21].

Consider the equation for hydrostatic equilibrium. In the atmosphere, we're going to use as independent variable the optical depth τ, so that

$$\frac{\mathrm{d}p}{\mathrm{d}r} = \frac{\mathrm{d}p}{\mathrm{d}r} \cdot \frac{\mathrm{d}\tau}{\mathrm{d}r} = -\frac{GM_r\rho}{r^2}$$

But recall the definition of optical depth (135) and absorption coefficient (6), so that

$$\frac{\mathrm{d}\tau}{\mathrm{d}r} = (-)\rho\bar{\kappa}$$

with $\bar{\kappa}$ a suitably averaged opacity coefficient. Since in the atmosphere the mass of the star is saturated $M_r = M_{\mathrm{tot}}$ and so is the radius, the equation for hydrostatic equilibrium becomes

$$\frac{\mathrm{d}p}{\mathrm{d}\tau} = \frac{g}{\bar{\kappa}}$$

where $g = GM_{\mathrm{tot}}/R_*^2$ is the surface gravity of the star. What about the other equations? The mass is saturated, so the 1st equation is irrelevant; for the same reason, also the 4th equation is irrelevant, for there is no energy production in the atmosphere[1].

So we're left with only the equation for energy transport and the EoS. Note that, in general, we cannot write an expression for radiative transfer as in (134), because we're not assured LTE to hold.

So what we're going to do is writing down an approximate relation for the temperature, that is going to depend on the optical depth and the effective temperature

$$T = T(\tau, T_{\mathrm{eff}})$$

One possible relation is the celebrated equation for the *grey atmosphere*

$$T(\tau) = \frac{3}{4}T_{\mathrm{eff}}^4\left(\tau + \frac{2}{3}\right) \tag{78}$$

which assumes LTE to hold and negligible frequency-dependence on the absorption coefficient (and thus for the opacity and optical depth as well). According to this equation, the photosphere, characterized by $T = T_{\mathrm{eff}}$, is to be found at an optical depth $\tau = 2/3$.

This approximation is sometimes used when we're interested in observing differential effects and we can be a little loose on having an accurate estimate of the effective temperature.

Another expression, that is still quite in use, is that found by [15], which is essentially a grey atmosphere modified so to include the absorption of some elements

$$T^4(\tau) = \frac{3}{4}T_{\mathrm{eff}}^4\left(\tau - 1.017 - 0.30e^{-2.54\tau} - 0.291e^{-30\tau}\right) \tag{79}$$

where the coefficients are the product of a fitting procedure.

It's clear that those who create hydrodynamic atmosphere models try to take into account every possible phenomenon that may occur (and be relevant) in the atmosphere, such as turbulence. Technically, especially in larger

1 Note that although in the atmosphere there is no energy produced by nuclear reaction, we may still have to consider the contribution of energy generation by the means of gravitational collapse. However, densities are very low in the atmosphere, making ϵ_g negligible.

stars with convective envelopes, it would be sensible to consider the contribution of *turbulent pressure* to the total balance

$$p = p_{\text{gas}} + p_{\text{rad}} + p_{\text{turb}}$$

As you should know if you've attended the Fluid Dynamics or the Astrophysical Processes course, turbulence is one hell of a beast to deal with. To give you an idea of how much complex it is to even build a half-functioning model of turbulence, here's Werner Heisenberg's take on the subject[2]:

> "*When I meet God, I am going to ask him two questions: why relativity? And why turbulence? I really believe he will have an answer for the first*."

If we really have to include turbulence in the picture, what's often done is invoking the Boussinesq's approximation and write a barotropic expression for the turbulent pressure

$$p_{\text{turb}} = \rho \bar{v}^2_{\text{turb}} \quad \bar{v}^2_{\text{turb}} = \beta c_s^2$$

where $\beta \lesssim 1$ is a free parameter that has to be estimated to find an accord. But is this worth the trouble? Probably not.

On top of not knowing how to deal with turbulence, we are not able, as yet, to give a precise description of the ambient temperature gradient in the atmosphere, which already enters our equations with a free parameter.

If you really want to make a meticulous job, don't bother stuffing your equations with all this stuff and get for yourselves a three-dimensional atmosphere model.

3.2.1 Starting the Integration

Our external boundary conditions are the star luminosity L_* and the effective temperature T_{eff}. Note, however, that the former doesn't help us much at this stage, since it still doesn't enter our equations in the atmosphere.

Let's assume we're starting with a very small optical depth $\tau = \tau_0 \ll 1$, so that the pressure (and the opacity) is small $p = \epsilon \ll 1$. So, assuming we're given an expression of the form $T = T(\tau, T_{\text{eff}})$, we can calculate T plugging in our values for $\tau = \tau_0$ and T_{eff}.

From the EoS, since we already knew $p = \epsilon$ and T we've just calculated, we obtain the density ρ. At this point, using all of this and the chemical composition, we can calculate the mean opacity $\bar{\kappa}$ and put everything in the equation of hydrostatic equilibrium and check if our solution is self-consistent. If not, we re-iterate the process (changing our starting values for τ and p) until the solution satisfies hydrostatic equilibrium within a given precision.

Now that we have all our variables determined on the external boundary, we can start integrating towards the inner regions. To do so, derivatives in our equations become incremental ratios and we solve mesh after mesh until we reach the base of the atmosphere (the boundary with the sub-atmosphere).

2 Yes, this is the same Werner Heisenberg that has laid the foundations of Quantum Mechanics.

Integrating through the sub-atmosphere, we have to joint the values of the n-th mesh of the region with the most external one of the stellar interior. So we'll have to match p_N, ρ_N, T_N, L_N and R_N.

As anticipated, in the stellar interior we're going to be integrating with respect to the mass M_r, and we'll be using as boundary conditions at the center of the star ρ_C, p_C, $L_r \approx 0$, $M_r \approx 0$.

Consider the equivalent formulation for the equation of hydrostatic equilibrium

$$\frac{\mathrm{d}p}{\mathrm{d}M_r} = -\frac{GM_r}{4\pi r^4}$$

In a given mesh we know the value of the right hand side of the equation, meaning we also know the value of the derivative. We can therefore linearize the relation for the pressure and (moving outwards) find

$$p(M_r + \Delta M_r) = p(M_r) + \frac{\mathrm{d}p}{\mathrm{d}M_r}\Delta M_r$$

As mentioned above, the derivative is known from the hydrostatic equilibrium. In principle we're not assured that the variation is exactly linear, so we're going to inevitably incurr into errors. What is usually done (automatically) is halving the mass step and check how the solution changes, eventually stopping when the variation gets under a given threshold.

Typically, the integration stops not in the center of the star (which would be non-sensical, computationally and physically speaking), but rather in a mesh nearby the center called *fitting mesh*. Once this special mesh is defined, we start integrating outwards from the center of the star with the aforementioned boundary conditions, stopping only when the fitting mesh is reached.

This leaves us with two sets of values at the fitting mesh, one obtained from fitting from the exterior inwards and one from the inside outwards. Clearly, there's no reason these two sets must turn out to be equal (computationally speaking), thus obtaining differences

$$X_i - X_e \neq 0$$

where X_i and X_e represent some physical property of the star as computed from, respectively, integrating outwards and inwards. Physically speaking, however, since the values are describing the same mesh, they must be coinciding.

To decide whether or not the difference is acceptable, we can set an upper bound to the maximum relative difference we're accepting

$$\frac{X_i - X_e}{X_i} \leq 10^{-4} - 10^{-3}$$

If that condition is not met, we change both the inner and outer boundary conditions and re-iterate the process, thus modifying the two sets of solution at the fitting mesh. The procedure will then be stopping when the difference between the solutions is negligible.

Moreover, we can check the dependence of our solutions from the boundary conditions we've chosen, calculating partial derivatives. For example, we may want to check the dependence of the inner solution for the pressure on the central temperature

$$\frac{\partial p_i}{\partial T_C} \sim \frac{p_i|_{T_C+\delta T_C,\, p_C} - p_i|_{T_C,\, p_C}}{T_C}$$

so that the difference between the inner pressure solution and the external pressure solution is

$$\Delta(p_i - p_e) = \left(\frac{\partial p_i}{\partial T_C}\right)\Delta T_C + \left(\frac{\partial p_i}{\partial p_c}\right)\Delta p_C - \left(\frac{\partial p_e}{\partial L}\right)\Delta L - \left(\frac{\partial p_e}{\partial T_{\text{eff}}}\right)\Delta T_{\text{eff}}$$

This variation must be able to cancel out the difference that we originally had in the fitting mesh. We're going to write similar expressions for all our variables. Sooner or later the procedure will hopefully converge and we'll find out the right boundary solutions.

The procedure we've just described is able to solve stellar structure equations *but* is not able to follow the "fine structure" of thinner structures within the star, for example a thin layer where Hydrogen is burned. This method serves just to build a first model of the star, paving the way for more complex and accurate methods, such as the one that is going to be described in the next section.

3.3 HENYEY METHOD

As mentioned in the passing at the end of last section, it is customary to use the approximate solution found with the fitting method at time t as a test solution for the *Henyey method* [12] [5] at time $t + \mathrm{d}t$.

The fitting method provides us with test-values for the k-th mesh; what we're going to do is checking mesh by mesh if the stellar structure equations are verified. We can expand derivatives as incremental ratios and write an expression, for each equation, with the form

$$f_{i,k} = \delta_{i,k} \neq 0$$

where $f_{i,k}$ is just a fancy way to indicate the i-th stellar equation in the k-th mesh and $\delta_{i,k}$ is the, hopefully small, result of the i-th equation in the k-th mesh. Clearly, after the first iteraction, it is unlikely that it is vanishing as we'd like it to be. In fact, in general $\delta_{i,k}$ is a function of all the physical properties of the star in the k-th mesh and of the $(k+1)$-th mesh. We can express it as

$$\delta_{i,k} = \mathcal{F}(p_k, T_k, R_k, L_k, p_{k+1}, T_{k+1}, R_{k+1}, L_{k+1})$$

What we're going to do is slightly changing the test-values for each mesh until $\delta_{i,k}$ is found within the desired accuracy. The change in $\delta_{i,k}$ can be expressed as the sum of all the variation induced by the little changes in parameters. Said X_i one of the physical properties, we can write

$$\Delta\delta_{i,k} = \sum_j \frac{\partial \delta_{i,k}}{\partial X_{j,k}} \Delta X_{j,k} + \sum_j \frac{\partial \delta_{i,k}}{\partial X_{j,k+1}} \Delta X_{j,k+1}$$

We have have 8 unknown variables, but four of them are always in common between two adjacent meshes. Since $i = 1, ..., 4$ and $k = 1, ..., N-1$, we thus have $4n-4$ equations and $4n$ variables to determine. We're in the need of adding boundary conditions.

The most obvious conditions to add are $L_C \sim 0$, $R_C \sim 0$ in the center of the star, so that $\Delta R_1 \sim 0$, $\Delta L_1 \sim 0$, and at the base of the sub-atmosphere $p_N = f_1(L, T_e)$, $T_N = f_2(L, T_e)$, $R_N = f_3(L, T_e)$, $L_N = f_4(L, T_e)$. Given these conditions, there are plenty of (rigorous) mathematical theorems that grants that the solution we're going to find this way is, both locally and globally, *unique* (within the requested accuracy, that should go without saying).

4 NUCLEAR REACTIONS

4.1 INTRODUCTION

Most observed stars, including the Sun, are powered by thermonuclear fusion. This term denotes nuclear reactions induced by the thermal motion of the ions, wherby lighter nuclei combine to form a heavier one.

If the combined mass of the interacting nuclei is larger than the mass of the produced nucleus, the mass difference is converted into energy according to the famous (to the point of being obnoxious) Einstein's relation.

The mass difference between the interacting protons and the product of their interaction stems from the properties of the nuclear binding energy E_B, which is the energy required to separate nucleons against their mutual attraction due to nuclear forces, and is essentially a measure of the stability of the nucleus.

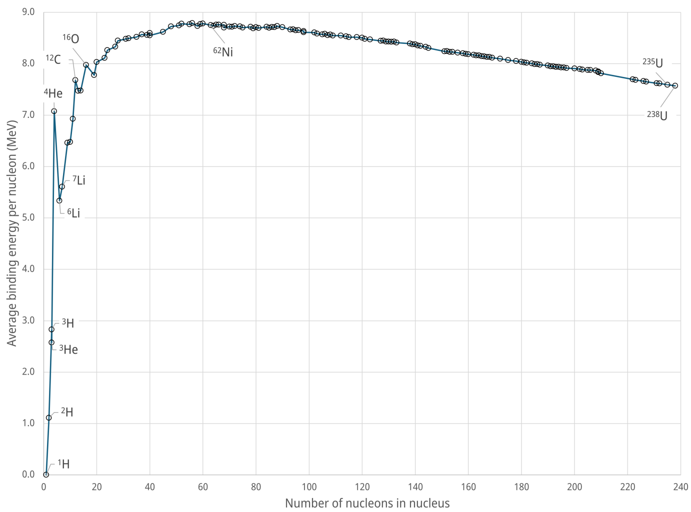

Figure 10: Binding energy per nucleon for a selection of nuclides. The nuclide with the highest value, ^{62}Ni, is noted.

The binding energy per nucleon, shown in Fig.10 is a most interesting quantity: Its value increases with A first steeply, then more gently, until it reaches a maximum of 8.79 MeV around $A = 56$, which corresponds to ^{56}Fe, and then drops slowly for heavier nucleons.

Most people claim, for reasons we'll soon find out, that Iron nuclei are the most stable ones, while it's false. It's actually ^{62}Ni nuclei that claim this figurative top position.

Elements up to Iron (read: Nichel) will therefore produce energy by thermonuclear fusion, whereas for higher values of A it is the splitting of heavier nuclei into ligther ones (*nuclear fission*) that does the deed.

We'll now focus on the determination of the nuclear energy generation coefficient.

4.2 ENERGY PRODUCTION RATE

Consider a reaction of the type

$$\mathrm{A} + \mathrm{b} \rightarrow \mathrm{C} + \mathrm{d}$$

in which the target nucleus A interacts with the particle b and forms nucleus C plus particle d. Three (or more) body interactions have a much smaller probability of happening and can therefore be neglected to a first approximation[1].

We define a cross section σ as the number of reactions per target nucleus per unit time, divided by the number of incident particles per unit area per unit time. By denoting with v the relative velocity of species A and b, the number of reactions r is given by $r = v\sigma(v)N_A N_b$, but since particles in stellar gas have a non-negligible velocity distribution, we'll have to integrate over the whole allowed range of velocities

$$r = N_A N_b \int_0^\infty v\sigma(v) f(v)\,\mathrm{d}v = \frac{N_A N_b}{1+\delta_{Ab}} \langle \sigma v \rangle \tag{80}$$

where δ_{Ab} is a Kronecker delta. Recalling that the mass fraction of a generic element k is given by

$$X_k = \frac{N_k m_\mathrm{H} A_k}{\rho}$$

we get the following expression for the reaction rate per unit mass

$$R_{Ab} = \rho \frac{X_A X_b}{m_\mathrm{H}^2 A_A A_b} \frac{\langle \sigma v \rangle_{Ab}}{1+\delta_{Ab}} \tag{81}$$

and the nuclear energy generation coefficient can be therefore written as

$$\epsilon_N = R_{Ab} Q_{Ab} \tag{82}$$

with Q_{Ab} being the amount of energy released by a single reaction. The value of Q_{Ab} can be determined from the difference between the sum of the masses of the interacting particles and the sum of the masses of the products

$$Q = \sum_r m_r(c^2) - \sum_p m_p(c^2) \tag{83}$$

Often nuclear masses are expressed in atomic mass units, and [21] points out that is worth recalling that 931.494 MeV is the amount of energy associated with one mass unit ($1.6605 \cdot 10^{-24}$ g). Moreover, if a positron is produced by the nuclear reaction, it is customary to add to Q_{Ab} its annihilation energy, whereas, in the case neutrinos are produced, their energy has to be subtracted from the total energy production budget.

In order to determine ϵ_N we still have to discuss how $\langle \sigma v \rangle$ can be determined. Matter involved in nuclear reactions is well approximated by a fully ionized perfect gas, and the particle velocities and kinetic energies follow the well-known Maxwellian distribution.

If the individual particles have a Maxwellian velocity distribution, it is possible to show that their relative velocities v will also follow the same

1 We're going to have to consider at least one of those if we want to produce heavier elements.

distribution. Since the corresponding kinetic energy is $E = \mu v^2/2$, where μ is the reduced mass of the interacting particles, the expression for $\langle \sigma v \rangle$ thus becomes

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma(E) E e^{-E/kT} \, \mathrm{d}E$$

but in order to complete the calculation we need further information about the dependence of $\sigma(E)$ on the energy E. This, of course, can be done by studying the process of thermonuclear fusion in more detail.

Whenever a pair of particles A and b interacts (that is, whenerver they come close enough that the attractive nuclear force tends to fuse them together) a compound nucleus C′ is formed, with atomic and mass number that is the sum of the respective numbers of its "progenitors".

Clearly, the nucleus C′ has no memory of how it was formed.

This compound nucleus is in a particular excited state for a given time τ, after which it'll break up in many possible ways, producing, among the various possibilities, the particles C and d.

The probability that our generic reaction $\mathrm{A} + \mathrm{b} \to \mathrm{C} + \mathrm{d}$ is really happening, depends on the product of the probability that the interacting particles come close enough to experience the effect of the attractive nuclear force, and the probability that this interaction produces the particles C and d.

In order to obtain the fusion of charged particles, they have to be able to get close enough that the short range forces dominate over the long range repulsive Coulombian forces. Nuclear attraction dominates for distances that are of the order of the nuclear size, so $\sim 10^{-13}$ cm. For larger distances, the repulsive electrostatic potential prevails

$$V_{\text{Coulomb}} = \frac{Z_A Z_b e^2}{r}$$

where Z_A, Z_b are the electric charges of A and b. Clearly, the electrostatic potential barrier can be overcome only if particles have a kinetic energy of the same order $kT \sim V_{\text{Coulomb}}$, which, for all but few cases, it's physically impossible.

Then how come nuclear reactions are indeed happening in the cores of stars?

As was investigated by G. Gamow [11], there is a small but quite finite probability of 'tunnelling' through the Coulombian potential barrier, even for particles of energy lower than V_{Coulomb}. It can be shown that the probability of tunnelling, in terms of the energy E of the relative motion of the colliding particles is

$$P_G = p_0 e^{-b/E^{1/2}} \quad b = \left(\frac{\mu}{2} \right)^{1/2} \left(\frac{2\pi Z_A Z_b e^2}{\hbar} \right) \tag{84}$$

Through this miracle, nuclear fusion can happen at temperatures much lower than predicted by classical physics.

At a given E the probability decreases for increasing Z_A, Z_b (the potential barrier gets higher), and progressively higher temperatures are needed for nuclear 'burnings' involving heavier nuclei.

In astrophysics it is customary to define the quantity

$$S(E) = \sigma(E) E e^{+b/E^{1/2}} \tag{85}$$

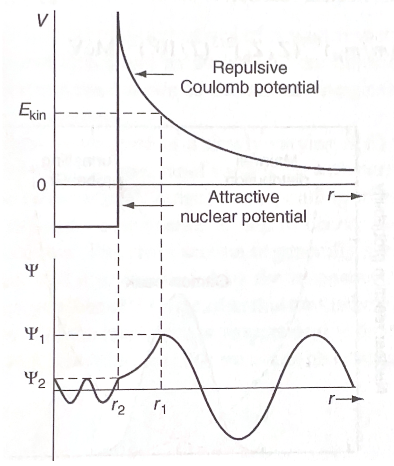

Figure 11: Illustration of the potential seen by a particle b when approaching a particle A with a kinetic energy E_{kin} and the corresponding wavefunction Ψ. Credits: [21].

called *astrophysical factor*, that is mainly sensitive to the nuclear contribution to the cross section. We may now rewrite the equation for $\langle \sigma v \rangle$ including $S(E)$

$$\langle \sigma v \rangle = \left(\frac{8}{\mu \pi} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty S(E) \exp\left(-\frac{E}{kT} - \frac{b}{\sqrt{E}} \right) \mathrm{d}E \tag{86}$$

In many cases of astrophysical relevance, the so-called non-resonant reactions, $S(E)$ is a slowly varying function of E in the range of interest. Note how in (86) the first term of the exponential represents the high energy wing of the Maxwellian distribution, while the second one represents the energy dependence of the tunnelling probability. Sometimes a scale energy E_{Gamow} is defined, so that $b^2 = E_{\text{Gamow}}$.

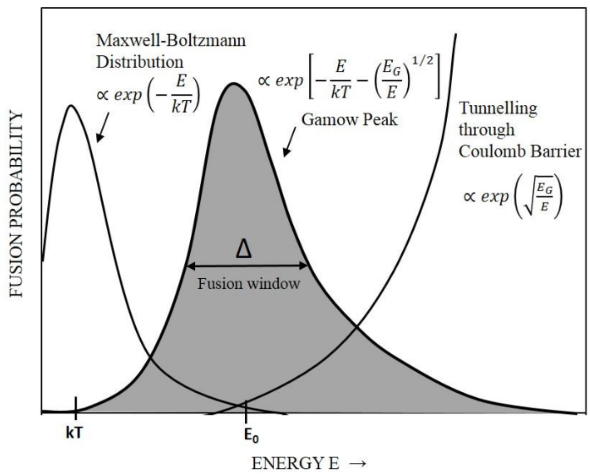

Figure 12: Illustration of the location and shape of the Gamow peak (not to scale) as compared with $e^{-E/kT}$ and $e^{-b/\sqrt{E}}$.

Their product, shown in Fig.12, produces a strongly peaked curve, with a maximum at the so-called *Gamow peak*, which also represents the most efficient energy for the nuclear reaction to occur.

The energy E_0 of the peak is

$$E_0 = \left(\frac{bkT}{2}\right)^{2/3} \approx 0.122 \left(\frac{\mu}{m_{\mathrm{H}}}\right)^{1/3} (Z_A Z_b)^{2/3} \left(\frac{T}{10^9\,\mathrm{K}}\right)^{2/3} \mathrm{MeV} \tag{87}$$

and the FWHM ΔE_0 is given by

$$\Delta E_0 \sim 0.237 \left(\frac{Z_A^2 Z_b^2 \mu}{m_{\mathrm{H}}}\right)^{1/6} \left(\frac{T}{10^9\,\mathrm{K}}\right)^{5/6} \mathrm{MeV} \tag{88}$$

It is in the Gamow peak energy range that $S(E)$ needs to be known in order to derive the reaction rate. Unfortunately, this energy range is at least a factor of 10 lower than the energies at which stellar reactions can be produced in laboratories; therefore, all we're able to do is extrapolate the measured $S(E)$ values to much lower energies if an empirical determination of $S(E)$ is sought.

As mentioned somewhere in the above, the astrophysical factor is often a slowly varying function of energy, and (86) is usually solved considering a second-order Taylor expansion for S and $e^{-b/\sqrt{E}}$ around E_0. If we approximate the exponential term as a Gaussian function[2]

$$I_{\max} \approx \exp\left(-\frac{E}{kT} - \sqrt{\frac{E_G}{E}}\right) \implies I_{\max} \approx I(E_0)$$

meaning

$$I_{\max} = \exp\left(-\frac{3E_0}{kT}\right) = \exp(-\tau)$$

This way we can write

$$I_{\max} = e^{-\tau} \exp\left[-\left(\frac{E - E_0}{\Delta/2}\right)^2\right] \qquad \Delta = \mathrm{FWHM}$$

Once plugged in (86), this provides

$$\langle \sigma(v) v \rangle = \sqrt{\frac{2}{\mu}} \frac{\Delta}{(kT)^{3/2}} S_{\mathrm{eff}}\, e^{-3E_0/kT} \tag{89}$$

with

$$\frac{S_{\mathrm{eff}}}{S(0)} = 1 + \frac{5kT}{36E_0} + \frac{S'(0)}{S(0)}\left(E_0 + \frac{35}{36}kT\right) + \frac{1}{2}\frac{S''(0)}{S(0)}\left(E_0^2 + \frac{89}{36}kTE_0\right) \tag{90}$$

The values for the astrophysical factor and its derivatives are obtained either from laboratory measurements (with the aforementioned limitations) or from theoretical quantum mechanical computations.

Let's see how we can deduce the (approximate) temperature dependence. Say T_0 a reference temperature, which may be the temperature for which the burning of a particular element is possible.

2 It is actually not a bad approximation, but the curve is slightly skewed, so we'll have to make some corrections afterwards.

We've shown that in general, the energy at Gamow's peak is dependent on the temperature

$$E_0 = \left(\frac{bkT}{2}\right)^{2/3} \qquad b = \frac{(2\mu)^{1/2}\pi e^2 Z_A Z_b}{\hbar}$$

Let us expand (86) around our reference temperature in the way presented below

$$\langle\sigma v\rangle_T \approx \langle\sigma v\rangle_{T_0}\left(\frac{T}{T_0}\right)^{\nu}$$

Taking the logarithms of both members

$$\ln(\langle\sigma v\rangle_T) \approx \ln(\langle\sigma v\rangle_{T_0}) + \nu\ln\left(\frac{T}{T_0}\right)$$

hence

$$\nu \approx \frac{\ln(\langle\sigma v\rangle_T) - \ln(\langle\sigma v\rangle_{T_0})}{\ln T - \ln T_0} = \frac{\mathrm{d}\ln\langle\sigma v\rangle}{\mathrm{d}\ln T}\bigg|_{T=T_0}$$

Let us express this in terms of the variable $\tau = 3E_0/kT$ we've used earlier when expanding the exponential term in (86).

You can show that

$$\tau \propto T^{-1/3} \implies T \propto \tau^{-3} \implies \ln T = C - 3\ln\tau$$

while for $\langle\sigma v\rangle_T$ we use (89). Let's explicit the various dependencies

$$\langle\sigma v\rangle_T \propto \frac{\Delta}{T^{3/2}}e^{-\tau} \qquad \Delta \propto T^{5/6}$$

which, after putting all together, implies

$$\langle\sigma v\rangle_T \propto T^{-2/3}e^{-\tau} \propto \tau^2 e^{-\tau} \implies \ln\langle\sigma v\rangle_T \propto D + 2\ln\tau - \tau$$

Now we can finally plug everything in the definition of ν. Calculating

$$\mathrm{d}\ln\langle\sigma v\rangle_T = \mathrm{d}\tau\left(\frac{2}{\tau} - 1\right) \qquad \mathrm{d}\ln T = -\frac{3}{\tau}\mathrm{d}\tau$$

$$\nu = \frac{\mathrm{d}\ln\langle\sigma v\rangle}{\mathrm{d}\ln T}\bigg|_{T=T_0} = -\frac{2}{3} + \frac{\tau(T_0)}{3}$$

meaning that

$$\boxed{\langle\sigma v\rangle \propto T^{-\frac{1}{3}(\tau_0 - 2)}} \tag{91}$$

For example, consider a reaction from the ppI chain

$$p + p \to D + e^+ + \bar{\nu}_e \implies \langle\sigma v\rangle \propto T^4$$

4.2.1 Resonant Reactions

So far we've been focusing on reactions whose astrophysical factor (85) was slowly varying with energy, the so-called *non-resonant reactions*. We've mentioned before that given two reactants, multiple products are possible, depending on the nature of the interaction (elastic, anelastic, etc).

$$
\begin{aligned}
\mathrm{A}+\mathrm{b} \to \mathrm{C}^* &\to \mathrm{A}+\mathrm{b} \text{ (elastic scattering)} \\
&\to \mathrm{A}+\mathrm{b}^* \text{ (anelastic scattering)} \\
&\to \mathrm{D}+\mathrm{e} \\
&\to \mathrm{C}+\gamma \text{ (radiative capture)}
\end{aligned}
$$

Note that if the energy of the center of mass added to the Q-value of the specific reaction is "close enough" to one of the energetic level of the compound nucleus C^*, then the reaction is *resonant*. We'll consider only *narrow* resonant reactions, for which the width of the resonance is much smaller than the resonant energy

$$\Gamma \ll E_R \tag{92}$$

It should come as no surprise that, around resonance, the interaction cross section becomes extremely larger, and it's no more the one we've found in (86), but it's rather the one found by Breit and Wigner

$$\sigma_{BW}(E) = \frac{\lambda^2}{4\pi} \frac{\omega \Gamma_1 \Gamma_2 (1+\delta_{Ab})}{(E-E_R)^2 + \left(\frac{\Gamma}{2}\right)^2} \tag{93}$$

where λ is the De-Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2\mu E}} \tag{94}$$

and Γ_1 is the probability amplitude of forming the compound nucleus, which is also equal to its decay probability; Γ_2 is the probability amplitude of decaying in the products of the reaction (say D, e), while Γ is the width of the excited level responsible of the resonance. Note that Γ is proportional to the total lifetime τ of the excited state according to

$$\Gamma = \frac{h}{2\pi\tau} \tag{95}$$

and Γ_1, Γ_2 have a similar definition, but considering the lifetime for the decay of the compound nucleus into the original interacting particles A+b, and into the products D+e, respectively. In the case of heavy compound nuclei many resonances are present in the range of stellar energies.

On the other hand, the factor ω is related to the spin of the reactants and to the (total) angular momentum of the compound nucleus

$$\omega = \frac{2J+1}{(2J_A+1)(2J_b+1)}$$

At this point we can write anew the reaction rate (80), but with a few caveats. First of all, now we're in the presence of a narrow resonance, meaning that if the Gamow peak E_0 is far from resonance E_R, we won't be able to observe the resonance.

On the other hand, if $E_R \sim E_0$, the resonance overrides all the other possible decay products, and thus we have to use (93) rather than the tunnelling probability.

We still have to average over the velocity distribution

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma_{BW}(E) E e^{-E/kT} \, dE \tag{96}$$

But since the resonance is narrow around E_R and, for extension, around E_0, we can bring the Maxwellian contribution out of the integral, evaluating them in E_R

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} E_R e^{-E_R/kT} \int_0^\infty \sigma_{BW}(E) \, dE$$

This integral can be calculated, thus finding

$$\langle \sigma v \rangle = \left(\frac{2\pi}{\mu kT} \right)^{3/2} \hbar^2 \omega \gamma e^{-E_R/kT} \tag{97}$$

where $\gamma = \Gamma_1 \Gamma_2 / \Gamma$. Unlike narrow resonances, *broad resonances* are more troublesome, since they have broad tails that can influence the interaction cross section even when the reaction is well out of resonance conditions.

4.2.2 Electronic Shielding

It is important to notice that in our previous discussion and the cross sections provided in the literature refer to reactions *happening in vacuum*. Instead, as I'm sure the sharpest among my four readers had been itching to point out for a while, ions in stellar interiors are surrounded by a sea of free electrons that tend to cluster near the nuclei and reduce their Coulomb potential.

Therefore an approaching particle will feel a Coulombian potential different from the case of an isolated positive charge. This shielding effect of the electrons increases the reaction rate $\langle \sigma v \rangle$ by a so-called *screening factor* f that will necessarily be a function of the matter density.

Consider the upper cartoon of Fig.11, and assume that the impinging particle on our target nucleus has an energy much lower than the Coulombian potential. The poor thing would not be able to interact effectively with the target nucleus and will "stop" at a distance known as *classical turning point*, which is easily computed

$$r_{TP} = \frac{Z_A Z_b e^2}{E_b} \tag{98}$$

In reality, however, the target nucleus has some electrons, that will be located at a certain distance from the nucleus. Within the electronic cloud, the Coulombian potential vanishes.

Then, if our impinging projectile had an energy such that $r_{TP} > r_A$, where r_A is the atomic radius, the resulting effect would be incredible since, because of the electronic shielding, the effective electrostatic potential would be identically zero.

Let us define an energy

$$U_e = \frac{Z_A Z_b e^2}{r_A}$$

so that if $E_b > U_e$, the projectile can break into the electronic cloud. This is indeed what happens, since U_e is rather small ($\sim 0.03\,\text{keV}$). If that is the case, then necessarily $r_{TP} < r_A$.

So how does the electrostatic potential changes? We can imagine the cloud as a symmetric source of potential around Z_A, so that

$$\Phi_A = -\frac{Z_A e}{r_A}$$

Therefore the total potential within the cloud decreases a bit

$$\Phi_{\text{tot}} = \frac{Z_A e}{r} - \frac{Z_A e}{r_A}$$

but we're interested in evaluating it at distances comparable to the range of nuclear forces, where our nuclear fusion interactions can take place.

This has the effect of lowering the Coulombian barrier

$$V_{\text{eff}} = \frac{Z_A Z_b e^2}{r_N} - \frac{Z_A Z_b e^2}{r_A}$$

The correction we've introduced is rather small, since $r_N/r_A \sim 10^{-5}$. In a sense, we could consider this effect as an increased energy of the impinging projectile $E_b \to E_b + U_e$. This clearly induces a slight increasing of the reaction rate by a factor f.

The shielding factor f is defined as

$$f = \frac{\sigma(E_b + U_e)}{\sigma(E_b)} \tag{99}$$

Albeit small, this effect is not at all negligible. From the definition (99) we can substitute the expressions for the cross sections (I'm dropping the b-subscript)

$$\frac{\sigma(E+U_e)}{\sigma(E)} = \frac{E}{E+U_e}\frac{S(E+U_e)}{S(E)} \exp\left(-\frac{2\pi Z_A Z_b e^2\sqrt{\mu}}{\hbar\sqrt{2(E+U_e)}} + \frac{2\pi Z_A Z_b e^2\sqrt{\mu}}{\hbar\sqrt{2E}}\right)$$

which, since $U_e \ll E$, we can Taylor-expand. First of all we can notice that the first two ratios out of the exponential are, at linear order, of order unity. Then, expanding the square root, we obtain

$$f \approx \exp\left(\frac{\pi Z_A Z_b e^2\sqrt{\mu}}{\hbar\sqrt{2E}}\frac{U_e}{E}\right)$$

which becomes more and more relevant for lower energies of the center of mass, which are those of astrophysical relevance, since the astrophysical factor starts to climb up back again.

To the reaction rates we've been calculating so far we have to add a correction to address the effect of electronic shielding in the stellar plasma. The idea is that, although the stellar plasma is globally neutral, whenever a impinging projectile comes close to an ionized nucleus, the two charged nuclei start repelling each other, whilst electrons tend to be attracted.

If that's the case, then we can imagine our stellar plasma as made of a myriad of charged nuclei that are getting "shielded" by electronic clouds at a certain distance from them, so that when a nucleus interact with another nucleus, the electronic cloud, which is just an electron over-density, trails behind it.

Say that n_i is the numeric density of the ions

$$n_i \approx \frac{\rho}{\mu_i m_{\mathrm{H}}} \approx \frac{1}{\frac{4\pi}{3}a^3}$$

with a the average distance between ions. We can invert the latter expression

$$a \sim \frac{1}{n_i^{1/3}} \sim \left(\frac{\mu_i m_{\mathrm{H}}}{\rho}\right)^{1/3}$$

In the inner regions of the stellar plasma, the density is quite large, meaning that the distance between two ions gets smaller.

We can define *strong shielding* the condition for which

$$\frac{Z_1 Z_2 e^2}{akT} \gg 1 \tag{100}$$

In such a scenario we can imagine that ions are so far away from each other that there's effectively only one ion per electronic cloud. This is a condition that can be verified only at very high densities, which are not those typical of the regions where nuclear reactions are active.

In stars, strong shielding is typical of regions where electrons are degenerate, but generally it's much more probable that, in regions of active fusion, the shielding is *weak* (which is exactly the opposite of the strong shielding we mentioned before) or an intermediate scenario that is known as *intermediate shielding*, which can be addressed only through numerical simulations.

Consider a *weak shielding* scenario, where the positions of ions is only slightly skewed with respect to the global absence of electric fields. Generally, the distance between the electronic cloud and a bare nucleus R is much larger than the distance among ions a.

In the surrounding of the target ion A with positive charge Z_A, there's going to be a non-zero electric field, which is going to induce a different distribution of ions and electrons from that that would be giving a vanishing net charge. In general we can write

$$\bar{n}_i \approx \frac{\rho X_i}{A_i m_{\mathrm{H}}} \to \bar{n}_e = \sum_i Z_i \bar{n}_i$$

assuming, for simiplicity, complete ionization in a region of active fusion. On average we know that

$$\Phi = 0 \quad \bar{\rho} = \sum_i Z_i e \bar{n}_i - e\bar{n}_e = 0$$

Instead, in the surrounding of each charged nucleus, there is a small, but non vanishing, electrostatic potential. Statistical mechanics tells us that

$$n_i = \bar{n}_i \exp\left(-\frac{Z_i e\Phi}{kT}\right) \approx \bar{n}_i \left(1 - \frac{Z_i e\Phi}{kT}\right)$$
$$n_e = \bar{n}_e \exp\left(\frac{e\Phi}{kT}\right) \approx \bar{n}_e \left(1 + \frac{e\Phi}{kT}\right)$$

Note that, despite the deviation from neutrality, the charge displacement is very small, so that the new numeric density of charges is not that much different from the one that grants neutrality.

The total electronic density is now

$$\rho = \sum_i Z_i e n_i - e n_e = \sum_i \left(-\frac{Z_i^2 e^2 \Phi \bar{n}_i}{kT}\right) - \frac{e^2 \Phi \bar{n}_e}{kT}$$

where we've used the global neutrality condition. Let's manipulate this last expression a bit so to extract the distance of the electronic cloud (*Debye radius*)

$$n = \bar{n}_e + \sum_i \bar{n}_i \quad \text{at } \Phi = 0$$

and we introduce the quantity χ classically defined as

$$\chi = \frac{\sum_i Z_i^2 \bar{n}_i + \bar{n}_e}{n}$$

Substituting what we've just found we obtain

$$\rho = -\frac{\chi e^2 \Phi}{kT} n$$

Now we'd like to understand how the electrostatic potential changes nearby the charged ion, so to determine how the electronic cloud is distributed. Recall Poisson's equation

$$\nabla^2 \Phi = -4\pi\rho \implies \frac{1}{r}\frac{\mathrm{d}^2(r\Phi)}{\mathrm{d}r^2} = -4\pi\rho$$

We can plug in the charge density ρ

$$\frac{1}{r}\frac{\mathrm{d}^2(r\Phi)}{\mathrm{d}r^2} = 4\pi\frac{\chi e^2 \Phi}{kT} n$$

At last, we can define the *Debye radius*

$$r_D = \left(\frac{kT}{4\pi\chi n e^2}\right)^{1/2} \tag{101}$$

so that our spherically symmetric Poisson's equation becomes

$$\frac{1}{r}\frac{\mathrm{d}^2(r\Phi)}{\mathrm{d}r^2} = \frac{\Phi}{r_D}$$

which can be solved for the electrostatic potential Φ

$$\Phi = \frac{Z_i e}{r} \exp\left(-\frac{r}{r_D}\right)$$

If we expand it for $r/r_D \ll 1$ we find out that

$$\Phi \approx \frac{Z_i e}{r} - \frac{Z_i e}{r_D} \tag{102}$$

which can be interpreted as the potential generated by a charged ion corrected for the presence of an electronic charge at distance r_D. We can also define a *Debye energy*

$$E_D = \frac{Z_i Z_j e^2}{r_D} \tag{103}$$

which is the limit energy for which the impinging projectile wouldn't be able to penetrate into the electronic cloud. Note that since the Debye radius is quite large, the associated energy is $E_D \ll E_0$. In "normal" stars[3] the Debye radius is much larger than

$$E_D = \frac{Z_i Z_j e^2}{r_D} \ll \frac{Z_i Z_j e^2}{r_0} := E_0$$

For example, for the Sun, at its center, the Debye radius is $r_D \approx 10^{-9} - 10^{-8}$ cm, while the ratio $r_D/r_0 \sim 50 - 100$.

As we've done before, we have to calculate the shielding factor and the associated $\langle \sigma v \rangle_S$

$$\langle \sigma v \rangle_S = \left(\frac{8}{\pi\mu}\right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma_{ns}(E + E_D) E e^{-E/kT} \, \mathrm{d}E$$

which we can integrate changing variable to $E' = E + E_D$

$$\langle \sigma v \rangle_S = \left(\frac{8}{\pi\mu}\right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma_{ns}(E')(E' - E_D) e^{-(E'-E_D)/kT} \, \mathrm{d}E'$$

Now, since $E_D \ll E_0$, we can neglect its contribution coming from the factor $E' - E_D$, thus leaving us with

$$\langle \sigma v \rangle_S \approx \left(\frac{8}{\pi\mu}\right)^{1/2} \frac{1}{(kT)^{3/2}} e^{+E_D/kT} \int_0^\infty \sigma_{ns}(E') E' e^{-E'/kT} \, \mathrm{d}E'$$

where we can easily recognize the shielding factor f, which comprises essentially of the exponential

$$f = e^{E_D/kT} \sim 1 \tag{104}$$

so that the shielded cross section is just a small correction on the non-shielded cross section

$$\langle \sigma v \rangle_S = f \langle \sigma v \rangle_{NS}$$

For typical solar values, given the Debye radius we've found earlier, we'd obtain a shielding factor for the pp reaction of order $f \sim 1.03$.

3 Read: non-strongly degenerate stars as white dwarves or neutron stars.

4.3 FUSION REACTIONS

Before we even begin looking into what happens when a star enters the longest stage of their evolution cycle, the main sequence, we'll consider briefly the reactions occurring in the phase known as *pre-main sequence*.

As you should already know, a star, whose nuclear reactions have yet to start, will balance the energy loss through their intrinsic luminosity depleting gravitational energy. This means that, as the star shrinks, it heats up, thus radiating away more energy and cooling down, ready to repeat the cycle over and over.

When temperature gets high enough ($\sim 10^6\,\mathrm{K}$), the combustion of the so-called *light elements* starts. The first that starts burning is Deuterion, which is already scarce ($X^{\mathrm{in}}_{D,\odot} \sim 2 - 4 \cdot 10^{-5}$) and gets burned quickly

$$^2_1\mathrm{D} +^1_1 \mathrm{p} \rightarrow^3_2 \mathrm{He} + \gamma$$

For temperatures just a little higher (respectively $2 \cdot 10^6\,\mathrm{K}$ and $2.5 \cdot 10^6\,\mathrm{K}$), two other reactions occur

$$\begin{aligned} &^6_3\mathrm{Li} +^1_1 \mathrm{p} \rightarrow^3_2 \mathrm{He} +^4_2 \mathrm{He} \\ &^7_3\mathrm{Li} +^1_1 \mathrm{p} \rightarrow 2\,^4_2\mathrm{He} \end{aligned}$$

Clearly, the ignition temperatures we've given are only approximative since they actually depend on the element abundance and density of the star. One thing in particular is curious about $^7_3\mathrm{Li}$: Only 1% of the abundance we observe in a Sun-like star (which is already not a lot, $X_{^7\mathrm{Li}} \sim 10^{-8}$) is produced during the BBN (*Big Bang Nucleosynthesis*). On average it is only burned in stars, while it's produced through *spallation on nuclei by cosmic rays*

$$^4_2\mathrm{He} +^{12}_6 \mathrm{C} \rightarrow^7_3 \mathrm{Li} + 2\,^4_2\mathrm{He} +^1_1 \mathrm{p}$$

On top of this, other reactions involving Berillium and Boron can be occurring if temperatures get high enough. For Berillium, assuming a temperature of at least $T \sim 3.5 \cdot 10^6\,\mathrm{K}$

$$\begin{aligned} ^9_4\mathrm{Be} +^1_1 \mathrm{p} &\rightarrow^4_2 \mathrm{He} +^6_3 \mathrm{Li} \\ &\rightarrow^8_4 \mathrm{Be} +^2_1 \mathrm{D} \rightarrow 2\,^4_2\mathrm{He} +^2_1 \mathrm{D} \end{aligned}$$

the last reaction in particular is strongly unstable, and $^8_4\mathrm{Be}$ immediately decays ($\tau \sim 10^{-16}\,\mathrm{s}$) with a double-$\alpha$ decay. Notice that normally $^9_4\mathrm{Be}$ is produced, for example, from reactions involving heavier elements

$$^1_1\mathrm{p} +^{14}_7 \mathrm{N} \rightarrow^9_4 \mathrm{Be} +^4_2 \mathrm{He} + 2\,^1_1\mathrm{p}$$

and has an overall initial abundance of $X_{^9\mathrm{Be}} \sim 10^{-10}$.

We've got reactions for the isotopes of Boron as well ($T \sim 4.5 \cdot 10^6\,\mathrm{K}$, $X_{^{11}\mathrm{B}} \sim 5 \cdot 10^{-9}$)

$$\begin{aligned} &^{11}_5\mathrm{B} +^1_1 \mathrm{p} \rightarrow 3\,^4_2\mathrm{He} \\ &^{10}_5\mathrm{B} +^1_1 \mathrm{p} \rightarrow^4_2 \mathrm{He} +^7_4 \mathrm{Be} \end{aligned}$$

It's curious to know that the abundance of $^{11}\mathrm{B}$ is four times larger than $^{10}\mathrm{B}$.

During the star's evolution along the Hayashi track (see Chapter 5), the star increases its temperature due to the virial theorem, and a radiative core will form at the center and grow in mass as the star evolves. When the radiative core appears, the star is no longer fully convective; this happens when

$$\nabla_{\text{rad}} < \nabla_{\text{ad}}$$

and the star will move out of the Hayashi track. The reactions we've presented are not relevant from the point of view of energy production, not even in this stage, but reduce practically to zero the abundances of the light elements involved (and do not appreciably affect the abundances of the elements they produce).

If the stars were fully convective during this stage, we would have a complete depletion of Lithium and Deuterium at the surface, since matter in a convective star is always well mixed. However, as we've mentioned en passant, the convective region will start to shrink, eventually; if the convective region is confined to the external and cooler region before all the ***insert light element here*** reservoir has been destroyed during the fully convective phase, some remaining nuclei will still be present in the convective envelope.

This means that the resulting surface abundances of these light elements depend on the interplay between the increase of the stellar temperature, the resulting efficiency of the nuclear reactions involed and the timescale for the shrinking of the convective envelope.

As a rule of thumb, we can say that, once the chemical composition is fixed, larger masses will have lower surface depletion of light elements, and the depletion increases along with the star's metallicity.

If this is the case, then why ${}^{6}_{3}\text{Li}$ is only 1% as abundant as theory predicts for Sun-like stars? Good question, we don't know. Apparently, it just continues to get burned also when the star enters the main sequence.

4.3.1 Hydrogen Burning: pp-chain

The H-burning mechanism is essentially the nuclear fusion of four protons into one ${}^{4}_{2}\text{He}$ nucleus, garnished with two positrons and two electronic neutrinos.

The energy production for the whole set of reaction we'll be going through in a moment can be shown to amount to 26.7 MeV, only decreasing to 26.1 MeV if we consider the energy subtracted by neutrinos. In any case, this amount of energy is still 10 times larger than that produced in any other nuclear reaction process occurring in stars.

Assuming burning of all protons into Helium, you'd produce (in a Sun-like star) $E \sim 6.5 \cdot 10^{57}\,\text{MeV}$. Since the luminosity of the Sun is $L_\odot \sim 2.5 \cdot 10^{39}\,\text{MeV}\,\text{s}^{-1}$, you'd obtain a timescale for nuclear burning of $\tau_{\text{nuc}} \sim 10^{11}\,\text{yr}$. Accounting for the fact that only 10% of the total Hydrogen is burned in the center, you get a more realistic timescale of $\tau_{\text{nuc}} \sim 10^{10}\,\text{yr}$.

The main channel of energy production in this phase is (for less-massive stars) the *p-p chain*.

The reaction networks involved in the so-called p-p chain are the following:

ppI ($T \sim 8 \cdot 10^6\,\mathrm{K}$)
$^1_1\mathrm{p} + ^1_1\mathrm{p} \to ^2_1\mathrm{D} + e^+ + \nu_e$
$^2_1\mathrm{D} + ^1_1\mathrm{p} \to ^3_2\mathrm{He} + \gamma$
$^3_2\mathrm{He} + ^3_2\mathrm{He} \to ^4_2\mathrm{He} + 2\,^1_1\mathrm{p}$

ppII ($T \sim 15 \cdot 10^6\,\mathrm{K}$)
$^3_2\mathrm{He} + ^4_2\mathrm{He} \to ^7_4\mathrm{Be} + \gamma$
$^7_4\mathrm{Be} + e^- \to ^7_3\mathrm{Li} + \nu_e$
$^7_3\mathrm{Li} + ^1_1\mathrm{p} \to ^8_4\mathrm{Be} \to 2\,^4_2\mathrm{He}$

ppIII ($T \sim 20 \cdot 10^6\,\mathrm{K}$)
$^3_2\mathrm{He} + ^4_2\mathrm{He} \to ^7_4\mathrm{Be} + \gamma$
$^7_4\mathrm{Be} + ^1_1\mathrm{p} \to ^8_5\mathrm{B} + \gamma$
$^8_5\mathrm{B} \to ^8_4\mathrm{Be} + e^+ + \nu_e \to 2\,^4_2\mathrm{He} + e^+ + \nu_e$

He-p
$^3_2\mathrm{He} + ^1_1\mathrm{p} \to ^4_2\mathrm{He} + e^+ + \nu_e$

Although it seems rather hard to digest at first sight, it's actually much simpler than it looks.

First of all, we know that these reactions can't possibly involve free neutrons, for there aren't any: Those few that had remained after the BBN have now decayed (10^{10} yr is much larger than $\sim 10\,\mathrm{min}$, isn't it?), so the only possiblity for our star to produce Deuterium is through Hydrogen burning, with a branching fraction $\Gamma_{\mathrm{pp}} \sim 99.8\%$[4].

The four reactions that we've presented are the four main ways $^3_2$He can be used to eventually produce $^4_2$He. The first reaction in the ppI chain requires that protons experience a β^+ decay, and can only occur if the protons are brought together by a nuclear collision. This process is governed by *weak* interactions and therefore has a low probability of occurring: Its nuclear cross section is in fact quite low ($\sigma_{\mathrm{pp}} \sim 10^{-23}\,\mathrm{b}$).

Until a higher temperature, roughly $8 \cdot 10^6\,\mathrm{K}$, is reached, the reactions producing $^3_2$He are more frequent than those consuming it, and as a consequence the abundance of $^3_2$He increases, only decreasing when successive reactions become effective. In a short time this element reaches an "equilibrium" abundance, which can be as high as earning it the title of *pseudo-primary element*.

The relative frequency of the ppII and ppIII depends strongly on temperature; in particular, the $^3_2\mathrm{He} + ^4_2\mathrm{He}$ reaction becomes competitive with respect to $^3_2\mathrm{He} + ^3_2\mathrm{He}$ for $T \approx 15 \cdot 10^6\,\mathrm{K}$. The reason for it lays mainly in (84) and

4 Actually, there's another reaction able to produce Deuterium, which is $2\,^1_1\mathrm{p} + e^- \to ^2_1\mathrm{D} + \nu_e$, which, you'll reckon, is quite unlikely to happen, since it's a three-body reaction on top of being mediated by electroweak interactions. The associated branching fraction is $\Gamma_{\mathrm{pep}} \sim 0.2\%$.

its dependence on the reduced mass μ: The reduced mass of ${}^3_2\text{He} + {}^4_2\text{He}$ is larger than that of ${}^3_2\text{He} + {}^3_2\text{He}$, meaning that the probability of tunnelling through the potential barrier is higher for the latter.

As a rule of thumb, you may say that with increasing temperature the importance of the ppII and ppIII increases with respect to the ppI if there is a sufficient concentration of ${}^4_2\text{He}$.

Key informations about our favorite energy production chains are summarized in Table 1.

	ppI	ppII	ppIII	He-p
Γ	$\sim 85\%$	$\sim 15\%$	$\sim 0.1\%$	$\sim 0.0001\%$
$T\,[K]$	$\sim 8 \cdot 10^6$	$\sim 15 \cdot 10^6\,\text{K}$	$\sim 20 \cdot 10^6\,\text{K}$	-
K.R.	Electroweak	Electronic capture	2α	Electroweak

Table 1: Table reporting key informations about the p-p energy production chain. "K.R." stands for Key Reaction, not the most significant, but rather the one that differentiate each chain from the others.

As per what we were saying in the above, the branching ratios for each reaction will be extremely sensitive to temperature, so the values reported in the table are to be considered for Sun-like stars currently burning Hydrogen in their core.

Note that neutrinos produced by different reactions will have, naturally, different energies, meaning different spectra will be produced. We can try to calculate an average flux of neutrinos in the following way

$$n_\nu = 2\frac{L_\odot}{26.1\,\text{MeV}} \approx 1.9 \cdot 10^{38}\,\text{s}^{-1}$$

which we can convert in a flux we can measure on Earth simply dividing by an appropriate power of the Earth-Sun distance (2)

$$\phi_\nu \sim \frac{1.9 \cdot 10^{38}}{4\pi D^2} \approx 6.7 \cdot 10^{10}\,\text{cm}^{-2}\,\text{s}^{-1}$$

The sharpest amongst my readers have surely noticed that there are some elements which are simultaneously involved in destruction and production processes (e.g., ${}^2_1\text{D}$ or ${}^3_2\text{He}$). These elements are usually called *secondary elements*.

It should come as no surprise that the variation of the abundance of element i will depend on all the creation and destruction reactions involving that element[5]. If you recall how we've defined the reaction rate (80), we can then say that the variation over time of the abundance (or number of particles) of a certain species will be

$$\dot{N}_i = r_i^C - r_i^D$$

Let us consider for example, the equilibrium equation for Deuterion

$$\dot{N}_\text{D} = \frac{N_p N_p}{2}\langle\sigma v\rangle_{pp} - N_\text{D} N_p \langle\sigma v\rangle_{\text{D}p}$$

5 The general equation is quite long and convoluted, and I see no point into copying it here. The interested reader can refer to [21], §3.17, equation 3.39.

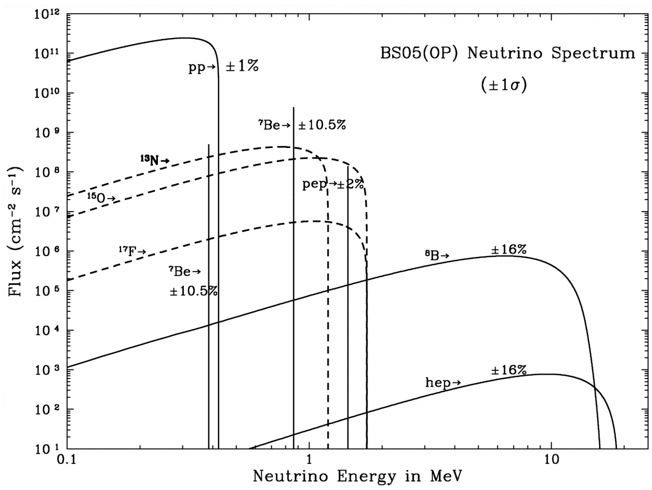

Figure 13: Solar neutrino energy spectrum for the solar model BS05(OP). The uncertainties are taken from Table 8 of [4].

Equilibrium will be reached when $\dot{N}_{\mathrm{D}} = 0$, so when

$$\frac{N_{\mathrm{D}}}{N_p} = \frac{\langle \sigma v \rangle_{pp}}{2\langle \sigma v \rangle_{\mathrm{D}p}}$$

What do we learn from this? The equilibrium abundance is *directly proportional* to the creation rate and *inversely proportional* to the destruction rate.

It is evident that if the abundance of ${}^2_1\mathrm{D}$ is larger than its abundance at equilibrium, then the destruction process prevails over the production one, and the chemical abundance will tend to evolve quickly towards its equilibrium configuration.

Under the assumption that $N_{\mathrm{D}} \gg N_{\mathrm{D,eq}}$ we can determine a timescale for the process, since the destruction process will be dominating

$$\frac{\mathrm{d}N_{\mathrm{D}}}{\mathrm{d}t} = -N_{\mathrm{D}} N_p \langle \sigma v \rangle_{\mathrm{D}p} \tag{105}$$

whose solution is of the type $N_{\mathrm{D}} = N_{\mathrm{D},0} \exp(-t/\tau)$, with

$$\tau = \frac{1}{N_p \langle \sigma v \rangle_{\mathrm{D}p}} \tag{106}$$

For typical temperatures at which the p-p chain is fully efficient, the value of τ is of the order of 1 s, meaning the equilibrium configuration is reached quite hastily.

From this we're learning also something else really important: With given initial conditions, the reaction chain can reach and (virtually) remain in equilibrium, but the reaction rate will be led by the slower reaction, which (106) tells us is that with the smaller cross section.

In the p-p chain, the reaction telling the time to the others is the very first one, which has, as we've seen, an abysmally small cross section

$$\sigma_{\mathrm{pp}} \sim 10^{-23}\,\mathrm{b} \quad S(0)_{pp} \sim 4.1 \cdot 10^{-22}\,\mathrm{keV\,b^{-1}}$$

which, for example, we can compare to the very following reaction, the one involving Deuterion, that has $S(0)_{\mathrm{D}p} \sim 2.1 \cdot 10^{-4}\,\mathrm{keV\,b^{-1}}$.

So there are 18 orders of magnitude of difference.

4.3.2 Hydrogen Burning: CN-NO

The CN-NO (bi)cycle is the combination of two independent cycles: The CN cycle and the NO cycle. The presence of some isotopes of Carbon, Nitrogen and Oxygen are necessary for either cycle to begin. Being both produced and destroyed during these cycles, those elements act as catalysts. The reaction networks involved are the following:

CN cycle

$$
\begin{aligned}
&{}^{12}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{13}_{7}\mathrm{N} + \gamma\\
&{}^{13}_{7}\mathrm{N} \rightarrow {}^{13}_{6}\mathrm{C} + e^{+} + \nu_e\\
&{}^{13}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{14}_{7}\mathrm{N} + \gamma\\
&{}^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{15}_{8}\mathrm{O} + \gamma\\
&{}^{15}_{8}\mathrm{O} \rightarrow {}^{15}_{7}\mathrm{N} + e^{+} + \nu_e\\
&{}^{15}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{16}_{8}\mathrm{O}^{*} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He} \quad (\Gamma \sim 99\%)\\
&\qquad\qquad\qquad\qquad\quad \rightarrow {}^{16}_{8}\mathrm{O} + \gamma \qquad (\Gamma \sim 1\%)
\end{aligned}
$$

This branch of the cycle has to follow the timescale of ${}^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{15}_{8}\mathrm{O} + \gamma$, which has the smaller cross section, followed by the first reaction of the cycle.

In the CN cycle the isotopes of C and N act as catalysts, and so behave as secondary elements. As a consequence, the cycle can start almost from any reaction if the involved isotope is present, and during a complete loop around the cycle the isotope is consumed and then produced again.

This, however, doesn't mean that the concentrations of the different isotopes will be unchanged as the final abundances depend strongly on the relative rates of the nuclear reactions in the cycle. Only at a high enough temperature ($T \approx 15 \cdot 10^{6}$ K) will the isotopes achieve their equilibrium abundance.

The branching ratio between the proton captures on ^{15}N, producing respectively ^{16}O and ^{12}C, is of the order of 10^{-4}, thus the amount of ^{16}O produced is practically negligible. Nevertheless, for slightly higher temperatures ($T \approx 20 \cdot 10^{6}$ K) the small production of ^{16}O unlocks a new branch for the cycle, one which transforms Oxygen into Nitrogen.

NO cycle

$$
\begin{aligned}
&{}^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{15}_{8}\mathrm{O} + \gamma\\
&{}^{15}_{8}\mathrm{O} \rightarrow {}^{15}_{7}\mathrm{N} + e^{+} + \nu_e\\
&{}^{15}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{16}_{8}\mathrm{O}^{*} \rightarrow {}^{16}_{8}\mathrm{O} + \gamma \quad (\Gamma \sim 1\%)\\
&{}^{16}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{17}_{9}\mathrm{F} + \gamma\\
&{}^{17}_{9}\mathrm{F} \rightarrow {}^{17}_{8}\mathrm{O} + e^{+} + \nu_e\\
&{}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{18}_{9}\mathrm{F}^{*} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \quad (\Gamma \sim 99\%)\\
&\qquad\qquad\qquad\qquad\quad \rightarrow {}^{18}_{9}\mathrm{F} + \gamma \qquad (\Gamma \sim 1\%)
\end{aligned}
$$

It should be noticed that for a typical PopI star chemical composition (i.e., about solar) the global amount of CNO elements is of order of 75% of the total metallicity

$$
X_{\mathrm{CNO}} \approx 0.75 Z \qquad \frac{X_{\mathrm{O}}}{X_{\mathrm{C}}} \approx 0.7
$$

We know, however, that the total abundance has to be conserved (unless particular kinds of burnings are involved, that is)

$$\sum_i N_i(t=0) = \sum_i N_i(t)$$

meaning only the relative abundances between elements can be changing. In particular, we know that the slowest reaction of the CNO cycle is the $^{14}\mathrm{N}(p,\gamma)\,^{15}\mathrm{O}$, so the most abundant element in the CNO cycle processed material is indeed $^{14}\mathrm{N}$. As a rule of thumb, the change of the CNO element caused by the different burning channels is

- CN cycle processed matter: C↓ N↑
- CNO cycle processed matter: C↓ N↑ O↓

Once the full CNO is operating, more than 95% of the cycle's elements are converted into $^{14}\mathrm{N}$, which also enables a new chain for unbound neutrons production.

At even higher temperatures, a new chain of reactions for processing Fluorine is unlocked

$$\begin{aligned}
&{}^{18}_{9}\mathrm{F} \rightarrow {}^{18}_{8}\mathrm{O} + e^+\nu_e \\
&{}^{18}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{19}_{9}\mathrm{F}^* \rightarrow {}^{15}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \\
&\qquad\qquad\qquad\quad\ \rightarrow {}^{19}_{9}\mathrm{F} + \gamma \\
&{}^{19}_{9}\mathrm{F} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{16}_{8}\mathrm{O} + {}^{4}_{2}\mathrm{He} \\
&\qquad\qquad\ \rightarrow {}^{20}_{10}\mathrm{Ne} + \gamma
\end{aligned}$$

Until temperature rises even higher ($T \approx 25 \cdot 10^6\,\mathrm{K}$) there is a *leakage* of un-burned Neon from the star. Only in very massive MS-stars and during the AGB[6] does a star suffice the necessary conditions to activate the Neon-Sodium cycle

Ne-Na cycle

$$\begin{aligned}
&{}^{20}_{10}\mathrm{Ne} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{21}_{11}\mathrm{Na} + \gamma \\
&{}^{21}_{11}\mathrm{Na} \rightarrow {}^{21}_{10}\mathrm{Ne} + e^+ + \nu_e \\
&{}^{21}_{10}\mathrm{Ne} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{22}_{11}\mathrm{Na} + \gamma \\
&{}^{22}_{11}\mathrm{Na} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{23}_{12}\mathrm{Mg} + \gamma \\
&{}^{23}_{12}\mathrm{Mg} \rightarrow {}^{23}_{11}\mathrm{Na} + e^+ + \nu_e \\
&{}^{23}_{11}\mathrm{Na} + {}^{1}_{1}\mathrm{p} \rightarrow {}^{24}_{12}\mathrm{Mg}^* \rightarrow {}^{20}_{10}\mathrm{Ne} + {}^{4}_{2}\mathrm{He} \\
&\qquad\qquad\qquad\qquad\quad\ \rightarrow {}^{24}_{12}\mathrm{Mg} + \gamma
\end{aligned}$$

which reflects into an increase of the abundance of Sodium and a decrease in the abundance of Oxygen, the so-called *Oxygen-Sodium anticorrelation*, which

6 *Asymptotic Giant Branch.*

is a distinctive characteristic of most globular clusters. At $T \approx 50 \cdot 10^6$ K, Magnesium is ignited, forming Aluminium

$$\begin{aligned}
&\textbf{Mg-Al cycle}\\
&{}^{24}_{12}\text{Mg} + {}^{1}_{1}\text{p} \rightarrow {}^{25}_{13}\text{Al} + \gamma\\
&{}^{25}_{13}\text{Al} \rightarrow {}^{25}_{12}\text{Mg} + e^{+} + \nu_e\\
&{}^{25}_{12}\text{Mg} + {}^{1}_{1}\text{p} \rightarrow {}^{26}_{13}\text{Al} + \gamma\\
&{}^{26}_{13}\text{Al} \rightarrow {}^{26}_{12}\text{Mg} + e^{+} + \nu_e\\
&{}^{26}_{12}\text{Mg} + {}^{1}_{1}\text{p} \rightarrow {}^{27}_{13}\text{Al} + \gamma
\end{aligned}$$

Note, however, that when temperature raises over a given threshold, protonic captures and α-captures are competitive to β^{+} decays. Two more protonic captures are given for isotopes of Aluminium

$$\begin{aligned}
&{}^{26}_{13}\text{Al} + {}^{1}_{1}\text{p} \rightarrow {}^{27}_{14}\text{Si} + \gamma\\
&{}^{27}_{14}\text{Si} \rightarrow {}^{27}_{13}\text{Al} + e^{+} + \nu_e\\
&{}^{27}_{13}\text{Al} + {}^{1}_{1}\text{p} \rightarrow {}^{28}_{14}\text{Si}^{*} \rightarrow {}^{24}_{12}\text{Mg} + {}^{4}_{2}\text{He}\\
&\qquad\qquad\qquad\qquad\quad \rightarrow {}^{28}_{14}\text{Si} + \gamma
\end{aligned}$$

which reflects the so-called *Magnesium-Aluminium anticorrelation*. This property we're not able to see in all clusters, from which we'd able to infer, in principle, the presence of a contamination in the cluster.

This is a good point to stop for a second (and take a deep breath) and ask ourselves who's burning Hydrogen how.

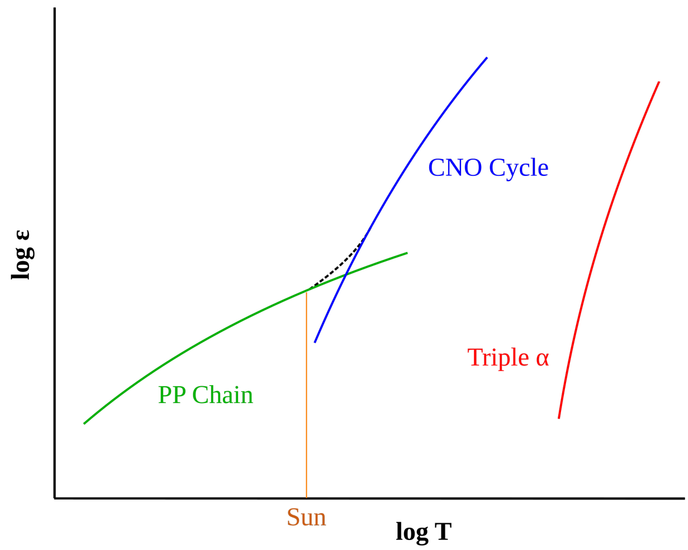

Figure 14: Logarithm of the relative energy output (ϵ) of proton-proton (p-p), CNO, and triple-α fusion processes at different temperatures. The dashed line shows the combined energy generation of the p-p and CNO processes within a star. Credits: Wikipedia.

It's possible to show that the energy production rate scales with $\sim T^4$ for the p-p chain, and with $\sim T^{17}$ for the CNO bi-cycle. When temperature gets over $\sim 20 \cdot 10^6$ K, the CNO takes over the p-p, as shown in Fig.14. At the end of the day, it all comes down to the mass of the star.

Lower Main Sequence stars (LMS), that have about solar chemical composition and are roughly in the range $M \lesssim 1.2 M_{\odot}$, will burn Hydrogen through the p-p chain, as will do *Very Low Mass Stars* (VLMS), which have masses $M \lesssim 0.3 M_{\odot}$ and are completely convective and very "cold".

Upper Main Sequence stars (UMS), that have masses $M \gtrsim 1.2 M_{\odot}$ will be burning Hydrogen through the CNO bi-cycle.

Note that as temperature climbs up to higher heights[7] under conditions such as those found in novae and X-ray bursts, the rate of proton captures can exceed the rate of β-decay, pushing the burning to the proton drip line.

The essential idea is that a radioactive species will capture a proton before it can β-decay, opening new nuclear burning pathways that are otherwise inaccessible. Because of the higher temperatures involved, these catalytic cycles are typically referred to as the *hot CNO cycles*. For example

H-CNO I

$$
\begin{aligned}
&{}^{12}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{p} \to {}^{13}_{7}\mathrm{N} + \gamma \\
&{}^{13}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \to {}^{14}_{8}\mathrm{O} + \gamma \\
&{}^{14}_{8}\mathrm{O} \to {}^{14}_{7}\mathrm{N} + e^{+} + \nu_e \\
&{}^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \to {}^{15}_{8}\mathrm{O} + \gamma \\
&{}^{15}_{8}\mathrm{O} \to {}^{15}_{7}\mathrm{N} + e^{+} + \nu_e \\
&{}^{15}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{p} \to {}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He}
\end{aligned}
$$

4.3.3 Helium Burning

All stars with initial total mass larger than $\sim 0.5 M_{\odot}$ are able to attain, in their interiors, the thermal conditions required for the onset of Helium burning. Unlike the Hydrogen burning stage, the physical processes at work in stars of different masses during the main core He-burning stage are quite similar.

The first and fundamental reaction in the He-burning process is the production of ^{12}C from the fusion of three Helium nuclei. This reaction is called the 3α reaction. This reaction usually occurs in two separate steps

$$
\begin{aligned}
&{}^{4}_{2}\mathrm{He} + {}^{4}_{2}\mathrm{He} \to {}^{8}_{4}\mathrm{Be} \\
&{}^{8}_{4}\mathrm{Be} + {}^{4}_{2}\mathrm{He} \to {}^{12}_{6}\mathrm{C} + \gamma
\end{aligned}
$$

Also, temperature is now high enough to allow α-captures on Carbon nuclei, producing Oxygen

$$
{}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He} \to {}^{16}_{8}\mathrm{O} + \gamma
$$

In a short time ($\sim 10^{-16}$ s), Berillium decays back into two α particles, thus making the probability of the second reaction extremely low and requiring longer timescales for the reaction to occur. However, as the interior temperature climbs up, the probability of the second reaction increases, due to these combined effects:

- the number of $\alpha + \alpha$ reactions increases, while the decay time for Berillium stands unchanged, thus significantly increasing the abundance of Berillium;

7 Poetical delirium.

- the nuclear cross section of the second reaction strongly increases.

A temperature of about $\sim 10^8$ K is necessary before 3α reactions start producing a sizeable amount of energy. Also, any increase in the density favors incredibly this reaction for

$$\epsilon_{3\alpha} \propto \rho^2 T^{41}$$

Even considering all of this, the measured abundances of Carbon and Oxygen in the Universe wouldn't be matching the results that the reactions we've written down would bear, because they're still strongly inefficient reactions.

In 1954, Fred Hoyle [14] predicted that in order to achieve the observed abundances of Carbon and Oxygen, the 3α reaction must be *resonant*.

That was indeed the case: the α-capture on Berillium nuclei has a broad resonance at approximately 300 keV (roughly to $\approx 2 \cdot 10^8$ K), meaning that the reaction gets influenced from the "tail" of the resonance, which would also explain how so much Oxygen is produced[8]

$$\frac{\mathrm{C}}{\mathrm{Z}} \sim 0.20 \quad \frac{\mathrm{O}}{\mathrm{Z}} \sim 0.47 \quad \frac{\mathrm{N}}{\mathrm{Z}} \sim 0.06$$

Since small stars which can achieve Helium burning can continue to do so for millions of years, we've now found a way to circumvent the instability hidden into elements with $A = 5, 8$, which prevented the formation of heavier nuclei.

Note that since the luminosity is fixed only by the *mass* of the star and from its *evolutionary stage*, various reactions might be competitive in order to balance the total luminosity of the star. So, for example, the contribution to luminosity might be distributed between the 3α and the $^{12}\mathrm{C}(\alpha, \gamma)\,^{16}\mathrm{O}$

$$L_{\mathrm{tot}} = L_{3\alpha} + L_{\mathrm{C}-\alpha}$$

with the latter being slightly favored because it has a lower Helium consumption rate (only one α instead of 3), which allows the star to live longer.

Production of free neutrons

Other important reactions are given

$$\begin{aligned}
&^{16}_{8}\mathrm{O} + ^{4}_{2}\mathrm{He} \to ^{20}_{10}\mathrm{Ne} + \gamma \\
&^{20}_{10}\mathrm{Ne} + ^{4}_{2}\mathrm{He} \to ^{24}_{12}\mathrm{Mg} + \gamma \\
&^{24}_{12}\mathrm{Mg} + ^{4}_{2}\mathrm{He} \to ^{28}_{14}\mathrm{Si} + \gamma
\end{aligned}$$

but the most notable feature of Helium burning is that the star is now able to produce *free neutrons*. Consider for example the following reaction chain

$$\begin{aligned}
&^{14}_{7}\mathrm{N} + ^{4}_{2}\mathrm{He} \to ^{18}_{9}\mathrm{F} + \gamma \\
&^{18}_{9}\mathrm{F} \to ^{18}_{8}\mathrm{O} + e^+ + \nu_e \\
&^{18}_{8}\mathrm{O} + ^{4}_{2}\mathrm{He} \to ^{22}_{10}\mathrm{Ne} + \gamma \\
&^{22}_{10}\mathrm{Ne} + ^{4}_{2}\mathrm{He} \to ^{25}_{12}\mathrm{Mg} + + ^{1}_{0}\mathrm{n}
\end{aligned}$$

8 Technically, Oxygen production is influenced by *three* resonant reactions.

For stars of solar metallicity, in which $Z_{\text{CNO}} \sim 0.01 Z_{\odot}$, we'd have, for example, a flux of $7 \cdot 10^{20}$ neutrons g^{-1}, but temperatures of order $3.0 - 3.5 \cdot 10^{8}\,\text{K}$ are needed. So, during the AGB phase, this chain of neutron production is possible only for $M \gtrsim 3M_{\odot}$ (and dominant for $M \gtrsim 5M_{\odot}$). Smaller stars produce free neutrons with a reaction involving Carbon

$$
\begin{aligned}
&{}^{12}_{6}\text{C} + {}^{1}_{1}\,\text{p} \rightarrow {}^{13}_{7}\,\text{N} + \gamma \\
&{}^{13}_{7}\text{N} \rightarrow {}^{13}_{6}\,\text{C} + e^{+} + \nu_e \\
&{}^{13}_{7}\text{C} + {}^{4}_{2}\,\text{He} \rightarrow {}^{17}_{O}\,\text{O}^{*} \rightarrow {}^{16}_{8}\,\text{O} + + {}^{1}_{0}\text{n}
\end{aligned}
$$

that is completed only at temperatures of order $\approx 8 \cdot 10^{7}\,\text{K}$, but not much higher since Helium burning would soon be starting.

Paradoxically, these reactions aren't easy to do not even during the AGB phase: Stars have the necessary temperatures, but, in those same regions, they don't have protons to spare. Other possible reactions involve the burning of heavier nuclei

$$
\begin{aligned}
&{}^{12}_{6}\text{C} + {}^{12}_{6}\,\text{C} \rightarrow {}^{23}_{12}\,\text{Mg} + {}^{1}_{0}\,\text{n} \\
&{}^{16}_{8}\text{O} + {}^{16}_{8}\,\text{O} \rightarrow {}^{31}_{16}\,\text{S} + {}^{1}_{0}\,\text{n} \\
&{}^{56}_{26}\text{Fe} + \gamma \rightarrow 13\,{}^{4}_{2}\text{He} + 4\,{}^{1}_{0}\text{n}
\end{aligned}
$$

where the latter is only active moments before a star goes supernova and photons are able to disintegrate Iron nuclei.

4.3.4 Heavier Elements Burning

Moving on to even higher temperatures, it is possible to ignite the combustion of heavier nuclei, such as Carbon. Since a temperature of order $T \sim 7 - 8 \cdot 10^{8}\,\text{K}$ is needed, many stars will never even get that far. The minimum mass to ignite Carbon is said M_{up} and is defined as such

$$
M_{\text{up}} = \begin{cases} \sim 7M_{\odot} & \text{metal poor halo stars, } Z \sim 10^{-4} \\ \sim 7M_{\odot} & \text{metallic disk stars, } Z \sim 0.015 \\ \sim 4M_{\odot} & \text{intermediate metallicity stars, } Z \sim 10^{-3} \end{cases}
$$

Not surprisingly, as progressively heavier nuclei are burned, more reactions products are possible. For Carbon

Carbon Burning

$$
\begin{aligned}
{}^{12}_{6}\text{C} + {}^{12}_{6}\,\text{C} \rightarrow {}^{24}_{12}\,\text{Mg}^{*} &\rightarrow {}^{20}_{10}\,\text{Ne} + {}^{4}_{2}\,\text{He} \quad (\Gamma \sim 50\%) \\
&\rightarrow {}^{23}_{11}\,\text{Na} + {}^{1}_{1}\,\text{p} \quad (\Gamma \sim 50\%) \\
&\rightarrow {}^{23}_{12}\,\text{Mg} + {}^{1}_{0}\,\text{n} \quad (\text{rare}) \\
&\rightarrow {}^{24}_{12}\,\text{Mg} + \gamma \quad (\text{very rare})
\end{aligned}
$$

Before reaching the temperature for Oxygen burning ($T \sim 1 \cdot 10^{9}\,\text{K}$), photons get energetic enough to photodisintegrate Neon

$$
{}^{20}_{10}\text{Ne} + \gamma \leftrightarrows {}^{16}_{8}\text{O} + {}^{4}_{2}\,\text{He}
$$

with the right hand side (so, Oxygen) being more favored. At these temperatures, α-captures are also possible

$$
\begin{aligned}
{}^{20}_{10}\mathrm{Ne} + {}^{4}_{2}\mathrm{He} &\rightarrow {}^{24}_{12}\mathrm{Mg} + \gamma \\
{}^{24}_{12}\mathrm{Mg} + {}^{4}_{2}\mathrm{He} &\rightarrow {}^{28}_{14}\mathrm{Si} + \gamma
\end{aligned}
$$

Finally we reach the temperature needed for Oxygen burning ($T \sim 1.4 \cdot 10^9$ K, $M \gtrsim 10 M_\odot$). As expected, multiple channels are available this time as well

Oxygen Burning

$$
\begin{aligned}
{}^{16}_{8}\mathrm{O} + {}^{16}_{8}\mathrm{O} \rightarrow {}^{32}_{16}\mathrm{S}^* &\rightarrow {}^{28}_{14}\mathrm{Si} + {}^{4}_{2}\mathrm{He} \quad (\Gamma \sim 45\%) \\
&\rightarrow {}^{31}_{15}\mathrm{P} + {}^{1}_{1}\mathrm{p} \quad (\Gamma \sim 45\%) \\
&\rightarrow {}^{31}_{16}\mathrm{S} + {}^{1}_{0}\mathrm{n} \quad (\Gamma \sim 10\%) \\
&\rightarrow {}^{32}_{16}\mathrm{S} + \gamma \quad (\text{very rare})
\end{aligned}
$$

Clearly, all other combinations of heavy nuclei are also possible, such as

$$
\begin{aligned}
{}^{12}_{6}\mathrm{C} + {}^{16}_{8}\mathrm{O} &\rightarrow {}^{24}_{12}\mathrm{Mg} + {}^{4}_{2}\mathrm{He} \\
{}^{12}_{6}\mathrm{C} + {}^{16}_{8}\mathrm{O} &\rightarrow {}^{27}_{13}\mathrm{Al} + {}^{1}_{1}\mathrm{p} \\
{}^{12}_{6}\mathrm{C} + {}^{20}_{10}\mathrm{Ne} &\rightarrow 2\,{}^{16}_{8}\mathrm{O}
\end{aligned}
$$

Now, the next in line after Oxygen would be Silicon. But there's a problem. The ignition temperature for Silicon burning is $T \sim 4.5 \cdot 10^9$ K, while at $T \sim 3 \cdot 10^9$ K Silicon gets photodisintegrated, as well as do other less bound nuclei, for example

$$
\begin{aligned}
{}^{28}_{14}\mathrm{Si} + \gamma \leftrightarrows\, &{}^{27}_{13}\mathrm{Al} + {}^{1}_{1}\mathrm{p} \\
&{}^{24}_{12}\mathrm{Mg} + {}^{4}_{2}\mathrm{He}
\end{aligned}
$$

This allows for a redistribution of chemical elements, favoring an overabundance of more tightly bound elements. At this temperatures are more probable α-captures up until the very most bound element (take a deep breath)

$$
\begin{aligned}
&{}^{28}_{14}\mathrm{Si} + {}^{4}_{2}\mathrm{He} \leftrightarrows {}^{32}_{16}\mathrm{S} + \gamma \\
&{}^{32}_{16}\mathrm{S} + {}^{4}_{2}\mathrm{He} \leftrightarrows {}^{36}_{18}\mathrm{Ar} + \gamma \\
&{}^{36}_{18}\mathrm{Ar} + {}^{4}_{2}\mathrm{He} \leftrightarrows {}^{40}_{20}\mathrm{Ca} + \gamma \\
&{}^{40}_{20}\mathrm{Ca} + {}^{4}_{2}\mathrm{He} \leftrightarrows {}^{44}_{22}\mathrm{Ti} + \gamma \\
&{}^{44}_{22}\mathrm{Ti} + {}^{4}_{2}\mathrm{He} \leftrightarrows {}^{48}_{24}\mathrm{Cr} + \gamma \\
&{}^{48}_{24}\mathrm{Cr} + {}^{4}_{2}\mathrm{He} \leftrightarrows {}^{52}_{26}\mathrm{Fe} + \gamma \\
&{}^{52}_{26}\mathrm{Fe} + {}^{4}_{2}\mathrm{He} \leftrightarrows {}^{56}_{28}\mathrm{Ni} + \gamma \\
&{}^{56}_{28}\mathrm{Ni} \rightarrow {}^{56}_{27}\mathrm{Co} + e^+ + \nu_e \rightarrow {}^{56}_{26}\mathrm{Fe} + 2(e^+ + \nu_e)
\end{aligned}
$$

The formation of ${}^{62}_{28}\mathrm{Ni}$, which would be the most tightly bound element, is greatly disfavored by the scarce number of reactions producing it. That's the reason why this crazed run for the most bound element stops only at ${}^{56}_{26}\mathrm{Fe}$.

4.3.5 Neutronic Captures

Up until now we've seen how elements up to Iron are produced all throughout stellar nucleosynthesis, and maybe you've began to wonder how even heavier elements like Silver, Gold and Platinum (if you're into riches) or Technetium, Uranium and Radium (if you're into radioactive stuff) are produced. The answer to all your questions was anticlimactically revealed in the title of this section: *Neutronic captures*.

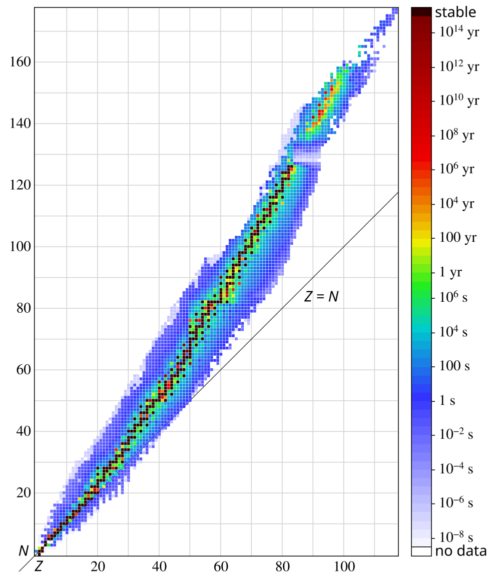

Figure 15: Isotope half-lives. The darker more stable isotope region departs from the line of protons $Z = N$ (neutrons) as the element number Z grows. Credits: Wikipedia.

As far as our understanding goes, there are two types of neutronic captures: *slow* (s) and *rapid* (r) captures. The distinction is essentially based on the comparison between the β^- decay half-life and the average timescale for neutronic capture in the case of an unstable element being produced.

Neutronic captures are usually happening during the AGB phase, which is millions of years-long and neutrons are supplied quite steadily all throughout the process. Other sources are typically SNe, during which we have an incredible amount of free neutrons available but that is more of a temporally-localized injection rather than a constant supply.

Since neutrons are, guess what, neutral, we don't have to worry about getting through the Coulombian potential as we've done earlier. If nuclei and neutrons are both thermalized (which is likely, if not certain) we can write

$$v_T = \sqrt{\frac{2kT}{\mu}} \qquad \sigma \propto \frac{1}{v} \propto \frac{1}{\sqrt{E}}$$

As we've done earlier, we now have to calculate (86) but considering only the Maxwellian contribution

$$\langle \sigma(v) v \rangle = \left(\frac{8}{\pi\mu} \right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma(E) E \exp(-E/kT) \mathrm{d}E$$

but since neutrons are neutral, the maximum reactivity is at $E_0 = kT$. Thus we can write an expression for the average interaction cross section of the form

$$\langle\sigma\rangle = \sigma(v_T) = \sigma_T$$

and

$$\langle\sigma(v)v\rangle = \text{const.} = \sigma_T v_T$$

Let us now calculate the average lifetime of a nuclide of mass A, so that we can compare it with the average timescales for neutronic captures.

The reaction we're interested in considering is

$$A(Z, N) + n \to (A+1)(Z, N+1)$$

We know that we can approximate $\dot{N}_A = -N_A(t)/\tau$ and, following its definition, we also have

$$\dot{N}_A = -N_n N_A \langle\sigma(v)v\rangle_{A+n}$$

Equating the two and solving for τ

$$\tau_n = \frac{1}{N_n \langle\sigma(v)\rangle_{A+n} v_T} \tag{107}$$

Note that in general the cross section $\langle\sigma(v)v\rangle_{A+n}$ is not constant with respect to A because of the presence of stable nuclides, in correspondence to which there is a "depression" because of the completed internal quantum states.

For s-processes, temperatures are those of Helium burning in shells, which are in the range $1 - 6 \cdot 10^8$ K.

As a reference nucleus we'll be taking ${}^{134}_{56}$Ba, which has a termal velocity $v_T \sim 3 \cdot 10^8\,\text{cm}\,\text{s}^{-1}$ and a cross section $\sigma_T \sim 0.1\,\text{b}$ at a temperature $T \sim 4 \cdot 10^8$ K ($kT = 30\,\text{keV}$). Thus

$$\sigma_T v_T \sim 3 \cdot 10^{-17}\,\text{cm}^3\,\text{s}^{-1}$$

If we want to make sure that a capture is indeed a s-capture, then we have to require $\tau_n \gg \tau_\beta$, with τ_β and τ_n respectively the timescales for β-decays and neutronic captures. Since the average timescale for β-decays in these conditions is $\tau_\beta \sim 3 \cdot 10^8$ s, the necessary neutron density is

$$N_n < 1 \cdot 10^8\,\text{cm}^{-3}$$

On the other hand, to make sure captures are of the r-type, we have to require $\tau_\beta \gg \tau_n$, with an average value of $1 \cdot 10^{-3}$ s, hence

$$N_n > 10^{20}\,\text{cm}^{-3}$$

In what remains of this chapter, we'll try to calculate the equilibrium abundance of (stable) nuclei that are forming due to neutronic captures, as well as explaining particular features in the abundances trend.

First of all, it's clear that if after a neutronic capture the resulting nucleus is stable, it won't be β-decaying, and will only be able to climb further on the N ladder. Consider

$$\dot{N}_A(t) = N_{A-1} N_n \langle\sigma v\rangle_{A-1+n} - N_A N_n \langle\sigma v\rangle_{A+n}$$

where the first term represents the contribution of neutronic captures, whilst the second that of decays. Since we'll consider here only s-captures, we can assume $\tau_s \gg \tau_\beta$, so that all unstable nuclei quickly decay into stable nuclei.

When equilibrium is reached, the abundance of nuclei of atomic weigth A is no longer varying, thus earning the label of secondary products. We can write the equilibrium condition as a function of the neutron flux $\Phi_n = N_n v_T$

$$\dot{N}_A(t) = \frac{\Phi_n}{v_T} N_{A-1} \langle \sigma \rangle_{A-1+n} v_T - \frac{\Phi_n}{v_T} N_A \langle \sigma \rangle_{A+n} v_T$$

so that if all nuclei are able to reach equilibrium and the flux of neutron is constant[9] we'd obtain

$$N_{A-1} \langle \sigma \rangle_{A-1+n} = N_A \langle \sigma \rangle_{A+n}$$

If we were to take a look at a plot[10] of the neutronic capture cross section as a function of A, we'd see that there are regions in which the product $N_A \sigma_A$ (dropping the bra-kets for convenience) is approximately constant in certain regions, while in others we're observing sharp drops. Why is it so?

Those sharp drops are always in correspondence with "magical numbers", which are essentially stable isotopes for which the neutronic capture cross section is strongly reduced. Hence the abundance of the associated secondary element increases, since the destruction processes are very weak.

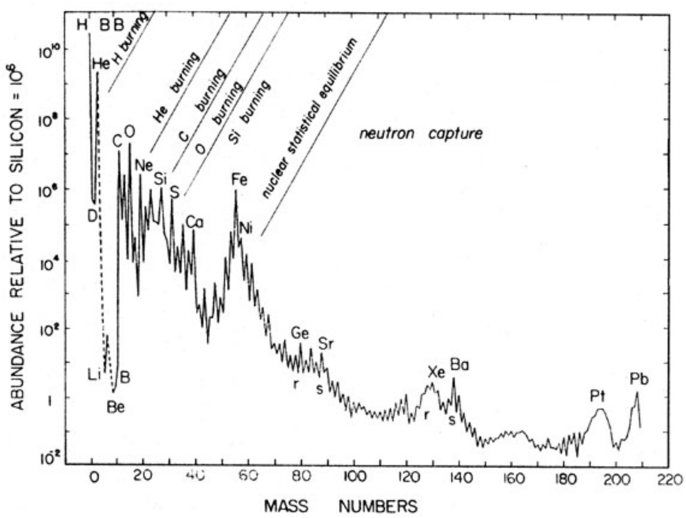

Figure 16: Numerical abundance of nuclides as observed in the Sun, normalized to 10^6 atoms of Silicon. Adapted from Cameron, 1920.

Note that for elements heavier than Iron there are some "bumps" in the curve shown in Fig.16, which appear in two kinds: Steeper s-peaks (e.g. that of Barium) and the rounder r-peaks (e.g. that of Xenon).

9 Note that this is an unreasonable assumption. The neutron flux can be assumed constant only during the AGB phase. When we're observing a star, however, we're observing its photosphere, and therefore we're seeing matter that has already been processed multiple times in plenty of other stars before and so it has been exposed to many different fluxes of neutrons in different stars for uneven amounts of time. Hence our initial claim.

10 I've yet to find one that is at least half-satisfying. For the time being, try to picture it in your head.

Finding an explaination for these particular shapes is actually very simple. For s-processes, the peaks are in correspondence of the aforementioned magic numbers: Neutronic capture on those nuclei is very hard because of closed shells and thus their abundance is very high (although still quite inferior to the equilibrium one).

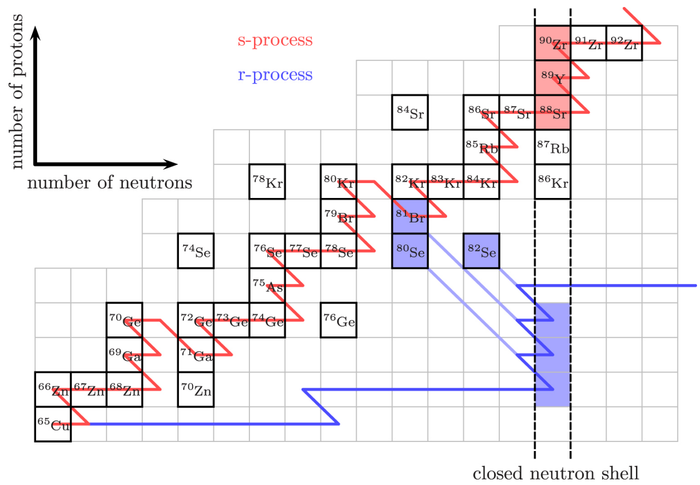

Figure 17: Schematic depiction of s-processes and r-processes. Credits: [13].

The other peaks are, surprise surprise, caused by r-processes. For this type of neutronic capture, the neutron flux is very intense but short-lived (roughly of the order of a couple seconds).

Consider an unstable element capturing neutrons. Since we're considering r-processes, the element isn't able to decay before capturing another neutron, and so it starts piling them up, until it reaches a number of neutrons that corresponds to that of a magic number. The capture cross section decreases abruptly and the nucleus is now able to decay.

The neutron flux hasn't disappeared, though. So the nucleus can capture a new neutron, reach the magic number and decay once more. When the neutron flux is finally over, the nucleus decays back into the "stability valley", but ends up with an atomic weigth A smaller than the closest magic number and therefore stops on the first stable isotope that is found.

This process thus leads to the formation of an element whose number of protons and neutrons is inferior to that of the closer magic number, hence the rounder peak we observe. That is because when the flux of neutrons is cut, not all elements will share the same configuration and will decay back to the first stable element at "hand" (see Fig.17 for a much clearer visual representation). The result is that there is a possible range of stable configurations rather than a single one.

Now you might be wondering: "If r-processes are exactly how we've described, nuclei shouldn't be able to go further than the first magic number they stumble upon. So how come they're not stopping there?"

It would be a very sensible question indeed.

The thing is that the more β-decays bring nuclei closer to the stability valley, the more neutronic captures become probable, hence, unless the neutron flux gets suddenly cut off, it is possible to capture another neutron, cross the magic number barrier and reach higher atomic weigths than the corresponding s-peak.

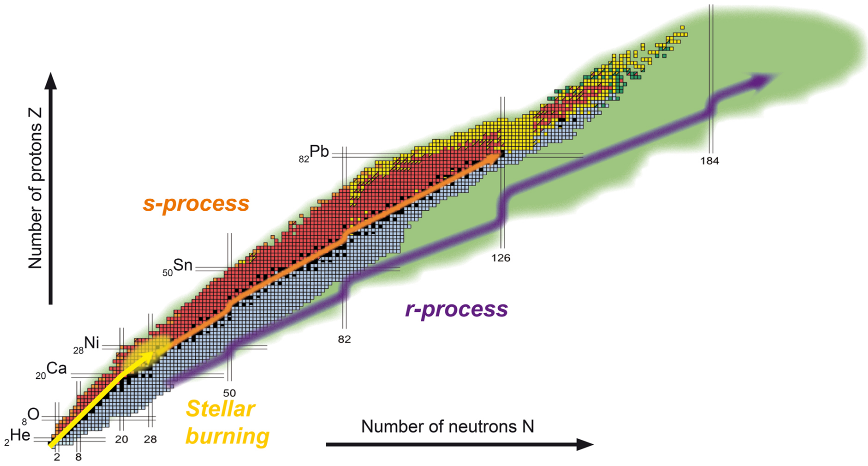

Figure 18: Nuclear chart showing the nucleosynthesis processes occurring during stellar burning (yellow), the s-process (orange) and the r-process (violet). Credits: this website.

Part II

STELLAR FORMATION AND EVOLUTION

5 STELLAR EVOLUTION

We know that stars are born into molecular clouds of atomic Hydrogen, Helium and so on. But for stars to ignite, molecular clouds have to first cool down, for example by radiating away the excess heat.

It's clear that in molecular clouds there's a complex interplay between cooling and heating mechanisms. The latter, for example, are powered by the passing of cosmic rays that ionize the atoms of the cloud, fast electrons that bump against other molecules and atoms thus heating them up, or maybe there are some nearby stars that are emitting X-rays, or high energy photons in general, that are able to heat up the cloud.

To cool down, the cloud is forced to emit in the IR since it is optically thick to the higher frequencies (optical, UV and so on). Emission in the IR range occurs mainly by the de-excitation of collisionally excited molecules returning to the ground state. Similarly, the cloud can cool down through radiative emission of dust, which is heated up by nearby stars and can be assumed to emit as a blackbody at a temperature of roughly $1000\,\mathrm{K}$[1].

If the cloud weren't cooling down, it would be kinda a problem since no stars would be able to form. In fact, consider the Jeans' mass

$$M_J \propto \left(\frac{T^3}{\rho}\right)^{1/2}$$

which is the limit mass for a gravitationally bound system to be stable against gravitational collapse. If the molecular cloud were to be shrinking adiabatically, it is possible to show that the Jeans' mass would increase with the density, thus leading to a monolithic collapse that would kinda suck for the purpose of forming non-popIII stars.

What happens is exactly the opposite. At first the cloud is in hydrostatic equilibrium[2], but then, due to some density perturbation, the cloud starts to shrink around its colder regions. This might happen for a whole series of reasons, for example the explosion of a nearby SN whose shockwave temporarily compresses part of the cloud inducing a collapse.

In this first phase the collapse is isothermal: The cloud is so sparse and the density so low ($n \approx 10 - 1000\,\mathrm{cm}^{-3}$) that the effect of opacity is essentially negligible, and thus the energy acquired through the gravitational collapse is immediately lost.

This thing that is shrinking is not a star, nor a protostar if we go by the strict meaning of the term[3], it is but a slightly-more-dense-than-average blob of matter. In this blob, density quickly increases as the blob shrinks, but the temperature stays roughly the same, hence the Jeans' mass decreases. The cloud incurs into fragmentation.

1 Dust cannot be emitting at higher temperatures simply because it starts melting.
2 We're neglecting the effects of rotations, magnetic fields etc. that would introduce a further complication that is beyond the aim of the present discussion.
3 A protostar would be a star as it grows up, when it is still accreting matter from the cloud.

As the blob shrinks, the inner regions become denser and denser, and only when it reaches density of order $10^{-13}\,\mathrm{g\,cm^{-3}}$ the blob becomes slightly opaque. Therefore it starts to retain part of the gravitational energy that would otherwise be depleted contracting adiabatically. It is in that moment that a first semblance of a core forms.

This first core is a conglomerate of matter (of order of $0.01 M_\odot$) in hydrostatic equilibrium and it's called *first Larson's core*. On this clump of matter there's a continuous infalling of other matter that leads to the formation of a shock front, where matter passes from supersonic to subsonic. In this region the kinetic energy of the infalling matter is released.

This first core, although in hydrostatic equilibrium, is unstable: As temperature increases, molecular Hydrogen dissociates ($\sim 2000\,\mathrm{K}$) and a new isothermal contraction begins: The excess energy released by the gravitational collapse is used to dissociate Hydrogen, thus leading to an expanding dissociation front from the hotter inner regions to the outskirts of the clump.

Now, this configuration is clearly unstable, for a little perturbation in density cannot still be matched by an analogous fluctuation in the pressure (remember that at this stage temperature isn't changing). Thus hydrostatic equilibrium is lost and the proto-core shrinks adiabatically once more, getting hotter and denser.

This process continues until the protostar is born and a *second Larson's core* is formed. This new core is in hydrostatic equilibrium and has a well-distinguishable structure from the rest of the cloud. The shock front is now on the surface of the core and matter is accreted with basically zero kinetic energy.

Note that although the system can be considered in hydrostatic equilibrium, it hasn't yet reached the conditions for thermal equilibrium. However, if we consider only what's within the shock front, we might as well be considereing that as a proper star.

At this stage the second Larson's core is about $1 - 50 M_{\mathrm{Jov}}$, so well-below the mass of the Sun, while the radius is still rather large, ranging from $1 - 10 R_\odot$, but there are still several criticalities on its determination, first and foremost the fact that the accretion rate is generally not known.

Consider that in the literature you might find values of the accretion rate $\dot{M}$ ranging from $10^{-7} - 10^{-3} M_\odot\,\mathrm{yr}^{-1}$, where the latter is for burst-type accretions. On top of that, we don't even know the temporal behavior of $\dot{M}$, so we can't really tell if the accretion is continuous, decreasing with time or is a localized burst.

We don't know how the accretion rate $\dot{M}$ varies with time but, moreover, we can't even see the star, since it's all nested within the cloud. Around the accreting star there is a region called *opacity gap*, where atomic Hydrogen gets dissociated by the energy released from the shock front near the core.

Note that atomic Hydrogen is essentially transparent, so in this region opacity is particularly low. Out of the opacity gap there's the rest of the cloud, not yet dissociated.

Of course, we don't even know the geometry that is followed by the accretion process: It is believed that the accretion must be necessarily spherical at first, only to later settle into accretion from a disk: The star's wind and radiation pressure sweep away part of the cloud, leaving only a thick disk from which the star continues to accrete.

At the end of the star's accretion process, what's left of the disk will be very thin and many believe that's how a protoplanetary disk is formed.

Despite this pretty pessimistic introduction there are also some good news. When the star reaches its total mass, its characteristics are essentially determined, apparently forgetting whatever the properties of the protostar were before that moment. This is great news: Apart from the first $10^5 - 10^6$ yr the star uses to accrete matter, stars all arrange themselves roughly in the same place in a HR diagram with similar characteristics.

This is what usually happens unless the accretion conditions are particularly extreme. Another quantity that determines the evolution during the accretion stage is the fraction of kinetic energy that is transferred to the star from infalling matter. We'll call the fraction that is transferred to the hydrostatic core α_{acc}.

Now if $\alpha_{\text{acc}} \gtrsim 14\%$, the accretion is *hot*, otherwise it's *cold*. Stars with cold accretion have extreme accretion conditions: During the accretion, the protostar shrinks too much, leading to strikingly different characteristics from other stars that have formed normally.

This characteristics, however, are in opposition to experimental observations, so some believe cold accretion isn't really a thing.

We set the *birth line*–when a protostar becomes effectively a star–at the moment when the protostar's accretion rate gets below $10^{-7} M_\odot\,\text{yr}^{-1}$. After the birth line, the star ufficially enters the *pre-main sequence* stage (PMS).

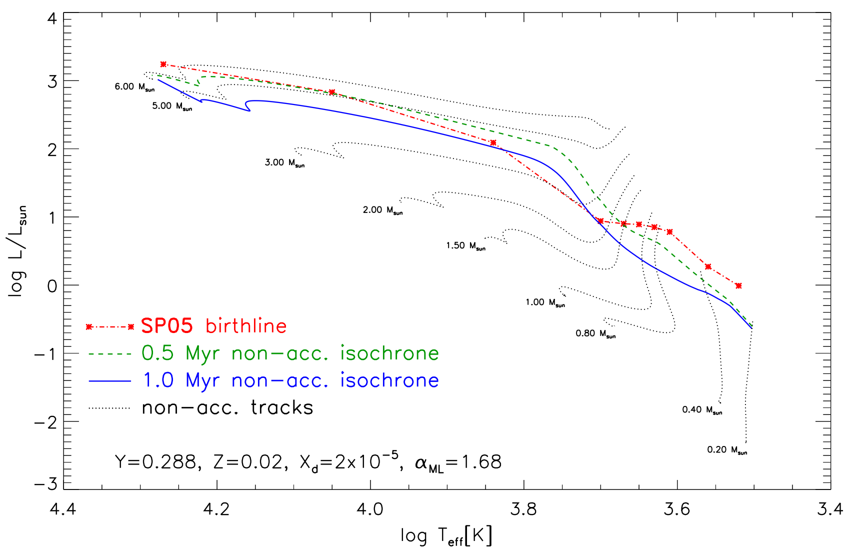

Figure 19

As we've perhaps mentioned in the last chapter, these stars are still quite cold, so there isn't much they can burn. The first element they will ignite is Deuterium, at a temperature of $\sim 1 \cdot 10^6$ K. This condition can be achieved both in the protostar phase and the pre-main sequence stage.

If Deuterium is ignited in the protostar stage, clearly it won't be stopping any time soon, since matter is still continuously falling into the star. This process continues until the star reaches its total mass and the molecular cloud is completely swept away.

If, instead, Deuterium is ignited during pre-main sequence stage, once it is completely depleted, the burning stops.

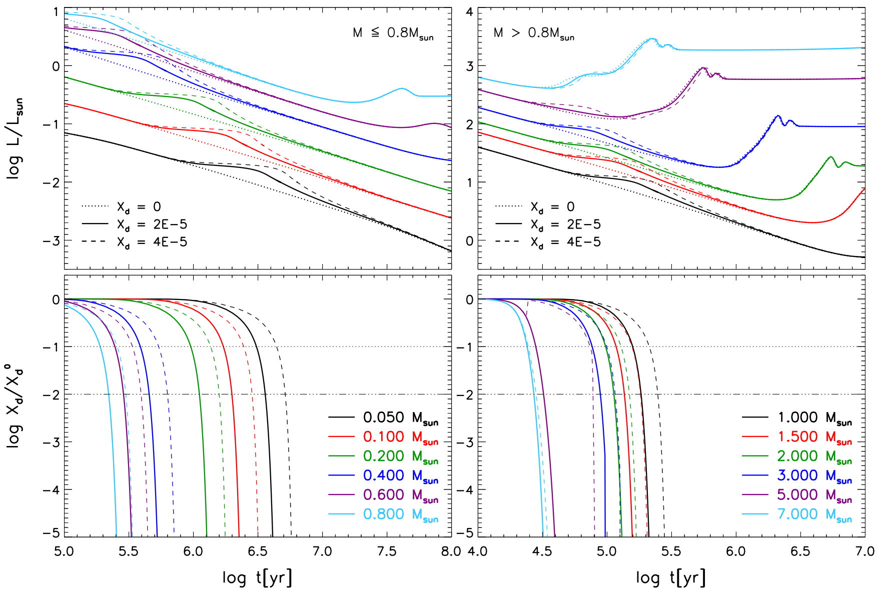

Figure 20: In the upper cartoons the evolution of luminosity over time for different initial abundances of Deuterium for masses $M \leq 0.8M_{\odot}$ (on the left), and $M \geq 0.8M_{\odot}$ (on the right). In the lower cartoons, the evolution of Deuterium abundance over time for different masses.

When a protostar "turns" into a star, it is still quite bloated, since during the accretion phase they have heated up and consequently expanded. But, they're still rather cold and thus quite luminous because they're all bloated up.

Since they're cold, the opacity is rather high. We know that opacity increases as the temperature decreases because of all the energy jumps, photoionization levels and so on. If the opacity is high, the radiative gradient is larger than the adiabatic gradient, and the star is, most of the times, completely convective. It might have a small radiative core in the inner, hotter regions, but as soon as Deuterium is ignited it vanishes, because something else happens at that point.

We know that the radiative gradient is proportional to the energy flux in cases like these. The star is convective because opacity is high. When Deuterium is ignited, the star heats up and the flux increases. The radiative gradient is still larger than the adiabatic one, so even if it has entered the pre-main sequence stage with a small radiative core, as soon as Deuterium is ignited it is lost.

The burning of Deuterium hinders the contraction of the star for a certain time and allows the star to "forget" everything that had happened before that moment. In the initial phase the star is completely mixed up so, even if at first it was dishomogeneous after all the complex accretion mechanisms, this dishomogeneity is lost.

One way or another, stars always pass through a stage where they are completely convective.

5.1 PRE-MAIN SEQUENCE

We've mentioned somewhere in the "few" introducing lines of this chapter that during the formation of a star, the yet-to-be protostar accretes matter from the surrounding molecular cloud. Such accretion might be distinguished into hot accretion or cold accretion[4]. Up until accretion stops, the luminosity of the star can be thought as made up of two terms

$$L_{\text{tot}} = L_{\text{acc}} + L_{\text{sup}}$$

where L_{acc} is the energy released through the accretion mechanism. The end of the protostar stage can be identified with the moment when $L_{\text{tot}} \approx L_{\text{sup}}$ and the accretion luminosity turns negligible. The speed of infalling matter is essentially equal to the escape velocity from the surface

$$0 = \frac{1}{2}mv_f^2 - \frac{GM_*m}{R_*} \implies v_f \sim \sqrt{\frac{2GM_*}{R_*}}$$

hence the kinetic energy associated with a clump of infalling matter Δm is just $E_{\text{k}} = \Delta m v_f^2/2$. The luminosity due to the accretion mechanism is just the variation over time of that kinetic energy (corrected with the efficiency of accretion α_{acc})

$$L_{\text{acc}} = (1-\alpha_{\text{acc}})\frac{\mathrm{d}E_k}{\mathrm{d}t} = (1-\alpha_{\text{acc}})\frac{G\dot{M}M_*}{R_*}$$

and a typical relation is given

$$L_{\text{acc}} = 61L_\odot \left(\frac{\dot{M}}{10^{-5}M_\odot\,\text{yr}^{-1}}\right)\left(\frac{M_*}{M_\odot}\right)\left(\frac{R_*}{5R_\odot}\right)^{-1}$$

As we've mentioned, at this stage the star is very bloated and cold, but its luminosity is rather high since

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4$$

where the large extension compensates for the low temperature. Since the star is cold, it is also very opaque, hence the radiative gradient is

$$\nabla_{\text{rad}} > \nabla_{\text{ad}}$$

and the star is basically everywhere convective. This is great news, because it means that the chemical composition of the star will be fairly homogeneous. On top of that, we know that, accretion or not, the newly born star will always be placed in the same spot in a HR diagram.

This is because the accreting stage is carried out in hydrostatic and (almost) thermal equilibrium, so the star will be regardless reaching a total

4 The latter in particular seems to be against experimental observations.

mass and complete convective-ness regardless of its previous accreting history (assuming it is not burst-like).

As our first model to describe a pre-main sequence star, we might avoid using the energy production equation (36) and assume adiabatic gradient everywhere but in the external regions, where superadiabaticity is not negligible[5]. At this stage, when nuclear reactions are still far from igniting, the evolution of the star is led by variations of the central temperature, that is highly influenced by the energy released during the gravitational collapse.

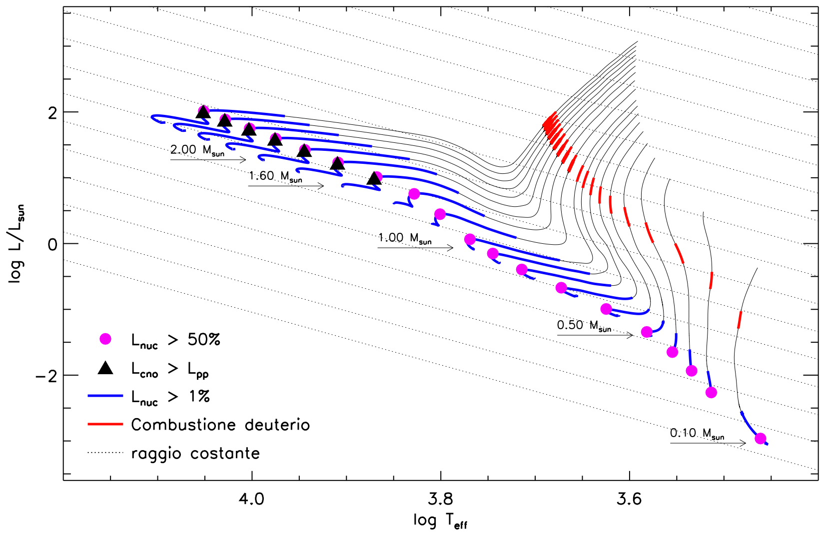

Figure 21: Evolution of different mass stars along the protostar and PMS stage. The red segments indicate the stage during which Deuterium is burned, while the blue segments indicate the stage where the luminosity generated by nuclear reactions is over 1%.

In his first works about the early stages of star evolution, Hayashi makes the following assumption

$$\frac{d \ln T}{d \ln p} = \text{const.} = A$$

and then varies A. What he observed is that increasing A, the effective temperature decreases. This is because, if we assume negligible superadiabaticity, the largest gradient that can settle in a structure is the adiabatic one, so

$$\nabla_{\text{amb}} \approx \nabla_{\text{ad}}$$

Thus, Hayashi proved that once the mass and chemical composition of a star are fixed there exists a forbidden region on an HR diagram, that is the region to the right of the line described by

$$\frac{d \ln T}{d \ln p} \sim 0.4$$

5 If you want to find an analytical model to describe a pre-main sequence star, you have to assume a constant adiabatic gradient. For example, assuming a monoatomic perfect gas that is also completely ionized, we'd find $\nabla_{\text{ad}} \sim 0.4$.

In the forbidden region, it can be shown that a star must have a temperature gradient greater than that limit. If a star is placed in the forbidden zone, with a temperature gradient much greater than 0.4, it will experience rapid convection that brings the gradient down. Since this convection will drastically change the star's pressure and temperature distribution, the star will be no longer in hydrostatic equilibrium, but will contract until it is. Eventually, the star is brought (literally) back on tracks.

We thus define *Hayashi's track* the limiting line along which all pre-MS stars will align, since they're both convective and in hydrostatic equilibrium. One striking feature of Hayashi's track is that, on an HR diagaram, it is almost vertical, meaning the star decreases its luminosity but keeps its effective temperature roughly constant all throughout its evolution.

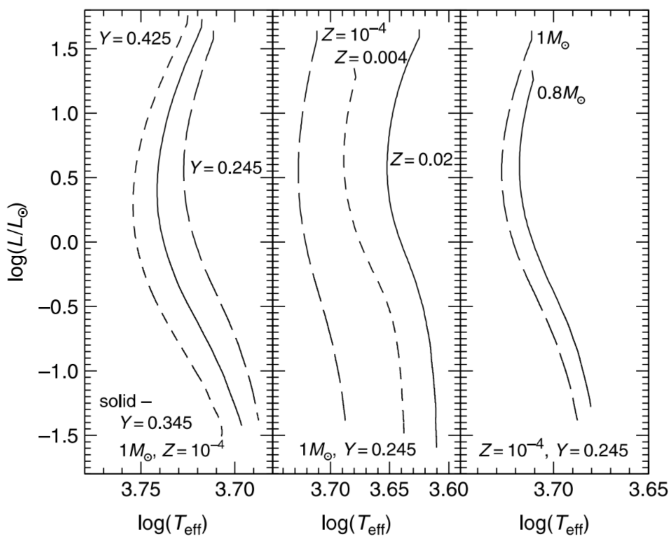

Figure 22: HR-diagram location of Hayashi tracks for various assumptions about the chemical composition and the stellar mass. Credits: [21].

This is because, given the low effective temperatures, the main source of opacity is that caused by the H^- (63). When the star shrinks, the temperature rises and so does the electron density and, consequently, the opacity. The radiative gradient quickly climbs up as well and so the change in temperature never really gets to the surface.

The evolution along the track is led by gravitational forces and so the timescale for the evolution is the Kelvin-Helmoltz

$$\tau_{KH} \propto \frac{GM^2}{RL}$$

but, since the mass is not changing anymore, it is effectively dependent only on the luminosity and the radius. Furthermore

$$L = 4\pi R^2 \sigma T_e^4$$

and since the effective temperature is constant along the track $R \sim L^{1/2}$, meaning that $\tau_{KH} \propto L^{-3/2}$. Hence the luminosity of the star is decreasing with time

$$L \propto t^{-2/3}$$

which is consistent with the star shrinking its radius along the track.

5.1.1 Analytical model

Although not particularly enticing, it is possible to find an analytical description for pre-MS stars. First of all we might describe the inner regions with a politropic equation

$$p = K_2 \rho^\gamma \quad \gamma = \frac{1}{1-\nabla_{\rm ad}}$$

which we might easily integrate as we've seen in earlier chapters. This, however, wouldn't work that well if we were to extend it all up to the atmosphere, because that's where our adiabatic convection approximation breaks down. A reasonable approximation to deal with the more external layers of a stellar atmosphere would be assuming radiative transport, using, say, the grey atomsphere approximation (78).

Even if the convection within the star reaches a layer above $\tau = 2/3$ (which is the demarcation line between the inner regions and the atmosphere in the grey atomsphere approximation), we can always perform the atmospheric integration down to the point where convection sets in and take the boundary conditions there.

If we assume

$$\frac{\mathrm{d}p}{\mathrm{d}\tau} = \frac{GM}{\bar{\kappa} R^2}$$

where the constant value of the opacity through the atmosphere is equal to $\bar{\kappa} = \kappa_0 p^a T^b$ (which is analogue to Kramers'). It is possible to show[6] that if we joint on a given boundary the solutions obtained by integrating the politropic equation and integrating over the atmosphere layers, we'd obtain a relation of this type

$$\log(T_{\rm eff}) = A \log\left(\frac{L}{L_\odot}\right) + B \log\left(\frac{M}{M_\odot}\right) + \text{const} \tag{108}$$

where A and B are two constants

$$A = \frac{0.75a - 0.25}{b + 5.5a + 1.5} \quad B = \frac{1.5a - 0.5}{b + 5.5a + 1.5}$$

Using realistic values for which the main source of opacity is the H^-, you'd obtain $A \sim 0.01$ and $B \sim 0.14$. The slope of the Hayashi track in the HR diagram is therefore

$$\frac{\partial \log L}{\partial \log T_{\rm eff}} \sim \frac{1}{A} > 10$$

meaning the Hayashi track is basically vertical; also, since

$$\frac{\partial \log T_{\rm eff}}{\partial \log M} \sim 0.14$$

its location has a mild dependence on the stellar mass, and it moves towards higher $T_{\rm eff}$ for increasing masses.

As soon as the star climbs down the Hayashi track, Deuterium is immediately ignited[7]

$${}_1^2\mathrm{D} + {}_1^1\mathrm{p} \rightarrow {}_2^3\mathrm{He} + \gamma$$

6 See for example [21] §4.3.1.

7 The required temperature, as we've seen, is of $T \sim 10^6$ K, has a Q value of ~ 5.5 MeV and an initial abundance $X_D \sim 2 \cdot 10^{-5}$.

The timescale for burning Deuterium is given by

$$\tau_D = \frac{1}{n_p \langle \sigma v \rangle_{\mathrm{D}+p}} \sim 10^6\,\mathrm{yr}$$

and the energy production rate is $\epsilon_D \propto T^{11-12}$. Note that if the burning of Deuterium is ignited in the protostar stage, this will go on as long as accretion feeds fresh Deuterium to the star.

It is also possible to show that an increase in metallicity (which causes an increase of the opacity) shifts the track to lower effective temperatures, whereas an increase of Helium at constant metallicity has the opposite effect. This can be roughly explained recalling the EoS for a perfect gas

$$p = \frac{\rho}{\mu m_{\mathrm{H}}} kT$$

If we assume the pressure to be fixed by the request of the star to be in hydrostatic equilibrium, an increase in Helium abundance increases the mean molecular weight and hence a higher temperature is needed to maintain equilibrium.

Meanwhile, in the external regions, the effect of convection is more relevant. In fact, if the mixing length coefficient α_{ML} decreases, the convective flux decreases (and the radiative one increases). Necessarily, the ambient temperature gradient has to ramp up and hence the effective temperature will decrease.

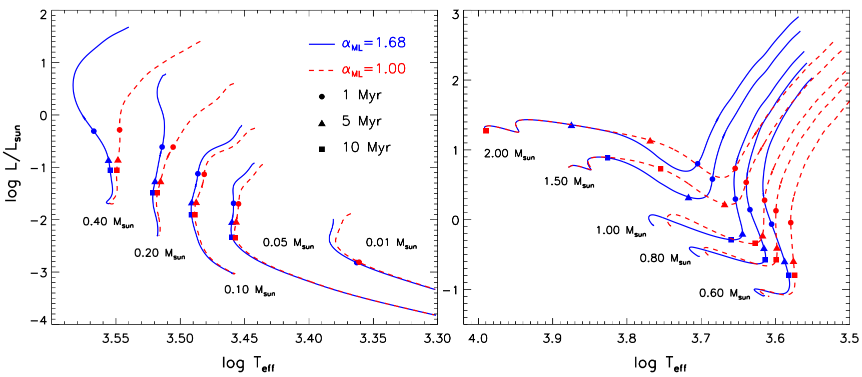

Figure 23: Effect of convection on the evolution of a star along the Hayashi track and further. Note how the great divergences introduced by convection early on are quickly smoothened away as the star progresses towards MS.

During the evolution along the Hayashi track, the star increases its temperature due to the virial theorem (40), and a radiative core forms at the center, growing in mass as the star evolves. This evolutionary phase is the so-called PMS (*Pre-Main Sequence*).

When the radiative core appears, the star is no longer fully convective, and it has to depart from its Hayshi track; this departure moves the track towards higher T_{eff} so that the Hayashi track acts as a rightmost boundary to the evolution of stars in a HR diagram. In fact, as we mentioned, a fully convective star *has* to lie on the Hayashi track.

This straying off the Hayashi track is almost horizontal when a sizeable radiative core is established, with a constant increase of the central temperature due to the virial theorem. This phase is sometimes known as *Henyey phase*. Here nuclear reactions have yet to begin (or grow relevant, depends on the points of view), and stars present different trends for luminosity depending on their mass

$$L_{\text{LMS}} \propto T_{\text{eff}}^{0.8} M^{4.4} \quad L_{\text{UMS}} \propto \frac{M^3 \mu^{-4}}{\bar{\kappa}}$$

Less massive stars (VLMS, $M \lesssim 0.3 M_\odot$) have other problems since they keep being convective and therefore stick to the Hayashi track[8].

We define *zero age main sequence* (ZAMS) the beginning of the MS, which has to be identified with the moment during which the star is entirely supported by Hydrogen burning and all secondary elements have reached equilibrium[9].

Consider for example UMS stars: As the star moves towards the equilibrium configuration, the core shrinks and the luminosity decreases. All production rates slow down to the pace of the rate of $^{14}_{7}$N and the energy production rate decreases consequently. To balance the now missing energy the star has to shrink, until $^{14}_{7}$N reaches equilibrium in a configuration of higher luminosity and effective temperature.

5.2 MAIN SEQUENCE FOR LMS STARS

For stars of about $1 M_\odot$ in mass and solar composition, the end of the Hayashi track takes place after about 10^7 yr, not long after the onset of the last fully-convective model. In fact, upon entering the Henyey stage the radiative core has extended up to 70% of the total mass.

For stars in which the pp-chain is dominant, the element that most postpones entering MS is $^{3}_{2}$He, whose burning reaction gets activated only at higher temperatures.

Near the end of PMS, due to the lack of $^{3}_{2}$He nuclei, the dominant reaction is exactly that of $^{3}_{2}$He production. As the nucelar processing proceeds, the amount of this element increases[10], thus increasing the number of reactions producing $^{4}_{2}$He. The large number of these reactions allow the star to complete the ppI branch and to produce more energy per burned Hydrogen.

During previous phases, the star was forced to reach a higher temperature and density in order to satisfy its energy needs, due to a only partially efficient pp chain. Since that problem now is no more, the star slightly decreases

8 Note that for stars in the range $M \lesssim 0.4 M_\odot$ the $^{3}_{2}$He becomes a pseudo-primary element, never reaching equilibrium.

9 Note that this might need a lot of time depending on the reaction cross sections. In the CN-NO bi-cycle, the main postponer of the beginning of MS is $^{14}_{7}$N, which has a tiny cross section.

10 Note that the abundance of $^{3}_{2}$He will be steadily rising in time, as well as rising as we move to the outskirts of the star, where the burning reaction is even more suppressed. Once a certain threshold is crossed, however, the abundance of $^{3}_{2}$He quickly drops, as the production rate is also suppressed due to lower temperatures.

its core density and temperature, fixing the number of nuclear reactions to the exact value required to match its energy needs.

During the phases preceding ${}^{3}_{2}$He equilibrium, Helium is burned only in a very spatially-limited region around the center. This means that for a short time the production is larger in the central part of the star, so energy production is more concentrated in the centre.

Due to the large energy flux $F \propto \nabla_{\rm rad}$, the central region becomes convective and a small, homogeneous, convective core establishes. This convective core, however, vanishes as soon as the abundance of ${}^{3}_{2}$He is increases in more external regions, as this causes a redistribution of energy generation over a larger area.

Beginning on the ZAMS, a LMS star burns Hydrogen in a *radiative region* whereas, due to the large opacity associated with the presence of partially ionized Hydrogen and Helium, the *outer regions are convective*. As you might imagine, Hydrogen is mainly burned in the inner regions, where conditions are more favorable, and we might expect its abundance to be slowly decreasing as the star evolves through MS. In Fig.24 the H chemical profile is shown for a $1M_\odot$ star at various stages, from the ZAMS to the exhaustion of Hydrogen in the core.

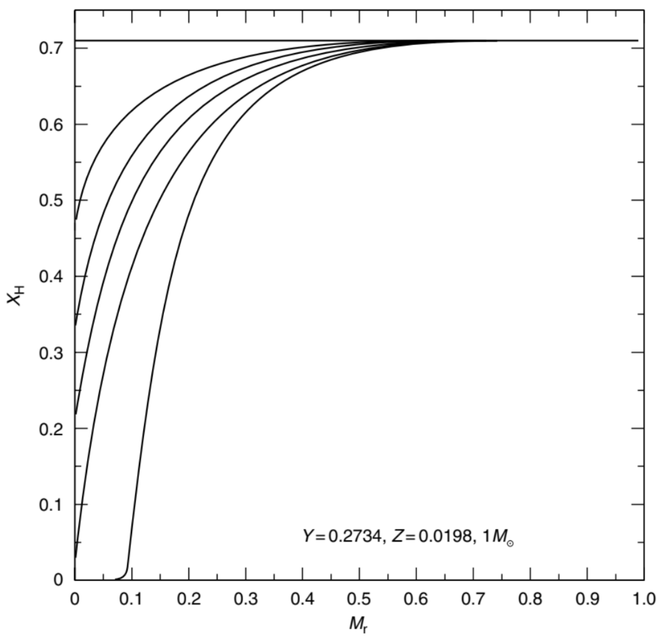

Figure 24: The chemical profile of Hydrogen in a $1M_\odot$ star at different stages during the core H-burning phase. Credits: [21].

During the conversion of Hydrogen into Helium the total number of free particles decreases and so does pressure. So, in order to remain in equilibrium, the star contracts ever so slightly and heats up a bit. The combined effect of the change in opacity (which decreases) in the stellar interior, the increase in mean molecular weight and temperature, all cause a slow monotonic increase of the luminosity. As per many things in this beatiful subject, a well-done image is worth more than a thousand of (my) words, so you might want to take a look at Fig.25.

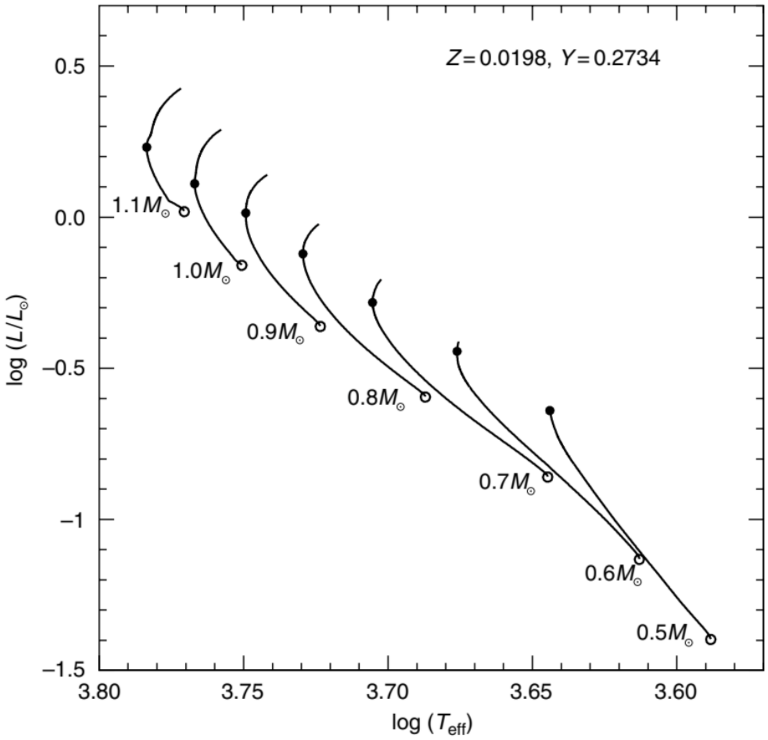

Figure 25: The HR diagram for LMS stars of different mass during the core H-burning phase. The solid dot marks the location of the turn off along each track. Credits: [21].

The central Hydrogen burning continues until, surprise surprise, the Hydrogen at the center is completely exhausted; this stage corresponds roughly to the hottest point on the evolutionary track in Fig.25, called the *turn off*, as it marks the end of the central burning phase for LMS stars.

Since the TO is easily identified, it is of pivotal importance to date clusters. If we were to indulge for a moment in technicalities, we might as well point out that in the surroundings of the TO the curve is fairly vertical. So, while it's pretty easy to identify the color (effective temperature) associated to the TO, it is usually much more difficult to determine the corresponding absolute magnitude (luminosity).

To circumvent the problem, we'd usually consider a slight displacement in color, about 0.05, and consider an average of the two luminosities, thus identifying the luminosity of the turn off up to a reasonable precision.

At the exhaustion of Hydrogen in the core, as the burning process already extends to a significant portion of the surrounding regions, the core slightly contracts (building up the conditions to ignite Helium) while H-burning continues in a shell around an inert He-core.

As a result of shell Hydrogen burning, heat is transferred to the surrounding envelope, which bloates up, thus decreasing the effective temperature. Opacity, on the other hand, increases, leading to an extension of the convective core. The sum of all these contributions makes the star "curve back" on the HR diagram towards the Hayashi track[11].

What about UMS stars?

11 Please note that this is just to give a sense of direction.

5.3 MAIN SEQUENCE FOR UMS STARS

The evolutionary path during the central Hydrogen burning phase for UMS stars is characterized by a monotonic increase of the luminosity and an almost steady decrease of the effective temperature. Why is that so?

The main effect of considering more massive stars is the significant increase in the interior temperature, that strongly favors energy production through the CN-NO (bi)cycle rather than the pp-chain. The strong temperature dependence has two main effects:

- the energy production is concentrated to the center;
- the concentration of energy production causes a steep increase of the radiative gradient towards the centre, as there is a very high energy flux in the innermost portion of the structure. Due to this, according to Schwartzschild's criterion (**??**), the interior of the star becomes convective and it will remain unstable against convection throughout the whole central H-burning phase.

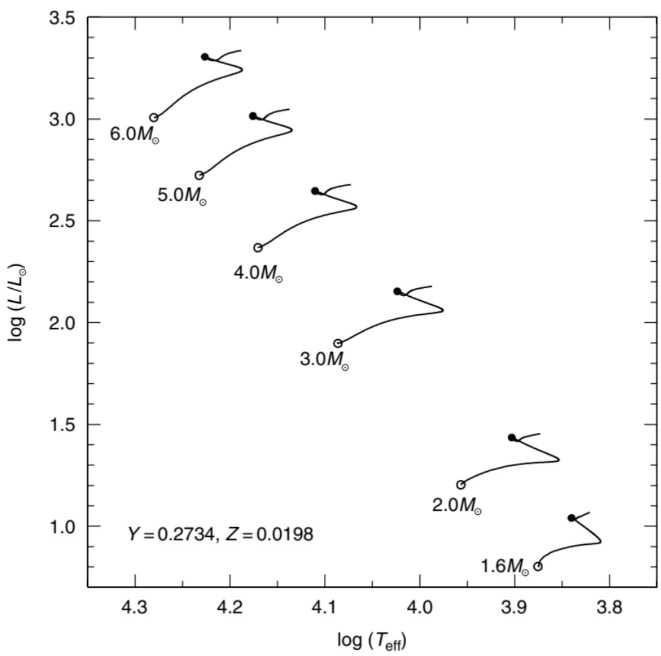

Figure 26: Evolutionary tracks of different intermediate mass stars during the core H-burning phase. The solid dot marks the evolutionary stage equivalent to the TO point. As we'll see, this phase is called *overall contraction* phase. Credits: [21].

Thanks to the presence of a convective core, Hydrogen is turned into Helium in a fully-mixed region. As a consequence, the chemical gradient inside UMS stars at various times during MS is completely different from that in LMS stars. The difference is most evident if you compare Fig.24 to Fig.27.

With increasing stellar mass, an additional effect related to the temperature increase is the increasing contribution to the total pressure coming from radiation pressure ($p_{\rm rad} \propto T^4$), which can get as high as 30% in high mass stars.

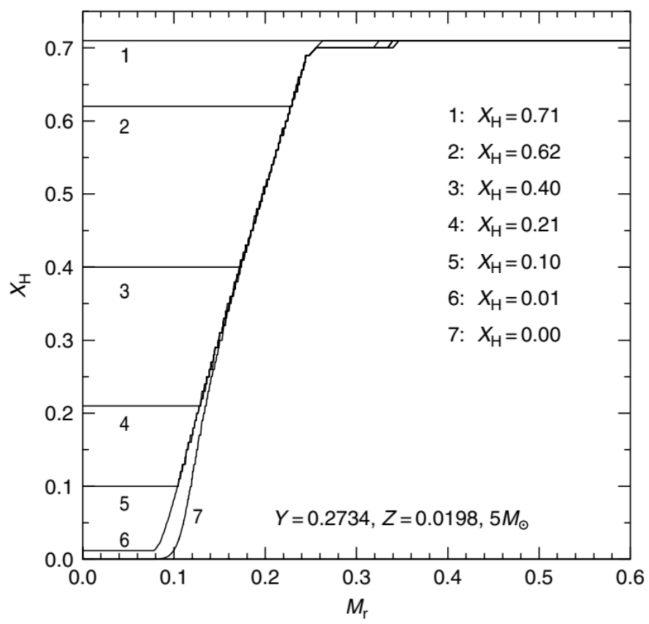

Figure 27: The chemical profile of Hydrogen in a $5M_\odot$ star at different stages. Credits: [21].

Not surprisingly, as the stellar mass increases, the size of the convective core increases accordingly, as a consequence of the larger temperature in the interior which leads to a larger energy flux. In addition, for massive stars, the decrease of the adiabatic gradient due to the increasing contribution of radiation pressure has a significant impact on the dimensions of the convective core.

At odds with LMS stars, UMS stars have a convective core and a radiative envelope. This is because the regions of partial ionization of Hydrogen and Helium are located too high up in the atmosphere (thus at too low a density) to affect the thermal properties of the envelope.

There's a problem we've briefly mentioned when we were discussing about convective motion that now makes a grand return as strong and problematic as ever: The problem of overshooting. Overshooting[12] poses a significant limitation to the determination of the extension of MS luminosity.

Let us now consider another major consequence of the presence of a convective core at the end of the central H-burning phase: When the abundance of Hydrogen inside the core drops below the 5% threshold, the remaining amount of Hydrogen is not sufficient for providing the energy necessary to maintain the structure in equilibrium, and so the star begins to contract. During this evolutionary phase, called *overall contraction*, the dominant energy source is gravitational, and so the evolutionary timescale is the Kelvin-Helmoltz ($\tau_{OC} \approx 10^8$ yr).

After this point, while the central region contracts, the outer layers expand and this occurence produces an increase in the stellar radius and a cooling of the external layers. This drastically increases the opacity of the envelope.

12 A quick reminder. The problem inherent with overshooting is that convective bubbles do not magically stop when trespassing the edge of the convective region, and thus the convective region "bleeds" a bit into the radiative one.

In massive stars, there is yet another phenomenon associated to convection which makes evolutionary properties near the end of core H-burning uncertain. When the abundance of Hydrogen in the core is significantly decreased, the convective core recedes, leaving behind a chemical abundance gradient. In the region outside the contracting core, the radiative flux increases as well as a consequence of the large energy flux produced by gravitational collapse, and therefore exceeds the adiabatic gradient.

According to (**??**), this region should become convective, but, if the region is mixed with the original convective core, its resulting Helium abundance should be larger than the initial one. Hence, as a consequence of opacity, the radiative gradient of these layers would decrease, inihibiting convection.

The solution to this problem is related to the choice of the most appropriate thermal gradient in a chemically inhomogeneous region, which usually comes down to the choice between the Schwartzschild criterion and the Ledoux criterion. It is a common procedure to adopt the Ledoux criterion (56), which usually inhibits the onset of convection in layers around the convective core.

5.3.1 The MS Dependence on Chemical Composition and Convection

We've seen someplace else, (44) for example, that along the main sequence we might expect a certain relation between the luminosity and the effective temperature (roughly $L \propto M^3 \propto T^3$). In a HR diagram we're clearly not seeing a straight line as we might have expected, bur rather some sequence of curves whose form depends on the mass the star. In general, we observe changes in inclination at the transition point between LMS and UMS (at $M \sim 1.2 M_\odot$) and where there is the formation of molecular Hydrogen in the atmosphere ($M \sim 0.5 M_\odot$). Another point of interest is given at $M \sim 0.15 M_\odot$, beyond which the contribution of electron degeneracy is no longer negligible.

If we change the chemical composition, perhaps increasing the abundance of Helium, we observe that

$$Y \uparrow \; T \uparrow \; \bar{\kappa} \downarrow \Longrightarrow L \uparrow \; T_{\text{eff}} \uparrow$$

meaning that at fixed color index (i.e., at fixed effective temperature) stars with more Helium are less luminous than average. If we change the metallicity of the star, the expected behavior is exactly the other way around

$$Z \uparrow \; \bar{\kappa} \uparrow \Longrightarrow L \downarrow \; T_{\text{eff}} \downarrow$$

hence at fixed color index, more metallic stars will be more luminous than average[13].

It should be expected that different stars will experience less severe temperature changes when trying to balance the effects of gravitational collapse. Consider for example a UMS. If the energy production rate is subpar with

13 Please note that, in principle, an increase in metallicity should induce an increase in the molecular weight and thus an increase in temperature. Changes in metallicity are, however, often negligibly small, so the dominant effect is that of the increase in opacity, which implies a decrease in luminosity and T_{eff}.

respect to the luminosity of the star, the structure will shrink, making temperatures increase. Since UMSs burn Hydrogen via the CN-NO cycle, which scales with high powers of the temperature, even a small change in central temperature would induce a large variation in the energy production rate.

It is for this reason that the curve representing the central temperature dependence on the mass is more steep for LMSs and barely rising for UMSs, as pictorially depicted in Fig.28.

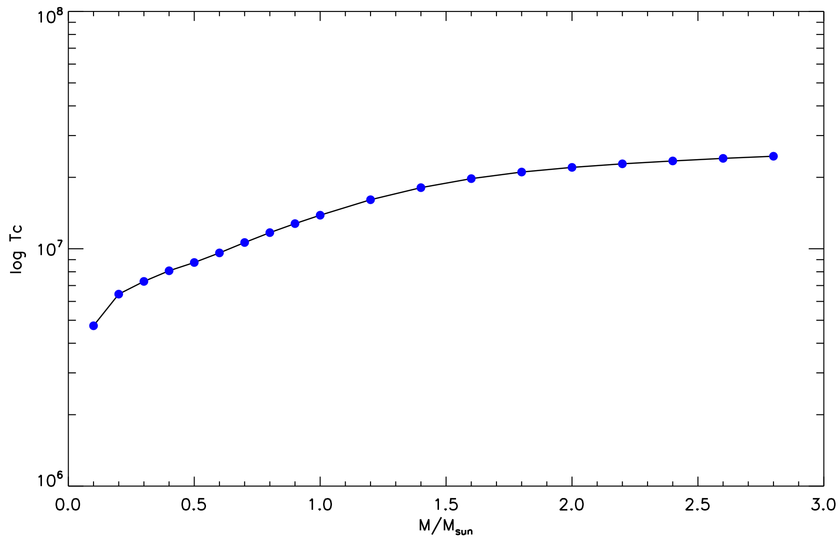

Figure 28: Curve representing how the central temperature of stars varies depending on the star mass. Note how the slope of the curve changes around $1 - 1.2 M_{\odot}$, where the transition between LMS and UMS is.

Starting from the ZAMS (*Zero Age Main Sequence*), the star begins to evolve, burning steadily Hydrogen in its core, and ultimately enters the MS. When this happens, the star shrinks (because of the "abrupt" decrease in Hydrogen) and central temperature climbs. The star will therefore progress to a new equilibrium configuration with a larger radius (because of the excessive energy production), and higher luminosity (because of the smaller opacity).

Please note that, in principle, *there is no correlation between the central temperature and the effective temperature*. The latter is dependent only on variation of luminosity and radius. This process and constant shift between equilibrium configurations goes on until Hydrogen is completely depleted in the core.

5.3.2 Main Sequence Fitting

Disclaimer: This part I'm mostly copying and cropping from [21], §9.2.6, to match with my own notes.

Figure 29 shows how the brightness of the lower MS of old SSPs[14] is unaffected by the age of the stellar population, even for old SSPs, because stars in this magnitude range are essentially still on their ZAMS location. It is only the initial chemical composition that determines the location of the

14 *Simple Stellar Populations*. They consist of objects born roughly at the same time in a burst of star formation of negligible duration, with the same chemcial composition. Good examples of SSPs are globular clusters and open clusters.

lower MS, that becomes redder for increasing metallicity and/or decreasing Helium mass fraction.

Once the Y and Z values are fixed, the lower MS can be used as a template and compared to the observed MS in an SSP with the same initial chemical composition. This is the so-called *MS-fitting method*.

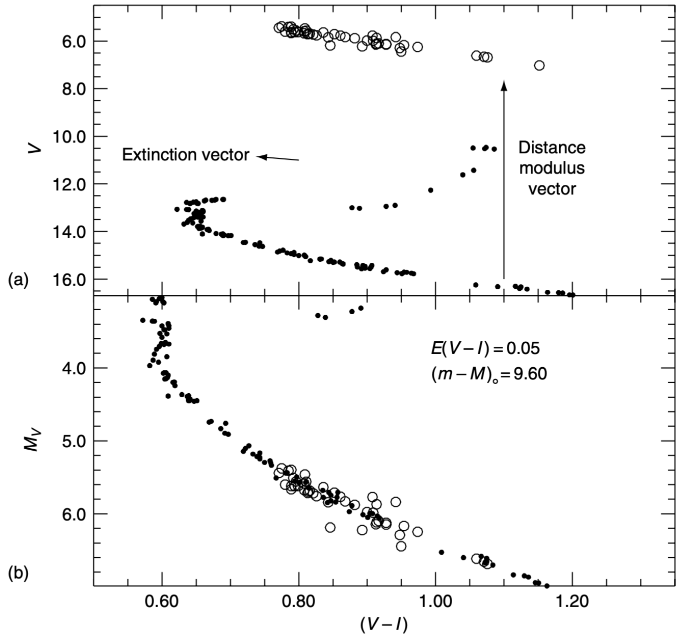

Figure 29: Example of MS-fitting distance determination applied to the old open cluster M67. (a) This displays the absolute magnitude and intrinsic colors of the template MS (large open circles) for the cluster metallicity ([Fe/H] $= +0.02$) and the observed magnitudes and colors of the cluster (small filled circles). The effect of extinction on the cluster CMD (E$[B-V] = 0.05$) and the effect of the distance modulus are also shown. (b) This shows the best fit of the (reddening corrected) cluster MS to the template one, and the derived distance modulus. Credits: [21].

In order to determine SSP distances using the MS-fitting method, we need the observed color-magnitude diagram of the SSP's main sequence in some filter combination and a template for the same chemical composition of the SSP, which may be either theoretical or empirical. In order to derive accurate distances, anything which may systematically affect the intrinsic and observed colors and magnitudes of either the cluster or the template MS must be accounted for (e.g., reddening).

In short, the MS fitting method reduces to computing the difference between the absolute magnitudes of the template MS and the apparent magnitudes of the observed one immediately, which provides the population distance modulus, hence the distance in parsec.

The choice of the template main sequence is obviously extremely important, and the preferred method is to build an empirical template MS instead of using theoretical isochrones, due to the existing uncertainties in the stellar $T_{\rm eff}$ scale and color transformations. In fact, an uncertainty of only 0.02 mag in colors (not due to the uncertainty in the reddening) translates into an uncertainty of ~ 0.10 mag in the derived distance modulus, and a consequent uncertainty of about $\sim 1\,\mathrm{Gyr}$ (which is gargantuan to say the least) in the ages obtained from the absolute magnitude of the observed TO.

An empirical MS is built by considering local field stars of known [Fe/H] with distances determined geometrically from parallax measurements, and known or negligible reddening. The main problem with this approach is that only a few of these stars with parallax distances have the exact metallicity of the SSP under scrutiny; therefore it is difficult to determine the appropriate template MS.

To overcome this problem one has to shift the position of many template field stars of various [Fe/H] values to the location they would have at the metallicity of the SSP.

This method can be particularly useful (although not always the most effective) to determine the distance of objects for which we cannot obtain precise parallax measurements.

Disk stars can also be used to calculate the Helium-to-metals enrichment ratio (16), which is necessary to determine the Helium abundance in stars we're not able to measure it[15]. One of the most common methods that were used in the past was trying to fit nearby stars with different Helium abundances as a consequence of the varying metallicity. For this purpose, the theoretical ZAMS was used, accepting the lower precision, on stars low in their evolution. To make sure stars were really at the beginning of the ZAMS and not at some random point in the midst, low mass (and thus faint) stars were generally chosen. The reason for this is that their MS is so long that their evolution is barely progressing.

As you can imagine, reducing to study faint stars wasn't that much easy, and as the surface temperatures decreased (as they correctly should) more complications arose. At the end of it, the theoretical template turned out to be "more blue" than the observations, likely because of the increasing relevance of molecules.

5.3.3 Constraints on the MS

As we've mentioned, not all stars forming from a molecular cloud will be able to ignite Hydrogen and shine bright like a new star, but they'll rather be cooling down and turn into brown dwarves, whose degeneracy pressure has halted the core contraction and effectively nullified their chance to become a star[16]. In the following we're going to consider an average density

$$\bar{\rho} \propto \frac{M}{R^3}$$

which we'll soon identify with the central density, and the dimensional analysis version of the hydrostatic equilibrium equation

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{GM(r)\rho(r)}{r^2} \rightarrow p_C \propto \frac{GM\bar{\rho}}{R}$$

in which we can plug in the definition of radius from inverting the expression for the average density, thus obtaining

$$p_C \propto M^{2/3} \rho_C^{4/3}$$

15 The interested reader might look at [7] for an example.
16 Electrons are the true career-killers.

Now we'd like to see if stars supported by a non-negligible contribution from degeneracy pressure can reach the necessary temperatures to ignite Hydrogen

$$p_C = k_{NR} n_e^{5/3} + n_i k T_c \quad n_e = \frac{\bar{Z}\rho}{\mu_i m_{\mathrm{H}}} \quad n_i = \frac{\rho}{\mu_i m_{\mathrm{H}}}$$

If we assume the gas to be made only of Hydrogen

$$p_C = k_{NR} \left(\frac{\rho_C}{m_{\mathrm{H}}}\right)^{5/3} + \left(\frac{\rho_C}{m_{\mathrm{H}}}\right) k T_c$$

Now, if we substitute in the hydrostatic equilibrium expression and explicit temperature

$$kT_C = c_1 M^{2/3} \rho_C^{1/3} - k_{NR} \left(\frac{\rho_C}{m_{\mathrm{H}}}\right)^{5/3} \implies T_C = \mathcal{F}(\rho_c)$$

If we compute the derivative of this expression with respect to the central density, we find out that the only stationary point is when $\rho_c \propto M^2$, which automatically implies

$$T_{C,\min} \propto M^{4/3}$$

If we fix the value of $T_{C,\min}$ with the one needed for H-burning to start (or complete), you'll surely notice that we've now found a minimum mass for which that process is indeed possible. The precise value, however, cannot be inferred from the present calculation.

Just as there is a minimum mass for a star to enter MS, there is a maximum mass for which a star can keep being a star. Such maximum value is due to the destabilizing effect that radiation pressure has on more massive stars. Consider the equation for hydrostatic equilibrium and the one for the radiation pressure

$$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{GM(r)\rho(r)}{r^2} \qquad \frac{\mathrm{d}p_{\mathrm{rad}}}{\mathrm{d}r} = \frac{4}{3} a T^3 \frac{\mathrm{d}T}{\mathrm{d}r}$$

The latter, if you recall how we've defined the temperature gradient, turns into

$$\frac{\mathrm{d}p_{\mathrm{rad}}}{\mathrm{d}r} = -\frac{\bar{\kappa}\rho}{c}\frac{L(r)}{4\pi r^2}$$

This we can equate to the gravitational force assuming, for the sake of the present discussion, negligible gas pressure. This yields a limit luminosity for which the system can be stable. Behold

$$L < L_{\mathrm{Edd}} = \frac{4\pi G c M}{\bar{\kappa}} \tag{109}$$

which is known as Eddington's luminosity. In the literature there is no common agreement to the maximum mass a star can have due to this criterion. Rough estimates put the limit to about $M \sim 150 M_\odot$, but sometimes estimates to double that number are found.

Note that, especially for very high mass stars, if during the accretion stage the star becomes to heavy and the luminosity crosses the Eddington limit, the poor cloud from which the star was accreting gets swept away from the intense radiation pressure, therefore hindering, if not blocking completely, accretion.

5.4 POST MS EVOLUTION

After the exhaustion of Hydrogen in the core, there remains an Helium core which doesn't provide in the slightest any contribution to the energy production. So, the Herculean task of supporting the whole star's structure has to be shouldered by the rich Hydrogen envelope which surrounds the lazy core. But before delving into the differences in evolution between low mass and higher mass stars, let us briefly consider what is called the *Schönberg-Chandrasekhar limit*.

5.4.1 Schönberg-Chandrasekhar Limit

As we've seen, regardless of the H-burning mechanism at work during the MS phase, at the exhaustion of the central Hydrogen, the star is left with an Helium core surrounded by Hydrogen-rich layers. At the base of this pile of layers temperature is high enough for Hydrogen burning.

Given that there is no burning inside the core and the temperature gradient is radiative (and rather small), its thermal stratification is isothermal.

With a pure analytical approach, Schönberg and Chandrasekhar investigated the equilibrium conditions of such configuration, assuming ideal gas EoS. Since temperature is (approximately) no longer varying in the core, variations in pressure must be matched by variations in density.

Note that the number of ions available to contribute to the density are significantly lower now that all the Hydrogen in the core has turned into Helium, thus presenting the star with the possibility of the equilibrium conditions not being met.

Schönberg and Chandrasekhar found out that the maximum mass an isothermal core can have for the star to be in equilibrium is given by

$$\left(\frac{M_{\text{core}}}{M_{\text{tot}}}\right)_{SC} = 0.37\left(\frac{\mu_{\text{env}}}{\mu_{\text{core}}}\right)^2 \tag{110}$$

At the end of the MS of a solar chemical composition object $\mu_{\text{env}} \sim 0.6$ and $\mu_{\text{core}} \sim 1.3$ (since it's essentially made of pure Helium); the SC limit is therefore set at about 8% of the total mass.

This means that if the Helium core is larger (or equal) than the SC limit, it must contract.

5.4.2 Post MS Evolution: UMS

In this mass range, the Helium core is generally larger than the SC limit (110). Numerical computations show that H-burning continues in an initially broad shell around the Helium core, which later thins down. Notice that, although the burning is now in a shell, the main burning channel is still the CN-NO (bi)cycle.

During this phase the core contracts very slowly and the envelope expands. Due to the expansion, the outer layers cool down and envelope opacity increases drastically, thus trapping more energy within the structure. Upon reaching this stage, the structure moves in the HR-diagram from the blue to the red side at almost constant luminosity.

This is called *Sub-Giant Branch* (SGB). The evolutionary rate for this phase is simply the Kelvin-Helmoltz, of order of $1 - 12\,\mathrm{Myr}$. This is phase is so short for massive stars, compared to MS and the next one coming, that the chance of observing objects in this phase is rather slim. As a rule of thumb, you can commit to memory the following

$$\frac{t_i}{t_j} = \frac{N_i}{N_j}$$

where t is the time passed in a given phase and N is the number of objects observed in that phase.

This phenomenon leads to the presence of the so-called *Hertzsprung gap* in the HR-diagram (literally a hole). Another gap of this sort you can expect, and find, during the overall-contraction phase.

During SGB, the stellar envelope becomes convective, as the outer layers cool due to their expansion. When the convective envelope reaches the base of the Helium core, the star exits the SGB and enters the *Red Giant Branch* (RGB), where it will remain as long as the conditions to ignite Helium are not met.

On a concluding note, for high and intermediate-mass stars, the central density at the end of the SGB is sufficiently low to prevent the onset of electron degeneracy, and therefore core contraction can cause an increase of the temperature of the interior.

5.4.3 Post MS Evolution: LMS

At odds with more massive stars, the transition to shell burning is not so fast for LMS stars. For these, the mass of the core is well below the SC limit, and even when it grows above this limit due to the production of Helium from shell H-burning, electron degeneracy in the core provides the necessary pressure to support the overlying envelope.

As we've seen, LMS stars still have a radiative core at TO, and as the central abundance of Hydrogen drops below a critical value, the maximum in the nuclear energy release ceases to be located at the center as it begins to move outwards.

During the SGB phase, due to dependence of the CN-NO cycle efficiency on temperature, the shell in which Hydrogen is burned becomes increasingly thinner as Hydrogen is depleted and temperature drops in the envelope[17].

Starting from a mass of about $M_{\text{shell}} \sim 0.2M_{\odot}$, the shell thins down to $\sim 0.001M_{\odot}$ at the end of the SGB and further decreases at the tip of the RGB.

All throughout this process, the star have yet to settle in a new equilibrium configuration for secondary elements. While this happens, the Helium core continues to grow and shrinks down, thus heating up and making the external envelope expand once more.

Opacity grows along with the radiative gradient, thus requiring the convective envelope to penetrate deeper into the star, eventually reaching the

17 The reason for this is that the energy produced from Hydrogen-burning is transferred to the envelope, which bloates and cools down, increasing the opacity.

shell where Hydrogen is burned. Note that, apart from a scant few exceptional cases, convection never breaches into ignited shells. In fact, this is greatly "discouraged" from Ledoux's criterion (56), since near the core there are quite extreme chemical gradients.

5.5 RED GIANT BRANCH

An important property of LMS stars during RGB evolution is the tight correlation between the surface luminosity and the mass of the electron degenerate Helium core. So important indeed that is has been bestowed a name, the *He core mass-luminosity relation* ($M_{\text{c,He}} - L$), which is essentially summed up in the following line

$$L_{\text{H}} \propto M_{\text{c,He}} \tag{111}$$

The physical rationale behind this expression is that the surface luminosity in this stage is almost entirely provided by the H-burning shell and its thermal properties, while, in turn, the nuclear burning rates are only determined by the mass and the radius of the star's Helium core.

The thermal properties of the envelope have no effect on the burning shell, because the mean density within the core is very high, while in the envelope it is not. The pressure gradient just above the border of the core is so large that the pressure decreases by several orders of magnitude as one moves outwards from the burning shell. This is also the reason why mass loss from the stellar surface, incredibly efficient in the RGB phase, does not affect the nuclear burning, the Helium core and the stellar luminosity. Mass loss is, in this phase, almost purely stochastic, caused by the formation of acoustic waves, convective cells and the potentially strong magnetic pressure, which can all be contributing to an average mass loss of order

$$\dot{M} \sim 10^{-9} - 10^{-8} M_{\odot}\,\text{yr}^{-1}$$

which can be roughly approximated as Gaussianely distributed.

The star evolves along the RGB increasing the mass of the Helium core, hence its luminosity and its radius. The effective temperature, on the other hand, remains roughly constant because of the H^- opacitive channel: The overall temperature in the envelope decreases, hence the electron density decreases along with the H^- opacity, thus regulating the changes in surface temperature.

For LMS stars the RGB is rather long[18], because they have low masses and are also relatively cold, so they need more time to reach the temperature needed for Helium burning. During this phase, the contraction of the core, on top of heating the whole structure, also induces the core to progressively degenerate.

This can potentially result in a problem: Stars with core masses $M_c^{\text{He}} < 0.5 M_{\odot}$ will never ignite Helium and will be slowly cooling, turning into pure-Helium white dwarves.

18 Although, you should commit this to memory, the RGB duration is not at all comparable to the MS phase. The life times in SGB and RGB for LMS stars are of order 10^8 yr, which are about 100 times smaller than the average MS timescale for stars in this mass range.

This is also supported by the fact that once the envelope thins down to about $\sim 0.06 M_\odot$ the Hydrogen shell shuts down, completely stopping the burning because the shell is too cold. The star will be thus contracting a bit and will eventually depart from RGB. On timescales longer than the actual age of the Universe, the white dwarf remnant will settle into a thermal equilibrium with the CMB.

When (and if) Helium is ignited in the core, a star will be sitting at the "tip" of the red giant branch (TRGB). Here, the mass of the core is essentially unchanged, while, as we've mentioned, the total mass might be significantly different because of severe mass loss processes.

5.5.1 First Dredge Up and RGB Bump

As convection penetrates deeper into the star, some of the He produced during the central H-burning phase is mixed in. Therefore, the surface abundance of He monotonically increases until the convection reaches its maximum penetration: This phase is called the *first dredge up*.

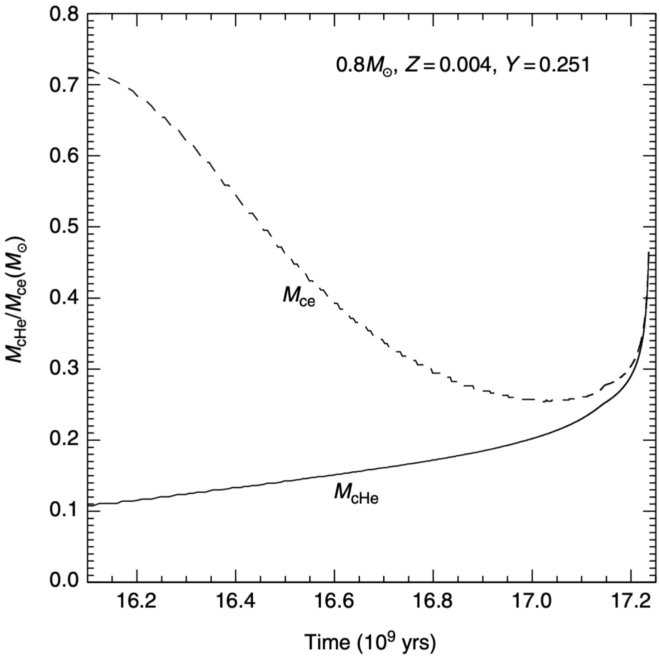

Figure 30: The behavior with time of the the base of the outer convective envelope and of the Helium core mass for a $0.8 M_\odot$ star. Credits: [21].

Other chemical species involved in the H-burning are also mixed in, such as $^3_2$He and the CNO elements. The major results are the Nitrogen overabundance on the surface ($\sim$ 100% increase) and Carbon and Oxygen seeing their average surface abundance reduced ($\sim$ 30% decrease for Carbon and a very slight change of the abundance for Oxygen).

Along the RGB (for both low-mass stars and more massive objects) the H-burning shell moves steadily towards more external mass layers that contain fresh Hydrogen, and the lower boundary of the convective envelope recedes towards the surface without overlapping with the shell. As the convective boundary recedes, a chemical discontinuity is left at the layer of maximum inward penetration of the envelope convection zone (cfr. Fig.31).

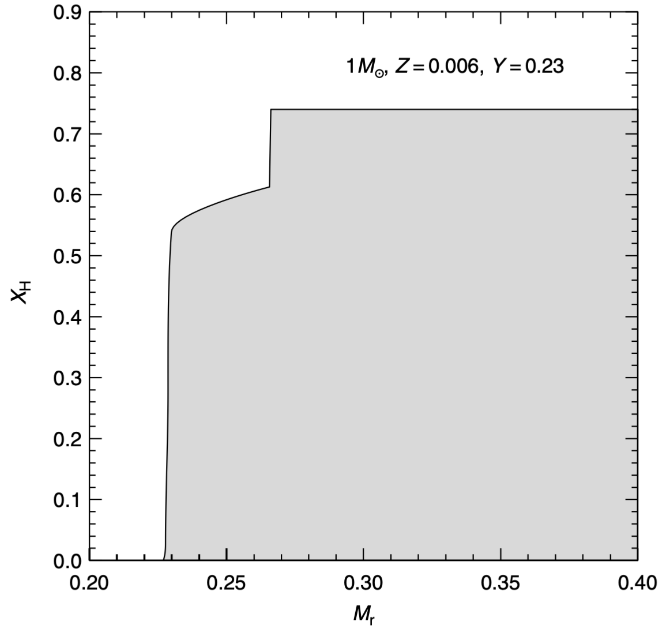

Figure 31: Hydrogen abundance profile within a $0.8M_\odot$ star, after the first dredge up. The bottom end of the convective envelope at its maximum extension corresponds to the abundance discontinuity at $M_r \sim 0.26$. Credits: [21].

When the H-burning shell encounters this discontinuity, the rate at which the star climbs the RGB temporarily drops and even reverses for a while. This behavior is due to the change in the H-burning efficiency caused by the increase of Hydrogen abundance, and hence a decrease of the mean molecular weight. In short, the star pops out of its former equilibrium condition.

After the shell source has crossed the discontinuity, the mean molecular weight remains at a fixed value[19] and the surface luminosity grows monotonically with increasing core mass. As a consequence, the star will cross three times the same luminosity interval (see Fig.32), and hence we can predict an increase of the star counts in this luminosity interval: The so-called *RGB bump*. The time spent during the RGB Bump phase is a significant fraction ($\sim 20\%$) of the total RGB lifetime.

Various Dependences of the RGB Bump Luminosity

$$Y \uparrow\ T \uparrow\ \bar{\kappa} \downarrow \Longrightarrow M_{CE} \downarrow\ L_{Bump} \uparrow$$

$$Z \uparrow\ \bar{\kappa} \uparrow\ T \downarrow\ T_{eff} \downarrow \Longrightarrow M_{CE} \uparrow\ L_{Bump} \downarrow$$

$$M \downarrow\ T \downarrow\ T_{eff} \downarrow\ \bar{\kappa} \uparrow \Longrightarrow M_{CE} \uparrow\ L_{Bump} \downarrow$$

Please note that the effectiveness of the first dredge up is closely related to the mixing length efficiency which, you should recall, regulates the relevance

19 Recall that the envelope was fully mixed thanks to convection.

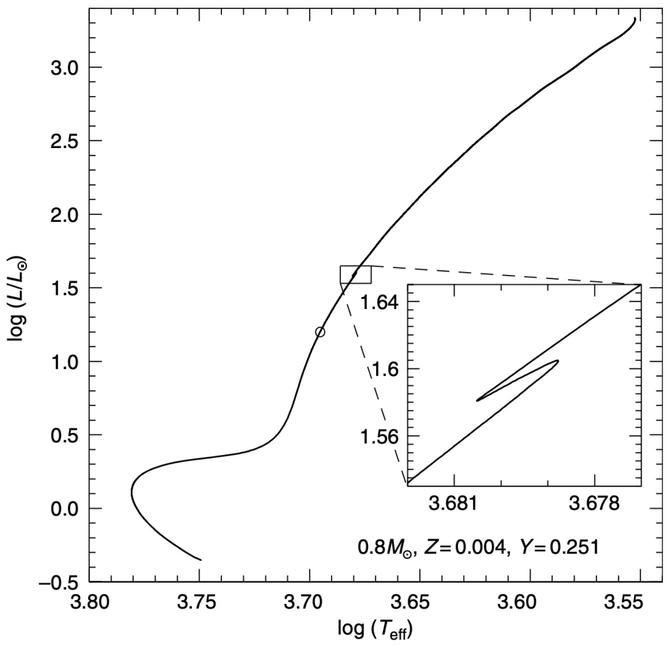

Figure 32: The HR-diagram of the same $0.8 M_\odot$ star from previous figures. The inset shows the trend of surface luminosity with the effective temperature when the H-burning shell crosses the chemical discontinuity produced by the first dredge up. The point along the evolutionary track corresponding to the first dredge up is marked with an open circle. Credits: [21].

of convection. Usually, the α_{ML} parameter is adjusted so to match the color of the RGB, thus introducing changes in the temperature gradient.

5.5.2 Helium Flash

We've seen that until the core meets the conditions to ignite Helium, its mass will be steadily increasing as long as Hydrogen is burned in shells. It won't be surprising then that the density of the core will be also increasing. The gravitational energy generation will thus be positive in the core. However, we should not forget that photons may interact with the surrounding hot plasma, producing neutrinos, which are likely to be escaping the structure without interacting[20] and subtract energy to the star. Within the core then the sum of energy production contributions might be negative

$$\epsilon_g + \epsilon_\nu < 0 \implies \frac{\mathrm{d}L_r}{\mathrm{d}M_r} < 0$$

so that in this region a *temperature inversion* develops, and the temperature maximum is no longer at the center, but rather in a shell. This also suggests that Helium will be initially ignited off-center.

An important point concerning the evolution of low-mass stars along the RGB is related to the electron degeneracy. During the RGB evolution the He core mass increases, its central density increases, and also the degree of electron degeneracy becomes increasingly stronger, causing a large decrease

20 Their free mean path is, after all, usually quite larger than the average stellar radius.

of the conductive opacity in the He core. This makes the conductive energy transport very efficient and, in turn, increases the cooling of the central portion of the He core. In spite of the energy losses from the core associated with the neutrino flux and electron conduction, the maximum temperature in the core increases monotonically.

This is because with growing $M_{c,He}$ the core contracts and the associated gravitational energy release heats up the layers below the H-burning shell–where the transition from degenerate to non-degenerate matter takes place–and, indeed, the whole He core. When the maximum temperature reaches a value $\approx 10^8$ K, He-burning is ignited. This occurs when the mass of the core is equal to $M_{c,He} \sim 0.48 - 0.50 M_\odot$, almost regardless of the initial total mass. This is a consequence of the fact that stars in this mass range develop very similar electron degeneracy levels.

Due to the highly degenerate state, the nuclear burning in the Helium core is unstable and this causes a thermal runaway at the tip of the RGB: The *Helium flash*. This marks the end of RGB evolutionary phase.

In the core, Helium is under conditions of very strong, partially relativistic electron degeneracy. We have already shown that under such thermal conditions, the gas pressure is not sensitive to temperature changes. When a nuclear burning ignites in a non-degenerate core, the new energy source causes a temperature increase and correspondingly, a pressure increase; to maintain hydrostatic equilibrium the core expands and cools, and this expansion prevents the continuous increase of the energy generation.

The same does not hold in the case of an electron degenerate core; the initial temperature rise is not followed by an immediate expansion so the rate at which the He-burning reactions (3α reactions) occur increases dramatically, with a continuous increase of the temperature, a thermal runaway.

During this runaway, there is a huge production of nuclear energy; in a few seconds an amount of energy of the order of $10^{10} L_\odot$ is produced. However, almost none of this energy reaches the stellar surface, as it is absorbed by the surrounding non-degenerate layers. The expansion of the layers just outside the shell where Helium ignites is the first factor which contributes to the moderation of the Helium flash strength. The second factor is that convection sets in as a consequence of the large energy flux, and this dilutes the produced energy over progressively larger mass.

The initial temperature's rise at constant density helps to remove the electron degeneracy at the point where Helium ignites. The degeneracy of the inner regions is subsequently removed through a series of much lower strength secondary flashes, until the Helium burning eventually reaches the star centre.

Note that during the flashes about 5% of the Helium in the core is turned into Carbon. The whole evolutionary phase from the start of the main Helium flash to the beginning of the core He-burning phase lasts about ~ 1 Myr.

Now, for UMS stars, something a bit different happens.

If you recall, higher mass stars enter the RGB phase with a relatively large Helium core[21] and are already quite hot. This means that they won't be

21 With respect to LMS stars, of course.

enjoying for long their stay in the RGB phase, for the conditions for He-burning won't take long to be met.

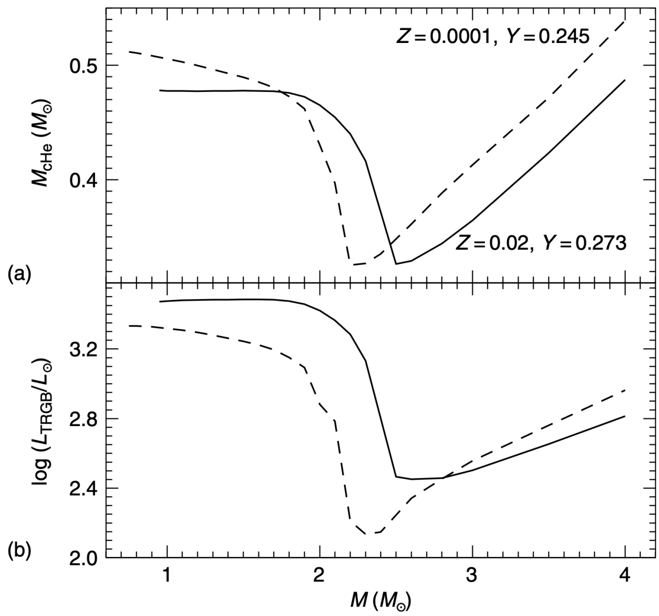

Figure 33: (a) The behavior of the mass of the Helium core and (b) of the surface luminosity at the onset of He-burning as a function of the stellar mass for two selected initial chemical compositions. Credits: [21].

Take a look at Fig.33. The value of $M_{c,He}$ does not depend strongly on the stellar mass for masses lower than $\sim 1.8 M_\odot$ at solar chemical composition (this limit depends strongly on the adopted chemical composition, significantly decreasing with decreasing metallicity and/or increasing He content). In fact, all stars with a mass below this limit develop very similar levels of electron degeneracy within the He core, and the mass of the He core has to reach almost the same value before He-burning ignites.

When the stellar mass value exceeds this limit, the electron degeneracy in the core is at a lower level as a consequence of the larger temperatures and lower densities. This has the effect that the luminosity of the tip of the RGB is almost constant for masses below the quoted limit, and strongly decreases at larger masses.

After a minimum has been attained, the value of $M_{c,He}$ starts to increase again with increasing total mass (the electron degeneracy is by now removed from the core) as a consequence of the increasing mass of the convective core during the core H-burning phase. The significant changes in the properties of the RGB, which are expected in stellar systems as a consequence of the differences of the age, and in turn of the masses evolving along the RGB, is called *RGB phase transition*.

Helium and Metals Dependence of the RGB Transition Mass

$$Y \uparrow \ T \uparrow \ M_{RGB}^{TR} \downarrow$$

$$Z \uparrow \ \bar{\kappa} \uparrow \ T \downarrow \ M_{RGB}^{TR} \uparrow$$

Stars with Helium core masses higher than the RGB transition mass will be non degenerate and will thus spend less time in the RGB phase.

Note, however, that more massive stars have a shorter lifespan. Therefore, the luminosity extension of the RGB will be consequently shorter. More importantly, for these stars Helium is ignited without a flash, for there's only partial or no degeneracy at all to remove. Still, the luminosity at ignition will be proportional to the mass of the core.

5.5.3 The Tip of the Red Giant Branch

The tip of the RGB marks the evolutionary phase corresponding to He-burning ignition through the He flash, and this occurs when the He core mass has reached a well-defined value. This means that the luminosity of the tip is in general a function of the Helium core mass at the Helium flash. Any parameter affecting the size of the Helium core at ignition affects the luminosity of the tip: For example, if the Helium core mass at the flash increases, the brightness of the RGB tip also increases.

Helium and Metals Dependence of the TRGB

An increase in the initial Helium content increases the interior temperature of a star through the increase in the mean molecular weight; this, in turn, decreases the electron degeneracy level in the He core during the RGB evolutionary phase. As a result, the star ignites Helium at a lower core mass and the luminosity of the tip of the RGB decreases.

$$Y \uparrow \; T \uparrow \; L_{\mathrm{H}} \uparrow \; M_{\mathrm{TRGB}} \downarrow L_{\mathrm{tip}} \downarrow$$

With increasing metallicity, for a fixed Helium content and stellar mass, the core mass at the Helium flash decreases. This is because at a higher metallicity the H-burning in the shell is more efficient; as a consequence the growth and, in turn, heating of the He core is faster. This way the thermal conditions for igniting He are reached at a lower Helium core mass.

$$Z \uparrow \; X_{\mathrm{CNO}} \uparrow \; L_{\mathrm{H}} \uparrow \; M_{\mathrm{c,\,He}} \downarrow \; L_{\mathrm{tip}} \uparrow$$

The luminosity of the tip of the RGB is not affected by changes in the efficiency of superadiabatic convection. This is because the thermal stratification within the core does not depend on the thermodynamical properties of the outer envelope.

Globular clusters, although rather well-populated, are often unable to display the TRGB (or the Helium flash) in a HR-diagram, given the short duration of the phase compared, for example, to the MS or the RGB.

On the other hand, the TRGB is very extended for low mass stars, thus becoming potentially really useful to date ancient stellar populations. As we'll have the pleasure to see in some of the last sections of this chapter, determining distance of objects of interest is often an Herculean task in astrophysics, which is usually made easier[22] with the introduction of *standard candles*. A condition for some phenomenon to be a good standard candle is that it has to be independent of the objects age.

Regarding the TRGB, this is usually true for lower mass stars. As we've seen, the bolometric luminosity of the TRGB is determined by the mass of the core at the Helium flash, once the initial chemical composition is fixed. Since low-mass stars ignite Helium all with similar core masses (slightly increasing for decreasing initial mass) the bolometric magnitude of the TRGB changes by a few hundred-ths of magnitudes in the age range between $\sim 4\,\mathrm{Gyr}$ and $12 - 14\,\mathrm{Gyr}$.

When approaching the RGB phase transition the magnitude of the TRGB increases sharply, since L decreases, due to the lifting of the electron degeneracy in the core, that causes Helium ignition to occur at significantly smaller core masses. On the other hand, once age is fixed, the magnitude of the tip decreases for increasing metallicity, in spite of the decrease of the Helium core mass at the TRGB.

The net effect is that, for ages larger than $\sim 4\,\mathrm{Gyr}$, $M_{\mathrm{bol}}^{\mathrm{TRGB}} \propto -0.19[\mathrm{Fe/H}]$ for metallicities well below solar, with an almost negligible dependence on the age. Considering now the TRGB magnitude in various photometric systems for ages above $\sim 4\,\mathrm{Gyr}$, one notices that in the I-band the dependence of the TRGB brightness on [Fe/H] is minimized. In fact

$$\frac{\mathrm{d}M_{\mathrm{bol}}^{\mathrm{TRGB}}}{\mathrm{d}[\mathrm{Fe/H}]} \propto -0.19$$

and from empirical determinations of bolometric corrections for bright RGB stars in globular clusters we obtain that

$$\frac{\mathrm{d}BC_I}{\mathrm{d}(V-I)} = -0.243$$

For $[\mathrm{Fe/H}] > -0.7$, the dependence of M_I^{TRGB} on [Fe/H] becomes very large so it cannot be used safely as a standard candle.

The first step towards obtaining distances is the detection of the TRGB. From what we know about stellar evolution, one expects TRGB stars basically to overlap with AGB[23] objects, that populate the color-magnitude diagram from below the TRGB level up to magnitudes brighter than the TRGB. However, AGB evolutionary times are shorter than RGB ones, and the luminosity function of bright red stars in an old SSP is expected to show a discontinuity in correspondence of the TRGB location.

The position of this discontinuity marks the level at which the contribution of the longer-lived (in comparison with AGB objects) RGB stars vanishes, and corresponds to the TRGB magnitude.

5.5.4 Theoretical Clusters: Isochrones

The theoretical color-magnitude diagram for a simple stellar population (like, for example, a stellar cluster) is called an *isochrone*.

22 The definition of "easy" is rather bland in cases like these.

23 Stands for *Asymptotic Giant Branch*. We'll deal with this phase in a while.

The computation of an isochrone is conceptually very simple: Consider a set of evolutionary tracks of stars with the same initial chemical composition and various initial masses; different points along an individual track correspond to different values of the time t' and the same initial mass.

An isochrone of age t is simply the line in the HR-diagram that connects the points belonging to the various tracks where $t = t'$. This means that when we move along an isochrone, time is constant whereas the value of the initial mass of the star populating the isochrone at each point is changing.

Once an isochrone of a given age and initial chemical composition is computed from stellar evolution tracks, it can be transferred to an observational color-magnitude diagram by applying to each point a set of appropriate bolometric corrections.

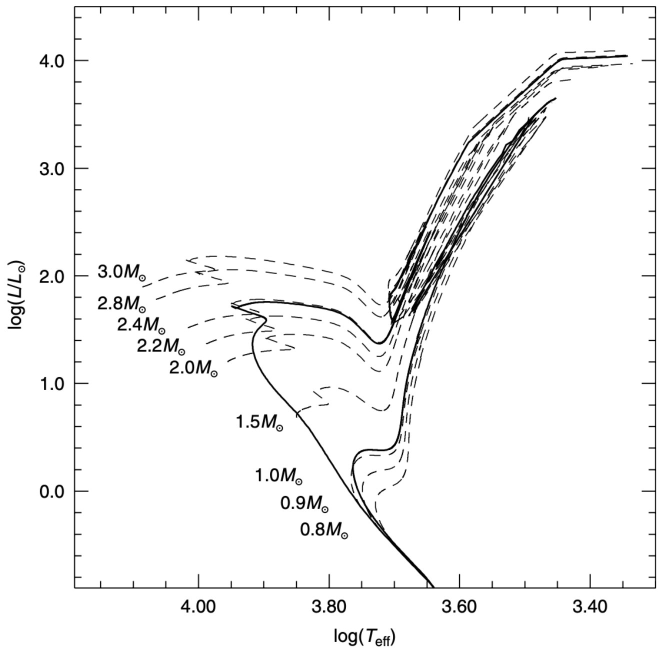

Figure 34: HR-diagram of selected stellar evolutionary tracks (dashed lines) with the same initial solar chemical composition and the labelled masses. The heavy solid lines display two isochrones for the same chemical composition and ages of 600 Myr (the brighter sequence) and 10 Gyr. Credits: [21].

It is clear from Fig.34 that the mass of the objects evolving at a given location along an isochrone increases when moving towards more advanced phases. This increase is due to the fact that lower masses age more slowly, and at a given time they are less evolved than higher-mass objects.

Another important property is that the isochrone MS is populated by a large range of masses (the lower limit being the minimum possible stellar mass) whereas the mass evolving along the RGB and successive phases is approximately constant, given that the isochrone stays very close to the evolutionary track of the mass at the MS end-point (the TO).

In short, if we increase the age of the cluster, we should expect to see less masssive stars at TO (and, consequently, fainter stars). This means that the TO can be used as a very precise age indicator for clusters, with a low dependence on the physical inputs received by the star. To use the TO as an age indicator you first have to find an independent way to determine the distance of the cluster. Once you've done that you can:

1. Measure the magnitude of the TO in some band;
2. Connect it to the age of the cluster using the theoretical relation between M_x^{TO}-age (with x some band) and thus find the age of the cluster.

Differently from globular clusters, which are usually nice and dandy, open clusters display a "parallel" track to the MS, which is that of the unresolved binaries: The fluxes of the two members of the binary get summed (thus the vertical shift in luminosity), while their masses will follow some kind of distribution that displaces the binary in color as well.

You can take as a rule of thumb the following chain of implications

$$\text{age} \uparrow \implies M_{\text{TO}} \downarrow \; L_{\text{TO}} \downarrow$$

Note, however, that decreasing the age of the considered cluster leads to an extension in luminosity of the reconstructed MS, but, as you can see in Fig.34, the morphology of the isochrone is significantly different.

5.5.5 Microscopic Diffusion

As we've mentioned somewhere in the above, the various elements' abundances can suffer severe changes along the radial direction because of diffusive mechanisms such as convection, gravitational sedimentation, temperature gradients and osmosis (concentration gradients). If several chemical species are present in the stellar gas, the net effect is to change the chemical stratification and therefore it is potentially important for the study of stellar evolution.

Pressure, temperature and chemical abundance gradients are the driving forces behind atomic diffusion. A pressure gradient and a temperature gradient tend to push the heavier elements in the direction of increasing pressure and increasing temperature, whereas the resulting concentration gradient tends to oppose the above processes.

The chemical evolution of the mass fraction X_i of an element i at a given mass layer M_r due to atomic diffusion can be written as

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}M_r}\left(4\pi\rho^2 X_i \omega_i\right)$$

where ω_i is some diffusive velocity for the i-th chemical species and is thus defined

$$\omega_i = D_p(i)\frac{\partial p}{\partial r} + D_T(i)\frac{\partial T}{\partial r} + D_c(i)\frac{\partial c_i}{\partial r}$$

where c_i is the concentration of the i-th species. The three D coefficients are obtained, for a given chemical and physical profile, by imposing the conditions of mass, momentum and energy conservation, together with charge neutrality.

For example, because of gravitational sedimentation, Helium and metals generally sink deeper into the stellar interiors on a timescale of about $\sim 1\,\text{Gyr}$, while Hydrogen tends to rise to the surface.

While sedimentation of heavy elements can only affect stars of mass in the range[24] $M \lesssim 1.2M_\odot$, the displacement of Hydrogen is mass independent and affects equally all stars. Let us not forget that convective motions have not suddenly vanished. Convection is actually a very good inhibitor of diffusive mechanisms, as the mixing timescales associated to convection are typically shorter than the diffusive ones.

In general, we've seen that the lower the total mass, the longer the star stays in the MS, and the larger the convective envelope manages to get. Because of this fact, diffusive mechanisms are able to peak in "effectiveness" in stellar interior for masses of about $\sim 0.8M_\odot$.

Is this the end of the story? Of course not. In the Sun, which is usually our preferential reference object, it has been observed that certain abundances aren't matching with the observations, for example at the *tachocline*[25], where we have to assume the presence of additional mixing mechanisms instead of considering only diffusion.

For example, during the SGB stars are expanding, and convective envelopes penetrate deeper into the stellar structure, on top of growing more opaque and larger. This means that convective motions can bring to the surface Helium and metals, while the surface abundance of Lithium usually drops[26].

What effects does this have on the star? There are mainly three:

1. $X_c \downarrow\ \tau_{\text{MS}} \downarrow\ L_{\text{TO}} \downarrow$;
2. $Y_c \uparrow\ \mu_c \uparrow\ T_c \uparrow\ L_{\text{TO}} \uparrow\ \tau_{\text{MS}} \downarrow$;
3. $X_{\text{sup}} \uparrow\ \bar{\kappa} \uparrow\ L_{\text{TO}} \downarrow\ \tau_{\text{MS}} \uparrow$.

so there are two effects working to reduce the luminosity at turn-off (and one to increase it) and two effects working to reduce the permanence in the MS. The final result is thus

$$L_{\text{TO}} \downarrow\ \tau_{\text{MS}} \downarrow$$

Assuming the presence of diffusive mechanisms, the abundance of Helium at ignition (after the first dredge up) in the external convective envelope drops, as a consequence

$$Y_{\text{HB}} \downarrow \Longrightarrow L_{\text{H}} \downarrow\ M_{c,\text{He}}^{\text{Flash}} \uparrow$$

since a longer RGB phase is needed. On the other hand, this would mean that the star would start the RGB phase hotter than without diffusion. The first effect, however, is dominant.

Another phenomenon working, this time, against diffusion is *radiative levitation*. Radiative levitation is an additional transport mechanism caused by the interaction of photons with the gas particles, which acts selectively

24 Merely because of timescales involved. Massive stars won't live long enough to experiment appreciable changes in their evolution because of the gravitational sedimentation of, say, Helium, whilst it might be particularly relevant for lighter stars.

25 "*The tachocline is the transition region of stars of more than 0.3 solar masses, between the radiative interior and the differentially rotating outer convective zone.*"

26 This is because there is virtually no $^7_3$Li in the stellar interiors, since it was completely burned in the PMS phase. As soon as the fresh Lithium from the surface is brought deeper into the star, it is immediately burned. Hence the drop.

on different atoms and ions. In simple terms, since within the star a net energy flux (albeit locally small, to preserve LTE) is directed towards the surface, photons provide an upward "push" to the gas particle with which they interact, effectively reducing the gravitational acceleration. Since we are dealing with interactions of photons with gas particles, it is clear that the efficiency of radiative levitation is related to the opacity of the stellar matter, in particular to the monochromatic opacity, and increases for increasing temperature (more energetic photons). The effect of levitation on the chemical abundances is simply an additional contribution to ω_i. Radiative levitation tends to increase the surface abundances, sometimes even increasing them *too much*.

What is usually done by those who think it might be worth spending their time watching their evolutive code run for an unfathomable amount of time is to consider an additional contribution caused by turbulence[27], which, most of the time, isn't really worth the trouble.

In general, within a convective region, the convective velocity is always much larger than the contributions of atomic diffusion and radiative levitation, so that the rehomogenization due to convection always prevails over atomic diffusion and radiative levitation. It's clear, however, that there must exist competitive mechanisms that are able to effectively inhibit both diffusion *and* radiative levitation.

5.6 THE HELIUM BURNING PHASE

Let us first consider the Helium burning phase for lower mass stars. Nuclear reactions relevant for this phase have already been discussed in a past chapter (Chapter 4), so in the following I'm going to assume the reader is somewhat familiar with the nuclear reaction that are going on.

5.6.1 Central Burning - LMS Stars

As we've already mentioned, during the RGB phase stars are affected by stochastic mass losses that can be roughly described by a Gaussian-like distribution. If the mass loss is not too severe, LMS stars manage to reach the TRGB and flash with a total mass that is approximately unchanged. But even if it weren't, it is worth pointing out that the luminosity of the TRGB depends only on the mass of the Helium core, which is never even affected by the mass losses of the external layers.

After the flash (and all the secondary flashes that might have occurred), Helium burning resumes in a more placid and quiescent way. Note that LMS stars never stop burning Hydrogen in shells, even when Helium is burning in the core.

The total luminosity of the star can be then considered as made of two contributions

$$L_{\text{tot}} = L_{\text{He}} + L_{\text{H}}$$

27 Hah! Who would have ever imagined?

Technically, Helium is burned in a smaller fraction of the core, but since the core is convective[28] Helium is *de facto* burned in the whole core. There is, however, a little shell of Helium just outside the convective core that, lacking the temperature to ignite, will remain inert until further notice.

At this stage, only the 3α and the ${}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He}$ reactions are active, and the luminosity due to Helium burning is clearly the dominant contribution. Now, since $L_{\mathrm{He}} \propto M_{c,\mathrm{He}}$ and all LMS stars get to the Helium flash with the same core mass, this stars will all have the same luminosity but different effective temperatures, which depend on the star's extension and hence on their total mass.

This defines an almost horizontal locus in the HR-diagram[29] that is called *Horizontal Branch*: The lowest mass envelopes are located in the hottest part of the branch, the location moving to cooler values with increasing envelope mass.

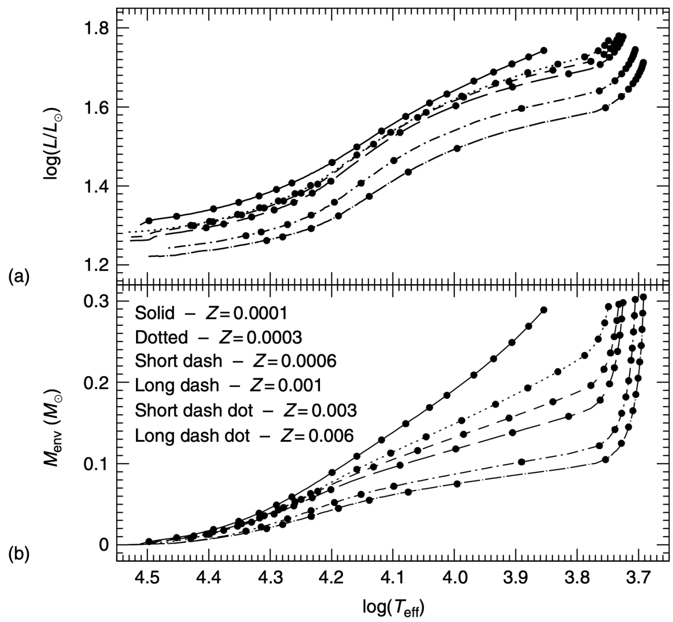

Figure 35: (a) The HR-diagram of the ZAHB for different values of the metallicity but the same RGB star progenitor, with $M = 0.8M_{\odot}$. (b) The corresponding trend of the envelope mass as a function of their effective temperature. Solid dots in both (a) and (b) mark the location of selected ZAHB models. Credits: [21].

Equilibrium models that burn He in a chemically homogeneous core, and H in a shell with chemical stratification similar to the one at onset of the Helium flash (in some sense this is equivalent to saying that the CNO elements are at their equilibrium in the H-burning shell) are called *Zero Age Horizontal Branch* (ZAHB) models.

The Horizontal Branch is only approximately horizontal because of the different masses the external envelope might have in different stars. In fact, a more massive envelope would induce an higher contribution to the total

28 It has to be. The energy production rate of the 3α reaction is proportional to T^{41}, meaning that huge radiative gradients are involved.

29 It is important to notice that the Horizontal branch will be almost horizontal only when expressed in terms of luminosity and not in some magnitude band.

luminosity from shell Hydrogen burning and an overall increase of the luminosity: More massive stars are then a little brighter.

Once in the HB, evolution is clearly not stopping. As Helium is burned, the star contracts and heat up. This time, only small variations are needed, for the high temperature dependence of the 3α does all the heavy-lifting. Note that although the contribution of Helium throughout this phase is unmistakably the dominant one, the energy produced per burned Helium nucleus is

$$\epsilon_{\mathrm{He}} \sim \frac{\epsilon_{\mathrm{H}}}{10}$$

and the associated burning timescales are

$$\tau_{\mathrm{He}} \sim \frac{\tau_{\mathrm{H}}}{100}$$

This is caused in particular by two facts:

1. The luminosity of the ZAHB is higher than the typical luminosity during the MS;

2. The region where the burning is happening during ZAHB is smaller than it was during MS.

The timescales for Helium burning would be even shorter if not for the constant support coming from the shell Hydrogen burning, that partly refills the core with fresh Helium and shoulders part of the needed energy production.

After the first dredge up the reaction products of the CN-NO (bi)cycle were brought to the surface. Note, however, that the sum of their relative abundances must remain approximately constant

$$X_{\mathrm{C}} + X_{\mathrm{O}} + X_{\mathrm{N}} \sim \text{const.} \quad X_{\mathrm{C}} \downarrow \; X_{\mathrm{O}} \downarrow \; X_{\mathrm{N}} \uparrow$$

Dependence on the Various Physical Parameters

With respect to other evolutionary phases, the HB doesn't span a great range in luminosity, as its luminosity extension is about ~ 0.1 mag wide. In view of the possible use of the ZAHB brightness as a distance indicator, it is important to know how the observational properties of the ZAHB react to any change in the most relevant parameters such as Helium content, metal abundance and any additional (non-canonical) process which can change the mass size of the He core at He-ignition.

- *The He content*: for a fixed metallicity, an increase in the initial Helium content, Y, causes a decrease in $M_{c,\mathrm{He}}$. This should cause a decrease in the ZAHB brightness. However, for ZAHB stars with massive enough envelopes (and so with efficient H-burning in the shell), the increase of the envelope He abundance causes a large increase in the H-burning efficiency, counterbalancing the decrease of the ZAHB brightness due to the change in $M_{c,\mathrm{He}}$.

  As a final result, with increasing He abundance, the blue part of the ZAHB (populated by stars with low-mass envelopes) becomes fainter, and the red part brighter. For fixed total mass, the stars become slightly hotter

$$Y \uparrow \; T_c \uparrow \; L_{\rm H}^{\rm RGB} \uparrow \; M_{c,{\rm He}}^{F} \downarrow \; \Longrightarrow \; L_{\rm H}^{\rm ZAHB} \uparrow$$

$$\left(\frac{\partial \log L_{\rm ZAHB}/L_\odot}{\partial Y}\right)_{M_c,Z} \sim 2.07$$

- *The metal abundance*: any increase in the metal abundance, at fixed Helium abundance, makes the ZAHB fainter. This occurrence is due to the combination of two effects: (1) the decrease of the value of $M_{c,{\rm He}}$. at the He flash, (2) the increase of the envelope opacity. For fixed total mass, the increase of the envelope metallicity makes the ZAHB location cooler

$$\left(\frac{\partial \log L_{\rm ZAHB}/L_\odot}{\partial Z}\right)_{M_c,Y} \sim -0.04$$

- *Non-canonical processes*: there are many physical processes such as rotation and non-canonical mixing processes, commonly not accounted for in standard evolutionary computations, which can produce sizeable effects on the ZAHB properties. Regardless of the detailed physical description of the process, we can easily predict, at least qualitatively, the effect on the ZAHB by simply considering the effect on the parameters characterizing a ZAHB star. Any physical process able to delay the ignition of the He flash at the TRGB, such as rotation, will cause an increase of the He core mass and, on average, a decrease in the total mass. This is because mass loss along the RGB then has more time to work. As a consequence, the ZAHB location will be brighter and hotter

$$\left(\frac{\partial \log L_{\rm ZAHB}/L_\odot}{\partial M_{c,{\rm He}}}\right)_{Z,Y} \sim 3.04$$

Clearly, the physical parameters we're accounting for are *not* constant along the whole stellar structure, and are not constant for different temperature ranges. What is customarily done is considering an approximately flat region of the HB, which is incidentally overlapping with the RR-Lyrae region ($3.73 \lesssim \log T_e \lesssim 3.83 - 3.85$).

At this point you might be wondering, can we use the ZAHB as a standard candle? As we've shown, we can determine the luminosity of the HB independently from the age of the (old) cluster we're observing: The mass of the core (within a reasonable range[30]) is only weakly dependent on the total stellar mass, hence from the age of the cluster[31].

Since the ZAHB is only weakly dependent on the age of the cluster, we can calibrate the difference in luminosity between the HB and the TO as a function of age: This is the so-called *Vertical Method*. A change of age at a given [Fe/H] changes the value of ΔV through the change of the TO brightness; for increasing ages, ΔV increases because the TO gets dimmer.

30 cfr. with Fig.33.

31 This last implication strikes as odd until you recall that the mass we observe at the Turn Off is (approximately) the same that we're going to observe at the TRGB, whose luminosity is in turn only dependent on the mass of the Helium core that is only weakly dependent on the total mass. Conclusion: the luminosity at the TRGB is weakly dependent on age.

This method works well mainly in the V band (or photometric bands at a similar wavelength) where the ZAHB is mostly horizontal. As a rule of thumb you can remember

$$\Delta V_{\mathrm{TO}}^{\mathrm{ZAHB}} \sim 0.1\,\mathrm{mag}\,\mathrm{Gyr}^{-1} \tag{112}$$

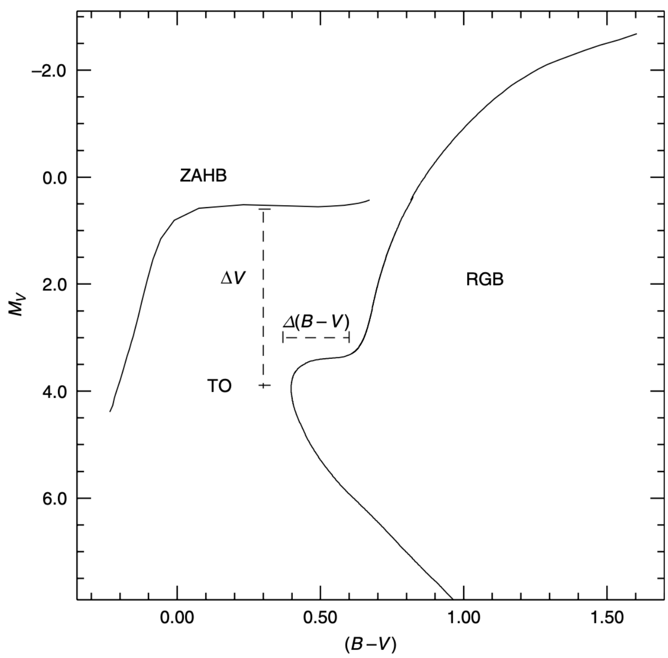

Figure 36: Graphical representation of the ΔV (vertical) and $\Delta(B-V)$ (horizontal) age indicators for old SSPs. Credits: [21].

Conversely, the *horizontal method* for age determination involves the comparison of the observed and theoretical values of the quantity $\Delta(B-V) = (B-V)_{\mathrm{RGB}} - (B-V)_{\mathrm{TO}}$, which corresponds to the color difference between the TO and the base of the RGB. This horizontal parameter is sensitive to age through the variation of $(B-V)_{\mathrm{TO}}$ with t, given that the RGB color is unaffected by age for old populations.

In order for this method to be of some use, however, we need an extremely high accuracy in both the observational determination and theoretical prediction of this quantity to keep the error on the estimated age low. Theoretical uncertainties (due to color transformations and treatment of superadiabatic convection) are surely higher than $0.01 - 0.02\,\mathrm{mag}$, and therefore the horizontal method is hardly used for absolute age determinations. We could still employ this horizontal method to determine age differences with respect to template clusters whose absolute age is well established using the ΔV technique.

The number of stars populating the HB depends on the average mass loss during the RGB phase, potentially impacting what sections of the HB we're be able to observe. Generally, the most used model to schematize mass losses during the RGB is the *Reimers model* which states

$$\dot{M} = 4 \cdot 10^{-13} \eta_R \cdot \frac{LR}{M} M_\odot\,\mathrm{yr}^{-1} \tag{113}$$

Given this model, all we have to do is essentially find the value of η_R that most reproduces the observed morphology of the branch ($0.2 \lesssim \eta_R \lesssim 0.8$). But the morphology of the HB doesn't depend only on the mass losses $\dot{M}$, but also on the metallic content of the considered cluster. At fixed age, there are three main effects we observe for increasing metallicity:

1. $Z \uparrow\ L_{\text{TRGB}} \uparrow\ \bar{\kappa} \uparrow\ T_e^{\text{RGB}} \downarrow\ R \uparrow \Longrightarrow\ \langle M_{\text{HB}} \rangle \downarrow \Longrightarrow$ the branch is more blue (at fixed η_R);

2. $Z \uparrow$ and if mass is fixed $\tau_{\text{MS}} \uparrow$, if age is fixed as well $M_{\text{TO}} \uparrow\ M_{\text{RGB}} \uparrow$ and at fixed η we also have $\langle M_{\text{HB}} \rangle \uparrow \Longrightarrow$ the HB is more red;

3. $Z \uparrow\ \bar{\kappa} \uparrow \Longrightarrow$ more red HB, $Z \uparrow\ L_{\text{H}} \uparrow \Longrightarrow$ more red HB

This is why metallicity is sometimes called the *first parameter* (age is the second). The "convolution" of all these effects results in a more red horizontal branch if we increase metallicity with fixed age and η.

What if we now fix η and Z but let the age change?

$$\text{age} \uparrow\ M_{(TO)} \downarrow \Longrightarrow\ M_{\text{RGB}} \downarrow \Longrightarrow\ \langle M_{\text{HB}} \rangle \downarrow$$

hence the HB turns more blue.

5.6.2 Hot Flashers

If during the RGB phase mass losses are severe enough to reduce the Hydrogen shell under a mere $0.06 M_\odot$, the unfortunate star will never reach the TRGB and will stray away from its original path, moving towards the cooling curve of pure Helium WD[32]. Note, however, that if the star exits the RGB before flashing, it is possible that during the contraction phase that is prior to the WD-cooling track, the core might heat up enough to ignite Helium. A series of flashes then brings the star back towards the ZAHB, but at a much higher temperature and a lower luminosity than those that would have been expected had the star gone through the "standard" iter.

These stars are known as *hot flashers*. Several had been observed with really large differences in the abundance of Helium and Carbon, most likely induced by severe flashes that might extend up to the most external layers.

5.6.3 Central Burning - UMS Stars

We now turn to considering the evolution of more massive stars along the RGB and beyond, when Helium is ignited in the core.

Owing to the absent (or partial) electron degeneracy within the core, the onset of He-burning is not characterized by the Helium flash, as in low-mass stars. Helium is ignited when the central temperature reaches $\sim 10^8$ K and central density $\approx 10^4\,\text{g}\,\text{cm}^{-3}$; this occurrence terminates, and reverses the upward climb along the RGB.

The star now begins an extended phase of He-burning in a steadily growing convective core. H-burning in a thin shell continues to provide the bulk

32 We've somewhat introduced them a few pages above.

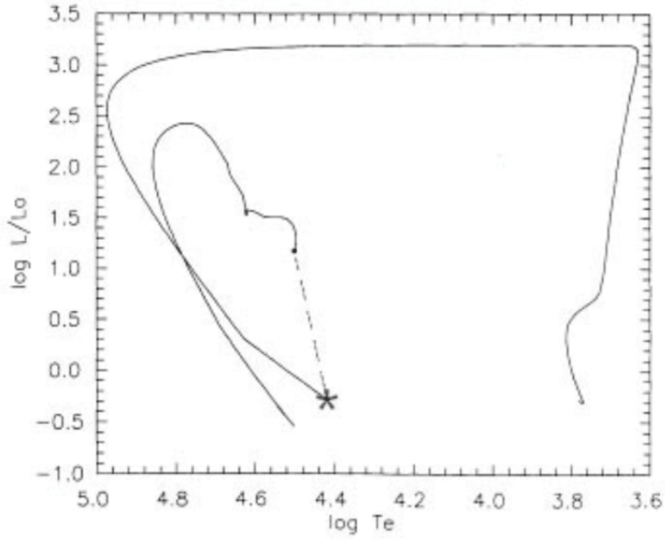

Figure 37: Evolutive trajectory for a "Hot Flasher" model. The * marks the Helium Flash ignition and the dashed line connects said point with the first quiescent burning model for Helium in the core. The following evolution in the quiescent Helium burning phase until the final cooling and settling into a CO WD is also shown. Credits: Unknown.

of the surface luminosity, and so the mass size of the He core continues to grow during this evolutionary phase.

The permanence of a UMS star in the RGB is positively correlated to the degeneracy of the core, a more degenerate core needing more time to have the degeneracy removed so that Helium can be ignited. Note that when ignition happens, the total mass of the star will be (roughly) the same that it was at TO[33]. The core He-burning lifetime is of the order of $\sim 20\%$ of the core H-burning lifetime, and it is $\sim 22\,\mathrm{Myr}$ for a $5M_\odot$ star and $\sim 4\,\mathrm{Myr}$ for a $10M_\odot$ star of solar chemical composition. The duration of the core He-burning phase is fairly large when you consider that the star is approximately two orders of magnitude brighter than during the MS phase, and that the specific gain of energy (per unit of mass of burnt fuel) is one tenth of that for H-burning. This is due to the very large contribution to the energy budget provided by the H-burning shell.

Consider now a stellar cluster. Since mass losses are essentially negligible, we'll find after the TRGB roughly N_{TRGB} stars

$$N_{\mathrm{TRGB}} = \frac{\tau_{\mathrm{RGB}}}{\tau_{MS}} N_{\mathrm{MS}} \ll N_{\mathrm{MS}}$$

all with approximately the same mass. What earlier we've called HB is now confined in just a tiny region, which is called *Helium clump* or *red clump*.

For ages between ~ 0.5 and $\sim 4\,\mathrm{Gyr}$ the He-burning phase is usually a red clump of stars (not only for metal rich SSPs) close to the position of the (depopulated) RGB sequence, because the RGB evolution is very fast and stars lose a small amount of matter. When ages are below $\sim 0.5\,\mathrm{Gyr}$,

33 Properly speaking, we should refer to the mass at the overall contraction. That said, the same argument we've presented for LMS stars holds: Mass losses are essentially stochastic and only marginally impact the total mass.

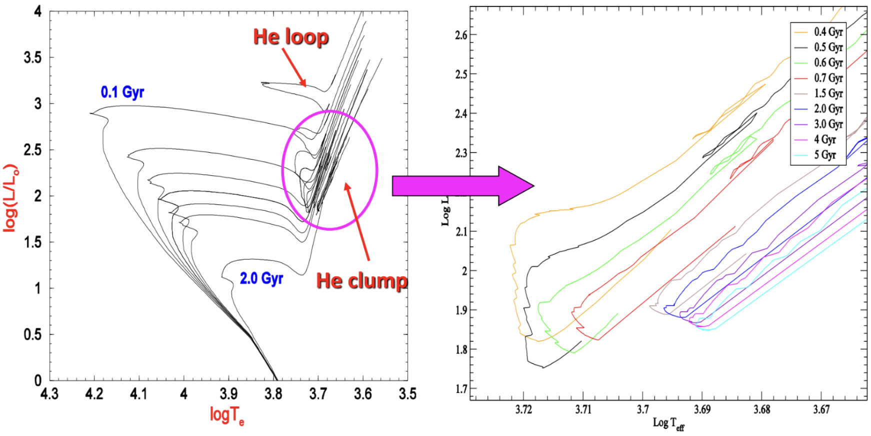

Figure 38: On the left cartoon, theoretical isochrones for stellar clusters displaying the whole MS, SGB and RGB phases. The Helium clump and the Helium loop are also visible. On the right cartoon, zoom on the red clump, highlighting its dependence on age. Credits: Professor S. Degl'Innocenti and the PISA group.

the Helium burning phase moves progressively to the blue side of the CM-diagram, because stars describe increasingly larger loops to the blue, and the longer evolutionary times are when He-burning objects are close to the blue end of the loops.

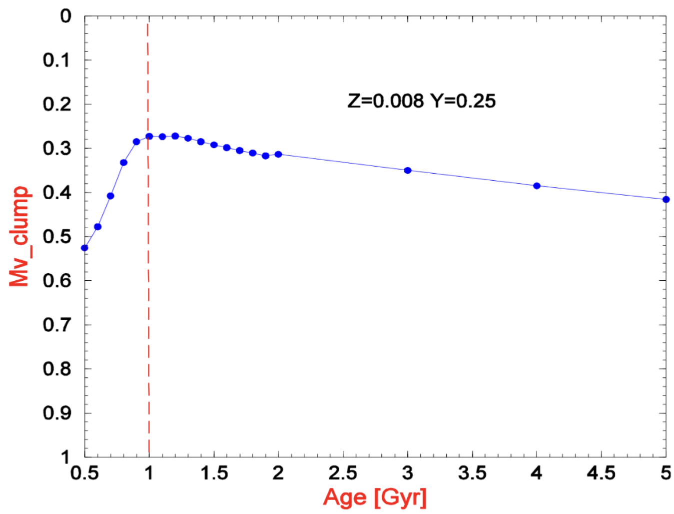

Figure 39: Magnitude in the Visual band of the red clump as a function of the age of the cluster. Note how only for ages higher than $\sim 1\,$Gyr the clump luminosity is approximately independent on the age. Credits: Professor S. Degl'Innocenti and the PISA group.

This suggests us that the red clump cannot really be used as a standard candle, especially not for stars younger than $\sim 0.5\,$Gyr ($M \gtrsim 5M_{\odot}$). Take a look at Fig.40.

Looking at the part of the track corresponding to the core He-burning phase (points E to I) one can easily note that after point F, the star moves from the RGB to the blue side of the HR-diagram. The bluest point, G is reached when the central abundance of He is $Y \sim 0.5$. During this period

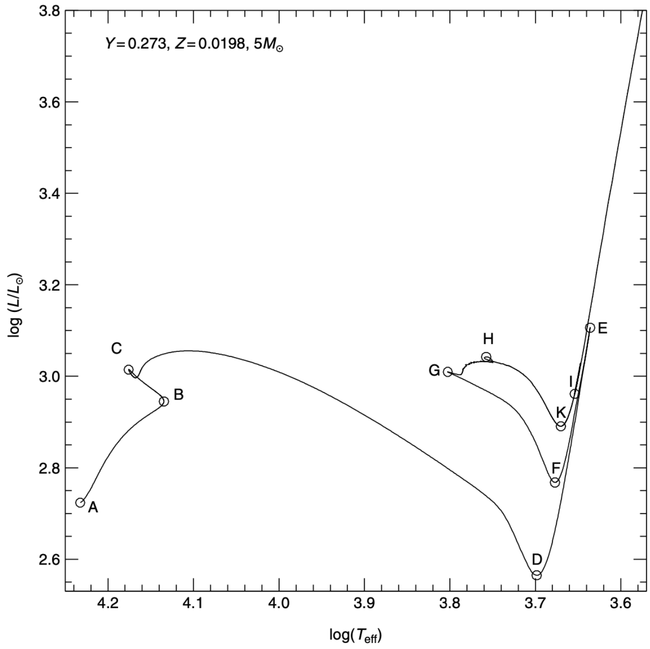

Figure 40: The evolutionary track of a $5M_\odot$ model during the Hydrogen and He-burning phases. The different evolutionary stages discussed in the text are marked A-K. Credits: [21].

before point G, the energy release in the H-burning shell has been steadily increasing. The maximum efficiency is achieved at point G, where the fraction of energy produced by the 3α reactions is of the order of $\sim 20\%$. After this point, the fraction of energy produced via H-burning decreases steadily.

The model then comes back toward the Hayashi line, so performing a loop in the HR-diagram, the so called *blue loop* (sometimes also called *Helium loop*). At point H, the core abundance of He is $Y \sim 0.02$, while $L_{3\alpha}$ accounts for $\sim 33\%$ of the energy produced by nuclear burning. At point K, the fraction of energy produced by the core He-burning process is $\sim 60\%$. The importance of the blue loop stems from the fact that it occurs during a long-lasting evolutionary phase, in which the star has a large probability of being observed.

This strongly extremized evolution is due to the severe contractions of the external envelope, which serves to increase the effective temperature of the star. The star then brings itself towards lower effective temperatures mainly for two reasons: Either the central Helium is greatly diminished (inducing a core contraction and a successive bloating of the envelope), or the active Hydrogen shell encounters the abundance discontinuity, which might have sunk deep enough into the structure. Note that only stars in the range $M \gtrsim 4 - 5M_\odot$ can show a blue loop behavior, but when they get too hot ($M \gtrsim 9 - 10M_\odot$) the loop disappeares. The core-Helium burning phase is so short that these stars immediately start moving towards redder regions of the HR diagram.

It is not surprising that determining the age of a cluster from looking at the blue loop is a huge mess. Thankfully, there's a region of the HR-diagram that partially overlap with the blue loop which is of crucial importante for age estimations and distance estimations, called the *instability strip*. Here

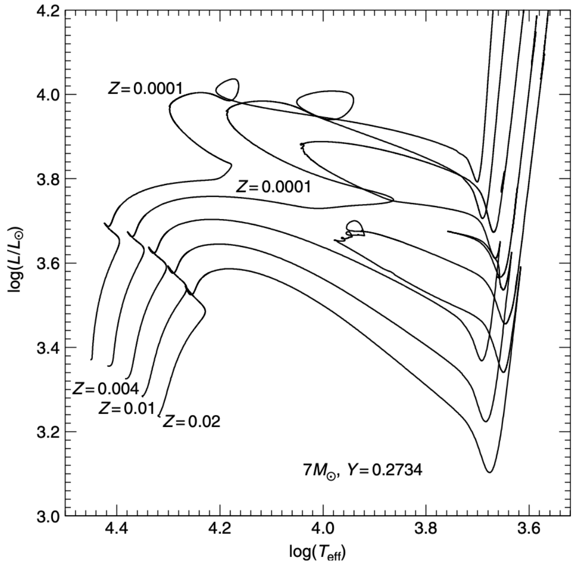

Figure 41: The HRD of evolutionary tracks of a $7M_\odot$ star for various values of the metallicity but with the same value for the initial He content. Credits: [21].

are located the *Cepheids variables*, that are stars affected by stable radial oscillations that induce periodic changes in brightness. For clarity purposes, however, we'll talk about variable stars in a dedicated section.

5.6.4 Mixing Processes

Inside the convective core, the burning of Helium into Carbon strongly increases free-free opacity, being $\kappa_{ff} \propto X_i Z_i^2$ (where X_i and Z_i represent here the abundance by mass and the atomic charge respectively, of the element i). Because of this opacity increase, the effect of chemical evolution on the radiative temperature gradient $\nabla_{\rm rad}$ overcomes those due to the changes of other physical quantities within the convective core.

As a result, $\nabla_{\rm rad}$ increases monotonically with time in the whole convective region, an occurrence that prevents any decrease of the mass size of this convective zone. You should also notice that convection occurs on a timescale much shorter than the nuclear burning timescale, which means that nuclear burning has no time to modify the chemical abundances in the convective core before mixing occurs.

The evolution with time of $\nabla_{\rm rad}$ within the star when convective overshoot at the border of the convective core is neglected is shown in Fig.42. We immediately notice the growing discontinuity of the radiative gradient at the boundary of the convective core, due to the changing chemical composition and, in turn, the radiative opacity, caused by the He-burning process.

In a realistic case, you should generally assume that a certain amount of overshoot (albeit possibly even very small) at the Schwarzschild border of the convective core has to occur. Due to the presence of the chemical discontinuity previously discussed and the increasing opacity inside the convective core (because of the conversion of He into C) the mixing of a radiative shell

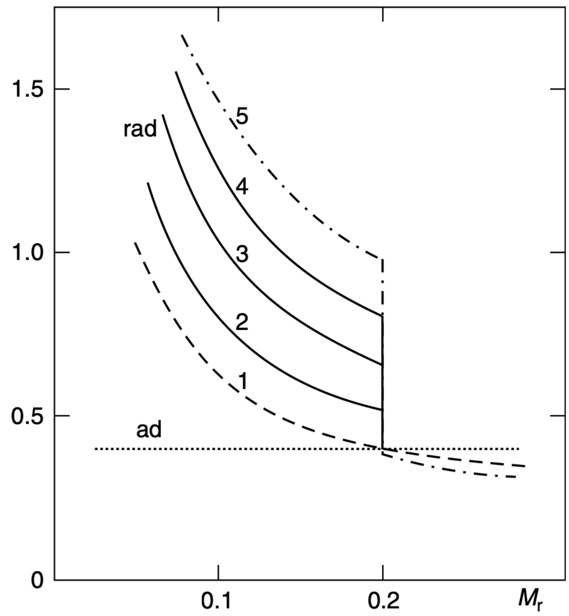

Figure 42: The growing discontinuity of the radiative temperature gradient at the boundary of the convective core for a core He-burning low-mass star. The numbers mark a time sequence (time is increasing going from stage 1 to stage 5). Credits: [21].

(surrounding the convective core) caused by the convective overshoot, causes a local increase of the opacity and, in turn, a convective boundary instability.

As a result, we can expect that a *self-driving mechanism*[34] for the extension of the convective core occurs in the star: Any radiative shell which is mixed as a consequence of the convective core overshoot will eventually become part of the convective core. Due to this process, at the boundary of the resulting enlarged convective core, the radiative gradient becomes equal to the adiabatic one.

After this early phase, during which the extension of the convective core closely follows the increase of $\nabla_{\rm rad}$, the radiative gradient profile shows a minimum. This is a consequence of the progressive shift outwards (within an Helium-rich region) of the convective boundary, due to the self-driving mechanism. The occurrence of this minimum depends on the complex behavior of the physical quantities involved in the definition of $\nabla_{\rm rad}$, such as opacity, pressure, temperature and local luminosity.

The mixing of a radiative shell produces a general decrease of the radiative gradient in the whole convective core. This effect is mainly due to the combination of the mixing of Helium-rich matter and of the change in the physical properties of the mixed shell. The radiative gradient will eventually decrease to the value of the adiabatic gradient at the location of the minimum.

The main problem in the computation of HB evolutionary models is related to the treatment of the intermediate convective zone, located between the minimum of the radiative gradient and the outer radiative zone. A full mixing between the convective core located inside the minimum and the ex-

34 In Italian: *meccanismo di auto-trascinamento del core*.

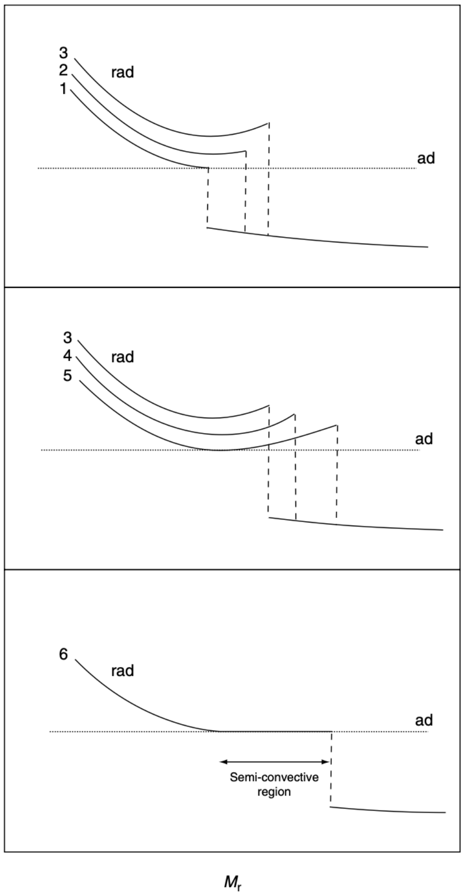

Figure 43: Qualitative behaviour of the radiative temperature gradient near the boundary of the convective core during the core He-burning phase, showing the time sequence of the events which result in the appearance of a semi-convective region. The numbers mark a time sequence (time is increasing going from stage 1 to stage 6). Credits: [21].

ternal convective shell cannot occur because at the minimum the radiative gradient is equal to the adiabatic one. Therefore, the convective shell outside the minimum is no longer mixed with the core.

The result of this decoupling between the convective core and the convective shell outside the minimum is the formation of an extended, partially mixed region, often called a *semi-convective region*, around the fully mixed core, between the minimum of $\nabla_{\rm rad}$ and the outer radiative zone. In principle, since in this region the following disequality generally holds

$$\nabla_{\rm ad} < \nabla_{\rm rad} < \nabla_{\rm ad} + \nabla_{\mu}$$

convection should be inhibited. What we observe, however, is that mixing happens either way.

In HB structures, the outer region is by itself stable against convection and it is the convective core that induces a partial mixing in the surrounding regions. For this reason, the partial mixing phenomenon that occurs in HB stars is often called *induced semi-convection*.

The effects of semi-convection on the evolution of HB models are: (1) the evolutionary tracks perform more extended loops on the HRD, (2) the central He-burning phase lasts longer, since the star has a larger amount of fuel to burn, (3) the mass size of the He-depleted core at the He-exhaustion is larger.

When the core abundance of He has been lowered by nuclear burning to about $Y \sim 0.05 - 0.10$, a convective instability can affect the convective core. In fact, when the core Helium abundance is lower than this limit, α-captures by ${}^{12}_{6}\mathrm{C}$ nuclei tend to overcome ${}^{12}_{6}\mathrm{C}$ production by 3α reactions, thus He-burning becomes mainly a ${}^{12}_{6}\mathrm{C} + \alpha$ production of Oxygen, whose opacity is even larger than that of ${}^{12}_{6}\mathrm{C}$. This causes an increase of the size of the semi-convective region and, in turn, fresh Helium is transferred into the core, which is now nearly He-depleted. Even a small amount of Helium driven into the core becomes very important in comparison with the vanishing amount of Helium otherwise present in the core. This enrichment of the He abundance in the core succeeds in enhancing the rate of energy production by He-burning, and thus the luminosity increases, driving an increase in the radiative gradient. As a consequence, a phase of enlarged convection zone is started, the so-called *breathing pulse*. After a pulse, the star adjusts to burning steadily in the core the fresh Helium driven there by the convection.

The main effect of breathing pulses would be a 20% increase in the duration of the HB, resulting in a larger mass for the CO-core[35] which would in turn result in larger WDs (which would take longer to cool). The effectiveness of these breathing pulses should be in reality much lower than what evolutionary models are predicting.

The ratio of evolutionary times between the subsequent Asymptotic Giant Branch phase (which we'll see shortly) and the HB is very sensitive to the occurrence of breathing pulses. From the observational point of view, the ratio of the star counts along these two phases, the so-called R_2 parameter, is a measure of the corresponding lifetime ratio

$$R_2 = \frac{N_{\mathrm{AGB}}}{N_{\mathrm{HB}}} \propto \frac{t_{\mathrm{AGB}}}{t_{\mathrm{HB}}} \tag{114}$$

Models without breathing pulses predict $R_2 \sim 0.12 - 0.15$, while inclusion of breathing pulses gives $R_2 \sim 0.08$. The value experimentally observed in ancient globular clusters is $R_2 \sim 0.14 \pm 0.05$, which suggests that the efficiency of the breathing pulse phenomenon is very low, if not straight up zero. The appearance of breathing pulses in theoretical stellar models may be related to the approximation of instantaneous mixing (rather than a diffusive one) when convection sets in, that could possibly break down in this late HB evolutionary stage.

With similar intentions at heart, it was defined the R parameter

$$R = \frac{N_{\mathrm{HB}}}{N_{\mathrm{RGB}}(L > L_{\mathrm{ZAHB}})} \tag{115}$$

as a proxy for an indicator of Helium abundance in a cluster. The logic behind it was that increasing the Helium abundance, the luminosity of the ZAHB increased consequently, while the duration in the RGB was essentially unchanged. That fact would reflect in a decrease of $N_{\mathrm{RGB}}(L > L_{\mathrm{ZAHB}})$.

35 Please note that it is note a core of Carbon Monoxide, but a core of Carbon *and* Oxygen.

Clearly, the R parameter cannot be used as an absolute indicator of Helium abundance, once again because of the various mechanisms that participate in determining the He-burning lifetimes in the ZAHB.

If we start including convective processes, like overshooting, we have to make some adjustments. First of all, overshooting increases the duration of the MS and the luminosity at the overall contraction. How does this influcence an age-fixed isochrone? That's simple, really. It just dilates the evolutive timescales, so we're expecting to see a star more massive than usual at the overall contraction.

In a (real) cluster of known distance, the luminosity of the OC is fixed whether we implement overshooting for our model or not[36], so if you want to fit the observational evidences, it's clear that the age of that cluster must be changing: It will increase if you're considering overshooting, it will decrease if not.

5.7 ASYMPTOTIC GIANT BRANCH

When the abundance of Helium becomes low enough, stellar tracks, regardless of the value of the initial mass, move on the HR-diagram towards a lower effective temperature and larger luminosity. This is the *Asymptotic Giant Branch* (AGB) which corresponds to the He-shell burning phase and shows a close similarity with the previous RGB phase.

The designation asymptotic comes from the fact that in low-mass AGB stars, i.e. less than $\sim 2.5M_\odot$, the effective temperature-luminosity relationship is very similar, albeit slightly hotter, to that of low-mass RGB stars. For more massive stars, the term asymptotic has no morphological significance.

After He-exhaustion, He-burning shifts to a shell around the CO core, whose mass-size increases as a consequence of the conversion of Helium to Carbon and Oxygen in the He-burning shell. The overlying H-burning shell, which has burnt outwards for some time, extinguishes due to the expansion and consequent drop of its temperature, caused by the onset of He-burning in the shell.

In low-mass stars ($M \lesssim 2.5M_\odot$), the onset of the He-shell burning induces a temporary drop of the surface luminosity and the star crosses the same region of the HR-diagram three times (as during the RGB bump phase). Why?

The conditions for shell-He ignition are of lower core density and temperature (with respect to the core-He burning). As a consequence, the luminosity because of He-burning decreases, while the luminosity of the H-shell increases (the shell heated up). Overall, the star heats up. While the star moves to more advanced stages, the luminosity of the He-shell increases once more, inducing the expansion of the adjacent layers, which in turn decreases the luminosity of the H-shell and the luminosity due to gravitational contraction. The total luminosity hence decreases. At a certain point, the total luminosity increases once more. As a consequence, there is a good

36 The star doesn't (and shouldn't) care about the model you choose. If you get it wrong, that's your fault, not its.

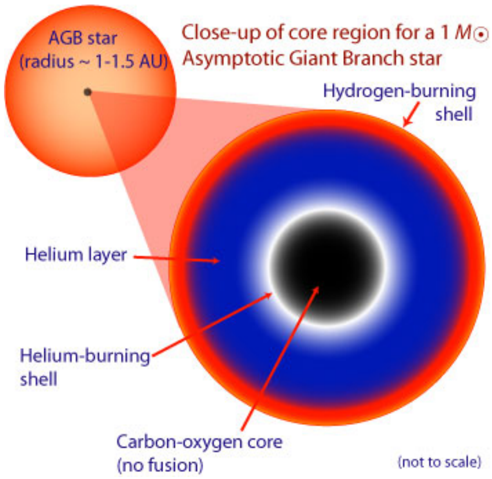

Figure 44

probability of observing an AGB star during this phase, which is called the *AGB clump*; this is indeed the case in well-populated old galactic stellar systems.

Stars with larger masses ($M \gtrsim 2.5M_\odot$) don't clump up. Since they have a large, hot CO-core, the He-shell is already hot enough to effectively support the whole star's weight, and the star experiences only negligible variations in luminosity.

The first part of the AGB phase is usually called *early AGB*[37]. The EAGB lays along an isoconvective track to the left of the AGB, hence the name asymptotic. In this phase, the total luminosity of the star is proportional to the luminosity of the Helium shell (since the H-shell has been turned off), which, in turn, is proportional to the mass of the CO-core: The more massive the core, the more it heats up. The excess heat is transferred to the adjacent Helium shell which shines a little brighter.

Please note that not all stars manage to transition to the AGB phase. Some never manage to expand the envelope and cool down and so never begin the climb up the AGB. This stars are sometimes called *AGB-manqué* and do only the EAGB.

While He-burning is progressively moving outwards inside the He core, and the mass of the CO core is increasing, an important convective episode occurs in stars more massive than $M \gtrsim 3 - 5M_\odot$, the precise limit being a function of the initial composition. In these stars, the large energy flux produced by the shell He-burning causes the base of the H-rich envelope to expand and cool, so that H-burning in the shell is immediately switched off.

When this occurs, the outer convection zone penetrates inwards, into the H-depleted zone. This process is known as the *second dredge up*. For lower mass objects, H-burning in the shell remains quite efficient, and prevents the outer convection from penetrating deeper into the star (therefore the second dredge up does not occur).

37 Properly speaking, the beginning of the early AGB phase is marked when the star is completely supported by the burning of the He-shell.

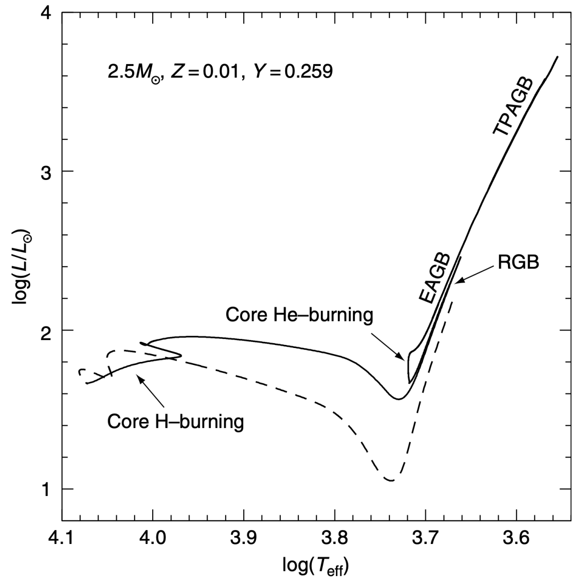

Figure 45: Evolutionary track of a $2.5M_\odot$ star from the PMS (dashed line) to the advanced TPAGB phase. Credits: [21].

In the dredged-up material, which can be as much as $\sim 1M_\odot$ for the most massive AGB stars, Hydrogen has been completely converted into Helium, and both ${}^{12}_{6}C$ and ${}^{16}_{8}O$ have been converted almost completely into ${}^{14}_{7}N$. In addition, the second dredge up has the effect of reducing the mass-size of the H-exhausted region, thus preventing the later formation of very massive white dwarfs.

The He-shell thins down until eventually turning off when the shell has reached the chemical gradient discontinuity between the He and H regions. At that point, the star contracts and heats up, igniting more external Hydrogen shells and piling up Helium on the switched off He-shell. When the shell heats up some more, it turns on once more. Generally, if the mass of the CO-core is $\sim 0.8M_\odot$, masses for the He-shell in the order of $10^{-3}M_\odot$ are enough to turn on the shell. As a rule of thumb, you might want to remember[38]

$$\frac{dM^{\mathrm{He}}_{\mathrm{shell}}}{dM^{\mathrm{CO}}_{\mathrm{core}}} \sim 50\frac{M_{\mathrm{shell}}}{M_\odot}$$

5.7.1 Thermal Pulses

During the early AGB phase, the He-burning shell moves outward in the Helium-rich envelope whose mass is not significantly increased since the H-burning shell is essentially inactive. When the He-burning shell approaches the H/He discontinuity, it dies down, and after a rapid contraction the H-burning shell becomes fully efficient to supply the energy necessary for the star's needs. This temporary stop of the He-burning shell marks the beginning of the *thermally pulsating AGB phase* (TPAGB) or *late AGB*.

38 I might have written it wrong. The sense is that for each $0.2M_\odot$ variation of mass in the core, the mass needed for the shell to turn on again increases by an order of magnitude.

As Hydrogen burns, the Helium ashes which are accreted above the degenerate CO core, are compressed and heated. When the mass of these ashes reaches a critical value (see the very last paragraph of the last section), Helium ignites and a thermonuclear runaway occurs. Thermonuclear runaway means in this case that the shell reacts to the new energy input with an increase of temperature that, in turn, increases the energy generation even more (*positive feedback*). This runaway occurs because of the small geometrical thickness of the He layer on top of the CO core, in spite of the low degree of electron degeneracy[39].

When the geometrical thickness of the shell is sufficiently small, as in the case of the thermally pulsing phase, this condition is not satisfied and the expansion of the shell induces an increase of the temperature (notice that in the case of the He flash the runaway proceeds at constant density).

At the peak of the flash, the rate of nuclear energy release can reach values as high as $L_{\rm He} \sim 10^7 - 10^8 L_\odot$. Most of this energy goes into heating up the nuclear burning layers causing them, and the layers above them, to expand against gravity. As a consequence of being pushed out to very low temperatures and densities, the Hydrogen burning in the shell is switched off.

Stellar models predict the formation of a convective zone extending from the He-burning shell up to the H/He discontinuity, as a consequence of the huge energy release during the thermonuclear runaway.

As the material in the convective shell continues to expand outwards, the shell source is now widely expanded and the condition for the onset of a runaway discussed above is no longer applicable.

The shell starts to cool and the rate of He-burning drops precipitously. In AGB stars of large enough core mass, say $M_{\rm CO} \gtrsim 0.7 M_\odot$, the decrease in $L_{\rm He}$ causes the convective shell to disappear. At lower masses, the base of the convective shell reaches into the zone where incomplete He-burning occurs, and so the products of He-burning (mainly Carbon) are mixed into the whole region located in between the He-burning shell and the H/He discontinuity. Eventually, a steady state is established between nuclear energy production and energy flow outwards.

The AGB star then continues through a quiescent Helium burning phase, which lasts until the total amount of material which is processed by 3α reactions equals the amount of material which was processed by the H-burning prior to the flash.

At this point, Hydrogen near the H/He discontinuity is reignited and the star embarks on another long phase of quiescent H-burning. When the mass of the He-rich layers reaches the critical value previously specified, another thermal pulse is initiated and this cycle is repeated many times[40].

During the inter-pulse phases, the H-burning shell, whose efficiency is determined by the mass-size of the H-exhausted core (hence the CO-core plus the He-shell), is able to provide all the energy necessary to compensate

39 You would expect a runaway in case of strong degeneracy, as in the case of central He ignition in low-mass stars, but here is the reduced thickness (thinness?) of the shell that produces the pulse, not the degeneracy itself.

40 Note that usually pulses have temporal durations rather variable, but that roughly sit in the range $10^3 - 10^6$ yr,

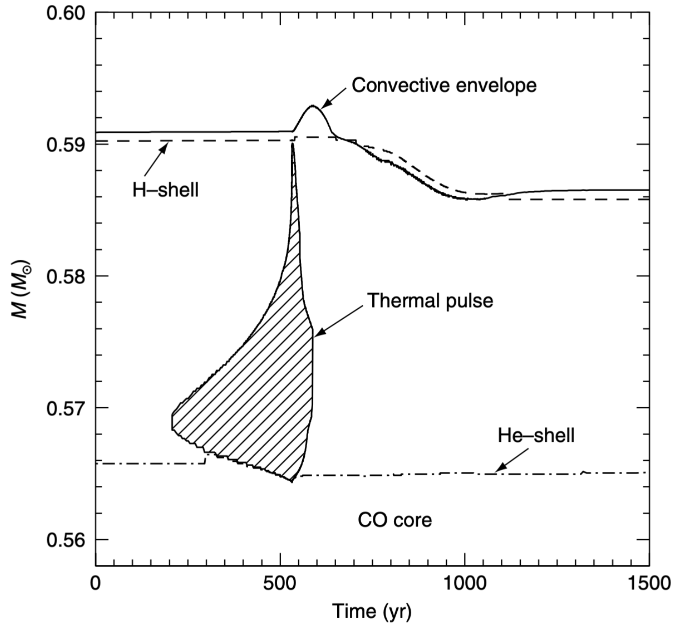

Figure 46: Evolution with time of part of the internal structure of a $2M_{\odot}$ star with $Z = 0.015$ and $Y = 0.275$, during the tenth thermal pulse; the locations of the H-burning shell, the He-burning shell and the base of the convective envelope are shown. The origin of the time coordinate has been arbitrarily shifted. The dashed zone shows the development of the convective region during the thermal pulse. Notice the occurrence of the third dredge up about 200 yr after the onset of the thermal pulse. Credits: [21] and O. Straniero.

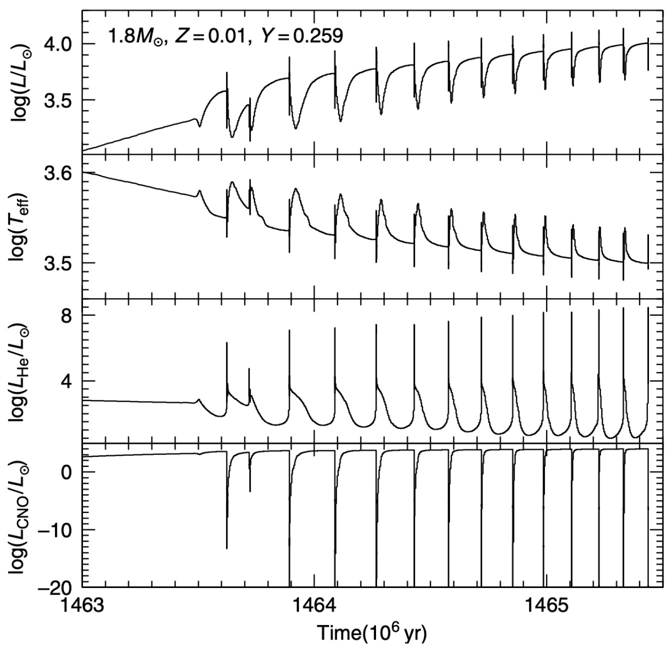

Figure 47: Evolution with time of the effective temperature, surface luminosity, luminosity associated with the He-burning in the shell, and the luminosity produced through the CNO cycle in the H-burning shell, during the first series of thermal pulses of a $1.8M_{\odot}$ star. Credits: [21].

for the surface energy losses. Therefore, as for during the previous RGB phase, there exists a direct correlation between the surface luminosity and the mass of the H-exhausted core.

Earlier we've mentioned the possibility of the occurrence of an inter-shell convective episode; you should note that, once the H-burning has been switched off after the He-shell burning reignition, the convection envelope can move inward in mass, crossing the H/He discontinuity.

If this is the case, a new dredge up occurs, the so-called *third dredge up*. If this dredge up is able to penetrate deep enough into the inter-shell region mixed during the previous thermal pulse, then Helium, products of He-burning (essentially Carbon) and heavy s-elements[41] are thereupon dredged up into the envelope and brought to the surface where we can observe them. As a consequence of recurrent third dredge up episodes, a Carbon-rich star[42] can be produced.

The third dredge up is driven by the expansion and subsequent cooling of the envelope which occurs during the thermal pulse. It penetrates deeper when the strength of the pulse is stronger. The strength of a pulse is fixed by the thermal conditions at the bottom of the He-rich layers, which are strongly dependent on the rate at which Helium is accreted by the H-burning shell.

Evolutionary computations show that there is a minimum envelope mass for the occurrence of the third dredge up. This has the consequence that stars with masses on the lower side (roughly $M \lesssim 1.2 - 1.5 M_{\odot}$) cannot experience the third dredge up since at the beginning of the late-AGB phase they already have a residual envelope mass which is too small. As a consequence, you shouldn't be expecting C-rich stars in very old stellar populations.

In more massive AGB stars, with $M > 6 - 7M_{\odot}$, a peculiar physical process can occur at the base of the convective envelope: the temperature can be so large ($T \sim 8 \cdot 10^7$ K) that significant burning can occur at the bottom of the convective zone, the so-called *hot-bottom burning* (HBB).

The consequences of the coupling between nuclear burning and convective mixing are many: (1) the surface luminosity increases significantly, breaking the core mass-luminosity relation; (2) the surface chemical composition is strongly affected. Carbon is converted into Nitrogen, preventing the formation of C-rich stars and an enhancement of Lithium can occur through the so-called *Cameron-Fowler mechanism* [6]. The general idea is that when the envelope is mixed down to temperatures of the order of 10^8 K, $^3_2$He is transformed very efficiently into $^7_4$Be, according to the reaction

$$^3_2\text{He} + ^4_2\text{He} \rightarrow ^7_4\text{Be} + \gamma$$

The produced Berillium is then transformed into Lithium through

$$^7_4\text{Be} + e^- \rightarrow ^7_3\text{Li} + \nu_e$$

The only Lithium that can survive, however, is the one that stays below its burning temperature of $\sim 2.5 \cdot 10^6$ K. The amount of surface Lithium preserved after the envelope retreats towards the surface is therefore determined by the balance between the $^7_4$Be transported out towards the surface

41 At the typical temperatures for this stage, neutrons chain reactions are active, thus the possibility of producing s and r elements through neutron captures is not negligible.
42 Not surprisingly called "Carbon Star".

by the convective mixing, and the ${}^{7}_{3}$Li produced at low enough temperatures that it is then transported back inward by the convection.

This mechanism may explain the empirical evidence that almost all the brightest *Long Period Variables* (commonly identified as bright TPAGB stars experiencing large amplitude radial pulsations) are super-Lithium rich objects, their surface Lithium abundance being about three orders of magnitude larger than normally expected.

5.7.2 Production of s-elements

Spectroscopical observations show that the surface of AGB stars is strongly enriched in s-process elements[43]. The key ingredient in activating the s-process reactions is the neutron source. In AGB stars, we can identify two major sources: The chain leading to α-captures on Neon nuclei and a modified version of the CN-cycle.

Let's address the first one. During the early phase of the convective episode occurring in the inter-shell region, the material is fully mixed and ${}^{14}_{7}$N, which is present in this region as a consequence of the previous H-burning process (whose abundance is almost equal to the sum of the initial abundances, by number, of the CNO elements), is converted into ${}^{22}_{10}$Ne by means of the following chain[44]

$$
\begin{aligned}
&{}^{14}_{7}\mathrm{N} + \alpha \rightarrow {}^{18}_{9}\,\mathrm{F} + \gamma \\
&{}^{18}_{9}\mathrm{F} \rightarrow {}^{18}_{8}\,\mathrm{O} + e^{+} + \nu_e \\
&{}^{18}_{8}\mathrm{O} + \alpha \rightarrow {}^{22}_{10}\,\mathrm{Ne} + \gamma \\
&{}^{22}_{10}\mathrm{Ne} + \alpha \rightarrow {}^{25}_{12}\,\mathrm{Mg} + {}^{1}_{0}\,\mathrm{n}
\end{aligned}
$$

If, during the thermonuclear runaway, the temperature at the base of the inter-shell region attains a high enough value ($T \sim 3 - 3.5 \cdot 10^{8}\,\mathrm{K}$) so that the last reaction of the chain can occur[45], providing a sizeable flux of neutrons. It has been demonstrated that this is the case for intermediate-mass AGB stars ($M \gtrsim 3M_{\odot}$). In smaller AGB stars with $M \lesssim 3M_{\odot}$ the base of the inter-shell region does not attain a high enough temperature to provide enough neutrons in this way.

In low-mass AGB stars, an alternative source of neutrons is provided by a modified chain for the CNO cycle, which is also the *only* source of free neutrons. In order to be effective, this reaction requires a temperature of $\sim 9 \cdot 10^{7}\,\mathrm{K}$, which can easily be attained in the inter-shell region. The chain is the following

$$
\begin{aligned}
&{}^{12}_{6}\mathrm{C} + {}^{1}_{1}\,\mathrm{p} \rightarrow {}^{13}_{7}\,\mathrm{N} + \gamma \\
&{}^{13}_{7}\mathrm{N} \rightarrow {}^{13}_{6}\,\mathrm{C} + e^{+} + \nu_e \\
&{}^{13}_{6}\mathrm{C} + \alpha \rightarrow {}^{16}_{8}\,\mathrm{O} + {}^{1}_{0}\,\mathrm{n}
\end{aligned}
$$

43 Brief reminder, s-elements are those elements beyond the iron peak, such as Sr, Y, Zr, Ba, La, Ce, Pr and Nd, which are produced through slow neutron captures (slow compared with β-decay).

44 I'm sure you know by now, but just in case $\alpha = {}^{4}_{2}$ He.

45 The first three reactions require a lower temperature, $T \sim 10^{8}\,\mathrm{K}$, so it reasonable to expect them to be always active in intermediate mass AGB stars.

Between two consecutive thermal pulses, some ${}^{13}_{6}$C is created at the top of the inter-shell region due to the reactions in the H-burning shell. Nevertheless, the burning of this ${}^{13}_{6}$C is not an efficient neutron source: in a region which has been processed by H-burning ${}^{14}_{7}$N is more abundant than ${}^{13}_{6}$C, and so any neutrons released by ${}^{13}_{6}$C would be immediately captured by ${}^{14}_{7}$N.

An alternative source of ${}^{13}_{6}$C is necessary in a region where ${}^{14}_{7}$N has already been exhausted. A possible way out is the following: In the inter-shell region, after the convective shell episode, there is a huge amount of ${}^{12}_{6}$C, and ${}^{14}_{7}$N is missing, having been converted into Neon. ${}^{13}_{6}$C is then produced via the nuclear chain we discussed earlier. How is it possible, however, to have sufficient protons in the He-rich inter-shell region after the convective shell episode?

A more promising phase for the formation of a ${}^{13}_{6}$C-pocket within the inter-shell, is the one corresponding to third dredge up and during the so-called *post-flash dip*, which is the period that immediately follows the third dredge up. At the time of the third dredge up, a sharp discontinuity between the H-rich envelope and the He- and C-rich inter-shell is produced, and this situation holds until H-burning is ignited again. If, during this time, a few protons can diffuse into the underlying layers, at the reignition of the H-burning shell the upper layers of the He-rich inter-shell region heat up and a ${}^{13}_{6}$C pocket can be formed, thus solving almost all our problems[46].

The Solar Standard Model (SSM)

It might be worth spending a few words in this secton to talk about the SSM and all the goodies that come with it.

The importance of studies of the Sun's internal structure lies not only in its intrinsic interest, but also in the possibility of testing some of the fundamentals of our theories, both physical and astrophysical. The Sun is the star for which, through helioseismology[47] and spectroscopic measurements, we can collect the largest amount of empirical data which can be directly compared to theoretical predictions.

Without going into detail, from the analysis of the spectra associated with solar non-radial oscillations, it is possible to determine accurately the depth of the envelope convective zone and the speed of sound at the transition radius between convective and radiative regions.

Coupling the empirical evidence with the other known properties of the Sun, i.e. its luminosity, $L_\odot = 3.842 \cdot 10^{33}\,\mathrm{erg\,s^{-1}}$, radius, $R_\odot = 6.9599 \cdot 10^{10}\,\mathrm{cm}$ and mass, $M_\odot = 1.9891 \cdot 10^{33}\,\mathrm{g}$, and the observed ratio Z/H, you could in principle work at obtaining the best stellar evolution model, based on the most up-to-date physics, able to reproduce the whole set of observational properties of the Sun at its present age ($t_\odot \approx 4.5\,\mathrm{Gyr}$): This would be a *Solar Standard Model* (SSM).

When computing an SSM, rotation and magnetic fields are assumed to have a negligible effect on the evolution. Furthermore, mass loss is almost

46 ...and creating some new ones. But that's another story.

47 Helioseismology studies the acoustic waves produced by oscillations within the solar structure, which are speculated to be induced and excitated by the convective cells lying underneath the surface.

universally neglected during the central H-burning phase of solar-like stars because the estimated mass loss rate is of the order of $10^{-14} M_{\odot}\,\mathrm{yr}^{-1}$.

Thus, the free parameters which, aside from some practical limits, are varied in solar ZAMS models until 4.5 Gyr worth of evolution yields the present-day Sun, are

- The abundance of heavy elements Z, which affects the structural evolution through the effect on the stellar opacity and on the efficiency of the CNO cycle;
- The Helium content, affecting the structure through the change in the mean molecular weight and, in turn, in the nuclear energy release, and in the opacity. Note that in constructing the SSM, the adopted initial values of Z and Y (Z_{in} and Y_{in}) are not independent;
- The value adopted for the free parameter α_{ML} in the mixing length theory. This affects the efficiency of convection in the envelope and, in turn, fixes the radius of the solar model.

Clearly, the model will aim to reproduce the observed properties within present day uncertainties, which, for some measurement in particular, are still rather large.

5.8 PULSATING VARIABLE STARS

When stars cross a number of well-defined regions in the HR-diagram, they are affected by stable radial pulsations. As a consequence, they change their brightness periodically and become variable stars. These peculiar regions of the HR-diagaram are called *instability strips*. The instability strip we'll consider in the following is related to the two most important classes of radially pulsating stars, the *RR-Lyrae stars* and the *Classical Cepheids*. These represent probably the most important standard candles for Population II and Population I stellar systems, respectively.

In either case, the pulsation involves only the more external layers of the star without affecting their interiors and the energy generation efficiency.

Detailed analysis shows that the Helium and Hydrogen ionization zones located near the surface (the total mass above the base of the He-ionization zone is only $\approx 10^{-7} M_{\odot}$ in a RR-Lyrae variable) are responsible for driving the pulsations, during which the stellar luminosity for an RR-Lyrae star can vary by a factor of two and the radius can change by $\approx 20\%$.

These ionization zones have to contain a sufficient fraction of the mass of the star in order to drive the pulsations efficiently, therefore they cannot be too close to the low-density surface layers; this explain the existence of the hot boundary of the instability strip, given that increasing the stellar T_{eff} moves the ionization regions towards the surface. On the other hand, when the stellar T_{eff} becomes sufficiently low, envelope convection sets in and the varying-opacity mechanism is no longer able to drive the pulsation efficiently; this is the reason for the existence of the cool boundary of the instability strip.

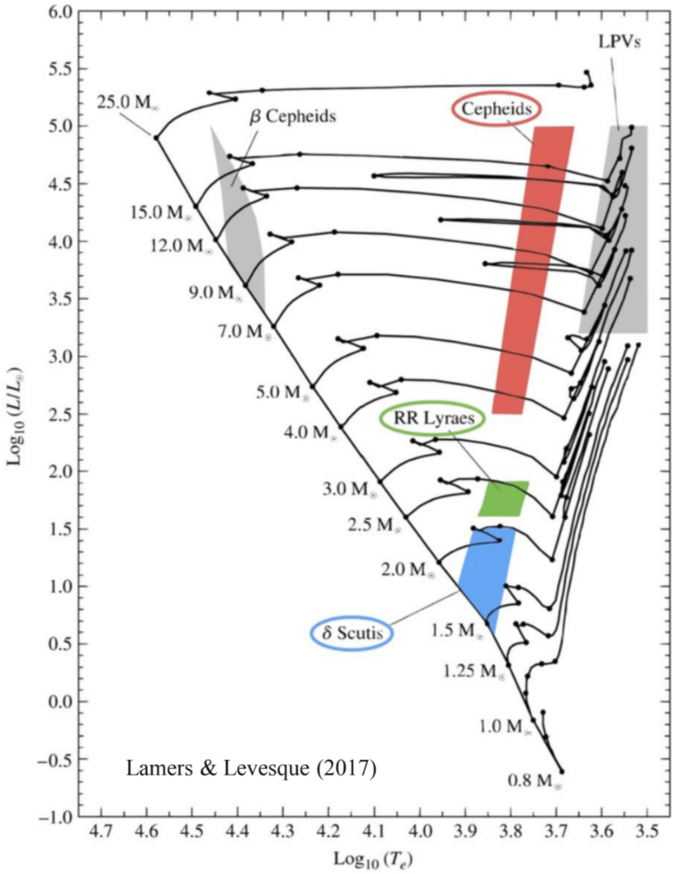

Figure 48: Location in the HRD of various pulsating stars. Cepheids, RR-Lyrae and δ-Scuti stars are located in a nearly vertical instability strip. The Long Period Variables are outside the strip.

During a contraction, the physical properties are changing in the following way

$$\chi \downarrow \ n_e \uparrow \ T \sim \text{const.} \ \rho \uparrow \Longrightarrow \ \bar{\kappa} \uparrow \ L \downarrow$$

where all symbols have their usual meaning aside from χ, that here refers to the ionization fraction of the layer. During the contraction, the external regions absorb the energy released from the compression and heat up just slightly, since most of the energy is used to ionize the material. Thus, more energy is retained thanks to the increased opacity, and a pressure able to bloat up the region is produced. Similarly, the temperature decreases only a little, since most of the energy needed for the expansion is obtained through the recombination of ions and electrons from the formerly ionized nuclei; inevitably, the electron density decreases $n_e \downarrow$ and the opacity decreases as well $\bar{\kappa} \downarrow$. The star is thus able to cool down, contract and repeat the cycle.

Note that the luminosity of the star is increasing only as long as the effective temperature remains approximately constant, so that as soon as it begins to drop (when the star expands over a certain threshold) luminosity decreases as well. This is why the point of maximum luminosity corresponds (roughly) to the point of maximum effective temperature rather than to the point of maximum spatial extension. A more detailed analysis would also tell you that it is also corresponding to the point where the star has maximum (absolute) radial expansion velocity.

With what timescale does this oscillation occur? Radial contractions are essentially free-fall collapses at heart, and so it would be logical to assume the process to follow the dynamical timescale given by the free-fall time (26) which is proportional to $\tau_{ff} \propto \rho^{-1/2}$, and hence to $\tau_{ff} \propto R^{3/2}M^{-1/2}$. Recall now how we've defined the luminosity

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4$$

We can invert it to find the radius and obtain for the free-fall time this important dependence

$$P \propto \tau_{ff} \propto L^{3/4} M^{-1/2} T_{\text{eff}}^{-3} \propto \sqrt{\frac{R}{g}} \tag{116}$$

where we've introduced the surface gravity $g \propto M/R^2$. The existence of this relation, connecting an empirical quantity such as the period of pulsation, with a structural property such as the mean stellar density, highlights the importance of stellar pulsations to stellar evolution theory; the analysis of radial pulsations provides a formidable tool for investigating stars, even more so since the measurement of the pulsating period isn't affected by possibly troublesome factors such as distance and reddening.

As a rule of thumb, you might want to remember the following properties that directly descend from (116)

$$L \uparrow \implies P \uparrow \quad T_{\text{eff}} \uparrow \implies P \downarrow \quad M \uparrow \implies P \downarrow$$
$$Z \uparrow \ \bar{\kappa} \uparrow \implies T_{\text{eff}} \downarrow \ L \downarrow \ M_{\text{strip}} \downarrow$$
$$Y \uparrow \ L \uparrow \implies T_{\text{eff}} \uparrow \implies M_{\text{strip}} \uparrow$$

Type	Pop	Period (days)	log ($L/L_\odot$)	Spectral Type	$\bar{Q}$(days)
δ Cepheids	I	2–60	2.8–4.6	F–G	0.04
δ Scuti	I	0.04–0.2	0.9–1.6	A–F	0.04
β Cepheids	I	0.1–0.2	4.0–4.7	B1–B2	0.03
W Virginis	II	1–20	2.0–2.9	F–G	0.06
RR Lyrae	II	0.3–1	1.5–1.6	A–F	0.04

Note. Data are from Cox (2000).

Lamers & Levesque (2017)

5.8.1 RR-Lyrae Variables

RR-Lyrae variable stars are found where the instability strip meets the horizontal branch, in a narrow region where the luminosity extension is fairly limited. RR Lyrae are then low-mass stars, which are crossing the instability strip during their core He-burning phase.

These variables have periods in the range $0.2 - 0.9$ days and light curve[48] amplitudes between $0.2 - 1.6$ mag in the B photometric band. They are classified into two groups: the *ab type* variables, characterized by asymmetric light curves of large amplitude, and *c type* variables with nearly sinusoidal light curves and small amplitude. The ab type RR Lyrae pulsate in the *fundamental mode*, and c type variables in the *first overtone mode*. There exists also a third group, the *d type* RR Lyrae, pulsating simultaneously in both modes.

48 *NdA*: The trend of the luminosity with time.

The hottest, blue edge, of the instability strip is located at an effective temperature of the order of $\sim 7200\,\mathrm{K}$ at the ZAHB luminosity level, a value which slightly decreases with increasing stellar luminosity. The reason for the existence of a blue edge (and of its change in effective temperature when increasing the luminosity) has to do with the fact that, for a given mass and luminosity, as the surface temperature increases, there is progressively less mass above the ionization zones and, in turn, their contribution to the pulsational driving decreases.

The red side of the instability strip is limited by the so-called red edge, located at $T_{\mathrm{eff}} \sim 5900\,\mathrm{K}$, whose existence is related to the presence of envelope convection in the stars; being an efficient energy transport mechanism, it acts to quench the process of pulsation.

It is also worth noting that both the width and the topology of the RR-Lyrae instability strip is marginally dependent on the stellar metallicity, at least in the heavy element abundance range appropriate for the old galactic stellar systems which host RR-Lyrae variables.

The most up-to-date relation between the quantities concurring to determine the (fundamental) pulsation period of an RR-Lyrae is

$$\log P_f = 11.242 + 0.841 \log\left(\frac{L}{L_\odot}\right) - 0.679 \log\left(\frac{M}{M_\odot}\right) - 3.410 \log T_{\mathrm{eff}} + 0.007 \log Z$$

while the period of the first overtone can be expressed in terms of the period of the fundamental as

$$\log P_{FO} = \log P_f - 0.13$$

5.8.2 Classical Cepheids Variables

From the point of view of stellar evolution Classical Cepheids (or Type I Cepheids) are intermediate-mass stars going through the stage of core He-burning. The evolutionary tracks of intermediate-mass stars can cross the instability strip up to three times. All stars cross the Cepheid instability strip during their expansion toward the Red Giant configuration, at the point of exhaustion of Hydrogen in the core, but this crossing is so rapid that there is almost no chance to observe a star pulsating as a Cepheid variable in this stage. So the Cepheid behavior can be observed only for those stars that experience a blue loop, extended enough to cross the pulsational instability strip.

Classical Cepheids are very bright objects, with intrinsic luminosity in the range $\sim 300 - 25000 L_\odot$. Their pulsational periods are mainly confined to the range $1 - 50$ days, with a few extreme examples of up to 250 days.

Both the shape and location of the Cepheid instability strip are significantly affected by metallicity[49]. For Cepheid variables as well, it is possible to derive pulsational relations connecting the period to the stellar mass, luminosity and effective temperature. At a fixed luminosity and mass, decreasing the effective temperature produces (according to this relation) an increase in

49 Typically, the larger the metallicity, the cooler and steeper is the location of the instability strip on the HR-diagram.

the pulsational period. Since an increase of the metallicity has the effect of shifting the Cepheid instability strip towards a lower effective temperature, and therefore longer periods, one should expect that the period-luminosity relation for Cepheid variables depends on metallicity.

The existence of this correlation can easily be understood by considering the dependence of the pulsational period on stellar luminosity, mass and effective temperature, and the prediction of evolution theory that there is, for a set chemical composition, a tight relation between the mass and surface luminosity of stars at the beginning of the core He-burning phase. As a consequence, for any set chemical composition, the brightness of a Cepheid variable is a function of its period and effective temperature.

Classical Cepheids as Standard Candles

A very important technique that can be applied to gauge the distance of young SSP makes use of the Cepheid period-luminosity $P - L$ relationships, whenever a number of Cepheid variable stars is detected in the SSP under scrutiny. The typical age range of SSPs harboring Cepheid stars is in the range $\sim 10 - 100\,\mathrm{Myr}$.

As already discussed, Cepheid stars display well-defined relationships between their average intrinsic luminosity along a pulsation period (hence average absolute magnitudes in various wavelength bands) and the period itself. The main idea of this method is to measure the periods (from their observed light curves) of the Cepheid stars in the observed SSP and fit these values to a standard $P - L$ relationship of the kind

$$M_A = a \log P + b \tag{117}$$

for the photometric band A employed. The use of $P - L$ relationships in two (or more) different photometric bands allows one to determine simultaneously the distance modulus and extinction as we've already discussed somewhere in an earlier chapter.

The preferred photometric band employed is the K band (in the IR range) where the variation in magnitude due to metallicity are roughly compensated for by bolometric corrections. A further advantage is that the K-band is also negligibly affected by interstellar extinction. As a general rule, the value of the $P - L$ slope becomes more negative moving towards longer wavelength filters, and the dispersion decreases.

Under these circumstances, stars roughly align along a line in the $F_K - P$ plane, from which you can easily obtain the distance through the distance modulus.

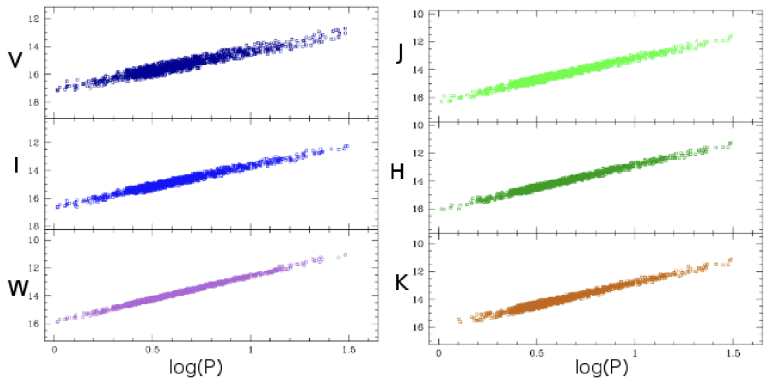

Figure 49: Dispersion of the $P - L$ curve for various photometric bands. Note of the infrared K band and the W band (which is a band optimized for the purpose) are the narrower of the ones considered.

5.9 TERMINATION OF THE AGB PHASE: FINAL DESTINY

And now, the end is near...

Frank Sinatra, *My Way*[50]

Before diving head-on into all the possible ending points of a star's life, it is perhaps sensible to spend a couple of line to introduce some concepts that are perhaps a little bit overdue now.

- *Massive Stars*: We call massive stars those that end the RGB phase with a fully non-degenerate He-core and that don't develop a degenerate CO core during AGB. These stars, in the range

$$M \gtrsim 10M_{\odot} = M'_{\mathrm{up}}$$

  will do all the burning we've described somewhere in Chapter 4 up to Iron, only to later die as Type II SNe;

- *Intermediate mass Stars*: We call intermediate mass stars those that have a non-degenerate He-core during the RGB but with a degenerate CO-core during AGB. These stars have masses in the range

$$M_{\mathrm{RGB}}^{\mathrm{TR}} \lesssim M/M_{\odot} \lesssim M'_{\mathrm{UP}}$$

- *Low Mass Stars*: We call low mass stars those in the range[51]

$$0.3 \lesssim M/M_{\odot} \lesssim M_{\mathrm{RGB}}^{\mathrm{TR}}$$

- *Very Low Mass Stars*: We call low mass stars those in the range

$$0.08 \lesssim M/M_{\odot} \lesssim 0.3$$

  which are the stars that are fully convective during MS.

50 The more I read the lyrics of this song, the more I realize you could easily transpose the whole meaning of the song to be referring to stars.
51 Note that they partially overlap with UMS stars.

Once the star settles into the thermally pulsating phase, in principle it can experience a huge number of thermal pulses, the number being limited only by the mass of the H-rich envelope or by the mass of the CO core.

If the CO-core mass exceeds the value $\sim 1.4M_\odot$ (the so-called Chandrasekhar mass limit) the ignition of Carbon burning can occur (at $T \sim 7-8 \cdot 10^8$ K, which requires $M \gtrsim M_{\rm up}$) and this marks the end of the AGB phase.

In reality, due to mass loss during the late AGB, we do not really expect that any star will reach the Chandrasekhar limit, as a consequence of the increase of the mass of the CO-core during the late AGB evolution, and ignite Carbon.

5.9.1 Low Mass Stars

It is now worth discussing briefly the evolution of low-mass stars from the end of the late AGB phase to the beginning of the following evolutionary phase, i.e. the white dwarf (WD) stage. As a consequence of radial pulsations (or any other mechanisms) the star loses a lot of its mass during the brightest part of the AGB. Once the mass of the H-rich envelope decreases below a critical value, the remnant star evolves rapidly towards hotter effective temperatures.

It continues to burn Hydrogen in a thin shell until the bluest point along its evolutionary track at almost constant luminosity. At this point, H-burning is switched off and both the H-rich envelope and He-rich layer contract rapidly. Depending on thermal conditions of the envelope, different evolutionary channels can be followed:

1. Nuclear burning definitively dies out and the remnant cools as a white dwarf;

2. Heating of the He-rich layers leads to an He-burning thermonuclear runaway in the He shell, the consequence of which is to carry the star back to the AGB. This is the so called *born-again* AGB scenario. During the subsequent quiescent He-burning phase, the star evolves to the blue side of the HR-diagram on a timescale approximately three times longer than that of the previous shell H-burning phase. At the end of this, the star becomes a white dwarf[52];

3. If heating of the H-rich envelope induces an H-burning thermonuclear runaway, the stellar remnant becomes a *self-induced nova*. To outline the final fate of a self-induced nova is not an easy task, due to the difficulties encountered in the numerical simulations.

   That said, it is possible to make some predictions: (a) if the process is very violent and dynamic, this can produce the ejection of the whole H-rich envelope and so the star becomes a white dwarf with no trace of Hydrogen on its surface; (b) if the process is only mildly dynamical and only a small portion of the H-rich envelope is lost, the mass of

52 Note that we've already talked about this in a previous section. I'm merely repeating it here because it is fitting of this section.

this envelope will be decreased by nuclear burning. After a phase of quiescent H-burning, the star will start to cool down the white dwarf cooling sequence.

It is possible that another H flash event occurs leading to a further reduction through nuclear burning of the H-rich envelope mass. This process could, in principle, be reiterated many times until there is too little Hydrogen left to burn. During the subsequent WD evolution, even a very modest wind could cause the loss of this layer.

Once again, it is worth mentioning that the evolutionary path followed by a star from the end of the AGB to the beginning of the cooling phase along the WD sequence is the one corresponding to the formation of Planetary Nebulae.

5.9.2 Carbon Ignition

As we've mentioned, only stars with masses larger than M_{UP} are able to ignite Carbon, with M_{UP} corresponding to the largest mass at which the electron degeneracy of the CO-core is high enough to prevent carbon ignition. Such mass is strongly dependent on the star's metallicity[53].

If the star is not massive enough, the core degenerates (intermediate mass stars) and undergoes a process alike that of low mass stars during the RGB: The mass of the CO-core steadily increases as long as Helium is burned in shells, increasing its density, but energy is subtracted by escaping neutrinos that may induce a temperature inversion, so that the maximum of the temperature is no longer at the center of the core.

For stars in the range $M_{\mathrm{UP}} \lesssim M \lesssim M'_{\mathrm{UP}}$ Carbon is ignited with a flash, *off-center*, in a strongly degenerate environment. The relevant reactions for Carbon burning were already introduced in Chapter 4, so you already know that Neon, Magnesium and Sodium are produced (aside from a large quantity of inert Oxygen that was already in the core from the He-burning phase). Most of ${}^{23}_{11}\mathrm{Na}$ is combined with protons to form ${}^{24}_{12}\mathrm{Mg}$. When Carbon is depleted the core will thus be made mainly of Oxygen, Neon and a rather smaller percentage of Magnesium. Stars with the mass in the range we're considering, however, are not able to ignite Oxygen. So what happens?

Supernovae Classification

Supernovae are broadly distinguished into two main categories and many sub-categories depending on observational properties of their emission spectrum.

- Type I: Do not present Hydrogen lines in the spectrum;
- Type II: Present Hydrogen lines in the spectrum because some was retained in the external layers of the envelope.

53 At solar metallicity $Z \sim 10^{-4}$, $M_{\mathrm{UP}} \sim 8M_{\odot}$, while for higher metallicity, say $Z \sim 10^{-3}$, the needed mass is lower $M_{\mathrm{UP}} \sim 6M_{\odot}$.

There are two main scenarios to consider, but for both we have to recall something that might have seemed like a mention en passant. Carbon ignition occurs off-center. This is crucial. It means that, *potentially*, Carbon burning might never reach the center of the star.

If the star is able to lose enough mass to be stripped of the little envelope it still held, it can turn into a WD with an Oxygen-Neon core or, if mass losses are less severe, into a stranger WD with a CO-core and a layer of Oxygen, Neon and Magnesium, which can be rather problematic for studies concerning SN Ia.

If mass losses do not prevent nuclear burning from happening, the density of the core might favor electronic captures on nuclei and the formation of neutrons[54]. This would have the effect of reducing the pressure that supported the star, facilitating the collapse of the core, that might ignite nuclear burning so violently to destroy the whole star. This type of SN is called *electron capture Supernova* (ECSN).

ECSNe should have rather different characteristics from those of normal Type II SNe (which are already rather varied), and its existence has been often questioned. Apparently, one had been detected in 2018.

5.9.3 Non-Carbon Ignition

For stars with masses lower than M_{UP}, ignition of Carbon is not possible, and their evolutionary path is once more determined by how much severe are mass losses.

If mass losses are high enough to shut down the shell He-burning, they do end up ending their lives as CO-cored white dwarves that will be cooling down for a long time. Otherwise, thermal pulses continue, feeding the core with more and more Carbon until it reaches Chandrasekhar's limit mass.

While the CO core mass increases because of the He-burning in the shell, it also contracts and becomes denser. The release of gravitational energy should heat the core, but the increasing densities in the core strongly favor neutrino energy losses, and the central portion of the core cools. However, with increasing CO core mass, the nuclear burning of Carbon is activated. When, and if, the CO core mass approaches the Chandrasekhar limit, ϵ_C becomes comparable to ϵ_ν. Ultimately, the Carbon burning energy release will overcome the neutrino energy losses, and a thermonuclear runaway occurs.

If this thermonuclear runaway occurs at the center of the star, we expect an explosive process that would cause its complete destruction (the so-called *Carbon Deflagration SN*). In the case where the Carbon burning starts in a shell far enough from the center, after one or more flashes Carbon burning quietly settles in the center of the star. The final fate of these stars is eventually to experience thermal pulses at a very high luminosity and finally to become white dwarfs with a degenerate O-Ne core.

Whether a star can develop an electron degenerate CO core of the order of the Chandrasekhar mass during its evolution along the AGB, depends not only on its initial mass but also on the efficiency of mass losses along the previous evolutionary phases. But, in the real world, mass losses significantly

54 This would happen before the core reaches the Chandrasekhar limit mass.

reduce the stellar mass so that very few, if any, stars with mass around M_{UP} (or slightly larger) are able to reach the conditions for the ignition of Carbon.

At present day, no Carbon deflagration SN has been observed.

5.9.4 Massive Stars

The evolution of stars more massive than M_{UP} ($M'_{\text{UP}} \lesssim M \lesssim 70 - 100 M_{\odot}$) during these early stages is quite similar to their lighter counterparts, the only exception being the sensitivity of the location on the HR-diagram to the mass loss (which can be huge for these stars) as well as the treatment of convection. In particular, the choice of the convection criterion affects the ratio between the time spent on the blue and the red side of the HR-diagram by these stars.

An additional peculiarity is that while, as in less massive stars, the mass of the He core is fixed both by the mass-size of the convective core during the central H-burning phase and by the amount of Hydrogen processed by the H-burning shell during the core He-burning stage, the mass of the CO core is essentially fixed by the size of the convective core during the central He-burning phase. This is because the evolutionary phases subsequent to the core H-burning stage are so fast that the He-burning shell has no time to advance significantly outwards. In addition, the chemical profiles of Carbon and Oxygen in the core are flat, being the result of the fully efficient mixing into the core during the central He-burning phase.

The core abundance of Carbon is a fundamental quantity that strongly affects all the successive evolutionary stages, since it essentially determines the amount of Carbon available for the core and shell C-burning. Needless to say, the amount of Carbon is strongly sensitive to the nuclear cross section for the $^{12}_{6}\text{C} + \alpha$ reaction and to the efficiency of all mixing processes which can occur inside the convective core.

The timescale needed for these stars to move from the red side of the HR to the blue side and vice versa, which is of the same order of magnitude as the Kelvin-Helmholtz timescale (26) for the envelope, is about $10^2 - 10^3$ yr, the exact value depending on the stellar mass and on its location on the HR-diagram. This has the important consequence that as the advanced evolutionary phases (as in successive to the core He-burning stage) are extremely fast, the surface evolution is, in some sense, frozen out, and so the surface luminosity and effective temperature do not change significantly until the star explodes. For this reason, the advanced evolutionary phases of massive stars can be described as the *neutrino-mediated Kelvin-Helmholtz contraction* of the CO core.

At the exhaustion of central Helium, the CO core immediately starts to contract so that gravitation can supply the energy necessary to support the star. During this phase, the core temperature of these massive stars exceeds the value $\sim 5 \cdot 10^8$ K, entering into a region of the density-temperature plane where neutrino losses from pair annihilation start to dominate the stellar energy budget. Radiative energy transport and convection are still important in determining the stellar structure, but it is the neutrino luminosity that mainly balances the energy generated by the gravitational contraction and by nuclear burning.

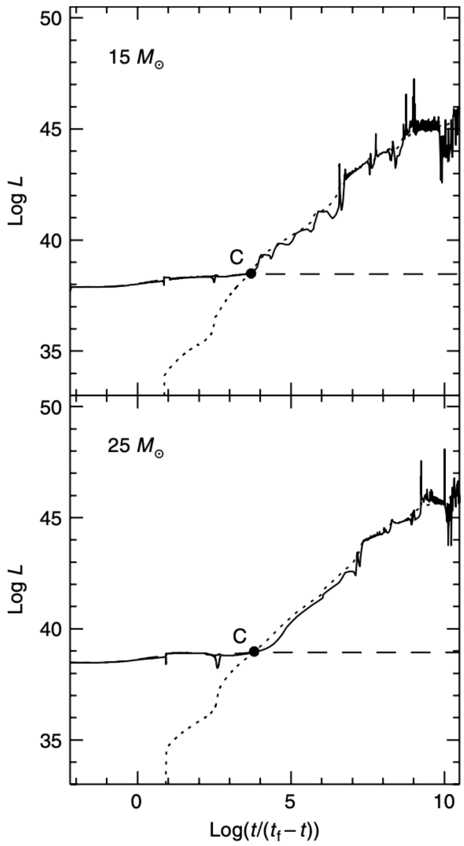

Figure 50: Behavior with time of various luminosities (in $\mathrm{erg\,s^{-1}}$) for two massive stars: the dashed line corresponds to the surface luminosity, the solid one to the energy produced by the nuclear burning, while the dotted lines refers to the neutrinos' energy. The solid dot marks the ignition of the core C-burning. Credits: [21].

Since the energy produced by nuclear burning has to supply the energy lost from the stellar surface, the nuclear luminosity initially follows the behavior of the surface luminosity, but from the ignition of Carbon it closely follows the neutrino luminosity. For many stars, the neutrino luminosity exceeds the standard photon (nuclear) luminosity. As a consequence, the star rapidly contracts to produce the missing energy through the work of gravitation.

At the end of the shell C-burning stage, before Oxygen is ignited in the core, there are the conditions for Neon to be photodisintegrated into Oxygen. A characteristic of this burning phase is that a convective core is always formed, regardless of the stellar mass, but it exists only for the short time during which the nuclear burning rate overcomes neutrino energy losses. Thus core Ne-burning occurs partially in a radiative region and lasts only for a short time. Since the central abundance of Neon is vanishing, Ne-burning shifts into a shell whose evolution is limited to the short time intervals between two consecutive core burning stages.

The star then goes through the Oxygen-burning reactions that we mentioned in Chapter 4, leading, most notably, to a large abundance of Silicon.

Core O-burning produces neutron-rich nuclei as a consequence of the large efficiency of weak interactions. The production of neutron-rich nuclei affect the so-called level of neutronization of the stellar matter, that is an indicator of how neutron rich the constituents of the gas are, and is related to the difference between the number density of neutrons and the number density of protons, both bound in nuclei and free. The higher the number density of neutrons with respect to protons, the higher the level of neutronization.

When Oxygen is exhausted in the core and the central temperature attains values of order $\sim 2.3 \cdot 10^9$ K Silicon-burning can start. Once again, please note that during Silicon burning there's nothing burning: Silicon-burning does not occur predominantly as a nuclear fusion reaction, but it occurs as a chain of photodisintegrations and α-additions that lead to the formation of heavier nuclei.

At the exhaustion of Si in the core, the burning moves into a shell surrounding the Si-exhausted core. This Si-burning shell usually experiences recurrent convective episodes, the first ones always extending at most up to the border of the previous convective core. However, the final episode(s) can extend beyond the border of the former convective core. This occurrence is very important as the maximum extension of this convective region fixes the location of the boundary between the part of the star that has experienced a strong neutronization process (inside the boundary) and the one that has not been affected significantly (if any) by this process (outside the boundary). This boundary determines the mass of the Iron core: this is a fundamental quantity for the subsequent explosive phase, determining the mass within which the shockwave loses the largest part of its energy through the photodisintegration of heavy elements.

Nuclear Statistical Equilibrium (NSE)

At the end of Si burning, all nuclear reactions between different elements become fully balanced by their inverses. The last reaction to achieve equilibrium is the 3α reaction, which finally occurs at a rate balancing that of Carbon photo-disintegration. When this occurs, the stellar material has achieved the *Nuclear Statistical Equilibrium* (NSE).

Classical Type II SNe: Core Collapse

The collapse of the Iron core can be understood considering the Virial theorem (40): A stable physical configuration is allowed until the specific heat ratio is larger than $4/3$. In the core of a massive star after reaching NSE, both photodisintegration processes and relativistic effects decrease the value of γ below the critical value $4/3$, and as a consequence the core becomes dynamically unstable and begins to collapse.

At temperatures of about $T \sim 8 \cdot 10^9$ K and densities $\rho \sim 4 \cdot 10^9\,\mathrm{g\,cm^{-3}}$ the photodisintegration of Iron occurs, inducing a large production of free neutrons. With the same mechanisms described at the end of Chapter 4, neutronic captures of the r-type begin to occur, with the consequent production of elements heavier than Iron. Perhaps a little bit surprisingly, it is not the photodisintegration of Iron that induces the collapse.

At the beginning, the core collapse is not a dramatic event. It occurs on a Kelvin-Helmholtz timescale due to the huge neutrino flux which carries away the core binding energy. However, soon after, the occurrence of two instabilities causes the collapse to accelerate strongly: (1) due to the increase in density, the neutronization process (electronic captures on nuclei) becomes increasingly efficient, this process removing free electrons which are the main contributors to the pressure

$$e^- + (Z, A) \rightarrow (Z-1, A) + \nu_e$$

and (2) photo-disintegration becomes more efficient, producing a large number of free α particles. The binding energy of the core resulting from this new composition, is lower than before, and so the star cannot gain enough energy from its contraction. As a consequence, the collapse accelerates.

The internal regions were entirely supported by the electrons degeneracy pressure so, when the electron density drops because of electronic captures, the supporting pressure decreases dramatically. At a certain point, when the core crosses a critical density threshold, the EoS changes. From degenerate electrons we have to start considering degenerate neutrons, which have a much less steeper density dependence

$$p_n \sim n_n^{2.5}$$

The main consequence of the bulk neutralization of the core is that the material becomes incompressible and the collapse of the central regions stops. As the compression phase ends abruptly, the core rebounds and the (supersonic) impact with the external layers that are continuing to contract produces a shockwave. In a perfectly elastic collision, the energy of the infalling outer portion of the core would bounce back after reflection to its position before the collapse. It seems possible that the rebounding core could be responsible for the expulsion of the whole stellar envelope, the so-called *prompt explosion scenario*. This scenario, however, is unlikely to be realistic.

As the shock wave moves through the infalling layers, it heats them up and induces photodisintegration; owing to this process the shockwave loses a considerable amount of energy for each $0.1M_\odot$ that is crossed. In addition, the emission of neutrinos from behind the shockwave is a quite efficient shockwave cooling process. For a successful explosion, an additional energy source is required.

It is now commonly assumed that this energy source is provided by the neutrino energy deposition (the neutrino-powered explosion). Due to the high density in the collapsing core, the mean free path of neutrinos becomes comparable to the core radius. This means that the neutrinos produced via the neutronization processes cannot escape from the star without interacting. With increasing density, the material in the collapsing core becomes more and more opaque to neutrinos, which experience multiple scattering processes.

In the inner core, the density is so large that the neutrino diffusion velocity becomes less than the velocity of the collapse. Under these conditions, the neutrinos cannot leave the star, and are thus trapped. The point at which the neutrino diffusion velocity equals the velocity of the collapsing core is defined as the *neutrino-trapping surface*: Below this point neutrinos are forced to deposit their own energy in the material, whereas in the more external layers they can diffuse outwards until they reach the so called *neutrino photosphere*.

This process is considered to be responsible for providing the outgoing shock wave with the energy necessary to produce an explosive event with the characteristics of a core-collapse, an alternative name to Classical Type II Supernova[55].

As the shockwave progresses to the external layers, it decreases their density and, consequently, their opacity, thus allowing the trapped neutrinos to escape. The neutronization process is no longer stalled, and actually resumes explosively. You should also note that the shockwave produced by the rebounding on the core is often energetic enough to nuclearly process external layers, inducing an explosive nucleosynthesis.

Know Your Neutrinos

At this stage in particular, it's important to know where your neutrinos are coming from and what are the main sources of production. As we have briefly mentioned somewhere else, there are five main channels of production of neutrinos in stars

$$
\begin{aligned}
e^- + (Z, A) \text{ or } e^- &\to e^- + (Z, A) \text{ or } e^- + \nu_e + \bar{\nu}_e \quad \text{Bremsstrahlung} \\
\gamma + e^- &\to e^- + \nu_e + \bar{\nu}_e \quad \text{Photoproduction} \\
\gamma &\to e^- + e^- \to +\nu_e + \bar{\nu}_e \quad \text{Pair annihilation} \\
\gamma_{\text{plasmon}} &\to \nu_x + \bar{\nu}_x \quad \text{Neutrinic oscillations} \\
(Z, A) + e^- &\to (Z-1, A) + \nu_e \quad \text{Neutralization}
\end{aligned}
$$

The most curious fact, perhaps, is that in the unimaginably dense conditions of the core, neutrinos start to actually interact with the surrounding matter, turning in a non-negligible souce of opacity. The main channels are listed below

$$
\begin{aligned}
\nu_e + n &\to p^+ + e^- \\
\nu_x + e^- &\to \nu_x + e^- \\
\nu_x + (Z, A) &\to \nu_x + (Z, A)
\end{aligned}
$$

the last one in particular, that is scattering of neutrinos on nuclei, is the major source of neutrinos opacity. Moreover, note that under this set of extreme conditions, neutrinos are actually forced to thermalize with the surrounding matter. Also, note that at the temperatures and densities we're considering, the aforementioned reactions go both the ways around, meaning that neutronization of the core might be stalled by the activation of the inverse reaction.

Roughly speaking, it is commonly estimated that stellar progenitors less massive than $\sim 25 M_\odot$ produce as a remnant a neutron star whose mass

55 The details of the shock propagation inside the structure are very complicated, due to the interplay between the neutrino energy flux and convection as well as the occurrence of mixing instabilities such as the Rayleigh-Taylor instability. It actually turns out that without considering a whole lot of other contributions, such as those mentioned above, the shockwave never exits the core.

should be about that of the Iron core at the beginning of the core collapse. More massive stars should produce black holes.

If the collapsing star produces a NS as a remnant, most of the gravitational energy is depleted as neutrinos. On average, the gravitational energy that is lost during the formation of a NS is

$$U_G \approx 3 \cdot 10^{53} \left[\left(\frac{M}{M_\odot} \right)^2 \left(\frac{R}{10\,\mathrm{km}} \right) \right] \mathrm{erg}$$

resulting in an average neutrinos' luminosity given by

$$L_\nu \approx \frac{U_G}{t_{\mathrm{diff}}} \sim 3 \cdot 10^{52}\,\mathrm{erg\,s^{-1}}$$

with t_{diff} the diffusion time of neutrinos through the structure. This also allows us to calculate the expected flux of neutrinos (of all kinds) during the formation of a NS. In fact, at NS-typical densities $\rho \gtrsim 10^{11} - 10^{12}\,\mathrm{g\,cm^{-3}}$, neutrinos have an average energy of $\langle E_\nu \rangle \approx 10\,\mathrm{MeV}$, meaning that

$$\frac{\mathrm{d}N_\nu}{\mathrm{d}t} \approx \frac{L_\nu}{\langle E_\nu \rangle} \sim 10^{57} \nu/\mathrm{s}$$

Note that the energy that is radiated away by photons is unfathomably lower, only about $L_\gamma \sim 10^{43} - 10^{44}\,\mathrm{erg\,s^{-1}}$, which is often higher than the average luminosity of the galaxy hosting the SN.

Let's crank up that mass a bit now, shall we?

Pair Instability SNe

For masses in the range $70 - 100 \lesssim M/M_\odot \lesssim 140$ can experience a different explosive process, one sustained by nuclear burning. After He-burning, due to the peculiar thermal condition (read: they're *really* hot), the production of electron-positron pairs can occur. This process significantly reduces the value of the specific heat ratio below the critical threshold for stability and a runaway collapse develops. The advanced burning processes described previously are not able to halt the collapse which has become dynamic. The final fate of these *pair-instability supernovae* is to experience a prompt explosion and to produce a black hole[56].

Pair instability SNe are actually divided into two cateogories, depending on the range of mass you're considering.

For stars in the range $70 - 100 \lesssim M/M_\odot \lesssim 140$, the star experiences *pulsational pair instabilities*. As a result of pair production, the radiation pressure supporting the core drops. As a consequence, the core contracts and heats up, inducing an explosive burning of Oxygen and, eventually, of Silicon. During this explosive burning, the star loses much of its envelope, but manages to settle into a stable configuration. This process might repeat many times, until Oxygen is exhausted and the star rapidly progresses up to the end of the Silicon burning and the formation of the Iron core. The star

56 It might be worth emphasizing that this kind of supernovae has been receiving remarkable attention in recent times, as it is a common belief that super-massive stars could be the major component of the primeval stellar population, the Population III.

then meets a similar end to that of its less massive counterparts. The star explodes[57] and leaves behind a BH.

For even higher masses[58], in the range $140 \lesssim M/M_\odot \lesssim 260$, stars experience *classical pair instability*. No remnants are left behind, and the whole stellar structure is destroyed by the explosive ignition of Oxygen.

Finally, for masses $M \gtrsim 260 M_\odot$ (assuming that they had actually existed), the star still experiences a strong pair production, but the core is so hot that it actually photodisintegrates itself. The star then understandably explodes. This kind of supernova would be called *photodisintegration supernova*.

Nucleosynthesis and SN Classification

At this point we might wonder, are Type II SNe enough to produce all elements heavier than Iron? The answer is that unfortunately they cannot. They're able to produce most of them, but never manage to produce elements like Gold and Silver, because the collapsing cores aren't really that neutron-rich as we might think.

The most probable source of those heavy elements is actually mergers of neutrons stars, like that, for example, detected by the LVK collaboration in 2017: GW170817 [1].

Let's take a quick look at the SNe classification. For the sake of simplicity and personal laziness, take a look at Fig.51.

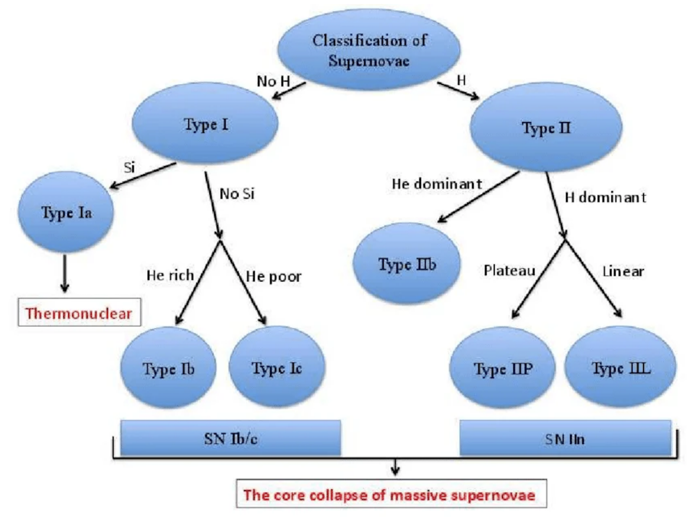

Figure 51: Supernova classification based on the presence of given elements features in the spectrum. Credits: [10].

Among Type I SNe, Type Ib and Type Ic are spiritually more similar to Type II SNe than Type Ia, but fell into the Type I category because they are produced by massive stars that lose all the H-enveloped through mass losses and superwinds and perhaps even the He-layers.

When the luminosity of a Type II supernova is plotted over a period of time, it shows a characteristic rise to a peak brightness followed by a decline.

57 During the explosion, the interaction with the envelope matter that had been expelled during the previous pulses produces a very bright light curve.

58 Note that this doesn't hold for every metallicity range. At solar metallicity $Z_\odot$, for example, only pulsational pair instability would be possible.

These light curves have an average decay rate much lower than the decay rate for Type Ia supernovae. Type II is furtherly divided into two classes, depending on the shape of the light curve. The light curve for a Type IIL supernova shows a steady (linear) decline following the peak brightness. By contrast, the light curve of a Type IIP supernova has a distinctive flat stretch (called a plateau) during the decline; representing a period where the luminosity decays at a slower rate.

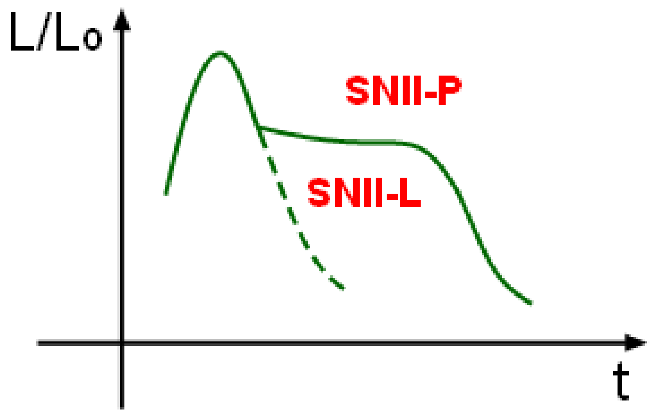

Figure 52: Morphological difference in the lightcurve between a Type IIL and a Type IIP SN. Credits: Wikipedia.

The decay the powers and characterizes the light curves is that of ${}^{56}_{28}\mathrm{Ni}$ and successive chains

$$
\begin{aligned}
&{}^{56}_{28}\mathrm{Ni} + e^- \rightarrow {}^{56}_{27}\,\mathrm{Co} + \gamma + \nu_e \quad (\tau_{1/2} \sim 6\,\mathrm{days}) \\
&{}^{56}_{27}\mathrm{Co} + e^- \rightarrow {}^{56}_{26}\,\mathrm{Fe} + \gamma + \nu_e \quad (\Gamma \sim 81\%,\ \tau_{1/2} \sim 77\,\mathrm{days})
\end{aligned}
$$

5.10 WHITE DWARVES

As discussed in the previous sections, stars with initial masses up to[59] $M \sim 6-8M_\odot$ lose all their envelope during the thermal pulses and continue the evolution with an electron degenerate core made almost exclusively of Carbon and Oxygen (whose stratification is determined by the nuclear transformations of the previous evolutionary phase) plus a residual non-degenerate envelope, with thickness of about $10^{-2}M_{\mathrm{tot}}$, where M_{tot} is the total mass of the white dwarf (WD). No nuclear energy processes are active, since temperatures are not high enough, and the stars evolve at approximately constant radius and decreasing luminosity.

As a first approximation, we may assume them to be entirely supported by the pressure of non-relativistically degenerate electrons, so that $p \propto \rho^{5/3}$.

59 The precise value depends on the initial chemical composition and the extension of the overshoot from the convective cores during the previous evolutionary phases.

From the Lane-Emden equation (48) we could find in principle a mass-radius relationship

$$R = 0.012M^{-1/3}\left(\frac{\mu_e}{2}\right)^{-5/3} \tag{118}$$

with M and R in terms of the correspondent solar unity. For the WD we're going to consider, we may assume $\mu_e \sim 2$.

Your typical WD, as we've mentioned, is that which ends up with a CO-core, as well with a thin shell of Helium ($\sim 1\% M_\odot$) and a residual of the H-rich envelope ($\sim 10^{-4} M_\odot$). Once the WD is formed, since no nuclear burning are active, it will start to cool down with approximately the same radius, since the only part that can effectively contract is the ionic (non-degenerate) component, which is minoritary.

Before moving on, let us just briefly point out that for WD the density scales with the mass squared

$$\rho \propto M^2$$

which can potentially be a problem with our initial approximation of a non-relativistic degenerate gas. In the real world, in fact, our approximation would hold only up to $\sim 0.6M_\odot$ (which, incidentally, is about the same mass of your average CO-WD).

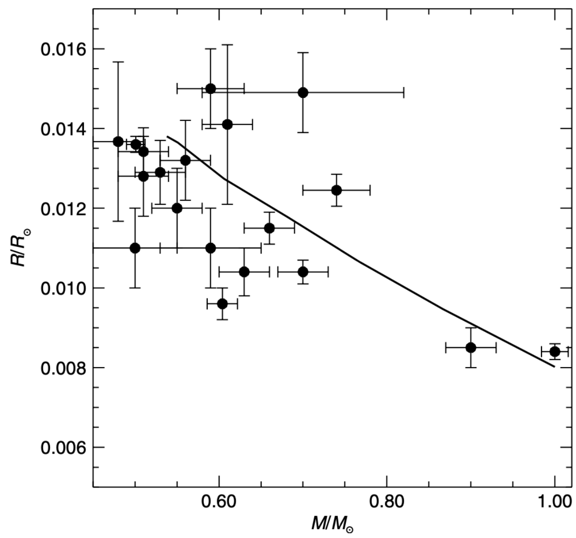

Figure 53: Comparison of the theoretical mass-radius relationship predicted by theoretical models with observations. With more recent data (perhaps those of the GAIA DR3) you could probably do better than this. But still- Credits: [21].

For larger masses[60], the stellar structure is partially supported by relativistic degenerate pressure and so we cannot use the easy politropic expression $p \propto \rho^{5/3}$, nor that for a fully relativistic gas $p \propto \rho^{4/3}$. An EoS able to describe this mixed configuration was found *analytically* by that madlad that was Chandrasekhar, but we won't talk about that. Just that in case of degen-

60 This is the case, for example, of Ne-O white dwarves.

erate gass which is not fully relativistic, the following approximate relation holds

$$R \approx 0.0012 R_{\odot} \left[\left(\frac{M_{Ch}}{M} \right)^{2/3} - \left(\frac{M}{M_{Ch}} \right)^{2/3} \right]^{-1/2} \tag{119}$$

Due to our imperfect knowledge of the mass-loss processes during the AGB phase (and, more generally, along all the various evolutionary phases) we cannot exactly predict the final WD mass produced by a progenitor with a given initial total mass M. In general, the CO core mass at the beginning of the late-AGB phase is a reasonable approximation since you do not expect a large increase of the core during the thermal pulses.

Still, even with its relatively low mass, the core has a very large thermal capacity, which continuously heats up the surrounding envelope. The transferred energy is then radiated away, inducing a cooling in the star.

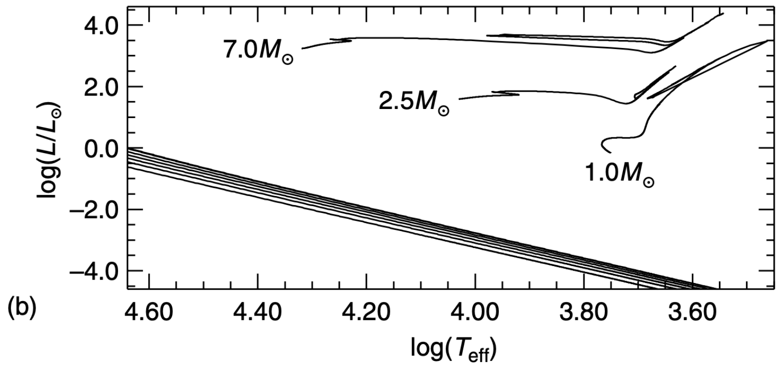

Figure 54: HRD location of WD evolutionary tracks for different values of the white dwarf's mass compared with previous evolutionary phases. Credits: [21].

The cooling occurs in regions of the HR-diagram that are at constant radius, and so WD are expected to align diagonally as shown in Fig.54.

These general properties of WDs are also valid when one takes into account the existence of the thin non-degenerate envelope. Different thicknesses of the envelope change the mass-radius relationship slightly at a given WD mass, but the effect is largely a second-order effect. Instead, when studying the evolution with time of the WD luminosities, the envelope plays a fundamental role, as we will see below.

According to the virial theorem, a star with a degenerate electron component and ideal ions radiates the thermal energy of the ions and lowers its temperature. The evolution of a WD is therefore often labelled as a cooling process, and we are now going to obtain an approximate relationship for the evolution of the WD luminosity with time.

But before moving on, let us briefly recall how the various WD are formed.

- CO-WD: are those stars with original mass larger than $M_{\rm UP}$ that never manage to ignite Carbon (and consequently, neither Oxygen). They have masses roughly equal to $\langle M \rangle \sim 0.55 - 0.6 M_{\odot}$;

- Ne-O-WD: as per the above, but with the difference that these stars ignite Carbon but not Oxygen. Note that only stars that ignite Carbon very far off-center are able to turn into Ne-O WDs. At this stage they're composed of 65% Oxygen and 25% Neon. Ne-O WD have a higher molecular weight in the envelope $\mu_i \uparrow$, which implies a lower density of ions $n_i \downarrow$ and then a lower energy to radiate away $E_{\text{th}} \downarrow$, even though they have larger masses overall. They have masses typically in the range $1 \lesssim M/M_\odot \lesssim 1.2$;

- He-WD: they are what remains of massive stars from RGB that experienced too-severe mass losses and never made it to Helium ignition. As so, they have very low masses, in the range $0.15 \lesssim M/M_\odot \lesssim 0.5$, and since they're molecular weight is lower, their thermal energy is higher. These class of WDs populate the tail of the MS for low mass stars.

5.10.1 WD Age-Luminosity Curve

We consider first the non-degenerate envelope and suppose it is radiative, so that the energy of the envelope is given by

$$E_{\text{th}} = \frac{3}{2}kTN_i \approx \frac{3}{2}kT\frac{M}{\mu_i m_{\text{H}}} \tag{120}$$

and that the opacity in the envelope is à la Kramers, so $\bar{\kappa} = \kappa_0 \rho T^{-3.5}$. We're going to assume that in the envelope the total mass and luminosity have already been achieved, and that the gas there follows the perfect gas EoS. If we divide the equation of hydrostatic equilibrium (with r as independent variable) by the equation of radiative transport we obtain

$$p\mathrm{d}p = \frac{4ac}{3}\frac{4\pi}{\kappa_0}\frac{GMk}{L\mu m_{\text{H}}}T^{7.5}\mathrm{d}T \tag{121}$$

This equation can easily be integrated from the surface to a generic point within the envelope using as surface boundary conditions $p \sim 0$ and $T \sim 0$, and assuming the luminosity L is constant throughout the envelope and equal to the surface value. The integration provides p as a function of T plus constants. After using the ideal gas EoS to replace p with ρ we finally obtain

$$\rho = \left(\frac{32\pi acGM\mu m_{\text{H}}}{8.5 \cdot 3\kappa_0 kL}\right)^{1/2} T^{3.25} \tag{122}$$

We consider the base of the envelope, i.e. the transition point to the electron degenerate interior, to be the point where the degenerate electron pressure equals the pressure of the ideal electron gas in the envelope (ions are an ideal gas at both sides of this transition point). This means that at the interface (the values of density and temperature at this point are denoted with the subscript b, μ_e is the electron molecular weight of the degenerate core)

$$\frac{\rho_b kT_b}{\mu_e m_{\text{H}}} = 1.0036 \cdot 10^{13}\left(\frac{\rho_b}{\mu_e}\right)^{5/3} \tag{123}$$

which we can use to solve for the luminosity in the previous equation, thus finding

$$\frac{L}{L_\odot} = 6.4 \cdot 10^{-3}\frac{\mu}{\mu_e^2}\frac{M}{M_\odot}\frac{1}{\kappa_0}T_b^{3.5} \tag{124}$$

This temperature T_b at the transition from degenerate core to envelope is approximately equal to the temperature at the center. In fact, due to the very low opacity of degenerate electrons the core is practically isothermal.

These simple properties of WD stars allow you to determine a good approximation of their cooling law easily. The virial theorem applied to degenerate objects tells us that the energy radiated away by a WD is equal to the rate of decrease of the thermal energy of its ions (we neglect here any contribution from the non-degenerate envelope). But the luminosity is

$$L = -\frac{\mathrm{d}E_{\mathrm{th}}}{\mathrm{d}t} \quad L \propto MT^{7/2} \implies \alpha T^{7/2} = \frac{\mathrm{d}T}{\mathrm{d}t}\left(\frac{3k}{2\mu_i m_{\mathrm{H}}}\right)$$

which we can integrate right away substituing the value of E_{th} we've obtained earlier. The integration of this equation between an initial time t_0 (when the WD stage starts) and a generic time t provides

$$t - t_0 = \frac{3k}{5\alpha\mu_i m_{\mathrm{H}}}(T^{-5/2} - T_0^{-5/2}) \tag{125}$$

where T_0 is the temperature of the core at t_0 and T the temperature at time t. When the WD has cooled down significantly, $T_0 \gg T$ and therefore

$$\Delta t = \frac{3k}{5\alpha\mu_i m_{\mathrm{H}}} T^{-5/2} \tag{126}$$

Inverting the luminosity-temperature relation (124) and plugging in the temperature as a function of the luminosity in the latter equation, we obtain

$$\Delta t \propto \left(\frac{L}{M}\right)^{-5/7} \approx \frac{4.5\cdot 10^7}{\mu_i}\left(\frac{L}{L_\odot}\frac{M_\odot}{M}\right)^{-5/7} \tag{127}$$

This approximated cooling law is called *Mestel law* and shows, for example, that higher WD masses evolve more slowly; this is easy to understand in terms of more ionic thermal energy stored because of the higher mass. Also, increasing the ionic molecular weight at a given M decreases the evolutionary times, since there are less ions in the star. This simple law predicts cooling ages of the order of $\sim 1\,\mathrm{Gyr}$ when $L \approx 0.001 L_\odot$.

Although the Mestel law is a good zero-order approximation of the real WD cooling law, complete models are more complicated. For once, we should in principle account for (a) corrections to the degenerate electron EoS since the temperature is $T \neq 0$, (b) what is it of the ions all-throughout the cooling.

To derive the Mestel law, ions have been approximated with an ideal gas with $c_V = 3k/2$ per particle. The specific heat c_V is obviously a crucial quantity for the determination of the cooling time, and we will now see that the simple approximation of the ideal gas is not good when the core cools down. Due to the steady decrease of the temperature, the ions in the core tend to move less freely because the Coulomb interactions play an increasingly major role in determining their thermal properties. For this purpose, it's useful to introduce the quantity

$$\Gamma = \frac{kZ^2e^2}{dkT} \quad d = \frac{1}{\left(\frac{4\pi}{3}n_i\right)^{1/3}} \tag{128}$$

where d_i is the average distance between the ions. During the cooling, the core gets cooler whereas the density is almost constant, hence Γ increases. When Γ becomes larger than 1 the ions tend to behave like a liquid, and with still decreasing temperatures, approaching $\Gamma \sim 180$, they form a periodic lattice structure that minimizes their total energy. This latter phenomenon is called crystallization, and when $\Gamma > 180$ ions behave like a solid.

The major change to our easy toy-model is in the specific heat. In fact, if temperature drops under the Debye temperature

$$\Theta_D \propto 4 \cdot 10^3 \rho^{1/2}\,\mathrm{K} \tag{129}$$

quantum corrections to the specific heat become very relevant.

In full quantistic regime $C_V \propto T^3$. There are, however, two other major effects. The first one is the release of latent heat during the crystallization process, which amounts to roughly $\sim kT$ per ion. This means that during the core crystallization extra energy is released, first at the centre and then at progressively more external core layers.

This latent heat (that in a full stellar evolution computation is included as an additional term to the energy generation coefficient ϵ) slows down the cooling of the WD with respect to the Mestel law, given that it is an extra energy contribution to the ionic energy budget.

There is an additional energy release during the crystallization process, due to the phenomenon of *chemical separation*. The physical principle behind this phenomenon is that, given a CO binary mixture with a given abundance ratio X_C/X_O in the gas and liquid phase, at crystallization the same abundance ratio cannot be maintained, because the two elements are not fully miscible in the solid phase. And so an instability due to the chemical gradients arises, and the whole region becomes convective.

If you recall, some time ago, we've given an expression for the production rate of gravitational energy[61]

$$\epsilon_G = -T \left[\partial_p S\Big|_{T,\mu} \dot{p} + \partial_T S\Big|_{p,\mu} \dot{T} + \partial_\mu S\Big|_{T,p} \dot{\mu} \right] \tag{130}$$

the contribution we're interested here is exactly the last one, which here is non-negligible. Clearly, thanks to these two new mechanisms, the cooling times of WD might be sensibly different from those predicted by (127), since in both cases we introduce either modifications to the energy that can be released (specific heat via crystallization) or to the rate that the energy is radiated away[62] (gravitational energy production rate via chemical gradient instability).

On a conclusive note, please notice that because of diffusive mechanisms and gravitational sedimentation, we're usually unable to see metallic lines in the atmosphere of WDs. Depending on the WD, even Helium might be subject to sedimentation. Three classes of WD are usually introduced:

- If Hydrogen lines are present in the atmosphere, the WD is said to be a DA type (but for a whole series of reason, it could be that its type might change);

---

61 Since I'm not a monster, I'll spare you the trouble of looking for it.

62 Actually, we should also keep in mind that in the more inner regions, there might be a non-negligible production of plasma-neutrinos, introducing yet another channel for energy dissipation.

- If the spectrum is Helium-dominated, the WD is said to be a DB type;
- If the spectrum is *approximately* a continuum, the WD is said to be DC type.

At this point, I believe I can put down my imaginary pen and let my wonderful four readers find out the rest[63].

The End?

63 Just kidding. There are still plenty of things that would be worth at least mentioning, but the course of Stellar Physics ends here, with 62 footnotes and a little bit shy of 200 pages.

Part III

APPENDIX

6 APPENDIX A: RADIATIVE TRANSFER

Most of our knowledge about the Universe is based on the electromagnetic radiation that reaches us from far far away. EM radiation is obviusly not the only way we can probe the Universe we live in but, in respect to neutrinos, cosmic rays or even gravitational waves, it's not a long stretch to claim it is by far the most understood.

It is most important then that an astrophysicist worthy of his (or her) name has a good grasp of the theory of radiative transfer and of its applications.

Apart from a few more key differences, I'll follow the description of radiative transfer of [8] and [20], but I won't fail to emphasize whenever I'll be doing otherwise.

Radiative transfer equation

In the presence of matter, it is not immediately obvious what changes may occur in the specific intensity as we move along a ray path. The aim of this section will be to eviscerate the matter.

Let's consider the following geometric construction

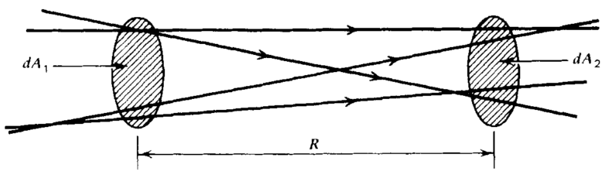

Figure 55: Geometrical construction for ray paths propagating in empty space. Credits: G. Rybicki, A. Lightman.

It won't take a lot of effort to convince yourself that in empty space the monochromatic intensity I_ν is actually conserved. In fact, from simply writing down the definitions and imposing the conservation of energy

$$I_{\nu_2}\,\mathrm{d}A_2\,\mathrm{d}t\,\mathrm{d}\Omega_2\,\mathrm{d}\nu = I_{\nu_1}\,\mathrm{d}A_1\,\mathrm{d}t\,\mathrm{d}\Omega_1\,\mathrm{d}\nu$$

the conclusion follows observing that $\mathrm{d}A_2\,\mathrm{d}\Omega_2 = \mathrm{d}A_1\,\mathrm{d}\Omega_1$.

If we consider an affine parameter of the form $\vec{x} = \vec{x}_0 + \hat{k}s$, we may as well write the previous results in a more familiar fashion

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = 0 \implies (\hat{k}\cdot\nabla)I_\nu = 0 \tag{131}$$

What changes if matter is present along the ray path? Clearly it will no longer be true that $(\hat{k}\cdot\nabla)I_\nu = 0$, but we're not that far off. All that we need is some little work on both terms.

How the right hand side of the equation should change is obvious: It needs to keep track of the "creation" and "destruction" of photons in the considered volume of spacetime.

The left hand side of the equation requires a little more care. Consider infinitesimal time and space displacements along the ray path, respectively $\mathrm{d}t$ and $\mathrm{d}\vec{x}$

$$\Delta E_\nu \, \mathrm{d}\nu = \left(I_\nu(\vec{x} + \mathrm{d}\vec{x}, t + \mathrm{d}t, \hat{k}) - I_\nu(\vec{x}, t, \hat{k}) \right) \mathrm{d}t \, \mathrm{d}\Omega \, \mathrm{d}A \, \mathrm{d}\nu$$

Taking a first order expansion in respect to the affine parameter s along the ray path yields

$$\left(\frac{1}{c} \partial_t I_\nu + \partial_s I_\nu \right) \mathrm{d}t \, \mathrm{d}s \, \mathrm{d}\Omega \, \mathrm{d}A \, \mathrm{d}\nu = \text{photon addition} - \text{photon removal}$$

This equation is a generalization of eq.131 for non-stationary radiative transport and in the presence of matter. It's about time we get to know what "lives" in the right hand side of the equation.

Monochromatic emission coefficient

For the moment, we'll define the *spontaneous* monochromatic emission coefficient j_ν as

$$\mathrm{d}E_\nu \, \mathrm{d}\nu = j_\nu \, \mathrm{d}V \, \mathrm{d}t \, \mathrm{d}\Omega \, \mathrm{d}\nu \tag{132}$$

which in general has a non-zero dependence on the emission direction. Sometimes the spontaneous emission coefficient is defined by the *emissivity* ϵ_ν (**please note** that often the two names are used almost interchangeably), which is the energy emitted spontaneously per unit frequency per unit time per unit mass

$$j_\nu = \frac{\epsilon_\nu \rho}{4\pi}$$

where ρ is the mass density of the emitting medium.

If we perform the decomposition $\mathrm{d}V = \mathrm{d}A \, \mathrm{d}s$, the contribution of spontaneous emission to the specific intensity is

$$\mathrm{d}I_\nu = j_\nu \, \mathrm{d}s$$

Absorption coefficient

Similarly, we can consider the energy that is absorbed from the radiation when passing through a medium. There exists various definitions; I'll use the one we gave in class and that is incidentally the one used in [8] and [20] as well.

We define the *absorption coefficient* α_ν through the following expression

$$\mathrm{d}I_\nu = -\alpha_\nu I_\nu \, \mathrm{d}s \tag{133}$$

If we use a microscopic model, then the absorption coefficient can be understood as particles with numeric density n presenting an effective absorbing area, the *cross section* σ_ν. The coefficient α_ν can thus be rewritten in terms of

$$\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$$

where κ_ν is called the mass absorption coefficient or the *mass-weighed opacity coefficient*.

Note that in eq.133, we consider "absorption" to include both "true absorption" and stimulated emission, because both are proportional to the intensity of the incoming beam. Depending on the entity of the contribution, the α_ν coefficient may be positive or even negative, giving raise to curious phenomena.

Making full use of what we've just defined, we can finally present the celebrated *equation of radiative transfer* (although in the notable absence of scattering)

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} = -\alpha_\nu I_\nu + j_\nu \tag{134}$$

which is actually fairly easy to solve when one of the two coefficients vanishes.

Emission only

We set $\alpha_\nu = 0$ and the equation may be solved by direct integration

$$I_\nu(s) = I_\nu(s_0) + \int_{s_0}^{s} j_\nu(s')\,\mathrm{d}s'$$

the result is not that interesting per se.

Absorption only

This time we set $j_\nu = 0$. The equation is easily solved this time as well

$$I_\nu(s) = I_\nu(s_0)\exp\left(-\int_{s_0}^{s} \alpha_\nu(s')\,\mathrm{d}s'\right)$$

In this case, it's rather common to write down the equation in terms of a new variable, the *optical depth* τ_ν

$$\mathrm{d}\tau_\nu = \alpha_\nu\,\mathrm{d}s \tag{135}$$

Given this definition we'll say that if

- $\tau_\nu \gg 1$: the medium is *optically thick or opaque*
- $\tau_\nu \ll 1$: the medium is *optically thin or transparent*

This has some crucial implications we'll be going through in a moment.

In the stationary limit, the equation of radiative transport may be written as

$$(\hat{k}\cdot\nabla)I_\nu(\hat{k},\vec{x}) = j_\nu(\vec{x}) - \alpha_\nu(\vec{x})I_\nu(\hat{k},\vec{x})$$

In terms of the *source function* $S_\nu = j_\nu/\alpha_\nu$ it becomes

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}\tau_\nu} = -I_\nu + S_\nu \tag{136}$$

which can be integrated to yield the formal solution

$$I_\nu(\hat{k},\tau_\nu) = I_\nu(\tau_{\nu,0})\exp(-\tau_\nu) + \int_{\tau_{\nu,0}}^{\tau_\nu} d\tau'_\nu\, S_\nu \exp(-(\tau_\nu - \tau'_\nu))$$

Assume for the moment that the matter through which radiation is passing has constant properties and has no background source. Then the source function S_ν is constant and the formal equation becomes

$$I_\nu = S_\nu(1 - e^{-\tau_\nu})$$

If the medium and is optically thin, then the equation is reduced to

$$I_\nu = S_\nu \tau_\nu = j_\nu L \tag{137}$$

by taking the Taylor expansion of the exponential term and calling L some typical length of the medium.

If, on the other hand, the medium is optically thick, we can neglect the exponential $e^{-\tau_\nu}$ to obtain

$$I_\nu = S_\nu \tag{138}$$

KIRCHHOFF'S LAW AND LTE

The most notable implication of eq.138 is if we consider the specific intensity coming out of a small hole on a box kept in thermodynamic equilibrium. We know that what's going to come out of there is the blackbody radiation

$$I_\nu = B_\nu(T)$$

but what if we were to put an optically thick object just behind the hole?

If the object is in thermodynamic equilibrium with the surroundings (and it *will* be, given an appropriate amount of time), then the radiation coming out of the hole will still be blackbody radiation. But eq.138 tells us that the source function will tend to be equal to the specific intensity, hence

$$S_\nu = B_\nu(T) \tag{139}$$

which actually puts a constraint on the possible values of the emission coefficient in terms of the absorption coefficient. This is exactly what is expressed in Kirchhoff's law

$$j_\nu = \alpha_\nu B_\nu \tag{140}$$

Let us briefly consider what we have just derived. Matter often tends to emit and absorb at specific frequencies corresponding to what are commonly called *spectral lines*. We would expect then both j_ν and α_ν to have peaks (or depression) around these lines. But Kirchhoff's law forces their ratio to be equal to a smooth blackbody profile.

Thus we can expect to observe two very different scenarios if the medium is optically thin rather than optically thick. In the former, the radiation emerging from the medium is essentially determined by its emission coefficient; since j_ν is expected to present peaks, so will the radiation spectrum, which will be appear in spectral lines, as shown in Fig.56 and Fig.57.

On the other hand, the intensity coming out of an optically thick body is its source function, which must be equal to the blackbody function. Hence we expect the medium to emit in a continuum, like a blackbody.

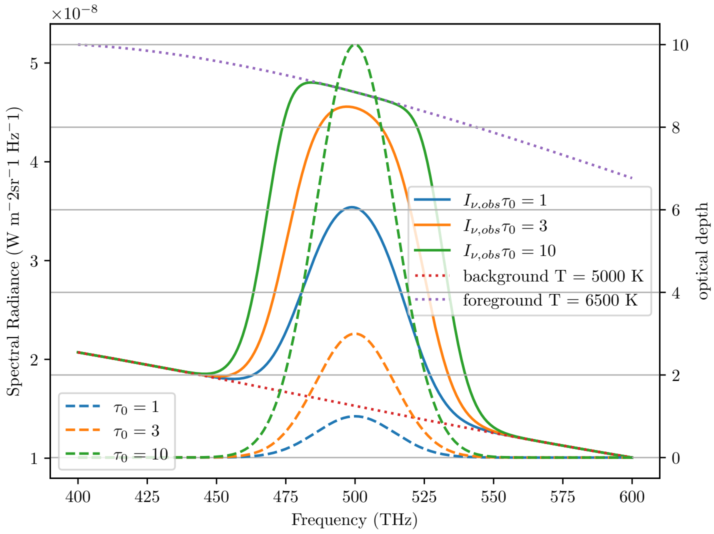

Figure 56: An example of emission features formation for different temperatures and different values of τ. Credits: Prof. Walter del Pozzo.

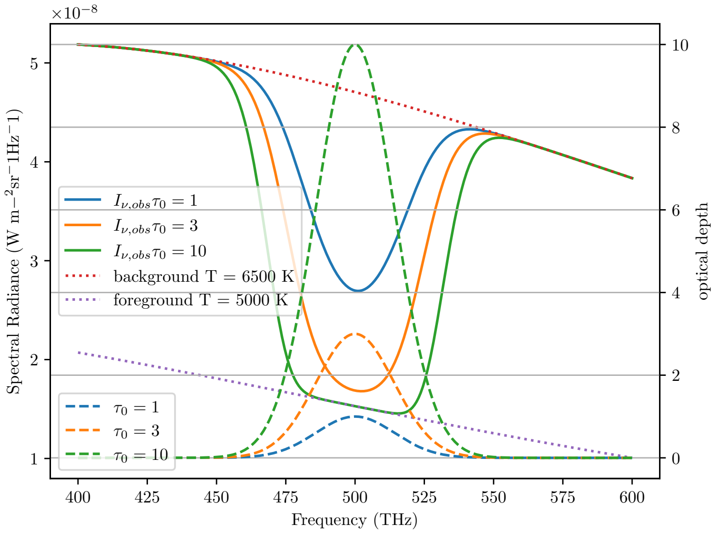

Figure 57: An example of absorption features formation for different temperatures and different values of τ. Credits: Prof. Walter del Pozzo.

All throughout this description, we've been assuming the medium to have constant properties, which has the perk of being a good approximation for many objects of interest, but still turns out to be a really poor one for many other objects. Stars, for example.

Ingenuously, we may expect stars to emit radiation like blackbodies, but they're not. Actually, stars present absorption lines, many, even, depending on the class of star. What we cannot assume in stars is them having constant properties, starting from temperature.

In fact, we could take a guess and claim that stars are in *strict* thermodynamic equilibrium. It would be a very bad guess indeed.

Local Thermodynamic Equilibrium (LTE)

Let's be honest: In a realistic situation, we *rarely* have strict thermodynamic equilibrium. If a body is in thermodynamic equilibrium, we can assume a number of important physical principles to hold, like the Maxwellian distribution

$$dn_v = 4\pi n \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right) dv \qquad (141)$$

where n is the total number of particles per unit volume and m is the mass of each particle. Similarly, we can expect certain laws to hold, like Boltzmann's law for occupation numbers

$$\frac{n_E}{n_0} = \frac{g_E}{g_0} \exp\left(-\frac{E - E_0}{kT}\right) \qquad (142)$$

and Saha's equation

$$\frac{N_{j+1} n_e}{N_j} = 2\frac{Z_{j+1}(T)}{Z_j(T)} \left(\frac{2\pi mkT}{h^2}\right)^{3/2} \exp\left(-\frac{\chi_{j,j+1}}{kT}\right) \qquad (143)$$

where n_e is the density of electrons and $\chi_{j,j+1}$ is the ionization potential. Saha's equation in particular is expected to be crucial in interpreting the effect that ionization has on the emission/absorption spectrum.

The proverbial "one-million-dollar-question" then is: When can we expect a system to be in thermodynamic equilibrium and when can we expect the previous principles to hold?

Even if the system initially does not obey the, say, Maxwellian distribution, it will eventually relax to it after undergoing some *collisions*.

Collisions are crucial in establishing thermodynamic equilibrium.

When collisions are frequent, the mean free path of particles will be small, and particles will interact more effectively. When this happens, we can expect the principles aforementioned to hold. Since we're physicists, vague sentences like "*the mean free path of particles will be small*" are destined to elicit a deep sense of unease and distress. How small does the free path have to be? One meter? Two micrometers? Below the Planck lengthscale?

When we've defined the absorption coefficient α_ν, the sharpest among my four readers total may have noticed that α_ν has the dimension of the inverse of a length. It is safe to assume that α_ν^{-1} may define some distance over which a significant fraction of the radiation would get absorbed by matter.

Such a "mean-distance" is defined in a homogeneous medium as

$$< \tau_\nu >= \alpha_\nu l_\nu = 1$$

Thus, if l_ν is sufficiently small such that the temperature can be taken as a constant over such distance, we can safely say that the useful relations we have defined earlier still hold, although only locally.

In such a fortunate scenario, known as *Local Thermodynamic Equilibrium* (LTE), all the important laws requiring thermodynamic equilibrium are expected to hold, provided that we use the local temperature $T(\vec{x})$.

In the interiors of stars, for example, LTE will prove to be a very good approximation, that will get progressively worse as we approach the "surface" of the star.

PARALLEL PLANE APPROXIMATION

One useful approximation that may be worthwhile to dedicate some of our time to is the *plane parallel atmosphere*, that will allow us to obtain notable results for describing how radiation travels through, say, the inner regions of the stellar atmosphere.

In the following, we're going to neglect the curvature of the stellar atmosphere and assume the various thermodynamic quantities to be constant over horizontal planes.

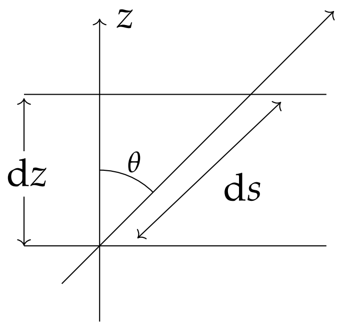

Figure 58: A ray path through a plane parallel atmosphere.

Using Fig.58 as a reference, we see that

$$\mathrm{d}s = \frac{\mathrm{d}z}{\cos\theta} = \frac{\mathrm{d}z}{\mu}$$

where we used the customary notation in astrophysics $\mu = \cos\theta$.

We shall consider a scattering free, stationary situation for the equation of radiative transport (134). Hence, due to planar symmetry, we expect the specific intensity to depend only on z and μ. For the sake of the current discussion, we perform a slight modification to the definition of optical depth, so that

$$\mathrm{d}\tau_\nu = -\alpha_\nu\,\mathrm{d}z$$

This way the equation of radiative transfer may be cast in the following form

$$\mu\partial_{\tau_\nu} I_\nu(\tau_\nu, \mu) = I_\nu - S_\nu$$

which has a formal solution easily computed

$$I_\nu \exp\left(-\frac{t_\nu}{\mu}\right)\Big|_{\tau_{\nu,0}}^{\tau_\nu} = -\int_{\tau_{\nu,0}}^{\tau_\nu} \frac{S_\nu}{\mu}\exp\left(-\frac{t_\nu}{\mu}\right)\mathrm{d}t_\nu \tag{144}$$

This is customarily solved considering two distinct intervals for μ: (I) $\mu \in [0,1]$ and (II) $\mu \in [-1,0]$. In case (I) we can assume the ray path to begin from a great depth inside the star, so that $\tau_{\nu,0} \to \infty$, while in case (II) we assume the ray to receive contributions beginning from the top of the atmosphere, where $\tau_{\nu,0} \approx 0$. For case (II), we're also assuming no radiation to be coming from *outside the star*[1].

Now we can assume LTE throughout the stellar atmosphere so that eq.140 is verified. The source function at some optical depth shall then be equal to

1 Please note that this condition may be not valid at all in close binary systems.

$B_\nu(T(\tau_\nu))$. For the source function at a nearby optical depth we can simply compute a Taylor expansion around the optical depth τ_ν

$$S(t_\nu) = B_\nu(\tau_\nu) - (t_\nu - \tau_\nu)\frac{dB_\nu}{d\tau_\nu} + o(t_\nu^2)$$

We can use this to solve (144), finding for both positive and negative values of μ a very important equation

$$I_\nu(\tau_\nu, \mu) = B_\nu(\tau_\nu) + \mu\frac{dB_\nu}{d\tau_\nu} \tag{145}$$

provided the point considered is sufficiently inside the atmosphere so that $\tau_\nu \gg 1$[2]. Using this simple result, we can compute the three momenta of the equation of transport

$$U_\nu = \frac{4\pi}{c} B_\nu(\tau_\nu) \tag{146}$$

$$F_\nu = \frac{4\pi}{3}\frac{dB_\nu}{d\tau_\nu} \tag{147}$$

$$P_\nu = \frac{4\pi}{3c} B_\nu(\tau_\nu) \tag{148}$$

The Grey Atmosphere

If we consider the absorption coefficient α_ν constant over all frequencies, then the atmosphere is called a "grey atmosphere". This implies that the value of the optical depth at some physical depth is constant for all frequencies. Under this assumption, we could solve

$$\mu\frac{\partial I}{\partial \tau} = I - S$$

I'll skip the explicit calculation (which is actually fairly easy for once) and present just the final result. Two more assumptions are to be made, however: The first is to assume *radiative equilibrium*, which roughly translates into requiring that there are no sources nor sinks of energy in the atmosphere, thus $\partial_\tau F = 0$; as a further semplification, we assume the *Eddington approximation* to hold everywhere in the atmosphere[3], so that

$$P = \frac{1}{3}U$$

It should be evident that this last equation is verified automatically in presence of an isotropic source of radiation, also in its frequency-dependent form.

We then come to the following conclusion

$$I_{obs}(\tau = 0, \mu) = \frac{3F}{4\pi}\left(\mu + \frac{2}{3}\right) \tag{149}$$

2 You can see [8], §2.4.1 for more detailed calculations.

3 I find most intriguing that Eddington's approximation essentially assumes that $T^\mu{}_\mu = 0$, with $T^{\mu\nu}$ the electromagnetic energy-momentum tensor if we are to treat electromagnetic radiation as a perfect fluid.

from which we deduce the equation for the *limb darkening*

$$\frac{I(0,\mu)}{I(0,1)} = \frac{3}{5}\left(\mu + \frac{2}{3}\right)$$

which roughly translates into saying that the radiation that we observe at the surface is the one equivalent at a source function S evaluated at $\tau = 2/3$. This is known as the *Eddington-Barbier estimation*.

A somewhat more general way to solve the problem is assuming the following functional relation for the specific intensity

$$I_\nu(\tau,\mu) = a_\nu(\tau_\nu) + b_\nu(\tau_\nu)\mu$$

and compute the three momenta proper

$$J_\nu = \frac{1}{2}\int_{-1}^{+1} I_\nu \,\mathrm{d}\mu = a_\nu \tag{150}$$

$$H_\nu = \frac{1}{2}\int_{-1}^{+1} I_\nu \mu \,\mathrm{d}\mu = \frac{b_\nu}{3} \tag{151}$$

$$K_\nu = \frac{1}{2}\int_{-1}^{+1} I_\nu \mu^2 \,\mathrm{d}\mu = \frac{a_\nu}{3} \tag{152}$$

Now we assume a stronger version of the *Eddington approximation* $K_\nu = J_\nu/3$ so that the two following expressions can be written

$$\frac{\partial H_\nu}{\partial \tau_\nu} = J_\nu - S_\nu \tag{153}$$

$$\frac{\partial K_\nu}{\partial \tau_\nu} = H_\nu = \frac{1}{3}\frac{\partial J_\nu}{\partial \tau_\nu} \tag{154}$$

The mixing of the two gives us a second order PDE that it's still (approximately) valid even in the outer regions of the stellar atmosphere.

RADIATIVE DIFFUSION APPROXIMATION

This section would probably be clearer if you take a brief detour to Chapter 3 to build some groundwork for scattering processes, but it still suits better the main subject of this chapter.

For the sake of coherence (and personal laziness) I'm going to talk about the radiative diffusion approximation right here, also because it makes use of the plane parallel approximation we just went through.

In Chapter 3 we will use random walk arguments to show that S_ν approaches B_ν at large *effective optical depths* in a homogeneous medium. Real media are seldom homogeneous, but often, as in the interiors of stars, there is a high degree of local homogeneity.

The equation of radiative transport in presence of scatteringz may be cast in a slightly different form

$$I_\nu = S_\nu - \frac{\mu}{\alpha_\nu + \sigma_\nu}\partial_z I_\nu$$

We shall then assume that over a distance l_* (the thermalization length) I_ν is constant and at zero-th order is $I_\nu^{(0)} = S_\nu^{(0)} = B_\nu$. Plugging this in the equation of radiative transfer gives us $I_\nu^{(1)}$ by a simple iterative procedure

$$I_\nu^{(1)} = B_\nu - \frac{\mu}{\alpha_\nu + \sigma_\nu} \partial_z B_\nu \tag{155}$$

With the simple redefining $\mathrm{d}\tau_\nu = -(\alpha_\nu + \sigma_\nu)\,\mathrm{d}z$, we can put eq.(155) in a form functionally equal to (145).

Let us now compute the flux F_ν using the above form for the intensity

$$F_\nu(z) = 2\pi \int_{-1}^{+1} I_\nu^{(1)} \mu \,\mathrm{d}\mu = -\frac{4\pi}{3} \frac{\partial_z B_\nu}{\alpha_\nu + \sigma_\nu} = -\frac{4\pi}{3} \frac{\partial_T B_\nu}{\alpha_\nu + \sigma_\nu} \partial_z T$$

Recalling the result

$$\partial_T \int_0^{+\infty} B_\nu \,\mathrm{d}\nu = \frac{4\sigma_{SB} T^3}{\pi}$$

we can define a *mean absorption coefficient* using the *Rosseland approximation for radiative diffusion*

$$\frac{1}{\alpha_R} := \frac{\int_0^{+\infty} \frac{1}{\alpha_\nu + \sigma_\nu} \partial_T B_\nu \,\mathrm{d}\nu}{\int_0^{+\infty} \partial_T B_\nu \,\mathrm{d}\nu} \tag{156}$$

If we integrate the monochromatic flux over the frequencies and make use of the Rosseland mean, we find a useful expression used in stellar structure models

$$F(z) = -\frac{16\sigma_{SB} T^3}{3\alpha_R} \partial_z T \tag{157}$$

ABOUT ME

Hi there, I'm Giacomo, sometimes known as Cael by a very small niche of people, the author of the notes you just finished reading.

I'm a Batchelor-graduate student at the University of Pisa now in the process of undertaking a major (Master degree) in Astronomy and Astrophysics at the same university. In my (meager) free time I'm an avid reader, gamer, anime-enthusiast as well as an independent author in the very process of publishing his first novel.

For those of you that may be interested I'll leave both the link to the novel's page as well to my instagram profile in the QRcodes right below.

Figure 59: On the left is the link to my instagram profile, on the right the one to my novel's page (Python really has a library for anything, huh?).

Come say hi if you fancy action-fantasy stories with magic, swords, some comedy to garnish and a lot of other stuff :)

At last but not the least, I want to thank all my collegues and fellow students that helped (and hopefully will help me again) during the drafting of this text, pointing out errors and occasions for further expansion.

If you'd like your names to be included and take your fair share of credits, feel free to contact me and I'll add your name right away in the few lines that are left of this page.

I'd like to give a special thank you to S. Grisostomi, which pointed out the sudden appearance of a whole McHelium Scottish clan and for pointing out the several arrows and dependencies I'd missed.

I hope my understading of the complex interplayings going on in our Universe has managed to facilitate your own understading.

If not, well... oopsie.

See you, space cowboys...

Giacomo

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